Alliance FTP Reference

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Chapter 1: Introduction

Alliance FTP Manager is an IBM i application that can help you automate the transfer of files, documents, and spool files between your IBM i and other systems. The other systems can be any other computer system that supports FTP (File Transfer Protocol) including Windows, Macintosh, UNIX (all versions), Novell servers, DEC VAX, Tandem, and even other IBM is. With FTP Manager you can send or receive a single file or multiple files in one session. You can even start an application on the remote system before or after the transfer to process your files.

Alliance FTP Manager lets you define and store scripts for FTP sessions on the IBM i. FTP scripts are sets of FTP commands you want to run in one FTP session with another system. The scripts let you indicate substitution parameters which makes them very flexible. One script can be used for many file transfer requirements. In addition to FTP scripts you can define and store many other aspects of an FTP session including EBCDIC to ASCII conversion requirements, user names and passwords, programs to run on the IBM i before and after a session, and logging options.

To help automate file transfer tasks Alliance FTP Manager provides both library scan and directory scan facilities. It can detect a file in a library or directory and automatically send it to a remote system. If a file has been received from a remote system it can automatically call your program to process the file and back up the file when complete. If you need to schedule file transfer activities you can define one or more schedules for an FTP session. Alliance uses the native IBM i job scheduling facility for scheduling tasks. With the library scan and directory scan facilities and job scheduling you can create a completely automated file transfer facility. Alliance FTP Manager also supports secure (SSL and ssh) transfers.

For spool file distribution you can use the output queue management capabilities to copy spool files to an IFS or shared document directory as text or HTML web files. You can also automatically send these files to a remote system using FTP. You can interface any number of output queues to the output management facility.

Alliance keeps two levels of history of FTP sessions. If enabled it will keep a detailed log of the FTP commands used for each session and the actual results of the commands. In addition it keeps a history file of all Alliance sessions. The history file retains important job status information including the job name, user, job number, ending status of the FTP session, and log information. It gives you a permanent record of the status of your FTP Manager sessions.

FTP Manager is part of the Townsend FTP Bundle, which gives you four other powerful tools to help you with your FTP tasks. The PGP Option is a powerful tool to help you protect your data. PGP encryption uses public/private key encryption technology. This is recognized to be the strongest encryption possible. The Alliance PGP Option includes all of the key management functions you need to create, verify, sign, distribute, and import keys from your trading partners. There are many automation components that make it easy to encrypt or decrypt data in unattended mode.

Alliance FTP Security helps you secure external access to your IBM i for FTP sessions. Alliance CrossData/400 helps you convert IBM i data into formats that can be used by remote systems. CrossData/400 can create CSV files for Microsoft Excel applications, or tab-delimited files for UNIX or DEC systems, or fixed length ASCII files for many external database systems.

The Alliance Application Monitor application can monitor the health of your IBM i and applications to report potential problems. When used together all four products provide an industrial strength FTP environment for your IBM i.

Alliance CrossData/400 and Alliance FTP Security are only available with the Alliance FTP Manager bundle. These products are not available on a stand-alone basis.

Chapter 2: Application Concepts

FTP (File Transfer Protocol)

FTP is an application that uses the TCP/IP Internet protocol to transfer files, spool files, and documents to and from remote systems. All modern computer systems include support for FTP with their communications facilities. The IBM i provides support for FTP as a part of the base IBM i operating system license. With FTP you can send a file on your IBM i to a remote system (including another IBM i), and pull a file from that system to your IBM i. In addition to file transfer, it includes support for ASCII/EBCDIC conversion, security, and remote execution of commands. Alliance FTP Manager uses the native FTP support of the IBM i for its automation functions.

FTP Manager architecture

FTP Manager is a set of interactive, batch, and API programs that let you easily automate FTP tasks. The interactive panels let you specify the characteristics of FTP sessions, run the sessions, review the results, and do problem analysis. The library scan facility is a batch subsystem that will automatically initiate the transfer of files when you copy or write a file to the library. The directory scan facility is a batch subsystem that will automatically initiate the transfer of files when you copy or write a file into an IFS directory. The output queue scan facility performs the same type of automated transfer of reports from output queues on the IBM i. The APIs (Application Program Interfaces) let you easily run FTP sessions from your CL or RPG programs. You can also schedule FTP sessions using the native IBM i job scheduling facility. With these interfaces you can deploy a complete solution for your FTP needs.

Spool file distribution

You can use FTP Manager to automatically distribute your IBM i reports. FTP Manager can copy reports to a shared folder in the QDLS directory or to an IFS directory in ASCII text or web HTML format. If you copy the files in HTML format you can specify the HTML header and trailer source for the resulting spool file. This lets you add your own graphics and links to the HTML documents produced by FTP Manager. If the target IFS directory is a part of your IBM i web server directory the files can be available to your web browser users.

FTP Manager high level APIs

The Application Program Interfaces provide you with an easy means of integrating FTP activity into your business applications. Your batch or interactive jobs can start an FTP session and receive the results of the session. The APIs provide a flexible means of adding FTP

automation to your existing business systems. All APIs are implemented as simple calls that can be made from CL, RPG, or Cobol programs. Sample code is provided in the application library.

ASCII / EBCDIC translation

FTP Manager can convert your EBCDIC data to ASCII for transferring files to an ASCII system, or convert ASCII to EBCDIC when you receive files from an ASCII system. Or you can specify no translation if you are communicating with another EBCDIC IBM system like the IBM i or IBM mainframe. You can use the IBM translation tables or define your own. IBM provides the QASCII table for EBCDIC to ASCII translation. This is the default for FTP Manager. IBM also provides a number of other translation tables for specific code pages. And you can define your own with the IBM i Create Table (**`CRTTBL`**) command. An example and template can be found in the `QTBLSRC file in the product library.

FTP Manager history records

Alliance FTP Manager keeps track of every FTP session you initiate through one of its interfaces. The records are stored in a permanent history file showing the session name, user, job, job number, date, time, and other information about the session. You can also directly link to the detailed log to see the commands that were executed. A maintenance program will purge these records as needed. The history records provide a permanent audit trail of your FTP activity.

FTP Manager application logs

Alliance FTP Manager incorporates a detailed application log that can help you analyze problems that may arise from your sessions. Logs are accessed through menu options or viewable in the **`ANLOGA`** file. You can use the interactive option to view or print the log, or you can create your own queries on the file. If you encounter problems with FTP Manager your software supplier may ask you to print the log and send the results for problem determination. Your programmers may find the log helpful in analyzing attempts to convert data that are not successful.

Library scanning

To help you with your data distribution needs FTP Manager includes a library scanning application that will automatically transfer files for you and put them in a remote location you specify. You can set up separate FTP session criteria for each file in the library. You can also process any files you've received into the library. After converting the file FTP Manager will delete or move the file to a library you specify. Additionally, FTP Manager will optionally execute a processing program you specify. You can use this program to process the file into your database. All actions taken by the library scan function are recorded in the history file. Please read Appendix B for instructions on how to set up and use the Library Scan functions. The library scan function supports SSL and SSH connections as well.

Scanning remote FTP servers

Alliance can scan a remote FTP server and automatically retrieve any new or changed files on the FTP server. You can use this auto scan function to retrieve all, new, or new & changed files. And you can use this if a secure session (SSL) is required. You can define the frequency with which the FTP server is scanned and which files to retrieve. You can specify the files get brought back into a library or the IFS directory. Please read Appendix E for instructions on how to set up and use the Auto Scan function.

IFS directory scanning

Many applications such as Lotus Notes, IBM web servers, and others use the IFS directory to store information. FTP Manager can automatically scan IFS directories and send files to a remote server or copy the data to an IBM i database file. ASCII to EBCDIC conversion, pre-processing, and post-processing options are included. Please read Appendix D for instructions on how to set up and use the Directory Scan functions. Directory scan function supports SSL and SSH connections as well.

FTP Manager error notification

FTP Manager incorporates a notification mechanism that will alert you to problems in FTP sessions. You can specify up to 3 message queues to receive notifications. This can include the operator's queue **`QSYSOPR`** or any three queues you want to specify. You can also provide up to 3 email addresses for notification via the SNADS/OfficeVision interface, and 3 SMTP email addresses for normal Internet and LAN delivery. The notification can be sent to OfficeVision/400 or JustMail/400, a Lotus or Exchange recipient, or an Internet SMTP recipient. Lastly, you can specify one data queue to receive error notifications so that you can implement automated error recovery procedures.

SNMP support

FTP Manager contains a complete SNMP sub-agent implementation and can send SNMP trap error messages to HP OpenView or other network management application that supports SNMP. You configure SNMP support on the IBM i and in FTP Manager and error messages will automatically be forwarded.

MQSeries support

FTP Manager supports sending and receiving data using one or more MQSeries queues. You can use Alliance FTP Manager as a service for any other platform that supports MQSeries including S/390 mainframes, all versions of UNIX, Windows NT, Web servers and applications, and wireless devices. You can configure the library scan function to send a file from the IBM i to a remote system using an MQSeries queue. You can also configure a job to receive data from an MQSeries queue and write the data to a library. Note that with MQSeries support you can use the IBM i as a server for any other platform for FTP automation.

Email notification support

FTP Manager contains a full SMTP email client to forward FTP confirmation messages to Internet addresses, pagers, and PCS devices. You can configure FTP Manager to notify you if a transfer is successful, if a transfer fails, or both. Up to three email addresses can be added to an FTP definition. You can also use FTP Manager to notify your customers, vendors or trading partners when a file transfer is complete.

File compression

FTP Manager includes the ability to compress files before transferring them to a remote system. Two types of compression are supported: Gzip for UNIX platforms, and Zip for PC platforms. You can select which types of compression you want to use when you define the FTP definition. Although Gzip compression is support by many PC Zip applications such as WinZip, the Zip implementation is more commonly used on PCs. Zip compression produces files with the standard `.zip` file name extension. The Alliance API program Unzip can extract files from a Zip archive for use on the IBM i.

PGP Encryption

FTP Manager includes the ability to encrypt files using the PGP (Pretty Good Privacy) encryption program. Files encrypted with PGP can be transmitted to other users in a secure fashion. The recipient will only be able to decrypt the data if they have the appropriate secret key. Similarly you can decrypt files that others send to you using your secret key. Before using PGP on the IBM i see the section below on PGP. It contains information on how to install the necessary directory support for PGP. PGP is an optional component of Alliance FTP Manager and is based on the commercial version of PGP from PGP Corporation.

EDI and XML Translation

Alliance FTP Manager interfaces with Alliance EDI Workbench/400 for ANSI X12 translation services. You can automatically generate EDI documents, or translate them to IBM i database files. For XML translation you can use the Alliance XML Workbench/400 product. Alliance XML Workbench/400 translates XML documents to IBM i database files, and converts IBM i database files to XML format. Please contact your software provider for information about these products.

ISO 8583 translation with IBM Advantis

Alliance FTP Manager contains Application Program Interfaces to translate ISO 8583 formatted messages to and from IBM i database files. Typically ISO 8583 files contain financial authorization messages. Alliance provides standardized input and output database files and conversion programs. See the sample code in the QCLSRC file for more information.

System operations

As you expand the communications connectivity with your trading partners, FTP Manager will become a mission critical application. Although the application is designed to function without operator intervention, the system provides operators with functions to assist them in using the

system and determining the cause and cure for problems. The menu AMOPER is provided for system operators and gives them access to the most common functions they may need to use.

GXS communications

Alliance FTP Manager can communicate with GXS (GE eXchange Services) over the Internet using secure SSL FTP. This means you can send and receive your EDI documents in a safe manner using a normal Internet connection. Alliance FTP Manager secures the connection using 128-bit encryption in the open systems SSL (Secure Sockets Layer) environment. This is the same security used in your web browser to protect sensitive data. In order to communicate with GXS you must have a GXS account and password. Contact GXS to set up your account before attempting to use Alliance for communications.

IBM Global Network (Advantis) communications

Alliance FTP Manager can communicate with IBM Global Network (Advantis, or Information Exchange) over the Internet using secure SSL FTP. This means you can send and receive your EDI documents in a safe manner using a normal Internet connection. Alliance FTP Manager secures the connection using 128-bit encryption in the open systems SSL (Secure Sockets Layer) environment. This is the same security used in your web browser to protect sensitive data. In order to communicate with IBM you must have an IBM account, user ID, and password. Contact IBM to set up your account before attempting to use Alliance for communications.

Limitations of FTP Manager

FTP Manager assumes and requires an FTP server on the remote system. It is not enabled for DBCS.

Chapter 3: Alliance FTP Manager Audit Trails and History

Audit trails

Alliance keeps a history file of all data conversion requests. You can view a list of conversions that have been performed on the IBM i. The audit trail is accumulated in a history file named **`ANHIST`** that you can query with IBM's query product or many third party query products.

The history file gives you a way of periodically reviewing use of FTP Manager on your system. You can identify potentially inappropriate uses of FTP Manager or attempts to gain unwelcome access to your IBM i database. You can also use the history file to analyze application usage and plan for growth.

Application logs

To support problem determination and support implementation of Alliance services, you can activate internal logging of the Alliance programs. When logging is active detailed information is written to the log file **ANLOGA** . You can view this log with the option to Display Application Log on the inquiry menu, or using any file display or query tool. The application log displays all input parameters to the Alliance control programs and all relevant Alliance options you have configured. If an FTP request fails the reason for the failure will be in the application log.

If you report a problem with Alliance, the support team will want the application log. You can print the results to a spool file and attach it to a support ticket for review.

Chapter 4: Before You Begin

IBM i release level

Before you begin you must verify that your IBM i is at the proper release level. You must be on release V5r4 or later. You should also be sure that you have applied the latest cumulative PTF tape from IBM.

Resources

The following publications may be helpful in understanding the concepts of cryptography and, specifically, Pretty Good Privacy (PGP) encryption.

- “Cryptography for the Internet,”, by Philip R. Zimmerman. Scientific American, October 1998. This article, written by PGP’s creator, is a tutorial on various cryptographic protocols and algorithms, many of which happen to be used by PGP.
- “Privacy on the Line,” by Whitfield Diffie and Susan Eva Landau. MIT Press, paperback 0262-541009. This book is a discussion of the history and policy surrounding cryptography and communications security. It is an excellent read, even for beginners and non-technical people, and contains information that even a lot of experts don’t know.
- “The Codebreakers,” by David Kahn. Scribner; ISBN: 0684831309. This book is a history of the codes and code breakers from the time of the Egyptians to the end of WWII. Kahn first wrote it in the sixties, and published a revised edition in 1996. This book won’t teach you anything about how cryptography is accomplished, but it has been the inspiration of the whole modern generation of cryptographers.
- “Network Security: Private communications in a Public World,” by Charlie Kaufman, Radia Perlman, and Mike Spencer. Prentice Hall; ISBN: 0-13-061466-1. This is a good description of network security systems and protocols, including descriptions of what works, what doesn’t work, and why. Published in 1995, it doesn’t have many of the latest technological advances, but is still a good book. It also contains one of the most clear descriptions of how DES works of any book written.
- “Crypto,” by Steven Levy. Penguin; ISBN: 0140244328. A good non-technical introduction to the work of public cryptography. It gives a concise history from Duffy’s discovery of public key encryption to more recent attempts to limit public encryption.

- “Applied Cryptography: Protocols, Algorithms, and Source Code in C,” by Bruce Schneier, John Wiley & Sons; ISBN: 0-471-12845-7. This is a good beginning technical book on how a lot of cryptography works. If you want to become an expert, this is the place to start.

Chapter 5: Installation

Installing for the first time

Authorization Lists:

You must sign on as `QSECOFR` and create an Authorization List called `ALLFTMOWN` and a User Profile called `ALLFTMOWN` to secure the product using the following commands:

```
CRTUSRPRF USRPRF(ALLFTMOWN) PASSWORD(*NONE) USRCLS(*SECOFR) INLMNU(

CRTAUTL AUTL(ALLFTMOWN) TEXT('Alliance AUTL for Ownership of FTM Pr
```

Next, you can add Users to the Authorization List by using the following commands:

```
CRTUSRPRF USRPRF(somebody) PASSWORD() AUT(*EXCLUDE)

ADDAUTLE AUTL(ALLFTMOWN) USER(somebody) AUT(*ALL)
```

You can add group profiles to the authority lists instead of individual users as well.

Installing from an Internet download

When you download the Alliance product from the website, you must unzip the product. Your software provider will supply a pass phrase for extracting the files. See the Readme.txt file in the download for instructions on how to copy the software to the IBM i using FTP. Once you transfer the Save file to the IBM i you will restore the library `ALLFTM100` :

```
RSTLIB SAVLIB(ALLFTM100) DEV(*SAVF) SAVF(QGPL/ALLFTM) MBROPT(*ALL)
```

Press ***Enter*** and then follow the prompts. Observe that all objects are restored.

Installing from CD Rom

From the CD, use the Load and Run (`LODRUN`) command to install the software:

```
Lodrun dev(opt01) dir('/')
```

A menu will be displayed that allows you to select the Alliance FTP Manager product for installation.

Installing an upgrade

If you are installing Alliance as an upgrade please refer to Appendix B for full upgrade instructions.

Migrating Alliance PGP from one IBM i to another IBM i

If you are moving Alliance with the PGP Option from one IBM i to another and you want to move the PGP keys to the new IBM i, please observe these procedures. (There is an FAQ #261 with specific commands and more details available on our support site).

On the Source IBM i:

1. Save the library **`ALLFTM100`** to a save file or tape.
2. If you are using encrypted spooled file support, save library **`ALLFTM100A`** to a save file or tape.
3. Save the IFS directory to a save file or tape. You can use the **`SAV`** command to save the objects in the /PGP/* directory. This will save all of your keys.

On the Target IBM i:

1. Restore the library **`ALLFTM100`** from the source IBM i.
2. If you are using encrypted spooled file support, restore the library **`ALLFTM100A`** from the source IBM i.
3. Restore the IFS directory /PGP from the source IBM i. You can use the RST command to do the restore.
4. Enter the new temporary or permanent code you received from your software vendor.
5. From the Alliance configuration menu use option **`13`** to work with PGP users. Make a list of all PGP users. Use option **`4`** to delete the PGP users. Then use option **`13`** and function key ***F6*** to add the PGP users for the new system. Do this even if the user names are the same as the old system. Adding the user creates the Authorization List for the PGP files in the IFS directory.
6. From the Alliance PGP menu use option **`2`** to view a list of PGP keys. Verify that the PGP keys are correct.

Trial and activation codes

If you are installing an upgrade or new release your trial or activation code will be restored when you run the update procedure. If you are installing Alliance for the first time, you will need a trial code and a license file for PGP from your software vendor.

Alliance library security

The owner of the Alliance library **ALLFTM100** and all programs and objects in the library is **QPGMR**. The public (***PUBLIC**) has all rights to the library and objects in the library. You should secure the library using normal IBM i security commands. There is also an option in the configuration menu to work with user access. This will allow you to restrict users from menus and options.

Chapter 6: Planning Your Implementation

Configuring Alliance FTP Manager is easy, but you will have to make some decisions first about what features you want to use and how to implement them. You should decide the following items before starting:

Separate libraries for files

You may want to create separate libraries to hold files. This can be helpful in avoiding errors in creating files in production libraries. Alliance FTP Manager will create any library you specify as an output library if it does not exist, but it is probably better from an administrative point of view if you create these libraries before using FTP Manager.

Output queues for report distribution

FTP Manager can process spool file reports from any output queue on the IBM i. One output queue is provided with the product named **ALLFTMOUTQ**. But you can use any existing output queue or create new ones for use by FTP Manager. FTP Manager uses the data queue interface of an output queue to process spool file reports. When you register an output queue to FTP Manager it changes the output queue to specify the FTP Manager data queue named **ALLFTMSPLQ** in the product library.

Please read Appendix F for instruction on how to set up and use the Output Queue Scan function.

Using library scan

Alliance FTP Manager includes a library scan facility that can detect the presence of a file in a library and automatically start an FTP session. You should identify the libraries you want to scan and define these to FTP Manager before you begin. It is recommended that you create separate libraries for the scan functions in order to avoid the accidental processing of files in an existing application library.

Please read Appendix C for a description and step-by-step set up for Library Scans.

Understanding job scheduling

Alliance uses the native IBM i job scheduling commands for FTP scheduling requirements. If you are not familiar with IBM i job scheduling you should review the available documentation from IBM.

IBM i support for SNMP

The IBM i includes support from IBM for SNMP agent functions. Alliance uses the IBM provided APIs to implement an SNMP subagent. An IBM i SNMP subagent acts just like an SNMP agent process. Alliance can send trap error messages to a network manager that implements SNMP manager functions. Applications like HPOpenView, IBM SystemView and others include this type of support. You must first configure and activate SNMP agent support on the IBM i before using the Alliance SNMP support. See the sections below on how to activate SNMP on the IBM i.

IBM i support for PGP

Alliance FTP Manager includes support for PGP encryption of files as a separate option. Before using PGP you must install the IFS directory support for PGP. See the section below on restoring the PGP directory structure and configuration files. We do ask that you complete the PGP Certification Guide when you start working with PGP.

MQSeries

If you are planning on using MQSeries to send and receive data you must first install MQSeries on your IBM i and configure it with the remote system. FTP Manager provides automated ways to send and receive data using MQSeries queues, but does not provide support for installing and configuring MQSeries on the IBM i. You should consult the MQSeries documentation for examples on how properly set up and start MQSeries support.

High availability considerations

Many Alliance customers using a high availability product such as Mimix, Vision, or Data Mirror. If you intend to mirror the FTP Manager product you should avoid mirroring the data areas that contain license keys. Check FAQ # 82 for complete information.

Some high availability products support journaling IFS directories and files. You should be aware that the files in the /PGP IFS directory are accessed, copied and renamed as a part of standard PGP encryption and decryption operations. If your HA product exclusively locks objects there can be conflicts with PGP encryption options. You should schedule the mirroring of objects in this directory when PGP functions are not being used.

Chapter 7: Platform-Specific Notes

Overview

The characteristics of FTP and file systems vary greatly from one platform to another. The following sections try to provide some information that will be helpful in using FTP with non-IBM i systems.

IBM MVS

For a remote IBM MVS system you can specify the name in normal DSName format. You must, however, enclose the target name within double and single quotes in order to remove the user ID from the first part of the path:

```
put mylib/myfile "'name1.name2.name3.name4.name5'"
```

If you do not enclose the MVS file name in quotes as above you may receive an error message indicating that the remote system cannot “store” or save the file. You will probably see the user ID pre-pended to the target file name. Using the quotes will eliminate this problem.

You may also need to specify the record length and block size of the transfer before you PUT a file to a mainframe. You can specify this with a command like the following:

```
site recfm=fb lrecl=100 blksi=1000
```

Your mainframe system administrator can give you the record size and block size values to use.

If you are transmitting a file from an IBM i database file to an MVS system you can use the following commands. Note the `SITE` command to set the record length, block size, and record type parameters for MVS.

```
EBCDIC
MODE B
CDUP
STR F
SITE LRECL=80 REC=F BLKSIZE=80
put ipasspgm/signfile.signfile   'EDI.APT.HKGP.BAFF(+1)'
quit
```

UNIX

UNIX file systems are stream oriented and file names are case sensitive. When you exchange data with a UNIX system be sure to provide directory information in exactly the format that it exists on the UNIX platform.

Most UNIX systems will have a directory called /tmp. It is designed to hold temporary files and is usually frequently cleared by the system administrator. But you can often use this directory for testing file exchanges with Alliance FTP Manager on the IBM i. A common put statement would look like this:

```
put mylib/myfile.mymember /tmp/myfile
```

Advantis Information Exchange

If you are using Advantis mailboxes for your EDI or non-EDI file transfer you should note that Advantis uses an internal file naming convention that is incompatible with the IBM i. However, you can enter the FTP site command to direct Advantis to switch to IBM i naming conventions:

```
site system 4
```

Logging on to the Advantis system also requires a special format for the user ID and password. You must enter the Account and User ID concatenated with a period for the FTP user ID followed by the password. For example:

```
account.userid password
```

If you are collecting files to receive in a specific class you can change to that class using the FTP CD command and treating it like a subdirectory. If you are receiving files to a class of ORDERS you would specify the CD command like this:

```
cd *.*/ORDERS
```

Be sure to obtain a copy of the Information Exchange FTP Quick Reference and Information Exchange FTP Reference manuals before attempting to connect to Advantis. These are available on the Information Exchange web site or by contacting your Advantis account representative.

Chapter 8: PGP Partner Considerations

Bank of America

If you will be exchanging files with Bank of America you will need to contact them to receive the procedures for receiving the Bank of America public encryption key, and to learn how to send them your RSA public key. Bank of America requires that all files you send them be encrypted, ASCII armored, and signed. The Alliance PGP Option supports these functions.

You will find sample configurations for Bank of America in the FTP Definitions, FTP scripts, and Library Scan definitions. In most cases you will be able to make minor changes to these sample definitions to start working with Bank of America.

Bank of America uses an FTP site with a special domain name. The instructions you receive from Bank of America will define this domain name. When you talk to Bank of America request the short document named “SENDING FILES TO Bank of America OVER THE INTERNET USING PGP”. This document is customized for each customer and will contain your user ID, password, and batch ID for the FTP transfer.

BankOne

If you will be exchanging files with BankOne you will need to contact them for the procedures for receiving the BankOne public encryption key, and to learn how to send them your public key. BankOne requires that all files you send them be encrypted. The Alliance PGP Option supports these functions. Note that BankOne may use the GNU Privacy Guard product to encrypt your files. Ask that they use the standard “.pgp” extension on encrypted files they send you.

Citibank

If you will be exchanging files with Citibank you will need to contact them for the procedures for receiving the Citibank public encryption key, and to learn how to send them your public key. Citibank requires that all files you send them be encrypted.

Definity

If you are exchanging health claims information with Definity Health you should contact them to receive the IP address of their server, their public key, and the user ID and password for the FTP server. They will send you their public key for any files that you need to encrypt for them. The Definity interface is very straight-forward and you should not have problems sending and receiving files.

Employease

If you will be exchanging files with Employease you will need to contact them for the procedures for receiving the Employease public encryption key, and to learn how to send them your public key. Employease requires that all Human Resources data that you send to them be encrypted. The Alliance PGP Option supports these functions.

Checkfree

Checkfree supports PGP encrypted transfers of eBill, remittance, and other transactions. You must register an external IP address for your IBM i and contact Checkfree to register these IP addresses. Once IP addresses have been registered you must create your PGP public key and send it to Checkfree. They will establish a directory for you to receive files.

Note that file names may contain the company or division number if you receive multiple files. Alliance FTP Manager can be configured to pull files from Checkfree on a periodic basis. Encrypted files are pulled to the IFS directory. IFS directory scan decrypts the files and transfers them to a library for processing. You can set up a user program to be called to automatically process the files if you wish.

Checkfree provides an optional web interface that can be accessed from the IBM i. The URL for the web interface requires a Triple DES encrypted PIN and User string in ASCII hex format. The DESCHK API is provided with FTP Manager to create the URL field you need. The DESCHKD API is provided for decrypting Checkfree data. Please consult the section on Application Program Interfaces for a description of this API.

Wells Fargo Bank

Wells Fargo Bank uses an older version of PGP, but you can still exchange encrypted files with them. You should be aware that you cannot encrypt a file for multiple recipients or use other special features of PGP. Wells Fargo may also require that you exchange files with them using FTP with SSL or SSH. Please contact support@townsendsecurity.com to work on this set up with you.

ZirMed

If you are sending medical insurance information to ZirMed you will need to contact their network support team to get a PGP public key. You will also need to tell ZirMed the IP address you will use to connect to their FTP server. This must be the external IP address from your Internet connection. They will enable this address to connect to their FTP server. ZirMed accepts PGP encrypted files to their web server at www.zirmed.com. You will find an example script and FTP definition in the Alliance FTP Manager configurations.

Note that ZirMed normally does not send files back using FTP. They have a web interface that is used to retrieve status information about the files you have transferred. If you need to receive files using FTP please contact their network support team to discuss.

The sample FTP script for ZirMed in Alliance FTP Manager will need to be modified to incorporate a change of directory. ZirMed will assign you a directory name for claims or uploads. You will need to add a CD statement to your script so that you upload to the correct directory.

VSP

If you are sending medical insurance information to VSP you will need to contact their network support team to get a PGP public key. VSP accepts PGP encrypted files to their FTP server. The FTP server is located at ftp.vsp.com.

Prudential

If you are exchanging benefits information with Prudential you will need to export your IBM i public key and send it to their network operations group. They will provide you with a user ID and password for their FTP server. You should be aware that the Prudential PGP implementation might remove the .pgp extension from the file name you receive. If you encounter this situation, you will need to rename the file as a part of the FTP GET operation to re-establish the .pgp file name extension.

Vanguard

If you are exchanging benefits information with Vanguard you should contact them to receive the IP address of their server, their public key, and the user ID and password for the FTP server. They will send you their public key for any files that you need to encrypt for them. The Vanguard interface is very straightforward and you should not have problems sending and receiving files.

TransUnion

If you are exchanging files with the TransUnion FTP server, you will need to use a non-standard FTP port and special User and Password script specification. In your script specify the following commands:

```
Open ftp.tuc.com 10021
User user-name pass-word
```

Note that you have to actually specify the “user” FTP subcommand to get the FTP server to accept your log in. TransUnion uses the Sterling CONNECT:Enterprise server which is used by several other financial institutions. However, this is the only interface that requires the special format.

Fidelity

If you are exchanging benefits information with Fidelity you should contact them to receive the IP address of their server, their public key, and the user ID and password for the FTP server. They will send you their public key for any files that you need to encrypt for them. The Fidelity interface is very straightforward and you should not have problems sending and receiving files.

Paymentech

If you are using the Alliance FTP bindable client program to send information to Paymentech you should be aware that Paymentech can abnormally terminate a transfer if the data in your file is not valid. An example of invalid data would be a sequence number out of order or missing. Be sure to monitor the return code (not FTP code) from the **`FTMRetr`** or **`FTMPut`** API to be sure there are no errors in the transfer. If you encounter errors you will need to contact Paymentech customer support for information on why the transfer failed.

GXS (GE eXchange Services)

You can use the SSL support in Alliance FTP Manager to communicate with the GXS network. You must first obtain a GXS account and password before attempting to use the Alliance communications support. Alliance implements automated transfer with GXS using the Library Scan facility to send files, and the Auto-scan facility to receive files. You can also integrate file transfers using the GXS API programs **`FTMGXSSND`** and **`FTMGXSRCV`** .

IBM Global Network (Advantis, Information Exchange)

You can use the SSL support in Alliance FTP Manager to communicate with the IBM Global Network. You must first obtain an IBM account, user ID and password before attempting to use the Alliance communications support. Be sure to request IEFTP support when setting up your account and user ID. Alliance implements automated transfer with IBM using the Library Scan facility to send files, and the Auto-scan facility to receive files. You can also integrate file transfers using the IBM API programs **`FTMIBMSND`** and **`FTMIBMRCV`** .

JP Morgan Chase

JP Morgan Chase requires files be sent to them encrypted with PGP & requires that you exchange files with them using FTP with SSL or SSH. Please contact support@townsendsecurity.com to work on this set up with you.

PGP compatibility considerations

The Alliance PGP support is based on the PGP Corporation commercial version of PGP. Any other PGP or eBusiness Server product from PGP Corporation for OS/390, UNIX (several varieties, Macintosh, VAX, etc. are fully compatible with the Alliance version. Note that most banks and insurance companies use the PGP Corporation product and its extended features.

Some versions of PGP do not support all of the features of the PGP Corporation product. OpenPGP and GNU Privacy Guard are such products. You may be able to exchange encrypted files with these products if certain extended features are not used. Conventionally encrypted files and files encrypted with a single public/private key pair are normally compatible. Certain encryption options used by PGP Corporation's PGP cannot be processed by OpenPGP and GNU Privacy Guard (GPG).

Megacryption creates a S/390 mainframe product based on the GNU Privacy Guard application. This product does not support compression. If you are sending encrypted files to a trading partner using Megacryption you must add the following statement to your PGP configuration:

```
Compress = off
```

Chapter 9: Firewalls and Proxy Servers

Overview

If your Enterprise uses a firewall and/or proxy server you may have to make some configuration changes to accommodate the connection through these units. Please refer to the following notes on known configuration requirements. If you encounter a problem that is not discussed in this section please contact our technical support department. We can often help resolve these issues in a timely fashion.

Raptor (Axent) firewall

The Raptor firewall can be configured to require a gateway user id and password to the firewall system before allowing you to connect through it to a remote site. Raptor firewall will prompt for a user id, password and account. At the login user ID prompt you must send the remote user name, remote system name (or ip address) and firewall user name enclosed in quotes in the following format:

```
"remote-user@remote-system firewall-user"
```

After supplying this user id Raptor will prompt for the password. You must supply the password for the remote FTP server. Do not use quotes around the password. After entering the password Raptor will prompt for the Account. You must enter the password for the Raptor firewall gateway user name. Again, do not use quotes around the gateway password.

You can use the following lines in the Alliance script to negotiate through the firewall.

```
open remote-system
"remote-user@remote-system firewall-user" remote-password  gwpasswo
```

Be sure to put quotes around the login user ID as in the example above. You can get more information about the Raptor firewall and using FTP at the Axent web site. Their link is:

```
        [http://www.axent.com](http://www.axent.com/)
```

On their web site follow the links to the Raptor firewall technical support section and the FAQs. There is an FAQ on using FTP. At the time this manual was written this was the link to FAQ on FTP:

```
        [http://www.raptor.com/cs/FAQ/ftpfaq.htm](http://www.raptor
```

Wingate proxy server

The WinGate proxy server requires a passive mode connection from the IBM i. IBM added support for passive mode connections in V4R2. You will not be able connect via the WinGate proxy server with any IBM i FTP client prior to V4R2.

IBM MVS FTP servers

In IBM i V4R2 IBM added support for passive mode FTP transfers as the default. Prior to this the default was PORT mode. However, certain versions of MVS FTP did not support passive mode connections. The symptom is that you can connect to the FTP server, enter a user and password, and issue an FTP PUT command, but you will receive an error after a time-out period of two or three minutes. The MVS system creates the file but it is empty. There is a patch available for the MVS system to correct this problem. There is also a work-around on the IBM i side to change the default behavior of the FTP client to not enter passive mode. You can get more information about the work-around on the IBM i APAR web site:

```
[http://as400service.rochester.ibm.com](http://as400server.
```

Search on “FTP”.

Socks servers

Although the IBM i FTP application does not implement Socks support directly, you can sometimes configure the FTP application to use a Socks server on your network. The following suggestions are a possible approach. This approach may not work for you.

You must have Client Access installed on your PC with Operations Navigator installed. Start Operations Navigator and select the Socks tab. Define the network or IP address you want to go to and the IP address of the firewall. After this configuration you must configure your firewall. Select Socks and define a daemon. The daemon configuration will prompt for something like: permit, from: as400-ip-address to: external-ip-address command B + C. After the daemon is started you should be able to use the IBM i FTP application.

Chapter 10: Getting Started

Overview

The following sections provide a step-by-step procedure for setting up Alliance FTP Manager. Some of the steps may not be necessary depending on your objectives outlined above.

Creating libraries for files

If you want to create special libraries for files you should create them now. You can use the following command as an example:

```
crtlib lib(transmit) text('FTP transmit library')
```

If you have several different types of FTP requirements or applications and you want to keep the files separate, create other libraries to hold files for each group.

Defining scan libraries

If you are going to activate automatic library scan and data file processing, you will probably want to create the scan libraries before you start. You can use the following command as an example:

```
crtlib lib(receive) text('Scan received files')
```

You must use the FTP Manager menu option to Work With Scan Libraries to define a library and the files to be processed. After you add the library you must add a list of files to scan for. When you add a file to the scan definition you tell FTP Manager how to process the file. There are other options to define for the FTP session, too. You should identify which files will be automatically processed and define them now. Please read Appendix C for instructions on how to set up and use the Library Scan functions.

Defining scan directories

If you are going to scan IFS or QDLS directories for automatic processing you should create them now. It is recommended that you not use the /tmp directory as some applications place temporary files in this directory. Be sure to specify adequate public authority on the directory so that FTP Manager can process the files. There is an FAQ explaining authority in the IFS directory. You can find it by doing search for IFS Directory Authorization and Ownership.

Please note that the /QDLS file system has special requirements. Before you can access the QDLS file system a user must be added to the system distribution directory using the Work With Directory Entries (**`WRKDIRE`**) command. The QDLS file system also has limitations on the length of file names. It is recommended that you use the IFS file system rather than the QDLS file system.

If you will be using the directory scan function you should be familiar with authority management on IFS files. The IBM i treats these files differently than library files. For example, programs cannot adopt authority to process files in the IFS file system. Please see the section on IFS File Authority before implementing directory scan and read Appendix D for instructions on how to set up and use the Directory Scan function.

Configuring FTP Manager

After defining all of your data conversion needs for the system, you should configure global options for Alliance FTP Manager. Use the Configure FTP Manager option on the main menu.

Installing PGP support

If you are going to use PGP (Pretty Good Privacy) support on the IBM i you must first install the IFS directory structure and configuration file and get PGP licensed. See the section below on PGP before using PGP support on the IBM i.

Checklist

The following checklist is provided to assist in the configuration and startup process. Some steps may not apply to you, so just skip those steps. See the sections above for more information on the following steps.

STEP	COMPLETE	DESCRIPTION
1		Review IBM i system requirements before starting. Page 26.
2		Are you installing an upgrade? If yes, rename the current **`ALLFTM100`** library to a new name. Refer to Appendix B for instructions for upgrade.
3		Restore library **`ALLFTM100`** from product media.
4		Are you installing an upgrade? Have you read Appendix B regarding an upgrade? If yes, run the Apply Release Upgrade option on the Installation menu now. This will bring over your licensing information and all of your configurations. And you are done with the upgrade.
5		If this is a new installation, enter your temporary or permanent activation code from your software vendor using the appropriate menu option. To display the menu: **`addlible ALLFTM100`** [Enter], **`go ammain`** [Enter], select option **`1`** . At this point you should be scheduled for a walk through with a Support Team member. They will work with you on the following steps for setting up scan jobs.

STEP	COMPLETE	DESCRIPTION
6		Identify or create the libraries you will use for FTP sessions. It's important to understand how the scan job logic works with files in the library so be sure to read Appendix C for guidance on setting up a scan job. You may want to do that before creating libraries for scanning.
7		Identify or create the directories you will use for FTP sessions. It's important to understand how the scan job logic works with files in a directory path so be sure to read Appendix D for guidance on setting up a scan job. You may want to do that before creating directories for scanning.
8		Configure Alliance FTP Manager global options. Use the option to Configure FTP Manager on the Alliance configuration menu.
9		To review history logs on the IBM i, use the Inquiry option from the main menu. If there are no errors the installation is complete. If there are errors please provide a job log from the upgrade to the Support team.
10		Are you going to use Alliance SNMP support? If yes, configure IBM i SNMP support and start the SNMP server jobs. See the section below on how to configure and start SNMP. You should also be sure that you have the latest security PTFs installed for TCP utilities.
11		Are you going to use the PGP Option? If yes, sign on as `QSECOFR` and install the PGP IFS directory and the supporting files. From the `AMMAIN` menu take option `1` for installation, then option `7` to install PGP files.
12		If you are using the spool file distribution facility, and you want to send reports as PDF attachments, you must install the PDF configuration and font files. From the installation menu take the option to install PDF files.

STEP	COMPLETE	DESCRIPTION
13		Are you going to use Alliance Email notification support? If yes, configure the Email application before using FTP Manager. You will need to know the mail server IP address or domain name. See the section below on how to configure SMTP mail.
14		Are you going to use the Alliance Biometric authentication control? If yes, use the option to install biometric support on the installation menu. WARNING: You must first configure biometric support and install administrative software on the security administrator's PC before you can use Alliance biometric support.

Automatically starting the Alliance subsystem

You can automatically start the Alliance FTP Manager by changing your IPL start up program **QSTRUP** . If you have not already modified the start up program you should begin by retrieving the source for the program **QSTRUP** from library **QSYS** . You can use the Retrieve CL Source (**RTVCLSRC**) command to do this:

```
RTVCLSRC PGM(QSYS/QSTRUP) SRCFILE(QGPL/QCLSRC) SRCMBR(*PGM)
```

Once you have the source restored you can use SEU or PDM to edit the source. Find the section of code in the CL program that starts the various IBM i subsystems. Insert the following lines into the program:

```
QSYS/STRSBS SBSD(ALLFTM100/ALLFTM100)
MONMSG MSGID(CPF0000)
```

Use the Create CL Program (**CRTCLPGM**) command to compile the source for the **QSTRUP** program. You can save the program in library **QGPL** or one of your user libraries. After you have created the modified program you must change the system value **QSTRUPPGM** to point to the new version of the program:

```
CHGSYSVAL SYSVAL(QSTRUPPGM) VALUE('QSTRUP QGPL')
```

The next time you IPL your system it will use this modified version of the start up program which will start the Alliance subsystem.

Chapter 11: Initial PGP Configuration

If you are not going to use the PGP Option for Alliance FTP Manager you can skip this section. If you plan to use the PGP encryption software, please take the steps below to install PGP files and add PGP users, then work with the Townsend support group to complete certification. You will be using these steps to create your private key and internal files needed by Alliance PGP support. You must complete these steps before attempting to use PGP.

PGP installation

Before you begin you must install the files used by the PGP encryption option. From the main menu take the option for Installation. Then select the option for installing PGP files. You must be `QSECOFR` or equivalent to perform this operation.

Adding PGP users

Before you can use the PGP options you must define PGP users to Alliance. From the main menu take the option for Configuration, then the option to Work With PGP Users. Add each user to the system. You must be `QSECOFR` or equivalent to perform this function. Note that the Alliance jobs that run in subsystem `ALLFTM100` use the `QPGMR` user profile by default. Be sure to add this profile to the list of PGP users if you will be using PGP with any scan job or other automation options.

PGP Certification

The Alliance PGP certification procedures are documented in a manual entitled **"PGP Certification Procedures."** After your walkthrough with a member of the support group, you can use this guide to create your keys, certify your installation, and view sample code on how to use Alliance APIs.

You should complete the certification procedures before you proceed further.

If you have completed certification please read Appendix G for direction for future key exchanges with trading partners. It will guide you through exporting your key and sending it to your trading partner, adding their key to your key ring and signing it, and any cautions you should be aware of.

> ***NOTE:*** Do not use options `7` (to check keys), `8` (to change keys), and `20` (edit PGP configuration) unless you first speak to someone in the support group about what these options do.

Chapter 12: Configuring IBM i SNMP

If you have not already configured SNMP on your IBM i and you want to have Alliance send error messages to an SNMP manager, you can use the following steps to configure and start SNMP support on the IBM i.

Please note that if you do not need SNMP support on your IBM i IBM recommends that you not start the SNMP support. You should review all security recommendations from IBM before implementing any TCP/IP service.

Before you begin you will need to know the IP address of the network manager and the community name to use. Consult with your network administrator for the IP address. Consult with your systems administrator for the community name to use. The community name is usually “public”, but may be assigned a different name.

Changing SNMP attributes

To configure SNMP use the Configure TCP/IP (**CFGTCP**) command. From the TCP/IP menu select the option to Configure TCP/IP Applications. Then select the option to Configure SNMP Agent and then select the option to Change SNMP Attributes. Enter the following information:

System Contact

Enter the name of the IBM i system administrator. You may also wish to add a phone number or operations support center contact number.

System Location

Enter the physical location of the IBM i system. You may want to provide short directions to the location of the IBM i.

Send authentication traps

Enter ***YES** if you want to send authentication errors to the network manager.

Automatic start

Enter ***YES** to have the SNMP agent support automatically started when the IBM i UPLs.

Object access

Enter ***READ** for the object access. For trap messages this is the only access that is needed.

Log set requests

Enter ***NO** for set request logging. If you are using other SNMP functions you may want to set this to *YES.

Log get requests

Enter **`*NO`** for get request logging. If you are using other SNMP functions you may want to set this to **`*YES`** .

Log traps

Enter **`*NO`** for trap logging. If you are having problems with SNMP trap messages you may want to set this to **`*YES`** .

Trap managers:

Manager internet address

Enter the IP address of the network manager. You can define more than one manager. For each manager enter the manager IP address and community name.

Community name

Enter the community name used by the network manager. The default is “public” and most network managers will work with this community name. Consult your systems administrator for the correct community name to use.

Translate community name

Enter **`*YES`** to translate the community name to ASCII. This is a requirement for any manager based on an ASCII platform.

Starting IBM i SNMP support

You must start IBM i SNMP support before Alliance can send trap messages to the network manager. Use the Start TCP/IP Server (**`STRTCPSVR`**) command to do this:

```
Strtcpsvr server(*snmp)
```

Chapter 13: Configuring SMTP Email

Alliance contains its own SMTP client for sending error messages and confirmation messages by email. You do not need to change any IBM i configurations to use the email support in Alliance. You only need to know the domain name or IP address of your corporate mail server. You system administrator or LAN administrator can give you this information. After you have the mail server information use the Alliance FTP Manager configuration options for email to define the mail server.

Chapter 14: IFS File Authority

For most FTP functions it is not necessary to use the IFS file system. You can send files directly from an IBM i library, and you can receive files to an IBM i library. However, you may find it necessary to move files through the IFS file system, for example: if you are going to use PGP decryption and need to receive PGP files, they must be received to an IFS directory for processing. IFS directory scan can help you automate decryption and processing of these files. Please see the remainder of this section for information on managing IFS file authority.

If you are going to use the Alliance IFS directory scan function you should be aware of IFS file authority and how it works. There are significant differences between library file authority and IFS file authority. For example, programs cannot adopt authority to access files in the IFS file system. You will need to plan carefully how you deploy FTP services and directory scan processes before you start to avoid file access problems.

Please note that there are subtle but significant differences in how different versions of IBM i manage authorities on IFS files. This affects how you plan and deploy Alliance directory scan.

Accessing the IFS file system

The Work With Links (**`WRKLNK`**) command is used to access the IFS file system. At a command line you can type **`WRKLNK`** and press ***Enter***. You will see a list of directories and files in the IFS file system. Your user profile may be set to a home directory of /home/USER (USER is your profile name). If this is the case you will only see directories below the /home/USER directory. You can change your current directory to the root of the IFS file system like this:

```
chgcurdir dir('/')
```

The **`WRKLNK`** command is used to view a list of directories, view sub-directories, and view and edit stream files. You will also use this command to change the ownership of directories and files, and to grant authority to files.

IFS directories

IFS directories are created with the Create Directory (**`CRTDIR`**) command. When you create a directory you assign data and object authorities. Data authorities are defined in a way that is similar to UNIX. That is, you assign Read, Write, and Execute (**`*RWX`**) authorities to the directory. A directory also has an owner and a primary group. Like IBM i library files, the owner of the directory is the user profile that creates the directory.

You can use the **`WRKLNK`** command and option **`9`** to view the authorities for a directory or file. You will also find function keys that let you change the owner of a file, etc.

IFS files

IFS files normally exist in a directory of the IFS file system. You should create a directory for PGP files that you receive and assign an owner to the IFS directory. We recommend not using the PGP directory that was installed with the product but create your own that you can work in. You should grant access to the directory for the user profiles that will place files in this directory.

When you use FTP to get a file into the IFS directory, or when a remote FTP client puts a file into the IFS directory, the file is owned by the user profile that is active at that time. If the file is retrieved to the IBM i using the FTP client, the file will be owned by the user profile that starts the FTP session. If the file is pushed to the IBM i be a remote FTP client, the file will be owned by the log-ser of the FTP session. The file will have the public authority that is defined for the directory. That is, if the directory excludes public authority the file in the IFS directory will exclude public authority.

If the file has no public authority and the user who creates the file is not the same user that runs the directory scan function, the file cannot be processed by Alliance and PGP decryption functions will fail. If the PGP file has all public authority the PGP decryption and processing will succeed.

It is ***NOT*** recommended that you grant public authority to directories and PGP files. You risk loss of confidential information if you do this. Instead, you should create a user profile for use by FTP. You should give the Alliance directory scan user authority to the directory where your PGP files will reside.

Directory scan user

The Alliance directory scan job (**`SCAN_DIRS`**) automatically starts when the Alliance subsystem **`ALLFTM100`** starts if it is set to be active. The user for this job is defined on the job description **`ALLFTMDIRS`** in the product library. Alliance ships with the user **`QPGMR`** defined for this job description. If needed you can change this user to a user profile better suited to access the IFS directories. You can do that by going to the configuration menu and taking **`1`** for Configure FTP Manager. The users are defined on the 2nd panel.

Archive directories

When directory scan finishes decrypting and processing a file it can move the original encrypted file to an archive directory that you define. You must use the Create Directory (**`CRTDIR`**) command to create this directory. It can have any name you like. Many Alliance users create a directory named “Archive” within the original path. Note that you must be sure that the directory scan process has authority to the archive directory in order to move files to it.

Recommendations

The following steps show one way to implement IFS directories and directory scan processing. There may be other ways to create the directories that are secure. When you deploy FTP, PGP, and Alliance directory scan you assume responsibility for insuring that your data is secure.

- Create a user profile for FTP access. You may wish to disable interactive use of this profile.

- Change the Alliance directory scan job to use the new FTP user profile using the **`Configure FTP Management`** option on the configuration menu.
- Create a directory for PGP files. Exclude *PUBLIC access. Restrict access to only the FTP user profile. The owner of the directory should be the FTP user profile.
- Add the FTP user as a PGP user on the Alliance configuration menu.
- On the configuration menu use the option to **`Work With Scan Directories`** to define a scan process.
- Enable directory scan using the option to **`Configure FTP Management`** on the configuration menu.
- Start the Alliance subsystem.
- Test the file processing.

You can view the results of directory scan PGP processing using the Inquiry menu to view PGP sessions and Application Logs.

> ***NOTE:*** The Alliance directory scan user can be different than the FTP user. Just add the Alliance directory scan user to the authority matrix of the IFS directory. Be sure that the Alliance directory scan user has *RWX data authority, and *ALL object authority.

Chapter 15: On Line Help

Every panel in Alliance has on-line help available. Press the ***Help*** or ***F1*** key on your keyboard to access help. You will be able to review a description of the panel's function, each field's function and use, and a list of active function keys and what they do.

Chapter 16: Installation

Main Menu

The main menu for Alliance FTP Manager is **`AMMAIN`** . You can access this menu by placing the library **`ALLFTM100`** in your library list and using the GO command:

```
addlible ALLFTM100      [Enter]
go ammain                       [Enter]
```

The following menu is displayed. All of the Alliance programs are available from this menu and sub-menus.

```
AMMAIN                    Alliance All-Ways Secure
                                Main Menu
Select one of the following:

  1. Installation                    30. SNMP menu
  2. Configuration                   31. Email menu
  3. Maintenance
  4. FTP sessions                  Alliance applications:
  5. Inquiry                         40. Alliance CrossData/400
  6. Reports                         50. Alliance FTP Security
  7. Support                         60. Alliance Application Monitor
  8. FTP Manager commands            70. Alliance EDI Workbench/400
  9. Related commands                71. Alliance XML/400
 10. PGP commands
 11. Sample code                     80. Alliance operations menu
                                     90. Sign off
 20. AES encryption menu
                                (C) PA Townsend & Associates, Inc., 1998, 2008
Selection or Command            Version 4.84
===>

F3=Exit    F4=Prompt    F9=Retrieve    F12=Cancel
F13=Information Assistant  F16=AS/400 main menu
```

Select an option to begin working with Alliance FTP Manager. If this is a first install of the product and you have done all the installation steps in option `1`, you should determine what process you want to configure first. There is documentation in the Appendix for all of the scan jobs. If you choose any of the following options, they must be installed and license keys installed in order to configure them.

`40. Alliance CrossData/400`

`50. Alliance FTP Security`

`60. Alliance Application Monitor`

`70. Alliance EDI Workbench/400`

`71. Alliance XML/400`

Installation

After you restore the FTP Manager product you must perform certain configuration tasks. Display the Installation menu for a list of all installation options. From this menu you can enter a trial or activation code, and upgrade from a previous release.

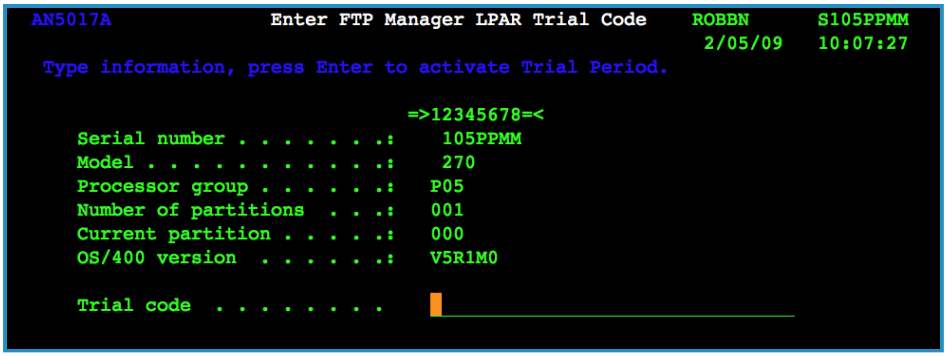
```
ANMNU1                    Alliance All-Ways Secure
                             Installation Menu
Select one of the following:

  1. Enter FTP Manager trial code          20. Apply a release upgrade
  2. Enter FTP Manager permanent code      21. View release notes
  3. Enter PGP option trial code           22. Grant user authority for DCM
  4. Enter PGP option permanent code
  5. Enter AES option trial code           30. Enter NGS-IQ trial code
  6. Enter AES option permanent code       31. Enter NGS-IQ permanent code
  7. Install PGP files
  8. Install PDF files                     90. Sign off
  9. Install AES files
 10. Install Biometric support
 11. Install SSH sFTP password support
 12. Install SSH utilities
 13. Install PGP CL 9 license
                                (C) PA Townsend & Associates, Inc., 1998
Selection or Command            Version 4.90
===>
```

Select an option to begin work.

Enter FTP Manager Trial Code

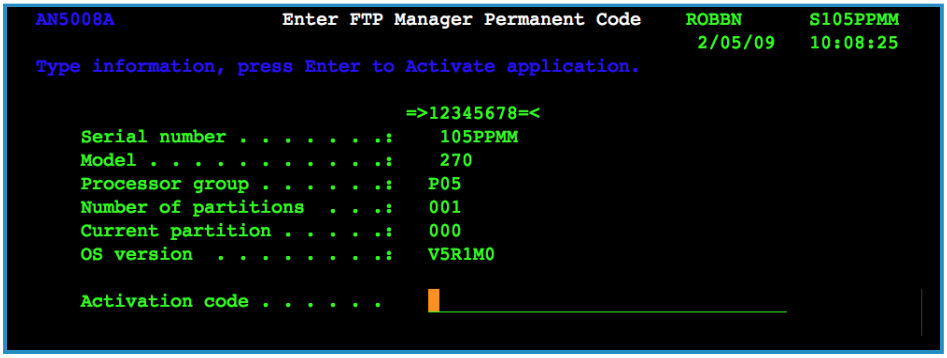
To use Alliance FTP Manager you must have a trial code or permanent code. A trial code was provided with your shipment and will be active for approximately 30 days from date of creation. Use this panel to enter the trial code.



`Trial code` : Enter the trial code provided by your software vendor.

Enter FTP Manager Permanent Activation Code

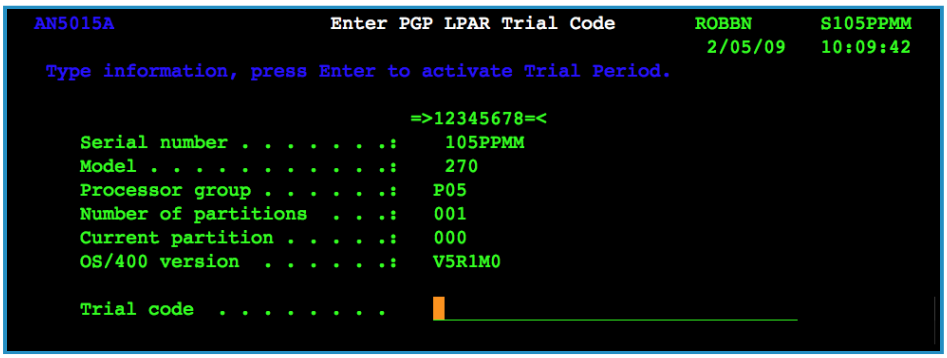
After you have licensed Alliance for your IBM i you will receive a permanent activation code. Enter the permanent code in this panel.



`Activation code` : Enter the permanent activation code provided by your software vendor. If you upgrade to a new system or model you will need to obtain a new activation code.

Enter PGP Option Trial Code

To use Alliance FTP Manager PGP Option you must have a trial code or permanent code. A trial code was provided with your shipment and will be active for approximately 30 days from date of creation. Use this panel to enter the trial code.



`Trial code` : Enter the trial code provided by your software vendor. This code is different than your Alliance FTP Manager trial code.

Enter PGP Option Permanent Code

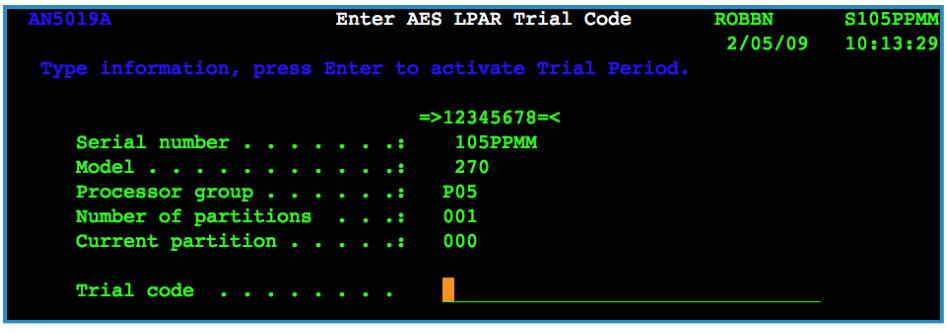
After you have licensed Alliance FTP Manager with PGP Option for your IBM i you will receive a permanent activation code. Enter the permanent code in this panel.



`Activation code` : Enter the permanent activation code provided by your software vendor. If you upgrade to a new system or model you will need to obtain a new activation code.

Enter AES Option Trial Code

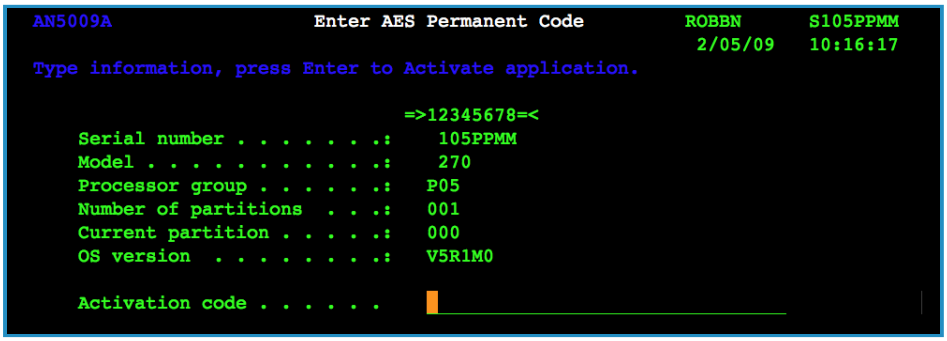
To use Alliance FTP Manager AES Option you must have a trial code or permanent code. A trial code was provided with your shipment and will be active for approximately 30 days from date of creation. Use this panel to enter the trial code.



`Trial code` : Enter the trial code provided by your software vendor. This code is different than your Alliance FTP Manager trial code.

Enter AES Option Permanent Code

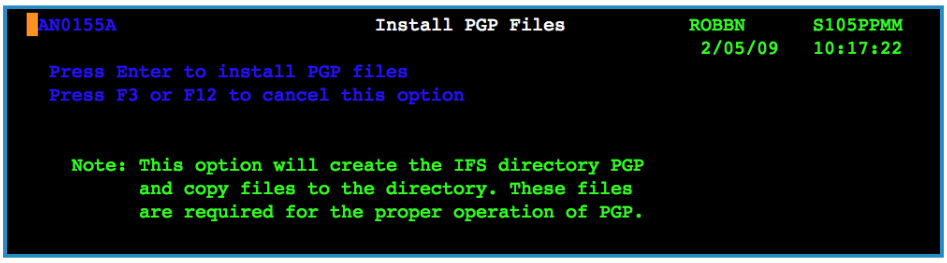
After you have licensed Alliance FTP Manager AES Option for your IBM i you will receive a permanent activation code. Enter the permanent code in this panel.



`Activation code` : Enter the permanent activation code provided by your software vendor. If you upgrade to a new system or model you will need to obtain a new activation code.

Install PGP Files

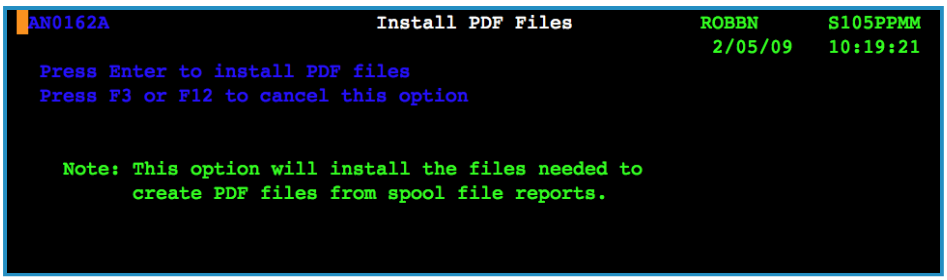
If you are going to use the Alliance PGP Option you must install the PGP IFS directory and related support files. This option will create the PGP directory and restore the files.



There are no input fields for this panel. Press ***Enter*** to install the PGP files. Press ***F3*** or ***F12*** to cancel this option.

Install PDF Files

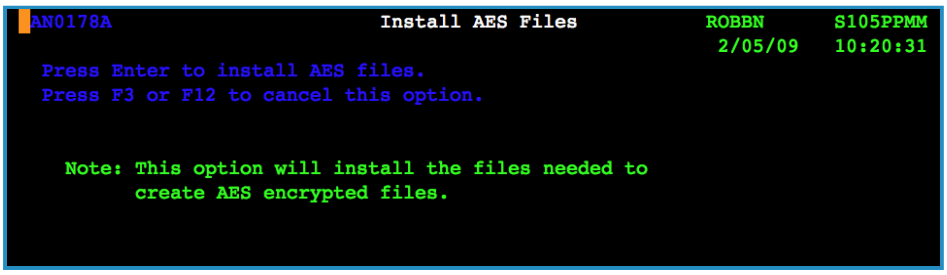
If you are going to use spool file distribution with the PDF option you must install PDF configuration and font files in the IFS directory. Use this option to install the files.



There are no input fields for this panel. Press ***Enter*** to install the PDF files. Press ***F3*** or ***F12*** to cancel this option.

Install AES Files

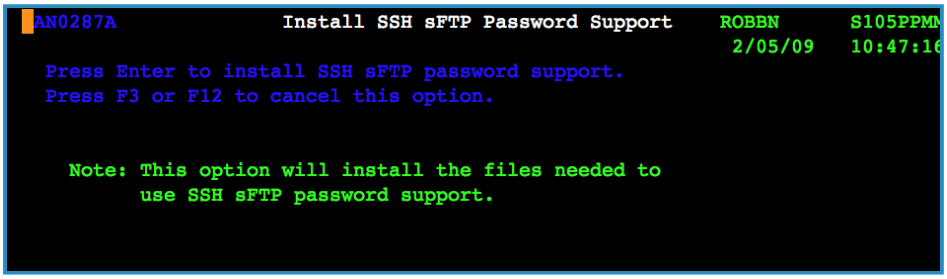
If you are going to use the AES encryption option you must install AES files in the IFS directory. Use this option to install the files.



There are no input fields for this panel. Press ***Enter*** to install the PDF files. Press ***F3*** or ***F12*** to cancel this option.

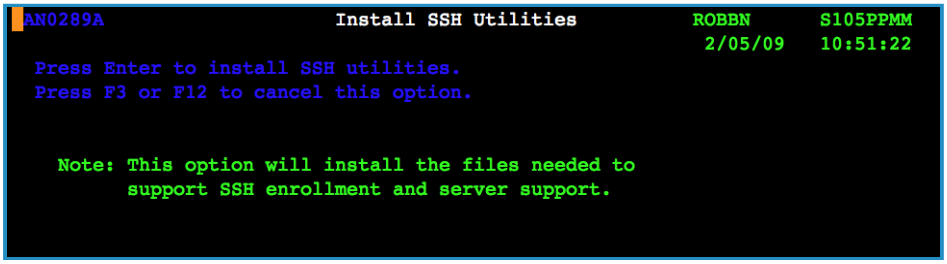
Install SSH sFTP Password Support

If you are going to use User ID and Passwords when you connect using SSH you must install sFTP Password Support in the IFS directory. Use this option to the files.

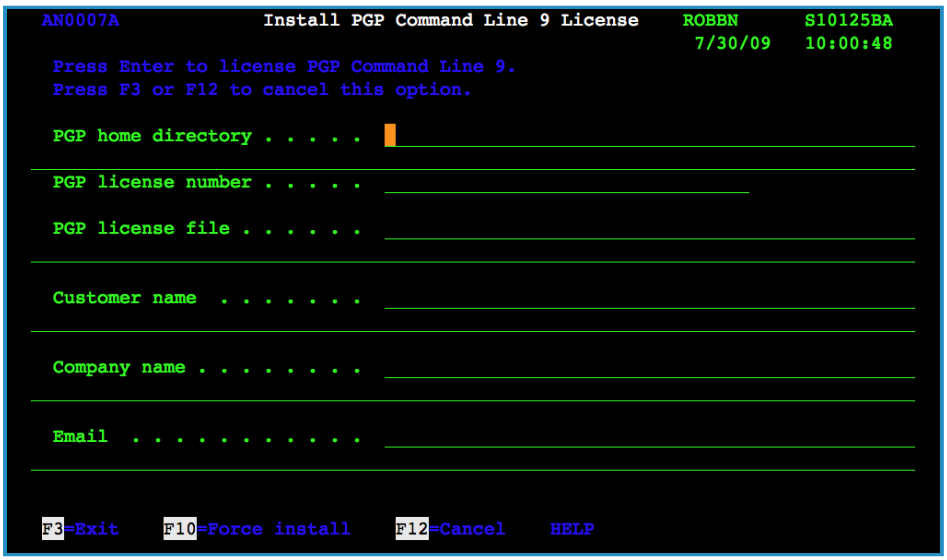


Install SSH Utilities

If you are going to use SSH sFTP install SSH files in the IFS directory. Use this option to install the files.



Install PGP CL 9 license



In order to use PGP you must licenses it first. This is 1 of 2 licenses you will receive from your software vendor for PGP. You will have received an email with the information to complete these fields.

Apply Release Upgrade

If you receive a release upgrade from your software vendor you will use this option to re-establish your application options and other configurations you developed on the earlier release. For complete instructions for upgrading please refer to Appendix B. When you select this option the following panel is displayed.



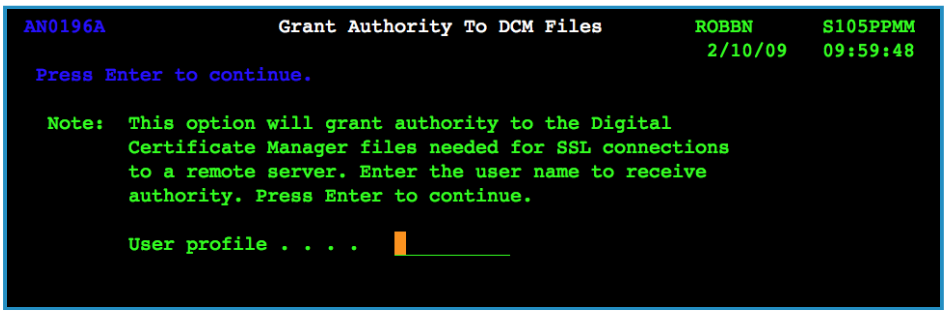
`Previous version library` : Enter the name of the library that contains the previous release.

View release notes

Taking this option will allow you to see release notes for FTP Manager. These are also available in the Solutions Tab of the support site by doing a search on “release notes allftm100”.

Grant user authority for DCM

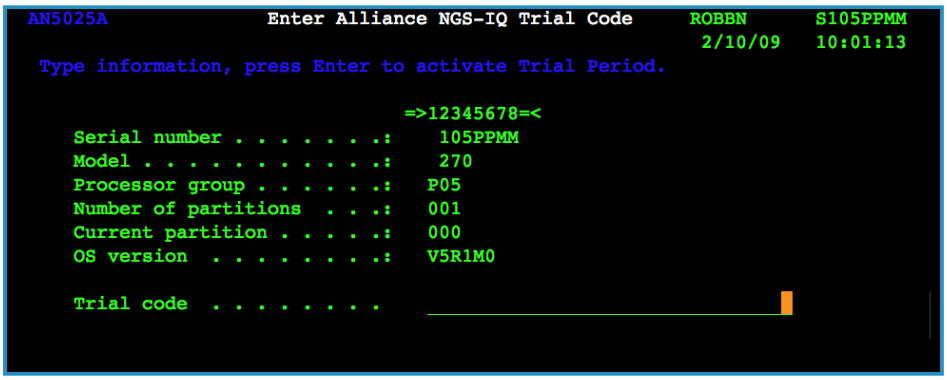
When you set up an SSL FTP connection and create a certificate with Digital Certificate Manager, you need to grant authority to an IBM i user for the DCM certificate files. For example, if you use library scan, directory scan, or auto-scan you would need to grant authority to the user defined in the Configuration menu to use the certificate files in the IFS directory. This option will automatically grant authority to all of the files need for DCM certificates.



`User profile` : Enter the user profile name to receive authority for the DCM certificate files.

Enter NGS-IQ trial code

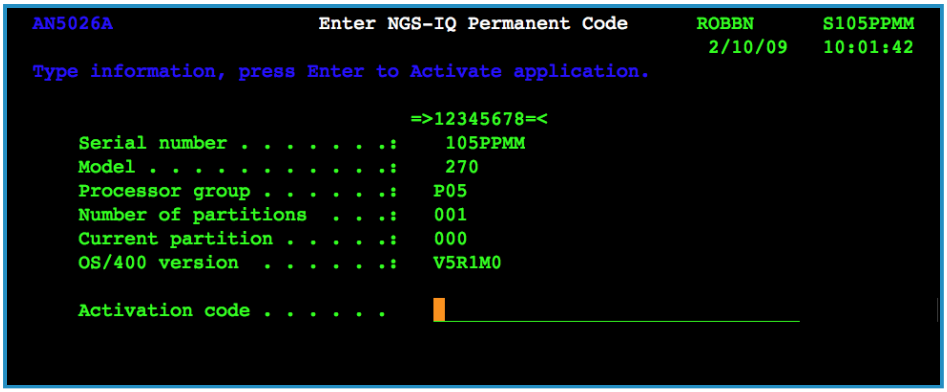
If you own New Generation Software and want to integrate Alliance FTP Manager with the commands, you must have a trial code or permanent code. A trial code should have been provided with your shipment and will be active for approximately 30 days from date of creation. Use this panel to enter the trial code.



`Activation code` : Enter the permanent activation code provided by your software vendor. If you upgrade to a new system or model you will need to obtain a new activation code.

Enter NGS-IQ Permanent Code

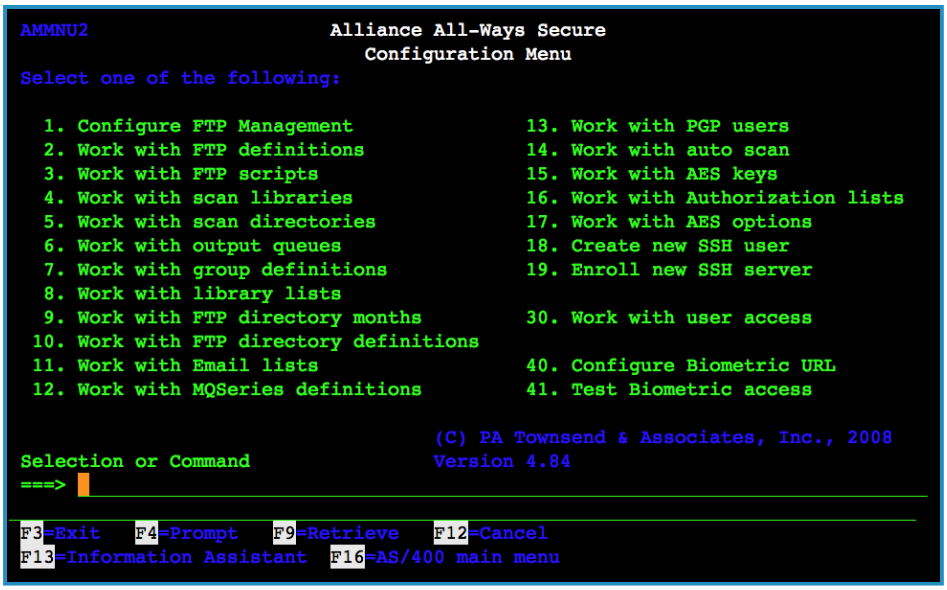
After you have licensed Alliance FTP Manager with NGS-IQ Option for your IBM i you will receive a permanent activation code. Enter the permanent code in this panel.



`Activation code` : Enter the permanent activation code provided by your software vendor. If you upgrade to a new system or model you will need to obtain a new activation code.

Chapter 17: Configuration

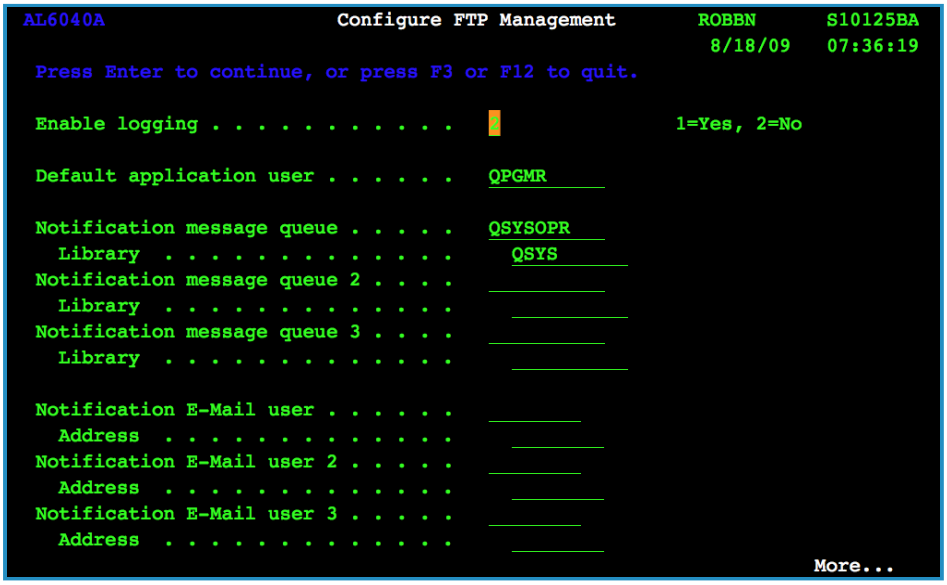
When you select option **2** from the main Alliance menu the following menu is displayed. You can configure all of the Alliance FTP Manager options and tables from this menu.



Select an option to begin work.

Configure FTP Manager

Use this panel to configure global options for Alliance FTP Manager. Keep in mind that any changes that take place here will only happen once the subsystem has been ended and restarted. All options are described in the help panels which you can access with ***F1***.



Enable logging: Enter **1** for **Yes** if you want to **Enable logging**. Enter **2** for **No** if you do not want to use logging. Logging can be helpful when you are attempting to analyze problems in FTP sessions. You may want to keep this option as **2** for **No** as log files get full

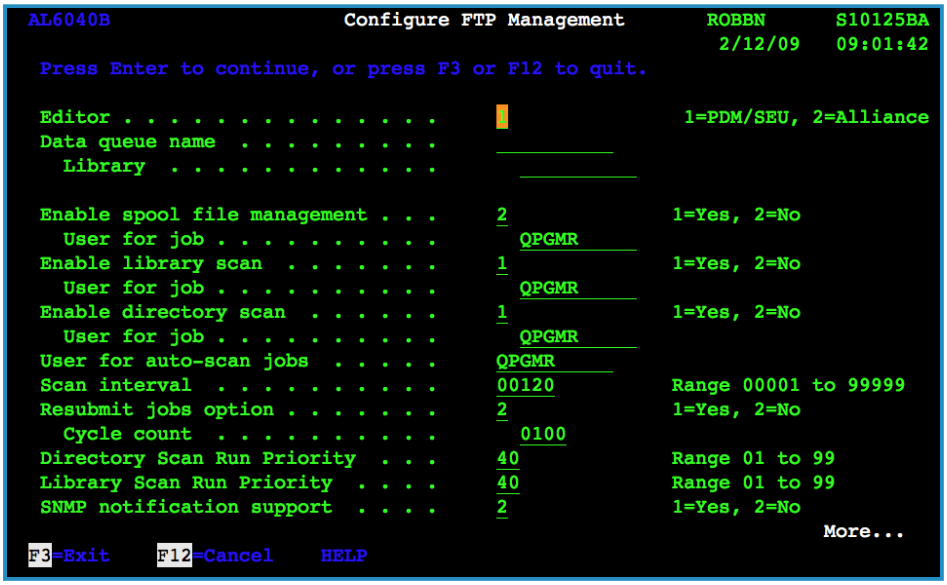
very fast and most issues can be resolved by examining the application logs.

`Default Application User` : This option allows you to specify a general user for jobs such as **`FTM_EXEC`** . You can still set some individual jobs to run under specific users.

`Notification message queue 1 through 3` : Alliance FTP Manager will send error and informational messages to the message queues you define. You must define at least the first message queue. You can define up to two additional queues for notification messages.

`Notification E-Mail user ID 1 through 3` : You can define up to three e-mail addresses to receive error and informational messages through the SNADS interface. These are entirely optional. If you want to use e-mail addresses for notifications you must first define these addresses in the System Distribution Directory (SNADS directory). The addresses may be OfficeVision or Internet recipients. For each e-mail address enter both a user ID and address.

Press ***Enter*** to go to the next screen.



`Editor` : Enter **`1`** to use **`PDM`** or **`SEU`** to edit source scripts. If you do not have an editor installed on your IBM i you can Enter **`2`** to use the Alliance editor. You will need one of these options to view and work with scripts.

`Data queue name and Library` : You can specify a **`Data queue name`** to receive copies of error messages generated by Alliance FTP Manager. The data queue interface is provided to enable you to automate your own problem handling routines. You must create application programs to process the messages in the data queue.

`Enable spool file management` : Enter option **`1`** if you are going to distribute spool file reports using FTP or email. Enter **`2`** for **`No`** if you are not going to distribute spool files.

If you selected **`Yes`** for **`Enable spool file management`** you must specify a user for the job. The default is **`QPGMR`** . You must have authority to the user profile to change it. Please refer to Appendix F for instructions on how to properly configure and send Spool Files.

`Enable library scan` : Enter option **`1`** if you are going to process files using library scan. Enter **`2`** for **`No`** if you are not going to work with library scan. Library scan is a facility that will automatically process files you copy into a defined library. Please refer to the Appendix C for instructions on how to configure Library Scan jobs for inbound and outbound files.

If you selected **Yes** for **Enable library scan** you must specify a user for the library scan job. The default is **QPGMR** . You must have authority to the user profile to change it. Please refer to Appendix C for instructions on configuring Library Scan jobs.

Enable directory scan : Enter option **1** if you are going to distribute files using IFS directory scan. Enter **2** for **No** if you are not going to work with directory scan. Directory scan is a facility that will automatically process files you copy into an IFS directory.

If you selected **Yes** for **Enable directory scan** you must specify a user for the directory scan job. The default is **QPGMR** . You must have authority to the user profile to change it. Please refer to Appendix D for instructions on configuring Directory Scan jobs.

User for auto scan jobs : Enter the user profile to use for the Auto-scan job. Auto Scan is designed to go to a remote server and retrieve files, placing them into a library or an IFS directory. The default user profile is **QPGMR** . You can change this to any valid profile. The job description **ALLFTMAUTS** will be changed for this job. The next time the Alliance subsystem starts the user for the job will be the new user. Please refer to Appendix E for instructions on configuring Auto Scan jobs.

Scan interval : Enter the **Scan interval** , in seconds, for library and directory scans. This is the amount of time that the scan will wait between jobs. This value does not apply to spool file processing as spool files are processed as soon as they go to “Ready” status. Nor does it apply to Auto Scan jobs since you set the scan times within the individual jobs.

Resubmit jobs option and Cycle count : If you want Alliance batch jobs to resubmit themselves periodically, specify **1** for **Yes** . If you do not want to use resubmit specify **2** for **No** . The default is not to resubmit Alliance batch jobs. If you have a problem with job message queue full errors you can specify Yes for this option and jobs will periodically resubmit themselves to avoid this problem. An alternative would be to set the system value **QJOBMSGQFL** to ***WRAP** .

The cycle count allows you to specify when jobs will be re-submitted when the number of files, or number of cycles, has reached this value.

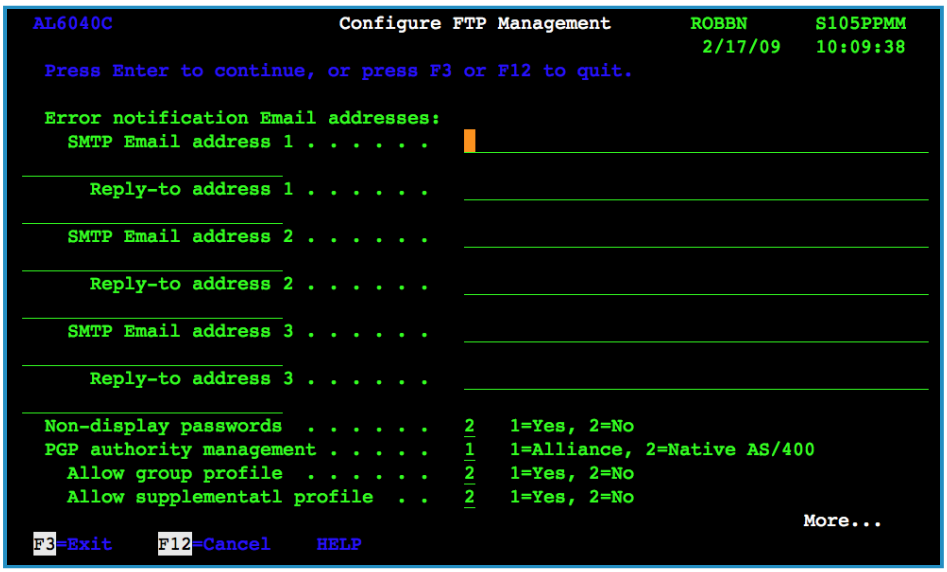
Directory Scan Run Priority : You can determine the run priority for your Directory scan. Enter a number from **01** to **99** . The default is **40** .

Library Scan Run Priority : You can determine the run priority for you Directory scan. Enter a number from **01** to **99** . The default is **40** .

SNMP notification support : Enter **1** for **Yes** if you want to use SNMP notification. Enter **2** for **No** if you do not want to use SNMP notification. You must first configure SNMP on your IBM i, and you must first configure FTP Manager for SNMP, before you can use SNMP notification.

If SNMP error notification is enabled Alliance will automatically start an SNMP subagent in the Alliance application subsystem when the subsystem starts.

Press ***Enter*** to go to the next screen.



SMTP Email address 1, 2, 3 : Enter up to three email addresses for error notification messages. These are the same messages that get sent to the message queues on the previous panel. You must enter the address in standard Internet SMTP format. For example: user@domain.com. If you enter an email address you must also enter a Reply-to address. Please note that many ISPs and mail systems will validate both the recipient and reply-to addresses and may not deliver the message if the domain is unknown or is blocked by spam filters.

Reply-to address : If you enter **SMTP email addresses** for notification you must also enter the **Reply-to address** . The **Reply-to address** does not have to be an address on the IBM i. For example, you can enter a **Reply-to address** on your Lotus Notes, Microsoft Exchange, or Internet email account. In many cases you may be able to use donotreply@domain.com.

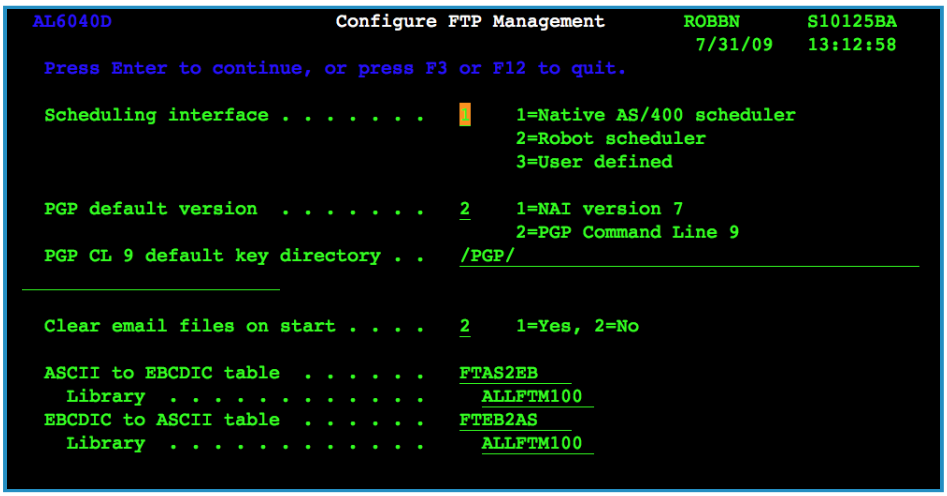
Non-display passwords : Enter **1** for **Yes** if you want to prevent viewing of passwords after they have been entered. Enter **2** for **No** if you want passwords to be visible. If you make the option **1** for **yes** then password options will display as blank and you will be unable to edit them until you change this option to **2** .

PGP authority management : Specify **1** for **Alliance** PGP file management. Specify **2** for **Native IBM i** PGP file authority management. If you specify **1** for **Alliance** PGP file management, Alliance will manage the ownership of PGP files in the IFS directory /PGP. If you specify **2** for **Native IBM i** management you must use IBM i commands to manage the authority of PGP files.

Group profile support : If you specified option **1** for Alliance user management specify **1** for **Yes** to include group profiles, or **2** for **No** to not include group profiles. If you select option **1** , Alliance will allow a user access to PGP functions if their group profile is explicitly authorized to PGP.

Supplemental group support : If you specified option **1** for Alliance user management specify **1** for **Yes** to include supplemental profiles, or **2** for **No** to not include supplemental profiles. If you select option **1** Alliance will allow user access to PGP functions if any profile in the supplemental group is explicitly authorized to PGP.

Press ***Enter*** to go to the next screen.



`Scheduling interface` : Enter **`1`** to use the **`Native IBM i`** scheduling application. Enter **`2`** to use the **`Robot scheduler`** application. When you select this option Alliance will call the RBT020 program to perform scheduling. Enter **`3`** to use a **`User defined`** interface for scheduling. If you enter option **`3`** you must create an application program named FTMSCHED in the Alliance product library.

`PGP default version` : You can now move your PGP version to Command Line 9 after running the license command on the Install menu.

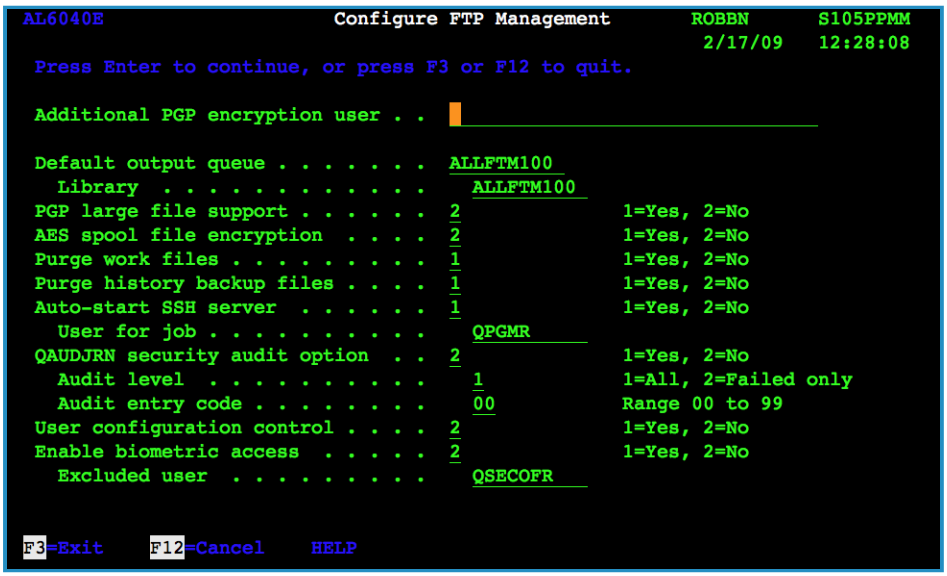
`PGP CL 9 default key directory` : Enter the directory path to the PGP keyfile. This can be any directory.

`Clear email on start` : Enter **`1`** for **`Yes`** to automatically clear the outbound email files when the Alliance subsystem starts. Enter **`2`** for **`No`** to not automatically clear these files. If you enter **`No`** for this option you can still maintain the email files using the normal Alliance maintenance options.

`ASCII to EBCDIC table and Library` : Enter the name of the **`ASCII to EBCDIC table`** for the Alliance FTP Manager client data conversion. The default is the **`FTAS2EB`** table in the product library.

`EBCDIC to ASCII table and Library` : Enter the name of the **`EBCDIC to ASCII table`** for the Alliance FTP Manager client data conversion. The default is the **`FTEB2AS`** table in the product library.

Press ***Enter*** to go to the next screen.



Additional PGP Encryption User : If you are using the Alliance PGP encryption option you can define an **Additional PGP encryption user** for encrypted files. If you specify your own PGP key for this value you can decrypt any file you encrypt and send to your trading partners.

Default Output Queue Name and Library : You can specify the **Default output queue** name and **Library** for the Alliance batch jobs. The default is **QPRINT** in library **QGPL** .

PGP Large File Support : Enter **1** for **Yes** if you want to use Alliance **PGP large file support** . Enter **2** for **No** if you do not need to use large file support. The default is **No** . If you need to process files larger than 4 gigabytes you can enable this option. You should keep it as **2** for **No** unless you need it.

AES Spool File Encryption : Enter **1** for **Yes** if you want to use the Alliance **AES spool file encryption** facility. Enter **2** for **No** if you do not want to use Alliance spool file encryption. Note that you must license the AES encryption support before you can use AES spool file encryption. **Purge Work Files** : Enter **1** for **Yes** if you want Alliance to automatically **Purge work files** from library **ALLFTM100W** when maintenance is performed (option **3** on the main menu). Enter **2** for **No** to not automatically purge work files. The recommended value is **1** for **Yes** .

Purge History Backup Files : Enter **1** for **Yes** if you want Alliance to automatically delete files that were moved to a backup library (such as **ALLFTM100B** or a user defined library) based on the date. This will take affect when the maintenance option is run (option **3** on the main menu). Enter **2** for **No** to not automatically remove the actual files. The recommended value is **1** for **Yes** .

Auto-Start SSH Server and User : Enter **1** for **Yes** if you want Alliance to automatically start the Secure Shell (SSH) server. Enter **2** for **No** to not start the SSH server (the default). You must first configure the SSH server before enabling this option. Please read the SSH documentation regarding SSH server configuration. The user default is **QPGMR** . The User for the job does affect keys & IFS directories when using SSH. Please refer to the SSH Quick Start Guide to understand the effects of user ID within SSH.

QAUDJRN security audit option : Enter **1** for **Yes** to enable security journal auditing. Enter **2** for **No** to not log events to the security audit journal. Before enabling this option you should create the **QAUDJRN** journal and related journal receivers. See the IBM Security

Reference manual for information on how to create the journal.

`Audit level` : Enter **`1`** to log all events in the journal. Enter **`2`** to only log access failure events.

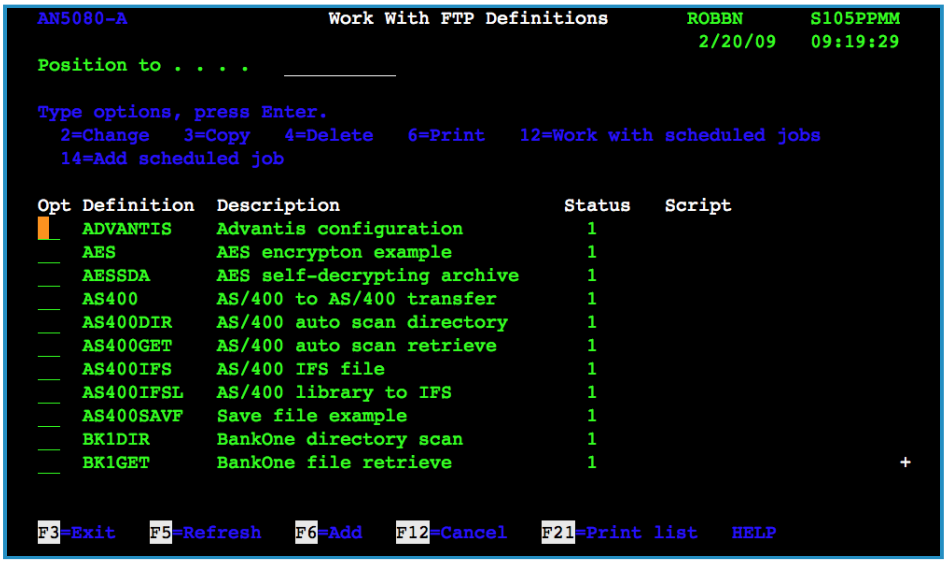
`Audit entry code` : Enter the journal entry code to use for journal records. Be sure not to use a code that IBM uses for system security events. The default value for the journal entry code is **`00`** .

`User configuration control` : WARNING: Enabling this option can result in disabling access to the Alliance FTP Manager product. You should first define all of the users and their access rights before enabling this option. (Option **`30`** . Work with user access on the configuration menu.) The security officer **`QSECOFR`** always has rights to change this option and define users. Enter **`1`** for **`Yes`** to enable user access control. Enter **`2`** for **`No`** to not enable user access control. This option takes effect immediately.

Work With FTP Definitions

An FTP Definition defines all of the parameters needed for an FTP session such as a library scan, auto scan, or directory scan. Before executing an FTP Manager session you must define these parameters using Work With FTP Definitions. And before creating the definition you must have a script in place. Please refer to the Appendix for instructions and direction on how to configure a specific scan job. Here you can create a new definition, change an existing definition, delete a definition, and print a definition or a list of all definitions.

Press ***Enter*** to go to the next screen.



`Position to` : To position the list to a specific record enter the key value in this field and press ***Enter*** or ***F16***.

`Option` : Enter the option corresponding to the function you want to perform. A list of valid options follows:

`2=Change` : Use this option to change the configuration of a definition.

`3=Copy` : Use this option to copy information about the item to a new name.

`4=Delete` : Use this option to delete an item from the list and all associated detail records.

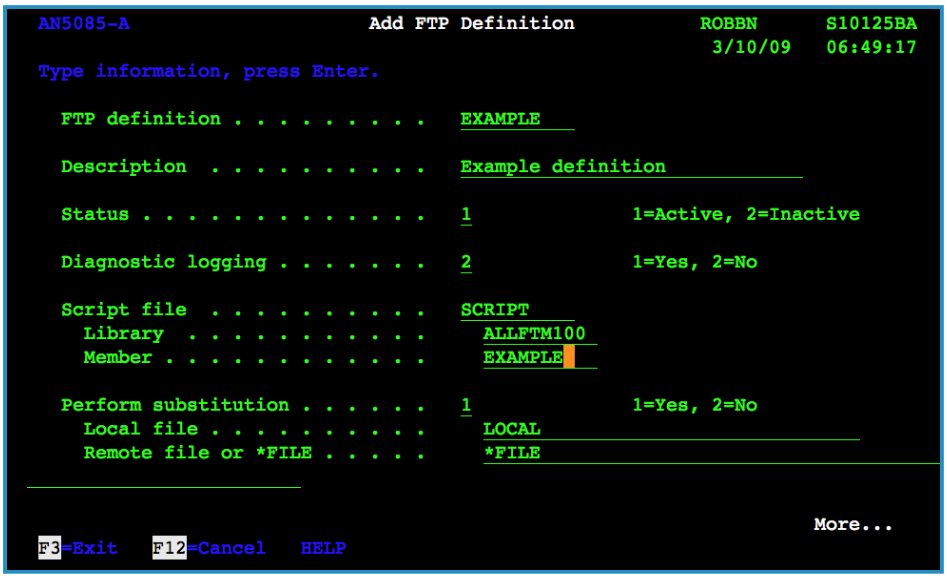
6=Print : Use this option to print a record. The printed report will be placed in the output queue that is the default for your job.

12=Work with scheduled jobs : Use this option to work with jobs you have previously scheduled.

14=Add scheduled job : Use this option to add a new job schedule. You can have as many schedules as you like for any particular FTP definition.

Change FTP Definition

Press ***Enter*** to go to the next screen.



Description : Enter a Description for this FTP session.

Status : Enter **1** for **Active** or **2** for **Inactive** . If a record is inactive the FTP session will not be performed by the library scan subsystem or interactive panels.

Diagnostic logging : Enter **1** for **Yes** if you want to enable Diagnostic logging for this FTP session. Enter **2** for **No** if you do not want to use diagnostic logging. You may want to keep this option as **2** for **No** as log files get full very fast and most issues can be resolved by examining the application logs.

Script file, Library, Member : A Script file is a source physical file that contains the FTP commands that you want to execute for the session. The command records may also contain substitution value that are resolved at the time of the session. A source file is provided by Alliance FTP Manager for your use named **SCRIPT** in the product library. You can, however, use your own source physical file for this purpose. A script must already have been created in order to move past this first panel.

Perform substitution : Enter **1** for **Yes** if you want FTP Manager to substitute values in the script file. Enter **2** for **No** if you do not want to use substitution. You want this **1** for **Yes** for any kind of automation such as one of our scan jobs or a scheduled transfer. If you specify **Yes** you must define at least one substitution value in the **Local file** , **Remote file** , User ID, and/or Password fields. This works in conjunction with the script substitution variables.

`Local file` : If you are using the definition in a program and want to perform substitution of the local file name you can enter the name in this field. This can be in the standard IBM i FTP format:

```
library/file
```

or IFS Directory format:

```
/directory/file
```

When using scan jobs, local file names are passed from the file action not the definition. When using scan jobs these option should be:

```
Local File…….. LOCAL
Remote file or *FILE …. *FILE
```

`Remote file` : If you want to run this through a CL or manually, and you want to perform substitution of the remote file name you can enter the name in this field. You must have a LOCAL path and file and this must be in the format that is appropriate for the remote system. For most PC servers (including Windows NT) you can specify the fully qualified path for the file:

```
c:\directory\myfile.txt
```

For a remote IBM i you can specify the file name in IBM i format:

```
library/file
```

For a remote IBM MVS system you can specify the name in normal DSName format. You must, however, enclose the target name within double and single quotes in order to remove the user ID from the first part of the path:

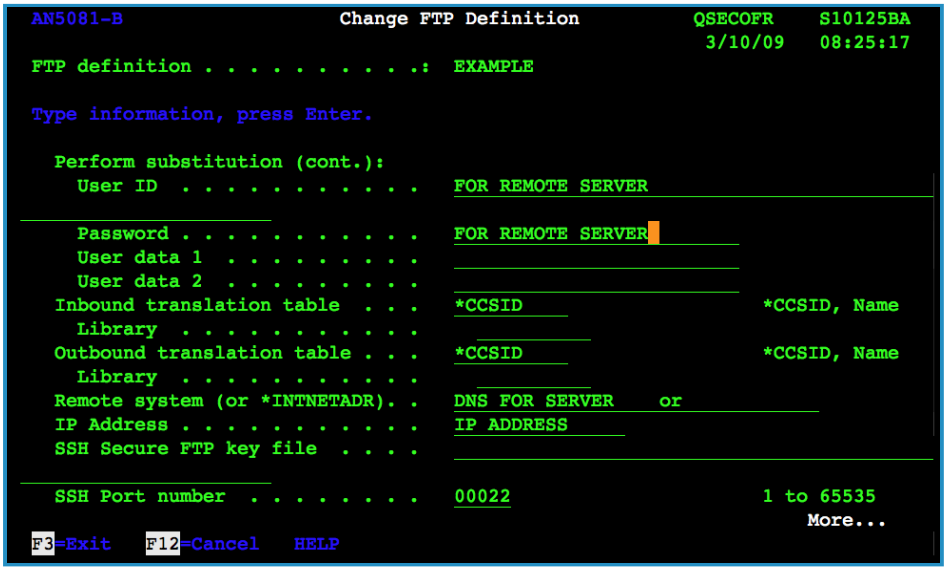
```
"'name1.name2.name3.name4.name5'"
```

When using scan jobs, and you want to specify a remote file name, the file name is determined in the script, not the definition. When using scan jobs these option should be:

```
Local File…….. LOCAL
Remote file or *FILE …. *FILE
```

Reviewing the documents in the Appendix will help you understand how these substitution parameters are used within the definition and script of scan job.

Press ***Enter*** to go to the next screen.



`User ID` : Enter a User ID for substitution if desired.

`Password` : Enter a password for substitution if desired. Note that if you have selected the configuration option to non-display password you will not be able to change this field. Use function key ***F7*** to change the password. See the example of this option below.

`User data 1` : Enter any data that you want to use for substitution in an FTP script. See the description of the &1 substitution process below.

`User data 2` : Enter any data that you want to use for substitution in an FTP script. See the description of the &2 substitution process below.

`Inbound translation table and library` : Enter the name of the Inbound translation table for this session. You can enter the value **`*CCSID`** . The CCSID parameter you specify in the following section will be used to determine the translation table. This translation table is used for the EBCDIC to ASCII translation.

`Outgoing translation table and library` : Enter the name of the Outgoing translation table for this session. You can enter the value **`*CCSID`** . The CCSID parameter you specify in the following section will be used to determine the translation table. This table is used for ASCII to EBCDIC translation.

`Remote system` : Enter the name of the Remote system. If you want to use an IP Address enter the value **`*INTNETADR`** in this field and an actual IP address in the following field. If you enter a name in this field it must exist in your local IBM i host table or be resolved by a DNS lookup.

`IP Address` : Instead of a system name you can enter the IP address of the remote system. If you want to use an IP address you must specify *INTNETADR for the Remote System field above. Enter the IP Address in dotted decimal format. For example:

```
192.10.4.8
```

`SSH Secure FTP key file` : If you are sending a file using SSH, and you have a secure key file to use for SSH keys you can enter that information here. If there is no entry, we will use the default key file.

`SSH Port Number` : If your connection is using the SSH protocol, enter the SSH Port number given to you for the server. The default port number is **`22`** .

F7 to change a password

This panel is used to display the password change dialog. If your system is configured to not display passwords you can use function key ***F7*** from the previous display to change the password.

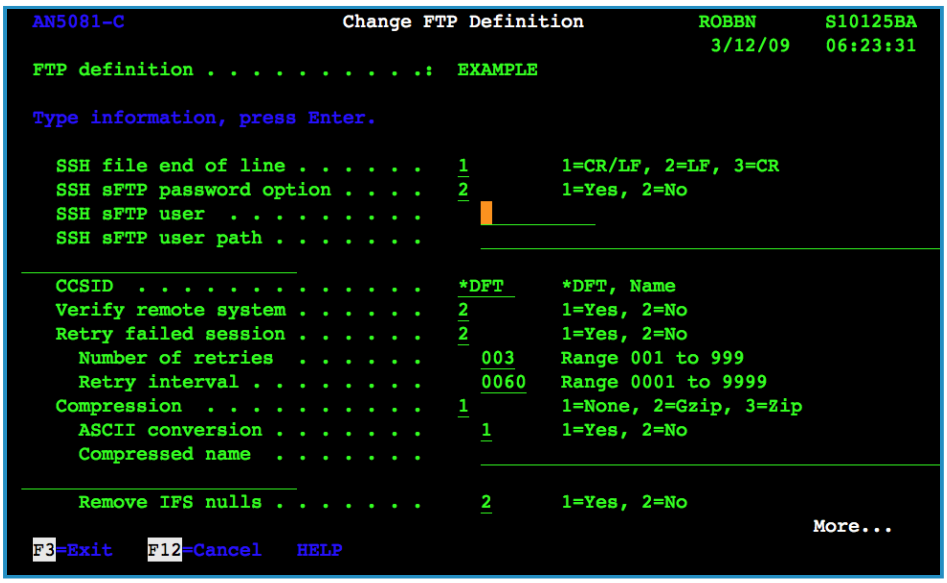


`Current password` : Enter the Current password for this definition. The password will not display as you type.

`New password` : Enter the New password you want to use for this definition. The password will not display as you type.

`New password (to verify)` : Enter the new password again to verify the entry of the password. The password will not display as you type.

Press ***Enter*** to go to the next screen.



CCSID : Enter a CCSID to use for this session, or the value ***DFT** for the current job default value. The CCSID

is used to determine how inbound and outgoing EBCDIC/ASCII translatic

*DFT` is the default.

Verify remote system : Enter **1** for **Yes** to verify a connection to a remote system before performing an FTP session. Enter **2** for **No** to not perform verification before starting an FTP session. If you encounter problems with remote systems being unavailable this option can help you avoid delays. When you specify **Yes** , Alliance will ping the remote system before starting FTP to verify that it can be contacted. This can save time in delays in FTP sessions. Be cautious when using this option as many servers block ping attempts and you will get an error or misleading information.

Retry failed session : Enter **1** for **Yes** to retry a session that fails. Enter **2** for **No** to not retry a failed session. If you enter **Yes** you must also specify the number of retries and the retry interval.

Number of retries : If you specified **Yes** to retry a failed session enter the number of times to retry.
Retry interval : If you specified **Yes** to retry a failed session enter the number of seconds to wait between retries.

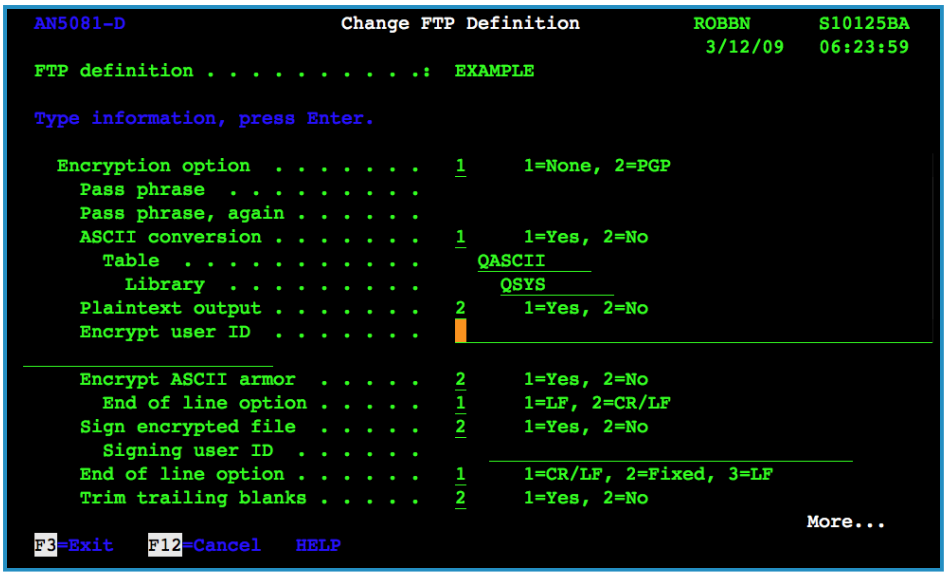
Compression : Enter **1** if you do not want to compress this file. Enter **2** for Gzip compression. Gzip is often used for compressing files on UNIX systems. Enter **3** for Zip compression. Zip compression is normally used on Windows and Macintosh systems.

ASCII conversion : Enter **1** for **Yes** if you want to translate the file to ASCII when using the compression option. This is the default for compressed files. Enter **2** for **No** if you do not want to translate the file to ASCII.

Compressed name : Enter the name of the compressed file. For a Gzip file this would normally be a file name with the ".gz" extension. For a Zip file this would normally be a name with a ".zip" extension.

Remove IFS nulls : Enter **1** for **Yes** to remove null characters at the end of an IFS compressed file. Enter **2** for **No** to not remove null characters at the end of a file.

Press ***Enter*** to go to the next screen.



Encryption option : Enter **1** for **None** (no encryption), **2** for **PGP** (Pretty Good Privacy). The default is **1** for no encryption.

Pass phrase : Enter a Pass phrase only if the encrypted file is to be signed by your public key. The pass phrase will be the pass phrase for your key. Only do this if your trading partner has asked for a signed file.

Pass phrase again : Enter the same pass phrase again. The pass phrase will not display.

ASCII conversion : Enter **1** for **Yes** or **2** for **No** . The default is **Yes** to convert the data to ASCII.

Conversion table and library : If you entered **Yes** for the **ASCII conversion** option you must enter a conversion **Table** name and **Library** . You can use the IBM table **QASCII** in library **QSYS** , or you can use the Alliance table named **ANEB2AS** in library **ALLFTM100** .

Plaintext output : Enter **1** for **Yes** if you want to use PGP Plaintext output. This is sometimes referred to as Canonical text output. Enter **2** for **No** if you do not want to use Plaintext output. This causes PGP to assume the plaintext is an ascii text file, not a binary file, and converts the plaintext to canonical text before encrypting it. Canonical text has a carriage return and a line feed at the end of each line of text.

Encrypt User ID : If you specified **Yes** to encrypt this message with PGP enter the user ID of the recipient's key.

Encrypt ASCII armor : Enter **1** for **Yes** if you want to encrypt using ASCII armor output format. Enter **2** for **No** if you want to use standard PGP binary output for the encrypted file. The default is **2** to use binary output.

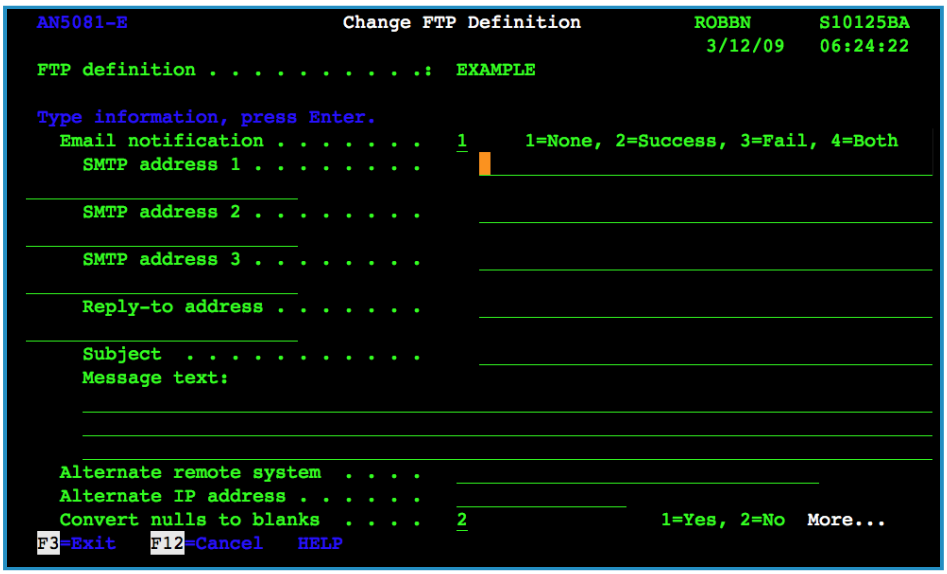
Sign encrypted file : Enter **1** for **Yes** if you want to sign the encrypted file. Signing an encrypted file provides for non-repudiation of the person encrypting a file (you). Specify **2** for **No** if you do not want to sign the file. If you specify **Yes** for this option you must provide a signing user ID Your key name) in the following field and your pass phrase in the first 2 options on this panel.

Signing user ID : If you specified Yes to sign the encrypted file enter the PGP User ID for the signature. This is normally the user ID of your PGP Private Key. Please note that if you sign the file with your private PGP user ID you must have exported and sent your public key to your trading partner.

End of line option : Enter **1** to provide a carriage return and line feed at the end of each database record in the encrypted file. Enter **2** to use a fixed length with no end of line characters.

Trim trailing blanks : Enter **1** for **Yes** to trim the trailing blanks from a record before encrypting with PGP. This is the default. Enter **2** for **No** to not trim trailing blanks.

Press ***Enter*** to go to the next screen.



`Email notification` : Enter **`1`** for no email notification. Enter **`2`** for notification only if the transfer is successful. Enter **`3`** for notification only if the transfer fails. Enter **`4`** to be notified if the transfer is successful or fails. You must first configure the FTP Manager email application before using notification. You can enter a pager number or PCS phone if your pager or PCS phone supplier supports email delivery (most do).

`SMTP address 1, 2, 3` : Enter the SMTP email address in standard Internet format. For example: user@domain.com. You can enter up to 3 addresses for notification. Note that you can also enter a list name in this field. You must first define the email list with the configuration option to Work With Distribution Lists.

`Reply-to address` : Enter the Reply-to address for the email address. This must be a standard Internet type of address.

`Subject` : Enter the text for the Subject of this message. This would normally be a description of the error.

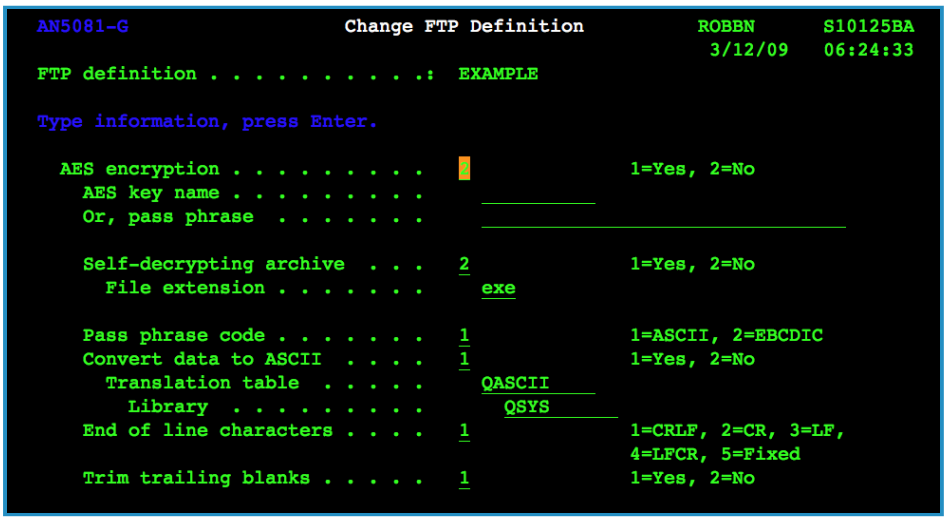
`Message text` : Enter up to three lines of Message text for this message. This can be any text you like. It might include information about the FTP transfer or instructions for handling an error condition.

`Alternate remote system` : If you want to use an alternate system name for the remote FTP location, enter the domain name for the site in this field. If you know the IP address of the remote system enter the value **`*INTNETADR`** . If the first system name (entered on a previous screen) or IP address cannot be used to connect to the remote site, this alternate name will be used. If you do not want to use an alternate site leave this field blank.

`Alternate IP address` : If you specified **`*INTNETADR`** for the remote system name, enter the IP address of the remote site in this field.

`Convert nulls to blanks` : Enter **`1`** to **`Convert nulls to blanks`** in the file before it is processed. Enter **`2`** for **`No`** to not convert nulls to blanks.

Press ***Enter*** to go to the next screen.



AES encryption : Enter **1** for **Yes** if you want to encrypt this file with Alliance AES encryption. Enter **2** for **No** if you do not want to encrypt this file.

> ***NOTE:*** You must first install the Alliance AES encryption option before you can encrypt files with AES.

AES key name : Enter an AES key name. If you use an AES key you must first create the key with the Alliance AES key configuration option.

Or, pass phrase : You can use a pass phrase instead of an AES key. Enter up to 32 characters of a pass phrase. This pass phrase can be created on the fly.

Self-decrypting archive : Enter **1** for **Yes** if you want to create the encrypted file as a Self-decrypting archive. Self-decrypting archives are executable on a Microsoft Windows platform. The recipient of this file does not need AES nor do they need an IBM i. All they need is the pass phrase with which the .sda was created. **File extension** : If you specified **Yes** to create the file as a Self-decrypting archive, enter the File extension for the executable file. This is normally **exe** , but you can specify a non-standard extension if you need to avoid problems with virus checking software. The recipient will need to rename the file to have an **exe** extension to execute it on a Windows platform.

Pass phrase code : Enter **1** to convert the pass phrase to **ASCII** . Enter **2** to leave the pass phrase in **EBCDIC** . If you are transferring the encrypted file to an **ASCII** system such as Windows you should use option **1** for **ASCII** .

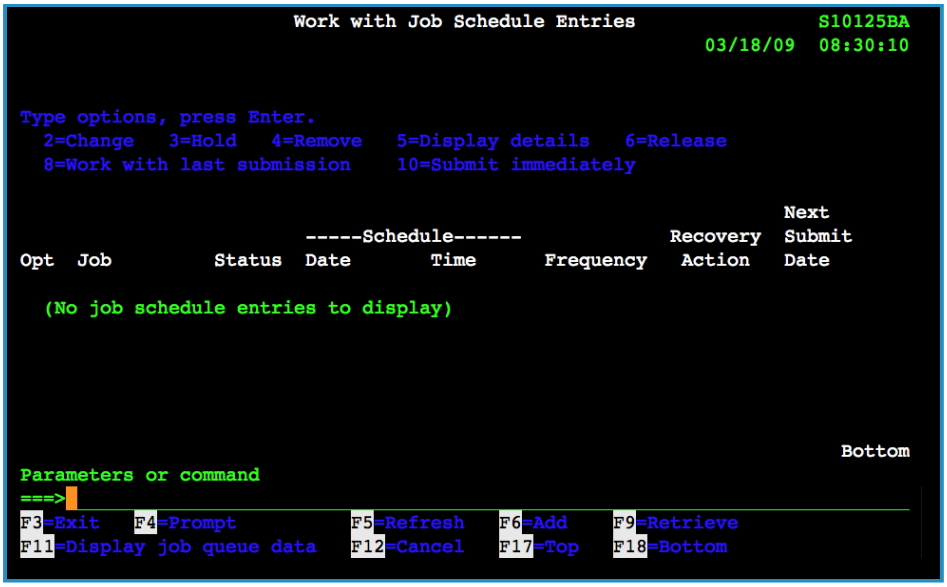
Convert data to ASCII : Enter **1** for **Yes** if you want the data to be converted to **ASCII** . Enter **2** for **No** if you do not want the data converted to **ASCII** . If the data is to be processed on a Windows PC or server you should chose **1** for **Yes** .

Translation table, library : If you specified to convert the data to **ASCII** enter the conversion table name and **Library** . The default is **QASCII** in library **QSYS** .

End of line characters : Enter **1** to use carriage return and line feed **End of line characters** . Enter **2** to use a carriage return. Enter **3** to use a line feed. Enter **4** to use a line feed and a carriage return for end of line characters. Enter **5** to use a **Fixed** format without end of line characters. **Trim trailing blanks** : Enter **1** for **Yes** to **Trim trailing blanks** from the end of DB2 records. Enter **2** for **No** to retain trailing blanks.

Work With Scheduled Jobs

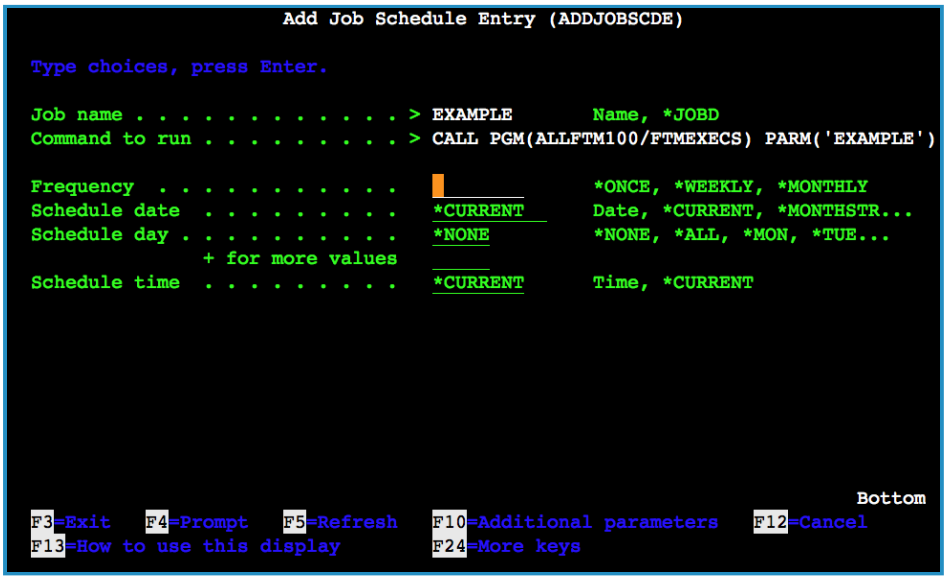
When you use option `12` from the list of FTP definitions a list will be displayed of all your schedules for this FTP definition. You can change or delete a scheduled job from this display.



Enter the schedule information you want for this FTP definition. For more information on job scheduling refer to the CLP Programmer's Guide for IBM i.

Add Schedule Entry

You can add a schedule entry with option `14` from the list of FTP definitions. When you select this option the IBM job schedule entry command is invoked allowing you to enter scheduling information.



You can schedule a job daily, weekly, monthly, on demand, or a combination of these schedules. You can enter multiple schedules for the same FTP definition. You can press ***F1*** for a description and a display of choices of each option.

Enter the schedule information you want for this FTP definition. For more information on job scheduling refer to the CLP Programmer's Guide for IBM i.

Work With FTP Scripts

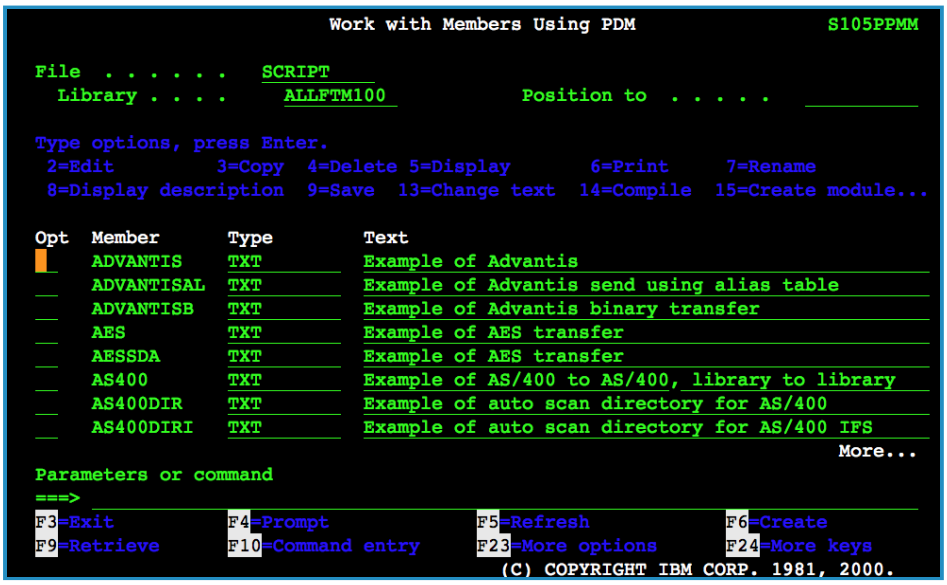
Selecting this option will start PDM or SEU to work with members in the **`SCRIPT`** source file in the application library. You can create or change a script for an FTP session. The script file contains the actual FTP commands that get executed during your FTP session. In addition, FTP Manager provides for substitution of the user ID and password, local and remote file names, and system date in the script file. See the comments below about rules for FTP scripts and substitution options.

If you do not have SEU or PDM on your IBM i you can use any IBM i editor to modify the source file. For using this option, go to the **`Configuration menu`** and enable the **`Editor`** option. (**`Main menu`** → **`option 2`** for **`Configuration`** → **`Option 1`** for **`Configure FTP Manager`** → Press ***Enter*** to go to the 2nd screen.

You can also create the source file on another IBM i and send it to this IBM i. Or, you can create FTP scripts on a PC and use FTP to send the script to the IBM i.

The default FTP script file is **`SCRIPT`** in the application library. You can use a different source file if you like. The script file, library and member are defined to the FTP definition.

From the configuration menu take **`Option 3`** for **`Scripts`** .



Select a source member to begin work.

FTP Script File Contents

The contents of the script file are FTP commands that you want to execute. The first line of the source file must be the user ID and password for connecting to the remote system. For example, if the user ID on the remote system is **`TOWNSEND`** and the password is **`JIBSAIL`** , you would enter the first line like this:

```
townsend jibsail
```

If the remote system uses case-sensitive user IDs and passwords be sure to enter them in the proper format.

After specifying the user ID and password you can enter any FTP commands that are valid on the IBM i and remote system. A simple script that sends a file from the IBM i to a UNIX platform might look like this:

```
townsend jibsail
put orders/ordmast /user/orders/database/ordmast.dat
quit
```

You can also specify substitution parameters for some of the elements of the script file. The substitution parameters will be supplied by the FTP Definition you create, or the API program you use to start an FTP session. See the following section for a description of substitution parameters.

FTP Script File Substitution

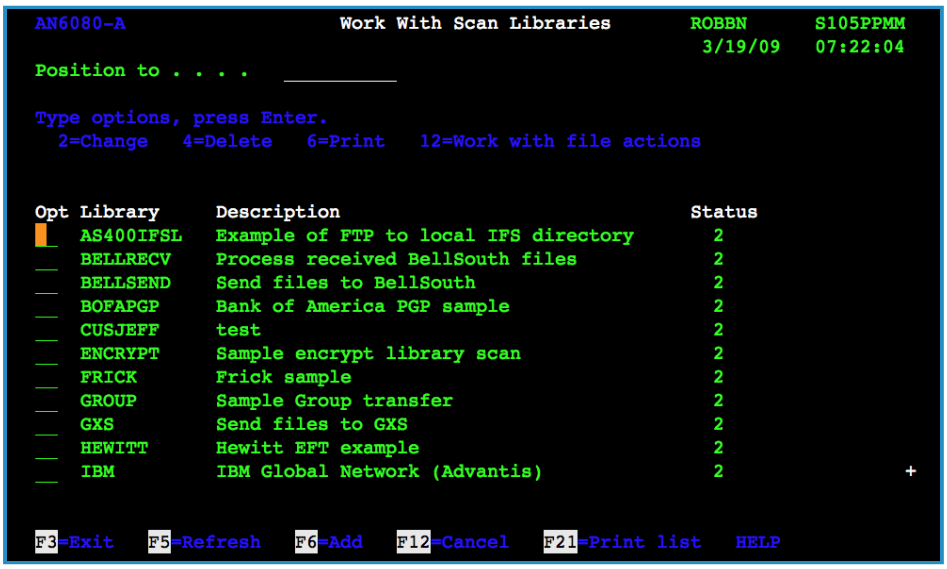
To make FTP scripts more flexible you can specify substitution parameters in the FTP script file. For example, you can specify that the local and remote file names be substituted from the FTP definition you create or from an API program that your application calls. This makes the FTP facility much more flexible. Instead of creating a script for every file you send to a remote host, you can create one script and specify that it use substitution for the file names.

Substitution is specified using a set of special characters. The special characters are always 3 bytes long and preceded with the ampersand character (`&`). You must always leave enough room in the script record for the substituted value. If you do not leave enough room FTP Manager will overwrite other FTP commands and the script will not work correctly.

Please read Appendix G for all the available substitution values.

Work With Scan Libraries

You can use this option to define libraries to scan for automatic FTP sessions. The library scan routine will search the list of libraries and process files one at a time. After defining a library with this panel you must use option `12` to define the files to be processed in the library and the actions to be taken. It is recommended that you do not define production libraries to scan. Rather, you should create libraries for scanning and define them to FTP Manager. This will help avoid unintentional errors processing FTP sessions. Please read Appendix C for direction on configuring Library Scan jobs.



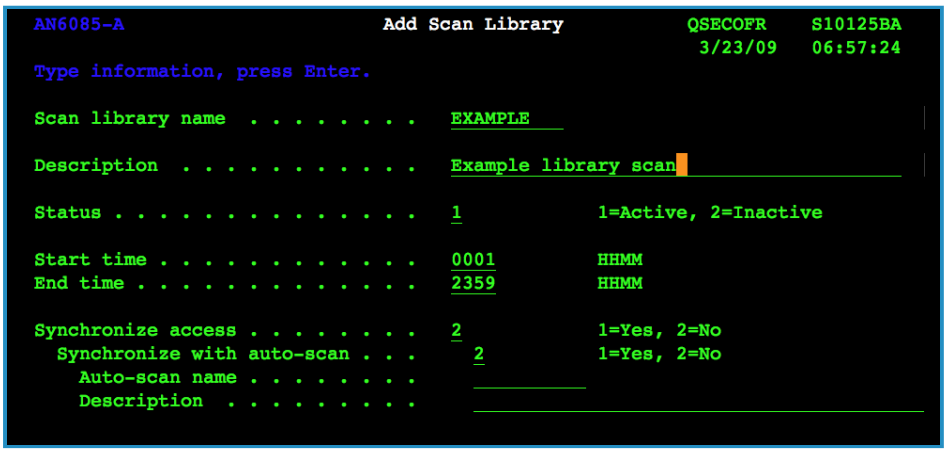
`Position to` : To position the list to a specific record enter the key value in this field and press ***Enter*** or ***F16***.

`Option` : Enter the option corresponding to the function you want to perform. A list of valid options follows:

OPTION	FUNCTION
2=Change	Use this option to change information about the item in the list.
4=Delete	Use this option to delete an item from the list and all associated detail records.
6=Print	Use this option to print a record. The printed report will be placed in the output queue that is the default for your job.
12=Work with file actions	Use this option to work with the files to be processed by FTP Manager.

Add/Change Scan Library

Use this panel to change information about a **`Scan library`** . You can change the **`Description`** and the **`Active`** or **`Inactive status`** of the **`Scan library`** .



`Description` : Enter a Description for this Scan library.

`Status` : Enter **`1`** for **`Active`** or **`2`** for **`Inactive`** . If you enter **`2`** for **`Inactive`** the library will not be processed by the Scan library function.

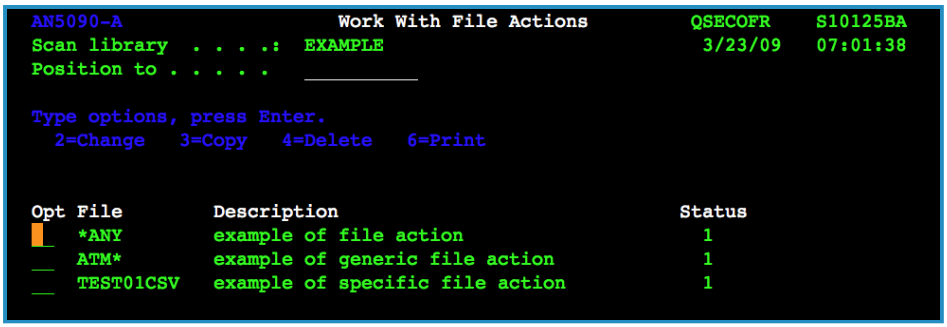
`Start time` : Enter the Start time for this library scan. The default is **`0001`** . You can control when library scan is active by setting a Start and End time.

`End time` : Enter the End time for this library scan. The default is **`2359`** . You can control when library scan is active by setting a Start and End time.

`Synchronize access` : Enter **`1`** for **`Yes`** to Synchronize access to the library with other Alliance applications. Enter **`2`** for **`No`** if you do not want to use access controls. If you are using Auto-Scan or other Alliance library operations you should enable this option to prevent collisions and object conflicts. If you are using Auto-Scan to get files from a remote server and library scan to place files to the same server, you can name the auto scan job here to synchronize the jobs.

Work With File Actions

After you add a library to the list of scanned libraries you must add file actions. File actions are defined based on the file name that is found in the library and specifies a set of actions to take for the file. A file will not be processed if the file action is not found for it. ***F6*** will add a new File Action to your Library Scan job.



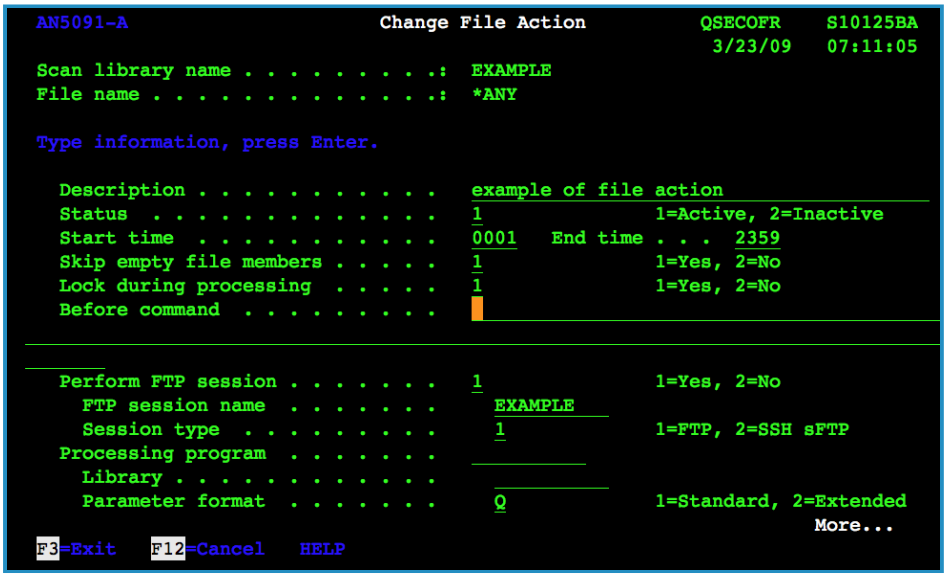
`Position to` : To position the list to a specific record enter the key value in this field and press ***Enter*** or ***F16***.

`Option` : Enter the option corresponding to the function you want to perform. A list of valid options follows:

OPTION	FUNCTION
2=Change	Use this option to change information about the item in the list.
3=Copy	Use this option to clone a file action. You will have to name it a different name.
4=Delete	Use this option to delete an item from the list and all associated detail records.
6=Print	Use this option to print a record. The printed report will be placed in the output queue that is the default for your job.

Add/Change File Action

Use this panel to change the actions you want to take for this file. These actions apply only to the file in the library that is being processed. You must enter an action record for each file in each library that you want to process.



`File name` : Enter the name of the file you want to process in the library, or **`*ANY`** . The file will be processed by library scan if it exists. You can name a file action with a specific file name, a generic name (e.g. **`ATM*`**) or **`*ANY`** . The library scan will find the first file in the library and see if there is a specific file action for that file. If it finds one it will process the file. If not, it will look for a generic file action that matches the naming of that file. If it finds one, it will process the file using that; if there is not generic file action, it will look for a **`*ANY`** file action and process it using that. If there is no **`*ANY`** it will go on to the next file and start the process again. It is for this reason we caution against using a library that has many files that will not be processed as you will take a big hit on CPU.

`Description` : Enter a Description for this file action.

`Status` : Enter **`1`** for **`Active`** if you want to process actions for this file. Enter **`2`** for **`Inactive`** if you do not want to process this file during the library scan. The status`takes effect immediately.

Start time : Enter a start time for this file action. The default is **0001** . You can control when files are processed by using a Start and End time.

End time : Enter an end time for this file action. The default is **2359** . You can control when files are processed by specifying a start and end time.

Skip empty file members : Enter **1** for **Yes** if you want to skip file members that are empty. Enter **2** for **No** if you want to process empty file members. In some cases an FTP client may create an empty file on the IBM i because it has no data to send. If you do not want to process these files specify **Yes** for this option. The default is **Yes** .

Lock during processing : Enter **1** for **Yes** if you want FTP Manager to place an exclusive lock on the file during processing. Enter **2** for **No** if you do not want FTP Manager to place an exclusive lock on the file. The default is Yes.

Before command : You can enter any single command to execute before the file is processed.

Perform FTP session : Enter **1** if you want to perform an FTP session for this file. This means sending a file to a remote server. Enter **2** if you do not want to start an FTP session for this file. If you specify **1** for **Yes** you must specify an **FTP session name** in the following field.

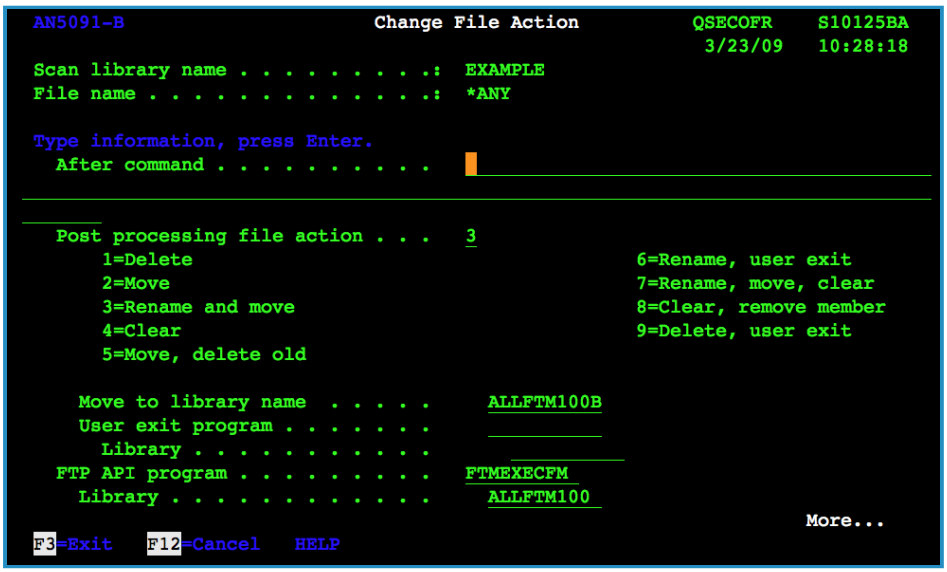
Processing program and Library : Enter the name of a program to process a received file. This program must exist at the time the file is received into the Library. FTP Manager will pass your program the parameters specified in the next option. This field is optional.

Parameter Format : Enter **1** for the **Standard** parameter format for the processing program. Enter **2** for the **Extended** parameter format for the processing program. **Standard** format will pass the name of the file, the library, and member. **Extended** format will pass the name of the file, the library, the member, all 3 User Data fields (to see these press ***Enter*** to see screen **AN5091-J**), the **File Action** name, the **Library List** , and the **Return Error** code. Your program can then process the file in any way you choose.

```
PARM(&FILE &LIBRARY &MEMBER &USERPARM1 +
  &USERPARM2 &USERPARM3 &ACTION &LIBLNAME +
  &ERROR)
```

For examples of these, look in **QCLSRC** . The standard parameter format example is **SAMPPROC** and the Extended Parameter example is **SAMPPROCE** .

Press ***Enter*** to go to the next screen.



After command : If you want to execute a command or call a program after the file processing is performed you can enter it here. The after command is executed immediately after the file is processed.

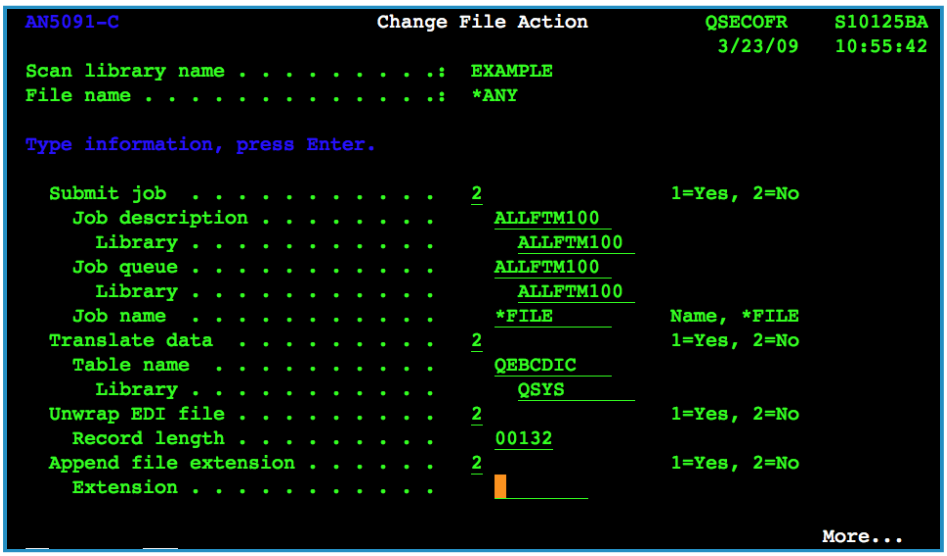
Post processing file action : This option refers to what you want to do with the file after it has been processed. Enter **1** if you want to **Delete** the file from the scan library. Enter **2** if you want to **Move** the file to a backup library. Enter **3** if you want to **Rename** the file to a unique name and then **move** it to a backup library. Enter **4** to **Clear** the file after processing. Enter **5** to **Move** the file to the backup library. If the file exists in the backup library it is deleted before the move. Enter **6** to **Rename** the file with a name you assign in a **user exit** program and move the renamed file to the backup library. Use option **7** to **Rename** the file, **move** it to a backup library, and **clear** the original file. Use option **8** to **Clear** the file then remove the member. Use option **9** to call a **user exit** program and **Delete** the file. See the notes below on the requirements for creating a user exit program. You must choose one of these options in order to prevent the file from being continually processed.

Move to library name : If you select post conversion option 2, 3, 5, 6, or 7 you must enter the name of a backup library. The library will be created if it does not exist.

User exit program and library : If you selected option 6, 7, or 9 for the post processing action you must enter the name of a program and its library for the user exit program. This is a program you create to assign a file name for the backed up file. You must assign a name that is unique and will not be found in the backup library. Your program must accept one 10 character field and return a valid file name in this field. See the sample **SAMPRTN** in RPG source.

FTP API program and library : Enter the name of the FTP API to execute to perform this transfer. This is normally **FTPEXECFM** , but you can construct your own API for special processing. Library scan will find the file, hand it to your program, and archive it. You are responsible for the transferring and error handling.

Press ***Enter*** to go to the next screen.



`Submit job` : Enter **`1`** to submit the processing for this file to batch. Enter **`2`** for **`No`** to process this file immediately. If you want to concurrently process files you can specify **`Yes`** for this option and submit the jobs to a batch subsystem. If you want to process files sequentially as they are found by library scan you should specify **`No`** for this option.

`Job description and Library` : Enter the name of a Job description and Library for the submitted job. The job description must exist before you attempt to process files.

`Job queue and Library` : Enter the name of the Job queue to use to submit the job for processing. The job will be submitted to this queue.

`Job name` : Enter the name of the submitted jobs or the special value **`*FILE`** . If you enter **`*FILE`** the submitted jobs will have the name of the file they are processing. **`*FILE`** is the default.

`Translate data` : Enter **`1`** for **`Yes`** if you want to translate the data in the file before processing. Enter **`2`** for **`No`** if you do not want to translate data before processing. If you specify **`Yes`** for translation you must enter a translation Table name and Library in the following fields. The translation option can be helpful if you are processing a file that has been received by FTP in binary mode.

`Table name and Library` : Enter the name of the table you want to use for translation. The default is **`QEBCDIC`** for **`ASCII`** to **`EBCDIC`** translation. You can enter any valid table name and library in this field.

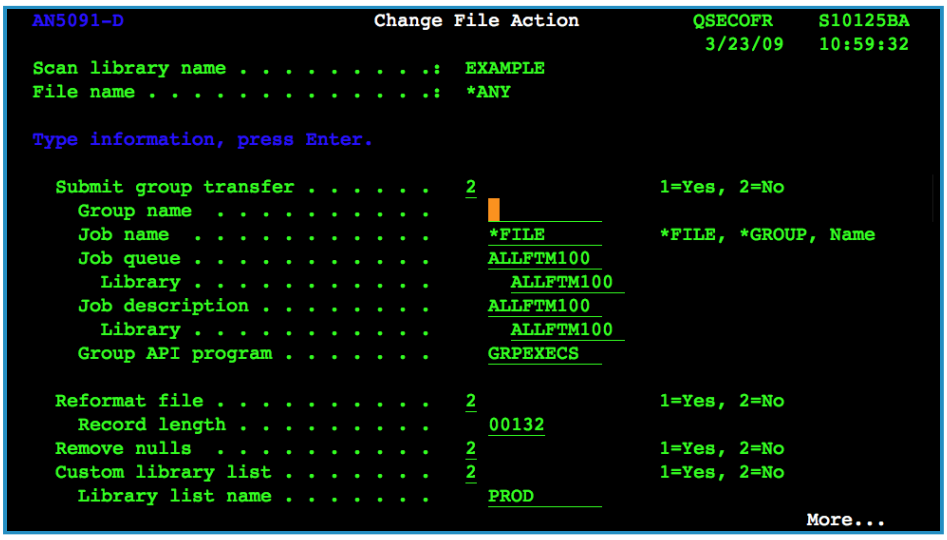
`Unwrap EDI file` : Enter **`1`** for **`Yes`** to unwrap an EDI file. Unwrapping the EDI data will put one segment per record in the output file. This can be helpful when integrating Alliance with an EDI translator. Enter **`2`** for **`No`** to not unwrap EDI data.

`Record Length` : If you specified **`Yes`** to unwrap EDI data, enter the **`Record length`** of the new file. This must be a value from **`00106`** to **`00256`** .

`Append file extension` : Enter **`1`** for **`Yes`** if you want to append a file extension to the remote file name. Enter **`2`** for **`No`** if you do not want to append an extension to the remote file name. This option can be helpful if you are using library scan to send Excel CSV, text, or HTML files to another system. If enabled the extension will be appended to the remote file name.

Extension : If you selected the option to append a file extension, enter the Extension in this field. It should be a valid file extension with the period. For example **.csv** .

Press ***Enter*** to go to the next screen.



Submit group transfer : Enter **1** for **Yes** if you want to process the file to a group of sites. Enter **2** for **No** if you do not want to send this file to a group of sites. The default is **No** . If you specify **Yes** you must specify an FTP group name that you previously defined to FTP Manager.

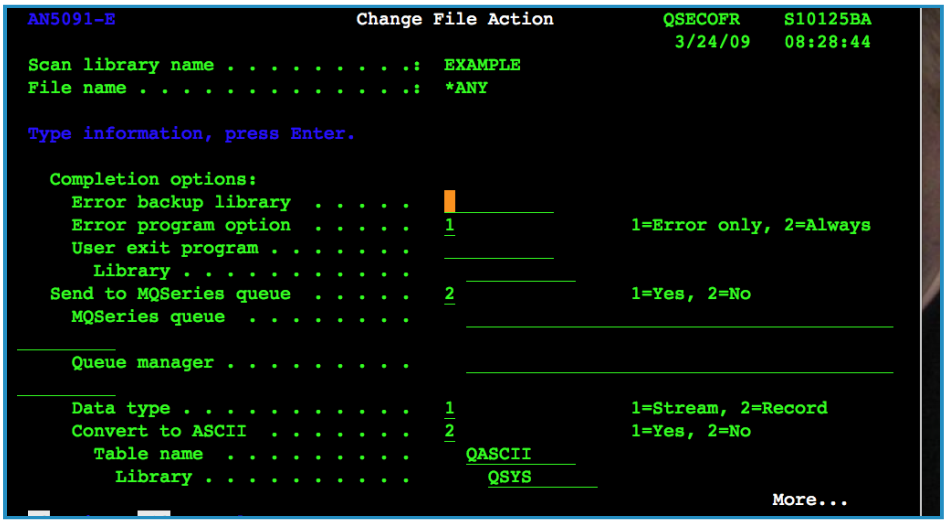
Group name : Enter the name of the FTP Group you want to use for the file transfer. The FTP Group must be defined to FTP Manager and contain one or more sites to send the files.
Job name : Enter the name of the job that is submitted to process the group transfer. You can specify the special value ***FILE** to name the job using the name of the file that is being processed. You can also specify the special value ***GROUP** to assign the job name as the name of the group.
Job queue and Library : Enter the name of the Job queue and Library to be used for submitting this group transfer. The job queue must exist before you process the group transfer. The default job queue is **ALLFTM100** in library **ALLFTM100** which is provided by FTP Manager.

Job description and library : Enter the name of the Job description and Library to be used for submitting this group transfer. The job description must exist before you process the group transfer. The default job description is **ALLFTM100** in library **ALLFTM100** which is provided by FTP Manager.
Group API program : Enter the name of the API (Application Program Interface) you want to use to process this group. The default is **GRPEXECC** which will process the transfers concurrently. You can also use **GRPEXECS** which will process the transfers sequentially. These two API programs are provided by FTP Manager. You can also specify your own program to process the group transfer.

Reformat file : Enter **1** for **Yes** if you want FTP Manager to reformat the record length of the file. Enter **2** for **No** if you do not want to reformat the record length of the file. The default is **No** . If you enter **Yes** for this option you must enter a record length in the next field.

Custom library list : Enter **1** for **Yes** if you would like to use a Custom library list for this file action. Enter **2** for **No** if you do not want to use a Custom library list. The default is **No** . If you enter **Yes** for this option you must enter a Library list name in the next field.

Press ***Enter*** to go to the next screen.



`Error back up library` : Enter the name of a library to receive the file if an error occurs in the library scan operation. The error might be on a file copy, call to a user processing program, FTP session, etc. If you define a library in this field the file being processed will be moved to this library.

`User exit program and Library` : If you specify a User exit program and Library, the program will be called if there is an error processing the file. See the source file **`QCLSRC`** for an example of an error user exit program (**`SAMPERREXT`**).

`Send to MQSeries queue` : Enter **`1`** for **`Yes`** if you want to send the file to an MQSeries message queue. Enter **`2`** for **`No`** if you do not want to send the file to an MQSeries queue. The default is **`2`** for **`No`** .

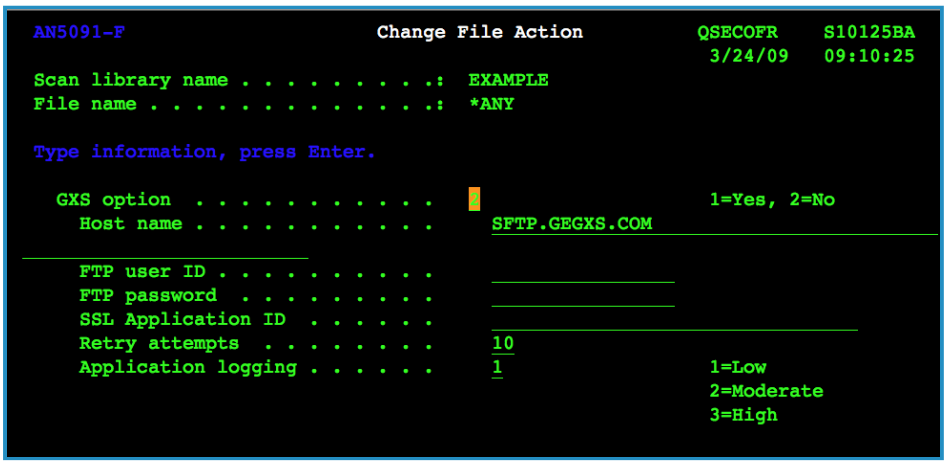
`MQSeries queue` : If you specified **`Yes`** to send the file to an MQSeries message queue, enter the name of the message queue in this field.

`Data type` : You can send data to the MQSeries queue in either **`Stream`** or **`Record`** format. Specify **`1`** if you want to use **`Stream`** format. Specify **`2`** if you want to use **`Record`** format. If you specify **`1`** Alliance will send all of the data in the file as one stream of data.

`Convert to ASCII` : Enter **`1`** for **`Yes`** if you want to convert the data to ASCII before sending. Enter **`2`** for **`No`** to send the data in EBCDIC.

`Table name and Library` : If you specified **`Yes`** to convert the data to ASCII enter the name of the table and library. You can use the IBM table **`QASCII`** in library **`QSYS`** , or you can use the Alliance table **`TMEB2AS`** in the product library. The Alliance table may handle conversion of special characters better than the IBM table.

Press ***Enter*** to go to the next screen.

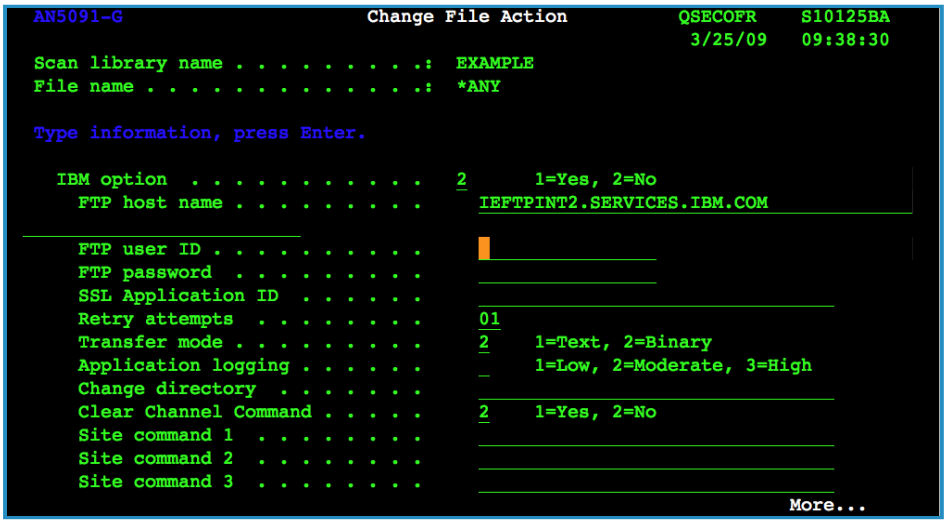


GXS option : Enter **1** for **Yes** if you want to send files to the GXS network using SSL FTP. Enter **2** for **No** if you do not want to send files to the GXS network.

FTP user ID : If you specified **Yes** to send files to GXS, enter the **FTP user ID** for the GXS FTP server. **FTP password** : If you specified **Yes** to send files to GXS, enter the **FTP password** for the GXS FTP server. **SSL application ID** : If you specified **Yes** to send files to GXS, enter the SSL application ID to use for the FTP server. The SSL application ID is created by Digital Certificate Manager. Please consult the Alliance Installation and User Guide for information on creating an SSL application ID and assigning a certificate to the SSL connection.

Retry attempts : Enter the number of Retry attempts for the file transfer. It is recommended that you choose a value of **02** or higher. You can adjust this value as you gain experience with the GXS network. The range of values is from **01** to **99** .

Press ***Enter*** to go to the next screen.



IBM option : Enter **1** for **Yes** if you want to send files to the IBM (Advantis) Global Network using SSL FTP. Enter **2** for **No** if you do not want to send files to the IBM network.

FTP user ID : If you specified **Yes** to send files to IBM, enter the **FTP user ID** for the IBM FTP server. The user ID is in the format Account.UserID. If your account is ABCD and your user ID is ABCD0001 the user ID for IE-FTP would be ABCD.ABCD0001.

FTP password : If you specified **Yes** to send files to IBM, enter the FTP password for the IBM FTP server.

> ***NOTE:*** The IBM network uses two levels of passwords. The first level is the network password, and the second level is the Information Exchange password. You may find it easier to manage your IBM account by setting the network and user ID to the same password. If you do this you cannot set the password to the user ID. For example, if your account is ABCD and your user ID is ABCD001, the password for both levels cannot be ABCD001. You can manage Information Exchange passwords at this URL: http://ieas.services.ibm.com

You can manage the network password at this URL:
http://globalnetwork.support.att.com

`SSL application ID` : If you specified **`Yes`** to send files to IBM, enter the **`SSL Application ID`** to use for the FTP server. The SSL Application ID is created by Digital Certificate Manager. Please consult the Alliance Installation and User Guide for information on creating an SSL application ID and assigning a certificate to the SSL connection. You will need to receive the PKIX certificate from IBM. Please see the Alliance installation instructions for IBM PKIX certificates.

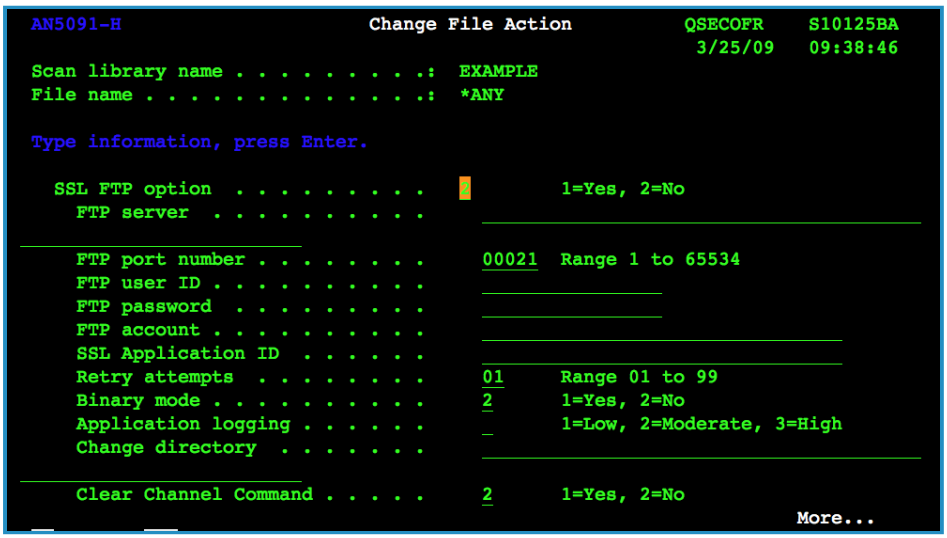
`Retry attempts` : Enter the number of Retry attempts for the file transfer. It is recommended that you choose a value of **`01`** or higher. You can adjust this value as you gain experience with the IBM network. The range of values is from **`01`** to **`99`** .

`Change directory` : Enter the target account, user ID and class for the file you are sending to the IBM Network. For example, if you are sending an EDI file to an account named PART and user ID of PART001, you would specify the following for the directory:

```
PART.PART001/#E2
```

Note that the #E2 class is used for ANSI X12 documents on the IBM network. The **`Change directory`** field is required even if you want to send documents to your own account.
Press ***Enter*** to go to the next screen.

When sending files using SSL FTP, please make sure you have DCM installed and configured and you have your Application ID. If you are not sure, or need help, please contact support for a conference call and walkthrough of SSL and DCM.



`SSL FTP option` : Enter **`1`** to enable an SSL FTP transfer of a file. Enter **`2`** to disable SSL FTP transfer.

`FTP server` : Enter the IP address or DNS name of the FTP server you want to connect to.

`FTP port number` : Enter the port number of the FTP server. The default is port **`00021`** . You can change the port number if the FTP server uses a non-standard port.

`FTP user ID` : Enter the user ID for the FTP server log in.

`FTP password` : Enter the password for the FTP server.

`FTP account` : If the FTP server requires an account name, enter the account name in this field.

`SSL Application ID` : Enter the Digital Certificate Manager Application ID to be used for this SSL connection.

`Retry attempts` : Enter the number of attempts to retry the session in the event the session fails. Enter a number from **`01`** to **`99`** .

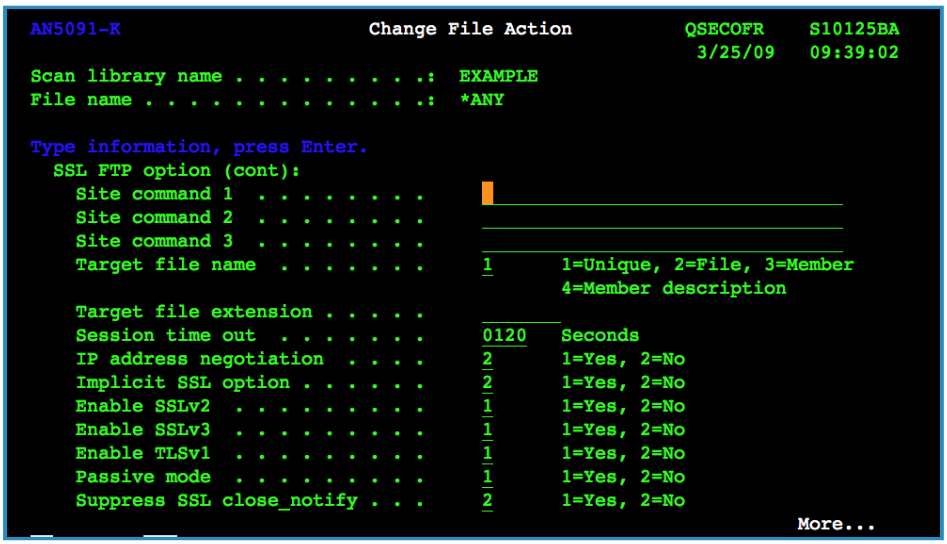
`Binary mode` : Enter **`1`** for **`Yes`** to transfer the file in binary mode. Enter **`2`** for **`No`** to transfer the file in **`ASCII`** text mode.

`Application logging` : Enter **`1`** for minimal FTP session logging. Enter **`2`** for intermediate level logging. Enter **`3`** for maximum logging of the FTP session.

`Change directory` : If you want to change to a different directory on the remote server before transferring the file enter the new directory name in this field.

`Clear Channel Command` : Enter **`1`** for **`Yes`** to send the Clear Channel Command (`CCC`) to the remote FTP server after logging in with the user ID and password. The `CCC` command will cause the FTP command channel to be non-secure and allow firewall negotiation. Note that some servers do not support the `CCC` command and will return an error if it is sent.

Press ***Enter*** to go to the next screen.



`Site command 1, 2, 3` : If the remote FTP server needs special **`SITE`** commands you can enter up to three site commands to be sent to the server.

`Target file name` : Enter **`1`** to use a uniquely generated file name for the remote file. Enter **`2`** to use the original file name in the IFS directory. Enter **`3`** to use the member name of the file. Enter **`4`** to use the description of the physical file member.

`Target file extension` : You can specify a special file extension of the remote file by entering the extension in this field.

`IP address negotiation` : Enter **`1`** for **`Yes`** to use IP address negotiation. In some cases a remote FTP server behind a firewall will respond to a port command with the internal address of the FTP server. When IP address negotiation is enabled Alliance will ignore the internal IP address of the FTP server and use the external IP address used to connect to the server. Enter **`2`** to not use IP address negotiation.

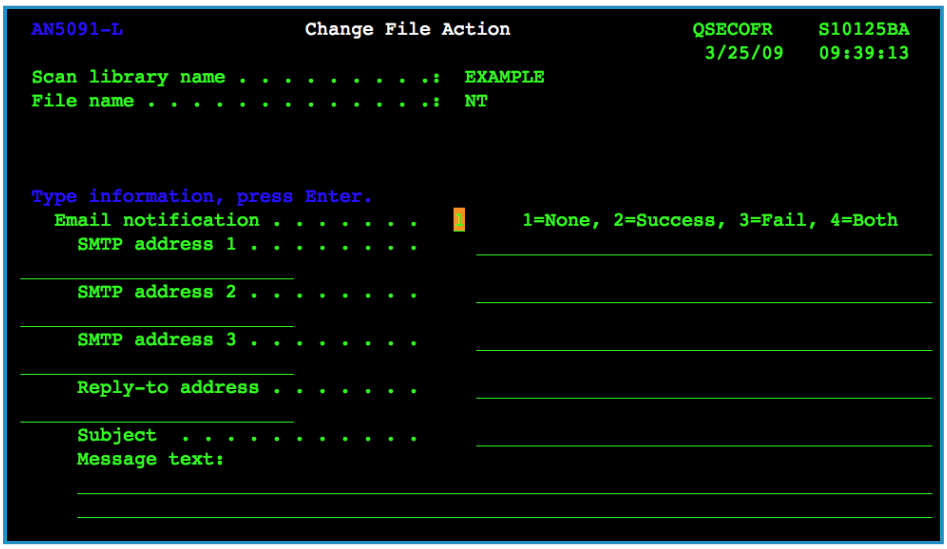
`Implicit SSL option` : Enter **`1`** for **`Yes`** to use implicit SSL connections to the FTP server. Implicit SSL negotiation is used by some older FTP servers that do not implement the RFC 2228 standard for SSL FTP.

`Enable SSLv2, SSLv3, TLSv1` : By default Alliance SSL FTP will enable all three methods of SSL encryption for the transfer. You can control which methods are actually used with these options.

`Passive mode` : Enter **`1`** for **`Yes`** to use FTP passive mode. Enter **`2`** for **`No`** to not use FTP passive mode. Using passive mode can help resolve firewall negotiation problems.

Press ***Enter*** to go to the next screen.

This panel is for sending emails specific to the SSL FTP transfer.



`Email notification` : Enter **`1`** if you do not want to use email notification. Enter **`2`** to send an email notification if the SSL session is successful. Enter **`3`** to send an email notification only if the SSL FTP transfer fails. Enter **`4`** to send an email notification regardless of the success or failure of the transfer. In this case the message will indicate the status of the transfer.

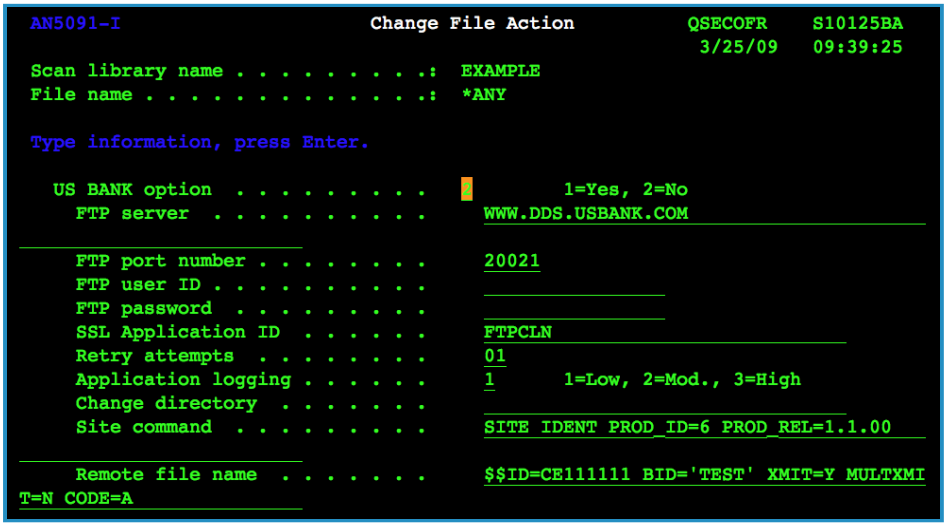
`SMTP email address 1, 2, 3` : Enter up to three email addresses for notification. You can use a distribution list as well.

`Reply-to address` : Enter the Reply-to address for the email message. This does not need to be an email address on the IBM i system.

`Subject` : Enter a subject for this message.

`Message text` : Enter up to three lines of message text for this message.

Press ***Enter*** to go to the next screen.



`US BANK option` : Enter **`1`** to enable an SSL FTP transfer of a file to US Bank. Enter **`2`** to disable US Bank SSL FTP transfer.

`FTP server` : Enter the IP address or DNS name of the US Bank FTP server you want to connect to.

FTP port number : Enter the port number of the FTP server. The default is port **20021** . You can change the port number if the US Bank FTP server uses a non-standard port.

FTP user ID : Enter the user ID for the FTP server log in.

FTP password : Enter the password for the FTP server.

SSL Application ID : Enter the Digital Certificate Manager Application ID to be used for this SSL connection.

Retry attempts : Enter the number of attempts to retry the session in the event the session fails. Enter a number from **01** to **99** .

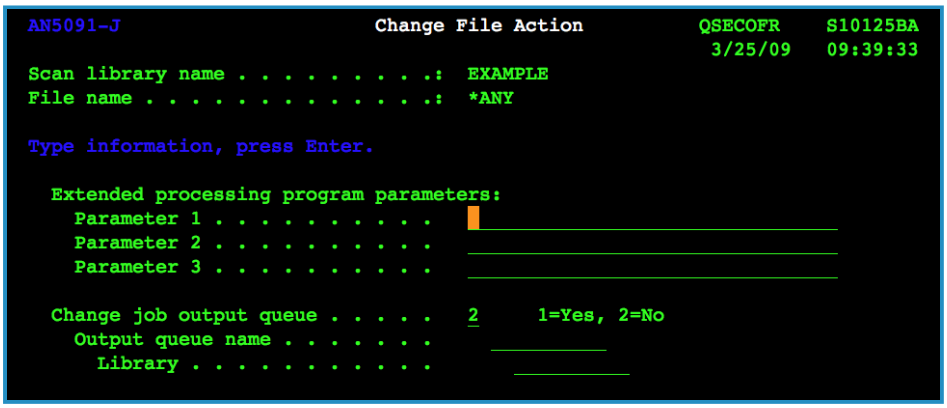
Application logging : Enter **1** for minimal FTP session logging. Enter **2** for intermediate level logging. Enter **3** for maximum logging of the FTP session.

Change directory : If you want to change to a different directory before transferring the file enter the new directory name in this field.

Site command : Enter the site command specified by US Bank.

Remote file name : Enter the remote file name format that you received from US Bank.

Press ***Enter*** to go to the next screen.

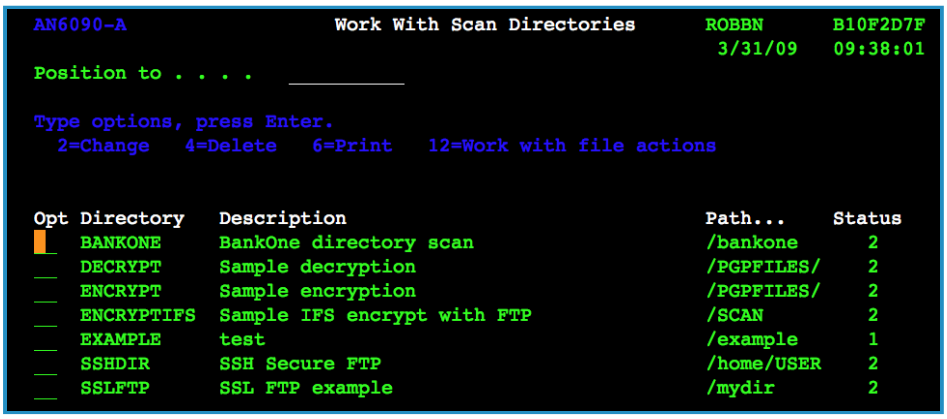


Extended processing program parameters : If you are using the extended parameter option for a Processing Program you can enter up to three user-specified parameters to pass to the program.

Change job output queue : Enter **1** for **Yes** to change the job output queue. Enter **2** for **No** to not change the default output queue for the library scan job. If you specify **Yes** for this option enter an Output queue name and Library.

Work With Scan Directories

You can use this option to define IFS or QDLS directories to scan for automatic processing. If you plan on using **QDLS** , please read FAQ #543 in the support site regarding how to set it up. The directory scan routine will search the list of directories and process files one at a time. After defining a directory with this panel you must use Option **12** to define the IFS files to be processed in the directory and the actions to be taken. It is recommended that you do not define production application directories to directory scan. Rather, you should create directories for scanning and define them to FTP Manager. This will help avoid unintentional errors processing directory processing.



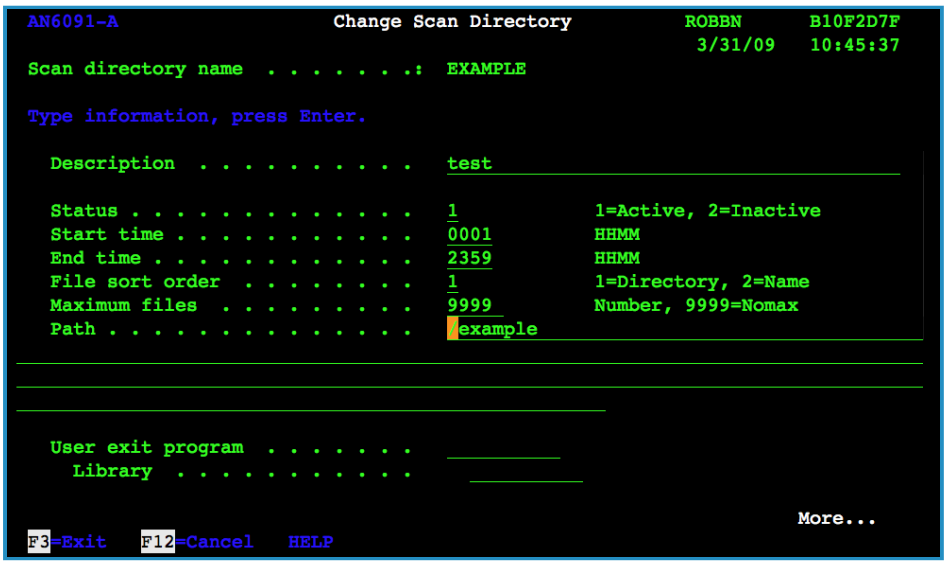
`Position to` : To position the list to a specific record enter the key value in this field and press ***Enter*** or ***F16***.

`Option` : Enter the option corresponding to the function you want to perform. A list of valid options follows:

OPTION	FUNCTION
2=Change	Use this option to change information about the item in the list.
4=Delete	Use this option to delete an item from the list and all associated detail records.
6=Print	Use this option to print a record. The printed report will be placed in the output queue that is the default for your job.
12=Work with file actions	Use this option to work with the files to be processed by FTP Manager.

Add/Change Scan Directory

Use this panel to change information about a `Scan directory` . You can change the description and the active or inactive status of the `Scan directory` .



`Description` : Enter a description **`9999`** for this scan directory.

`Status` : Enter **`1`** for **`Active`** or **`2`** for **`Inactive`** . If you enter **`2`** for **`Inactive`** , the directory will not be processed by the scan directory function.

`Start time` : Enter a Start time for this IFS directory scan. The default is **`0001`** . You can use the Start and End time to control when IFS directory scans occur.

`End time` : Enter an End time for this IFS directory scan. The default is **`2359`** . You can use the Start and End time to control when IFS directory scans occur.

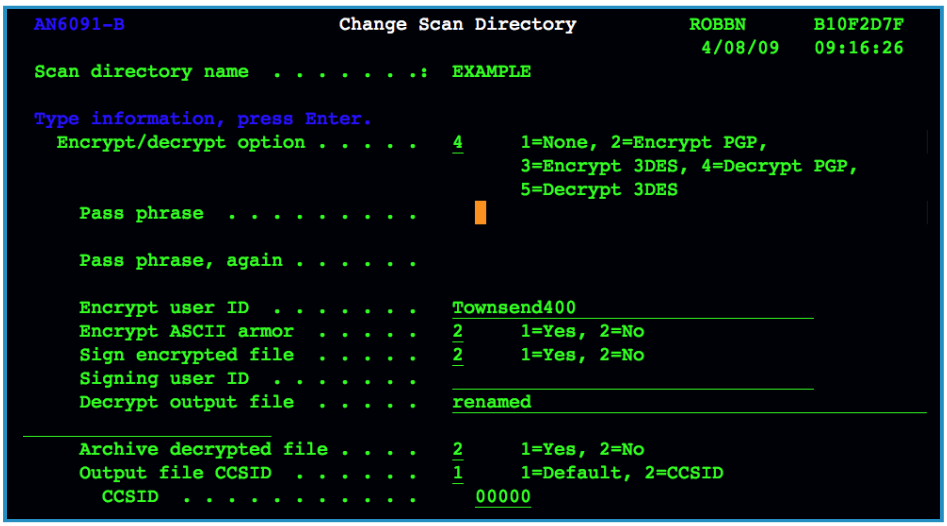
`File sort order` : Enter **`1`** to process files in the order they are retrieved from the directory. Enter **`2`** to process files sorted by name.

`Maximum files` : This allows you to specify a maximum number of files on each cycle. Or you can specify **`9999`** for **`No maximum`** . You may want to specify a maximum number if you have a very large number of files to process and choose to do them in smaller batches.

`Path` : Enter the full path for the directory. Enter the leading slash to indicate that the directory is specified from the root. You do not need to specify the trailing slash. For shared folder directories add the leading **`/QDLS/`** directory name.

`User exit program and Library` : You can register your own program to process an IFS file by entering the program name and library in these fields. If you specify an exit program, Directory scan will pass your program the file and archive it. You are responsible for everything else.

Press ***Enter*** to go to the next screen.



Decrypt option : Enter **1** for no decryption. Enter **4** to decrypt this IFS file with PGP.

Pass phrase : If you selected to decrypt you must enter the pass phrase or password to use with the decryption key. You must enter the password twice.

Encrypt user ID : If you are decrypting a file using PGP enter the **user ID** for key. If you do not know the recipient's user ID you can use the PGP View Key (**PGPKEYVIEW**) command. Note that key user IDs are case sensitive.

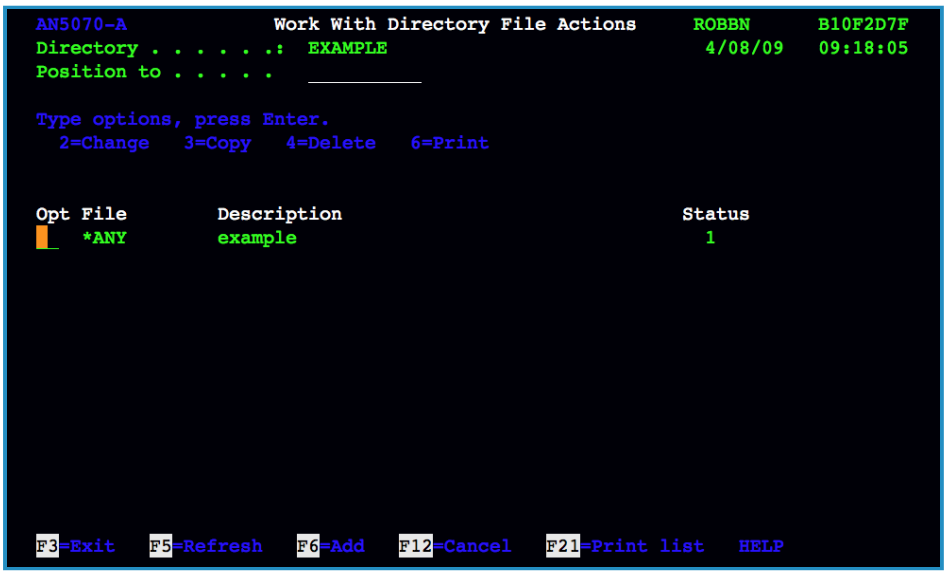
Decrypt output file : When you decrypt a file you can specify an override name for the decrypted output file. Normally PGP creates the output file with the same name as the encrypted file without the **.pgp** or **.asc** extension. In some cases you may want to override the default output name. You can put the name in this field. Do not specify any path information, just the actual name of the file.

Output file CCSID : Enter the value **1** to use the default **CCSID** of the decrypted file. Enter the value **2** to specify the **CCSID** you want to use. If you are making decrypted files available to users with Windows Networking support you should use option **2** and the **CCSID** of **1252** . This will provide the best character interface for Windows Networking directories.

CCSID : If you specified **2** for the output file **CCSID** enter the numeric **CCSID** you want to use. Alliance will convert the output file to this **CCSID** . You must enter a valid **CCSID** . Consult the IBM National Language Support document for a list of CCSIDs.

Work With Directory File Actions

After you add a directory to the list of scanned directories you must add file actions. File actions are defined based on the file name that is found in the directory and specifies a copy action for the file. The file can be copied to any library on the IBM i. The name of the file in the target library can be the same as the file in the directory, can be automatically generated, or can be modified for IBM i name standards.



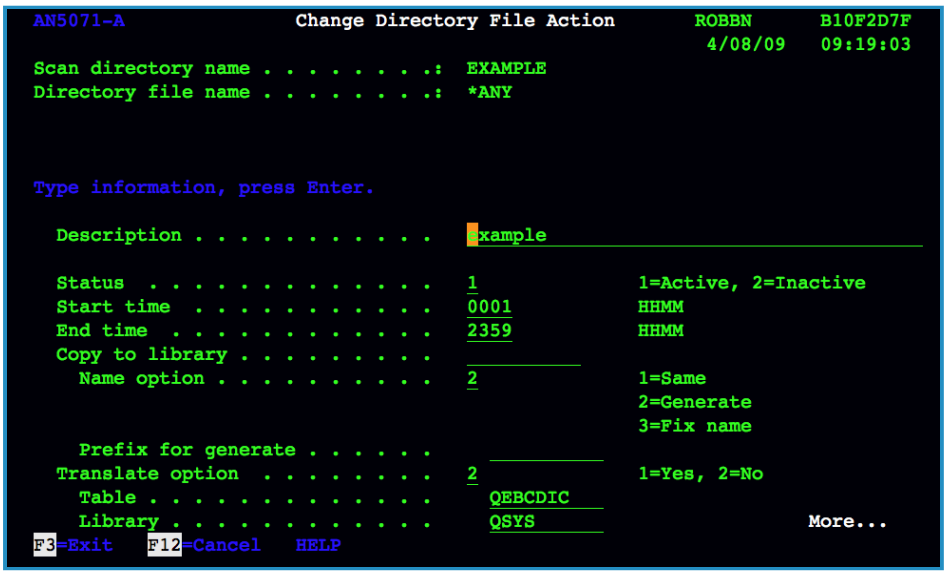
`Position to` : To position the list to a specific record enter the key value in this field and press ***Enter*** or ***F16***.

`Option` : Enter the option corresponding to the function you want to perform. A list of valid options follows:

OPTION	FUNCTION
2=Change	Use this option to change information about the item in the list.
3=Copy	Use this option to copy all the information in this file action to create a new one. You will have to name the newly created file action a different name.
4=Delete	Use this option to delete an item from the list and all associated detail records.
6=Print	Use this option to print a record. The printed report will be placed in the output queue that is the default for your job.

Add/Change Directory File Action

Use this panel to change the actions you want to take for this directory file. These actions apply only to the file in the directory that is being processed. You must enter an action record for each file in each library that you want to process. You can look at Appendix D for details around setting up a Directory Scan and how to work with File Actions.



`Directory File name` : Enter the name of the file you want to process in the directory, a generic name, or **`*ANY`** . If you enter **`*ANY`** for the file name, all files will be processed using this definition, unless a specific file action or generic file aciton exists for that file.

`Description` : Enter a **`Description`** for this file action.

`Status` : Enter **`1`** for **`Active`** if you want to process actions for this file. Enter **`2`** for **`Inactive`** if you do not want to process this file during the library scan. The status takes effect immediately.

`Start time` : Enter a start time for this IFS directory file action. The default is **`0001`** . You can use the start and end time to control when files are processed.

`End time` : Enter an end time for this IFS directory file action. The default is **`2359`** . You can use a start and end time to control when files are processed.

`Copy to library` : For inbound processing, enter the name of the library into which to copy this directory file. The library must exist before the copy operation takes place.

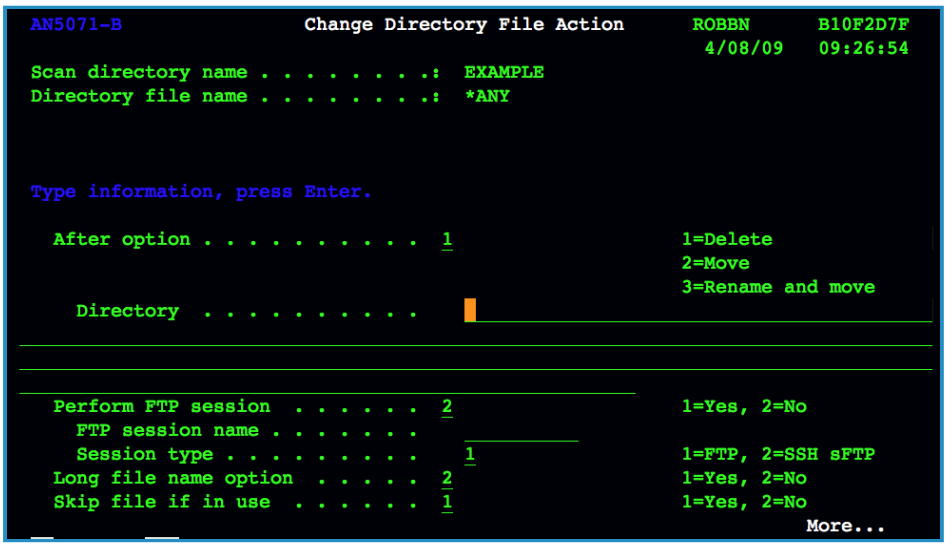
`Name option` : Enter **`1`** to use the same name as the IFS file. If you use this option you must be sure the IFS file name is a valid IBM i file name. Enter **`2`** to have Alliance automatically generate a name for the file. The name will be unique and will be a valid IBM i file name. Enter **`3`** to attempt to use the IFS file name after it has been adjusted for IBM i file name conventions.

`Prefix for Generate` : If you specify a name option of generate you can specify a prefix for the name. Enter the prefix followed by spaces. If an **`*`** is entered into the first position of the prefix field, up to the next 9 characters will be used to give it a specific name, not just a prefix. If the name is a full 10 characters, then an **`*`** is not needed in the first position.

`Translate to EBCDIC` : Enter **`1`** for **`Yes`** to translate the file to **`EBCDIC`** . Enter **`2`** for **`No`** to not perform data translation.

`EBCDIC table and library` : If you specified **`Yes`** to translate to **`EBCDIC`** , enter a table name and library. The default **`QEBCDIC`** table is provided by IBM.

Press ***Enter*** to go to the next screen.



`After option` : This option determines what you want to do with the directory file after it has been processed. Enter **`1`** if you want to delete the file from the scan directory. Enter **`2`** if you want to move the file to a backup directory. Enter **`3`** if you want to rename the file to a unique name and then move it to a backup library. We will rename it with an **`Fxxxxxxxxx`** name. You must choose one of these options in order to prevent the file from being continually processed.

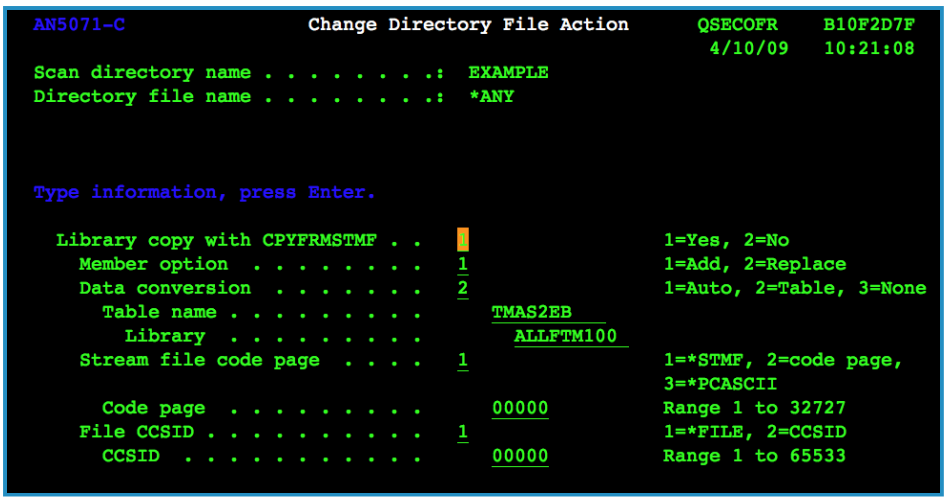
`Directory` : If you choose option **`2`** or **`3`** you must enter the name of a backup directory. The directory will be created if it does not exist.

`Perform FTP Session` : Enter **`1`** for **`Yes`** if you want to transmit this file with an FTP session. Enter **`2`** for **`No`** if you do not want to use FTP. The FTP session name is the name of the definition you have created for the send. And if you are using SSH sFTP you must signify this with an option **`2`** .

`Long File Name Option` : Enter **`1`** for **`Yes`** if you want to use long file name processing. Enter **`2`** for **`No`** to use standard file name processing. If the length of the directory and file you are processing exceed 64 bytes you should choose the long file name option.

`Skip File if in Use` : Enter **`1`** for **`Yes`** if you want to avoid processing files that are in use. Enter **`2`** for **`No`** to not skip these files.

Press ***Enter*** to go to the next screen.



Copy to library : You can choose to convert files into a library using the **CPYFRNSTF** command by entering **1** for **Yes** . If you want to use another method, choose **2** for **No** . If you choose **2** for **No** , Alliance will ignore the remaining option on this screen.

Member option : Enter **1** to **Add** data to a member. Enter **2** to **Replace** data in a file member.

Data conversion : Enter **1** for **Automatic** data translation. The code page of the stream file and the **CCSID** of the target file are used to do the data conversion. Enter **2** to use a conversion **Table** . If you select this option you must enter a table name and library in the following field. Enter **3** to not perform any data translation. The copy will be a binary copy. The default is option **1** to automatically convert the file.

Table name and Library : If you selected option **2** to use a conversion table enter the table name and library. The conversion table can be an IBM supplied table, a table provided by Alliance, or a table you create. See the source file **QTBLSRC** in the product library for an example of source used to create a conversion table.

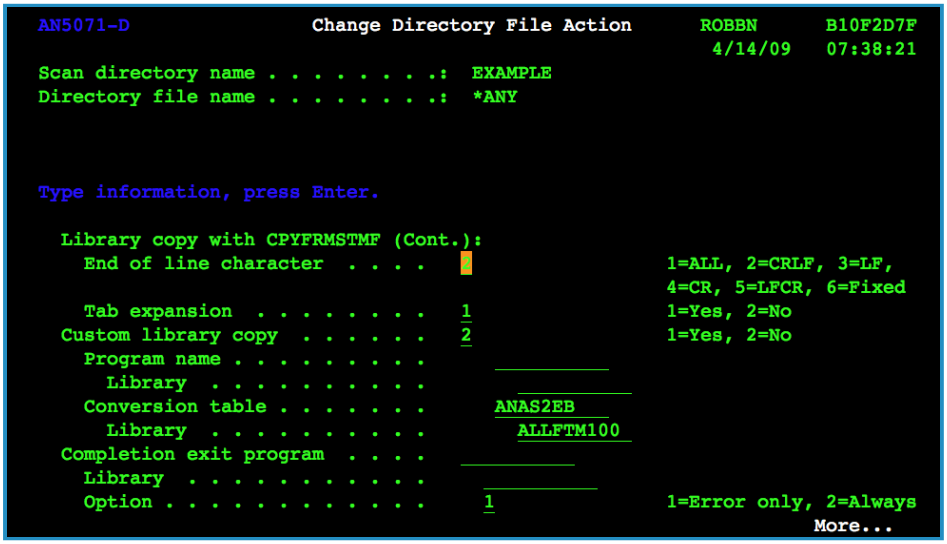
Stream file code page : Enter **1** to use the code page of the file in the IFS directory. This is the default. Enter **2** to use a specific code page. If you specify option **2** you must enter a code page number in the following field.

Code page : If you selected option **2** for the code page you must enter a code page number in this field.

File CCSID : Enter **1** to use the **CCSID** (Coded Character Set ID) of the target file. Enter **2** to specify the code page for the target file. If you select option **2** you must specify a **CCSID** in the following field.

CCSID : If you specified **2** for the previous option enter a **CCSID** in this field.

Press ***Enter*** to go to the next screen.



End of line character : Enter **1** to use all standard end of line character sequences. Enter **2** to use Carriage Return and Line feed for the end of line character. Enter **3** to use a Line Feed character to indicate end of line. Enter **4** to use Carriage Return for the end of line. Enter **5** to use Line Feed and Carriage Return for the end of line. Enter **6** to assume a fixed line length for the IFS stream file.

Tab expansion : Enter **1** for **Yes** to expand tabs. Enter **2** for **No** to not expand tabs.

Custom Library Copy : Enter **1** for **Yes** to use a custom copy program. Enter **2** for **No** to not use a custom copy option. You can either use **CPYFRMSTF** or **Custom library copy** option.

Program name and library : If you specified **Yes** to use a custom copy program enter the **Program name** and **Library** . Press ***F1*** to get a full description of how each of the of the options work. You also need to enter a conversion table. The default is **ANAS2EB** in library **ALLFTM100** .

Completion user exit program and Library : You can specify a user program and library in this field. The program will be called if an error occurs in the scan directory operation. The program will be passed the IFS file name, error code, FTP Manager log ID, and PGP log ID. Your program must accept the following parameters:

IFS path (character, 256)

IFS directory (character, 256)

IFS file name (character, 256)

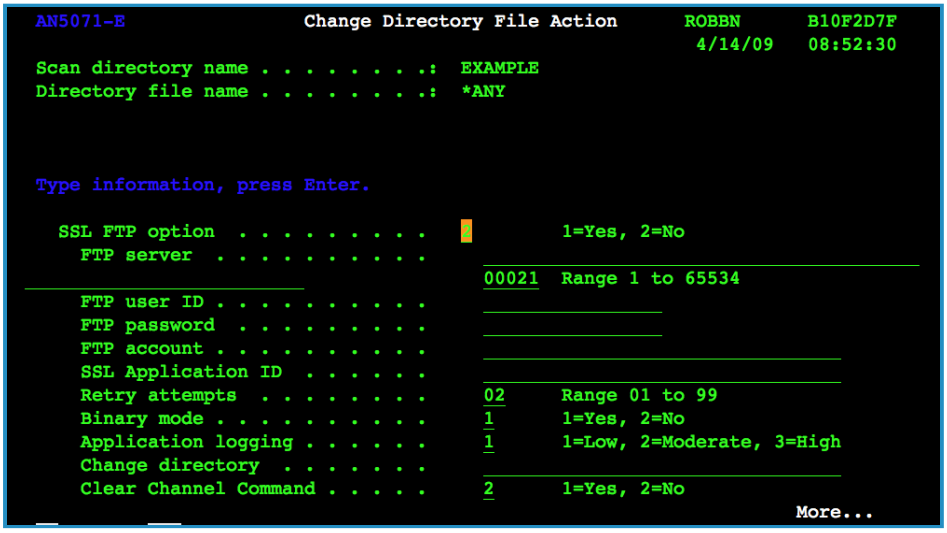
Error code (character, 4)

Log ID (character, 10)

PGP Log ID (character, 10)

You must take care in your user exit program to not halt on an inquiry message. If your application generates a halt it will interrupt normal directory scan processing.

Press ***Enter*** to go to the next screen.



SSL FTP option : Enter **1** to enable an SSL FTP transfer of a file. Enter **2** to disable SSL FTP transfer.

FTP server : Enter the IP address or DNS name of the FTP server you want to connect to.

FTP user ID : Enter the user ID for the FTP server log in.

FTP password : Enter the password for the FTP server.

FTP account : If the FTP server requires an account name, enter the account name in this field.

SSL Application ID : Enter the Digital Certificate Manager Application ID to be used for this SSL connection.

Retry attempts : Enter the number of attempts to retry the session in the event the session fails. Enter a number from **01** to **99** .

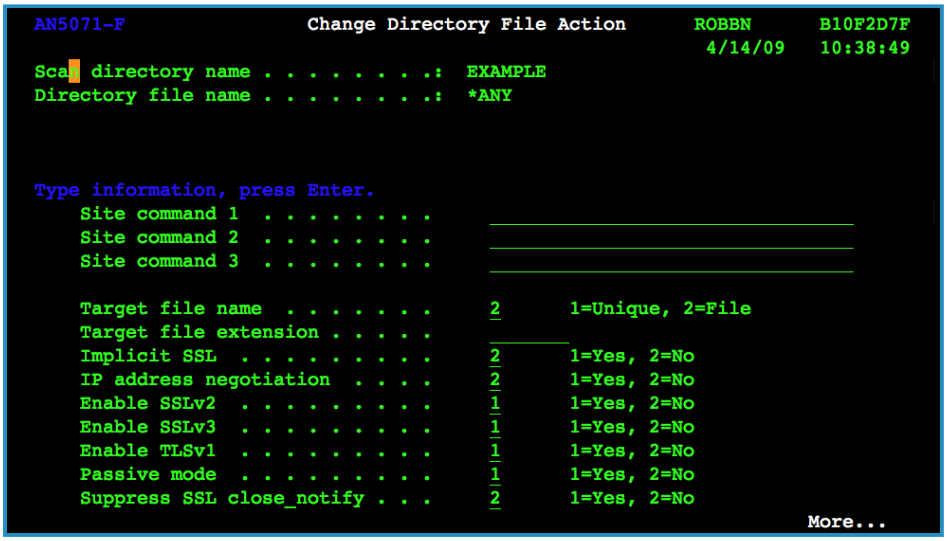
Binary mode : Enter **1** for **Yes** to transfer the file in binary mode. Enter **2** for **No** to transfer the file in **ASCII** text mode.

Application logging : Enter **1** for minimal FTP session logging. Enter **2** for intermediate level logging. Enter **3** for maximum logging of the FTP session.

Change directory : If you want to change to a different directory before transferring the file enter the new directory name in this field.

Clear Channel Command : Enter **1** for **Yes** to send the Clear Channel Command (CCC) to the remote FTP server after logging in with the user ID and password. The CCC command will cause the FTP command channel to be non-secure and allow firewall negotiation. Note that some servers do not support the CCC command and will return an error if it is sent.

Press **Enter** to go to the next screen.



Site command 1, 2, 3 : If the remote FTP server needs special **SITE** commands you can enter up to three site commands to be sent to the server.

Target file name : Enter **1** to use a uniquely generated file name for the remote file. Enter **2** to use the original file name in the IFS directory.

Target file extension : You can specify a special file extension of the remote file by entering the extension in this field.

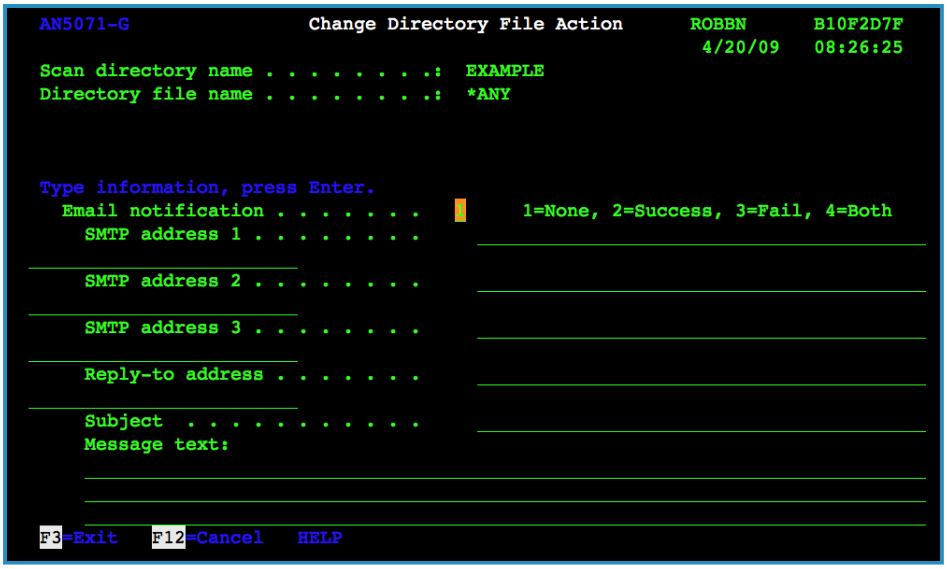
IP address negotiation : When a remote FTP server does not support the Clear Channel Command and the transfer is in passive mode, the remote FTP server can send a connection IP address that is an internal IP address. This is usually due to the inability of the remote firewall to detect the passive command response and adjust the IP address. When enabled Alliance will ignore the IP address in the response from the server and use the original IP address of the FTP server to make the data channel connection.

Enable SSLv2, SSLv3, TLSv1 : By default Alliance SSL FTP will enable all three methods of SSL encryption for the transfer. You can control which methods are actually used with these options.

`Passive mode` : Enter **`1`** for **`Yes`** to use FTP **`Passive mode`** . Enter **`2`** for **`No`** to not use FTP **`Passive mode`** . Using **`Passive mode`** can help resolve firewall negotiation problems.

`Suppress SSL close_notify` : In some cases the **`close_notify`** cannot be turned off on the remote FTP server. Alliance can handle the encrypted frame that is returned with **`close_notify`** .

Press ***Enter*** to go to the next screen.



`Email notification` : Enter **`1`** if you do not want to use Email notification for the SSL session. Enter **`2`** to send an **`Email notification`** if the SSL session is successful. Enter **`3`** to send an Email notification only if the SSL FTP transfer fails. Enter **`4`** to send an Email notification regardless of the success or failure of the transfer. In this case the message will indicate the status of the transfer.

`SMTP email address 1, 2, 3` : Enter up to three email addresses for notification.

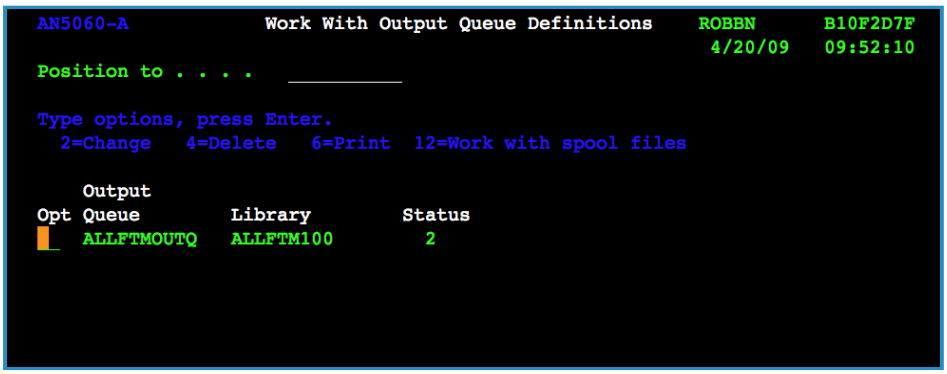
`Reply-to address` : Enter the reply-to address for the email message. This does not need to be an email address on the IBM i system.

`Subject` : Enter a subject for this message.

`Message text` : Enter up to three lines of message text for this message.

Work With Output Queues

Use this option to define output queues for spool file processing. When you add an output queue to the FTP Manager list of output queues it can be processed for automatic distribution of reports. The spool file can be copied to an IFS directory, passed to a user exit program, send to a remote site via FTP, or sent to the NSA Report.Web product for publishing on a web site. You can read Appendix F for direction on how to configure and use Output Queues.



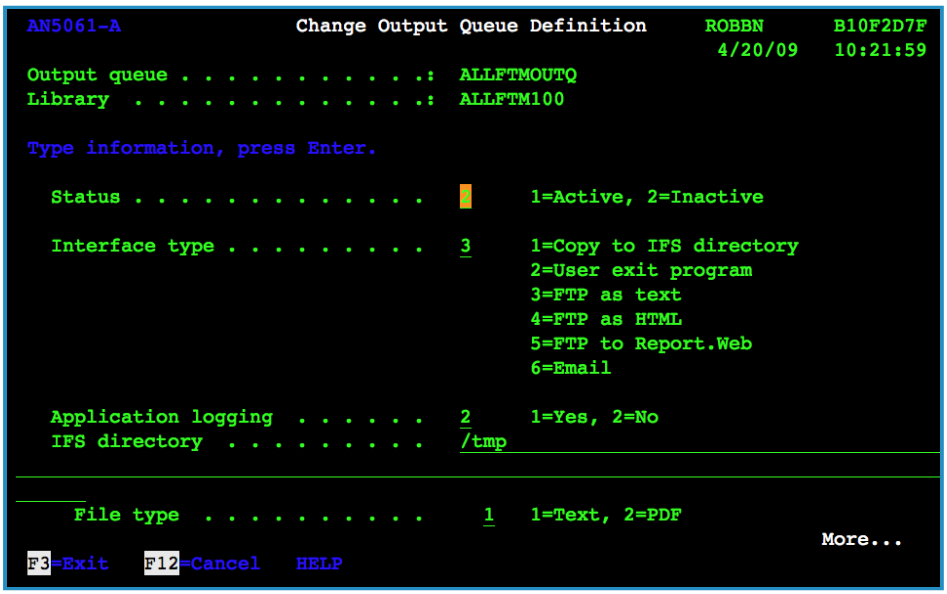
`Position to` : To position the list to a specific record enter the key value in this field and press ***Enter*** or ***F16***.

`Option` : Enter the option corresponding to the function you want to perform. A list of valid options follows:

OPTION	FUNCTION
2=Change	Use this option to change information about the item in the list.
4=Delete	Use this option to delete an item from the list and all associated detail records.
6=Print	Use this option to print a record. The printed report will be placed in the output queue that is the default for your job.
12=Work with spool files	Use this option to work with spool files for this output queue. You can define individual spool files or generic processing parameters.

Add/Change Output Queue

Use this panel to add or change information about an output queue. You can change the description, status, and processing options for the output queue. Note that when you add an output queue to FTP Manager the data queue definition of the output queue is changed to the FTP Manager data queue **`ALLFTMSPLQ`** in the product library.



Status : Enter **1** for **Active** status, or **2** for **Inactive** status. Spool files will only be processed from an output queue that has an **Active** status.

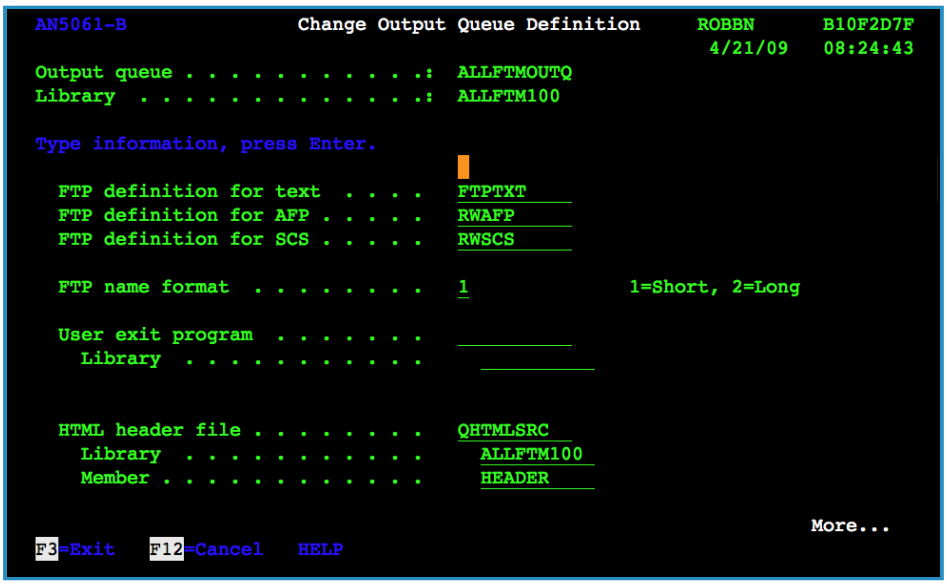
Interface type : Enter **1** to indicate that the spool files should be copied to an IFS directory. The resulting file will be an **ASCII** text file with embedded top-of-form characters for page breaks. If you choose this option you must fill in the IFS directory path and file type below. Enter **2** to indicate that a **User exit program** should process the spool file. When you select option **2** you must create and specify your own program to process the file. Enter **3** to indicate that you want FTP Manager to send the spool file to a remote system using FTP. You must specify a text FTP definition on the following panels. Enter **4** to indicate that you want FTP Manager to send the spool file to a remote system in HTML format. The HTML header and trailer will be used that you specify on the following panels. Enter **5** to indicate you want to process the spool files with the NSA **Report.Web** product. You must be using the FTP Manager version that supports the **Report.Web** interface. The files will be transferred to the NT Server that is running the **Report.Web** product. This option supports publishing Advanced Function Print reports to a web server. Enter **6** if you want the file sent via email.

Application logging : Enter **1** for **Yes** to use **Application logging** . Enter **2** for **No** to not perform any application logging. You should only enable application logging when you are trying to perform problem analysis. The logs can become quite large if the option is left enabled.

IFS directory : If you specified interface type **1** you must specify an **IFS directory** to receive the spool file reports. Specify the complete path with a leading forward slash character (**/**). Do not specify the trailing slash.

File type : If you specified interface type **1** and an IFS directory, specify the type of file for the IFS file. Enter the value **1** for a text type of file. Enter the value **2** for a PDF type of file.

Press ***Enter*** to go to the next screen.



FTP definition for text : If you specified interface type **3** on the previous panel you must specify the FTP definition for the transfer. The default is **FTPTXT** and is supplied by FTP Manager. You can create your own FTP definition or use the default. The default should be adequate for most uses.

FTP definition for AFP : If you specified interface type **5** on the previous panel you must specify an FTP definition to use to transfer Advanced Function Print (AFP) files to the **Report.Web** server. The definition **RWAFP** is provided by FTP Manager. If you do not use **RWAFP** you must first create a different definition and specify it here.

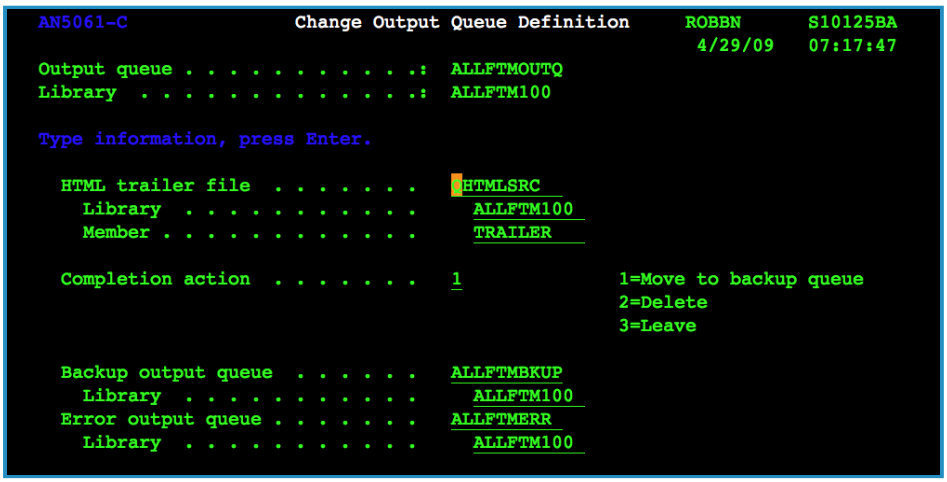
FTP definition for SCS : If you specified interface type **5** you must also specify an FTP definition to use to transfer SCS (SNA Character Stream) data to the **Report.Web** server. SCS data is the normal green-bar style output from IBM i report files. The default is **RWSCS** and is provided by Alliance FTP Manager.

FTP name format : Enter **1** for the **Short** format name. This name uses the spool file name plus an extension. Enter **2** for the **Long** format name. This name uses the entire spool file, spool file number, and job information to create a name. It is guaranteed to be a unique name.

User exit program and Library : If you specified an interface type of **2** on the previous panel you must specify the name of a program to process the spool file. You program will be called with the spool file and job information.

HTML header file, Library, and Member : If you specified an interface type of **4** on the previous panel you must specify a source file, library and member that contains the HTML header you want to use. FTP Manager provides a file named **QHTMLSRC** in the product library. The member **HEADER** is provided as a default. You can modify this source member, or provide a name of your own. The member provided by FTP Manager provides a simple web interface to standard reports.

Press ***Enter*** to go to the next screen.



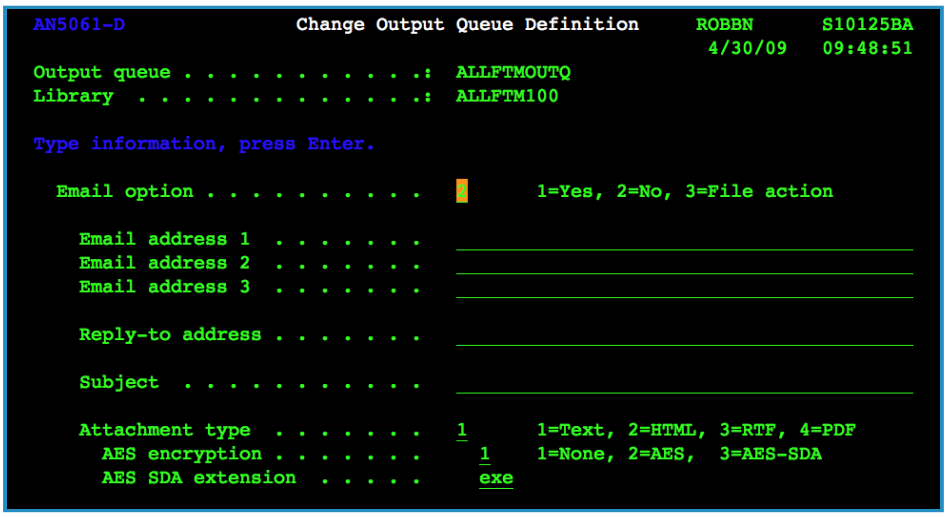
HTML trailer file, Library, and Member : If you specified an interface type of **4** on the first panel you must specify a source file, library and member that contains the HTML trailer you want to use. FTP Manager provides a file named **QHTMLSRC** in the product library. The member **TRAILER** is provided as a default. You can modify this source member, or provide a name of your own. The member provided by FTP Manager provides a simple web interface to standard reports.

Completion action : Enter **1** to move the spool file to a back up output queue after processing. This is the default. The back up output queue is **ALLFTMBKUP** in the product library. Enter **2** to delete the spool file after processing. You should be absolutely sure you do not want to save the spool file before selecting this option. Enter **3** to leave the spool file in the output queue after processing.

Backup output queue and Library : Enter the name of the output queue for receiving the spool file after processing. This parameter is only used if you specify a Completion Action that requires a backup queue. The default is **ALLFTMBKUP** in library **ALLFTM100** . You can specify any output queue for this parameter. The output queue must exist. You should be careful not to specify an output queue that is being processed by FTP Manager as you may create a loop of processing activity.

Error output queue and Library : Enter the name of an output queue to be used for receiving spool files when an error condition occurs. When Alliance encounters an error processing a spool file it will move the spool file to this output queue. This will occur even if you have specified **Delete** or **Leave** as the **Completion action** . The output queue must exist. The default is **ALLFTMERR** in library **ALLFTM100** .

Press ***Enter*** to go to the next screen.



`Email option` : Enter **`1`** for Yes if you want to send spool files in this output queue to email recipients. Enter **`2`** for No if you do not want to send the spool file as email. Enter **`3`** for **`File Action`** if you want to selectively send spool files using the file action definitions. This last option lets you select which spool files in the output queue you want to send via email. If you choose option **`1`** or **`3`** in this field you must fill out the email address fields and the **`Reply-To address`** field.

`Email address 1, 2, 3` : If you chose option **`1`** or **`3`** in the previous field you must enter at least one address here. You can enter up to three email addresses to send to. These must be valid SMTP email addresses. Note that you must first configure Alliance to send email before using spool file distribution. Also, you may need to enable SMTP mail, router, and gateway options to send email from the IBM i. You can also specify an Email distribution list in the first Email address field. The distribution list must have first been created using the configuration option for distribution lists.

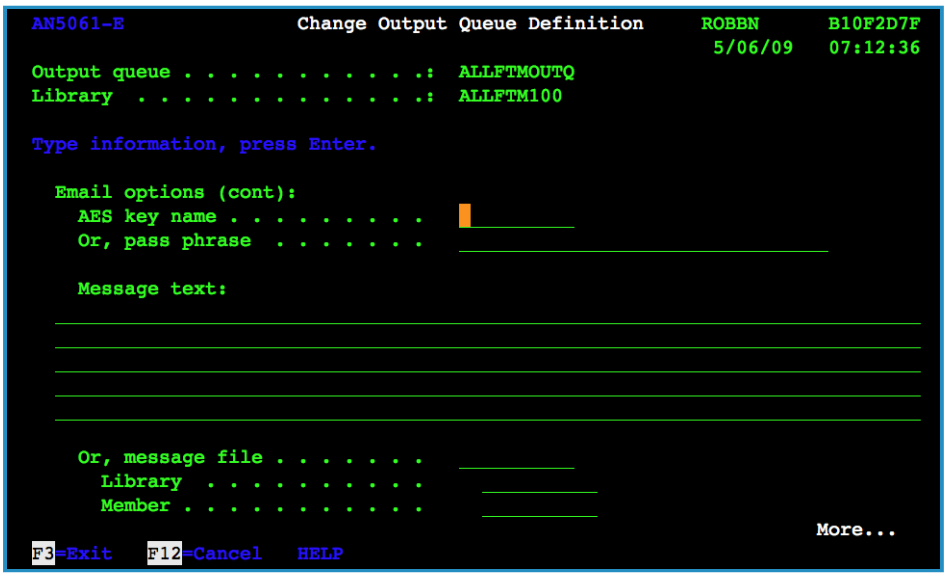
`Reply-to address` : If you chose option **`1`** or **`3`** in the **`Email option`** field, you must enter the Reply-to address for the email. This may be an IBM i email address, or may be an email address on another mail system.

`Subject` : If you chose option **`1`** or **`3`** in the Email option field , you must enter a Subject for the email message.

`Attachment type` : If you chose option **`1`** or **`3`** in the Email option field, you must choose an Attachment type.

`AES encryption` : Enter **`1`** for no encryption. Enter **`2`** to encrypt with AES. Enter **`3`** to encrypt with AES and create as a self-decrypting archive and enter the extension in the following field. Note that if you encrypt with AES the recipient will need the Alliance Windows AES application to decrypt the file. The Alliance Windows AES application is located on your Alliance product CD.

Press ***Enter*** to go to the next screen.

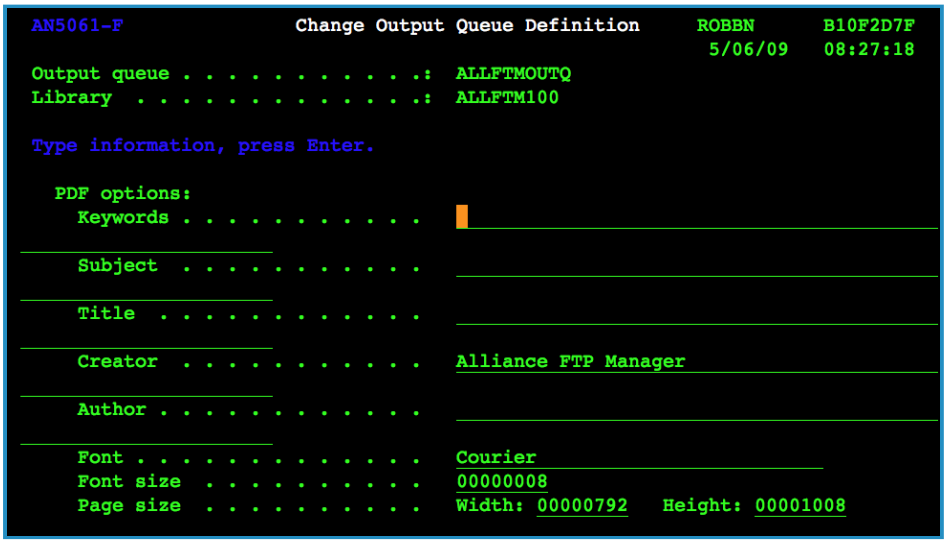


`AES key name` : Enter an AES key name for encryption.

`Or, AES pass phrase` : Instead of an AES key name you can use a pass phrase for encryption. **`Message text`** : Enter up to five lines of message text for the email message.

`Or, Message file, Library, Member` : If you need more than five lines for the message text you can put the message text in a file and enter the file name here. Enter the file, library and member name. You should limit the length of message lines in the file to 75 characters to remain compatible will all email applications.

Press ***Enter*** to go to the next screen.



PDF options:

`Keywords` : Enter the keywords that are to be used with this PDF attachment.

`Subject` : Enter the subject for this PDF attachment.

`Title` : Enter a title for this PDF attachment.

`Creator` : Enter the name of the creator for this PDF attachment.

`Author` : Enter the name of the author of this PDF attachment.

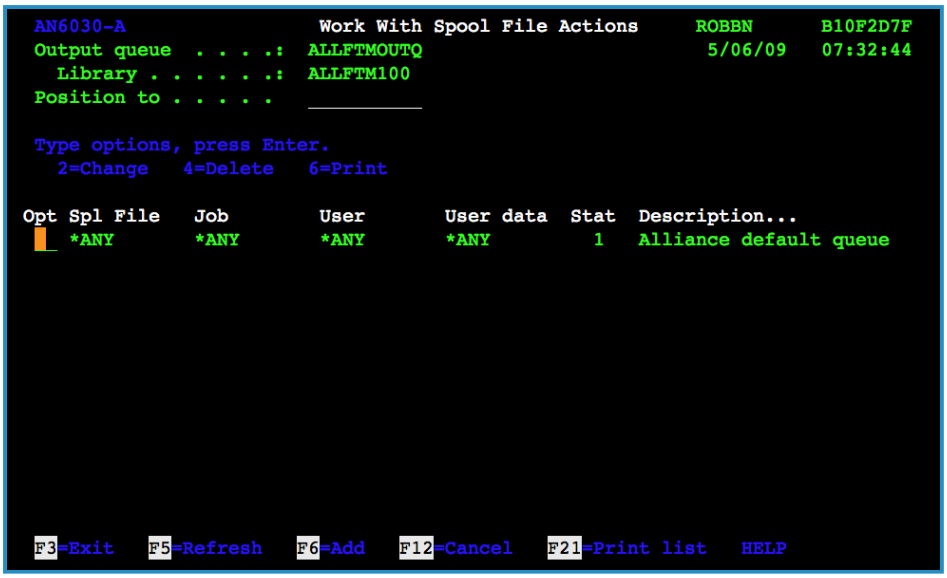
`Font` : Enter the font to be used to create this PDF attachment.

Font size : Enter the font size to be used to create this PDF document.

Page size : Enter the page width and height in twips.

Work With Spool File Actions

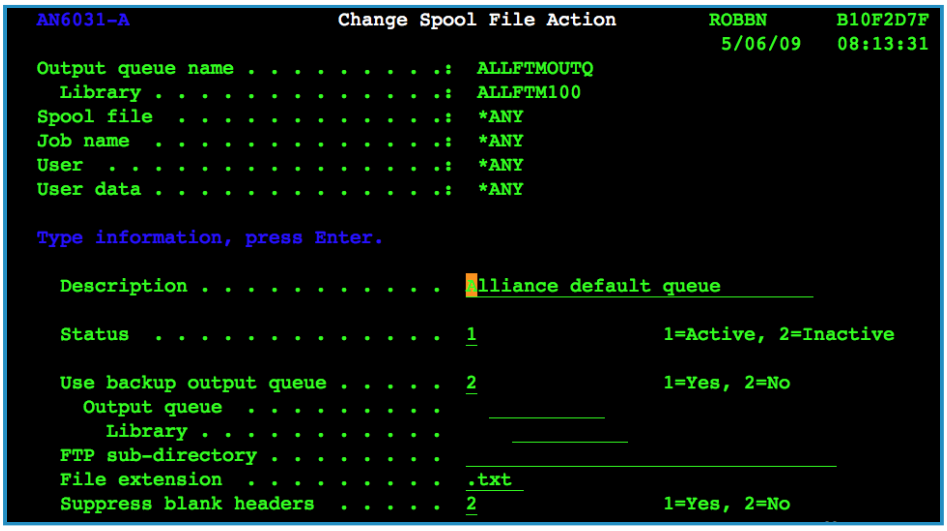
This panel is displayed when you use option **12** from the **Work With Output Queues** panel. You can view a list of spool file actions, add a new spool file action, or delete a spool file action. A spool file entry will not be processed for this output queue unless there is a specific entry for the spool file, or the spool file meets the criteria defined for a generic ***ANY** entry.



Use option **2** to change an existing entry. Use option **4** to delete an entry. Use option **6** to print the details of an entry. Use function key ***F6*** to add a new entry.

Add/Change Spool File Action

Use this panel to add a new entry or change an existing entry.



Description : Enter a description for this spool file action.

Status : Enter **1** for Active or **2** for Inactive. A spool file action with an Inactive status will not be used to determine if a spool file will be processed.

Use backup output queue : Enter **1** for Yes if you want to use a different backup queue than the one defined for the output queue. Enter **2** for No if you do not want to specify a backup output queue. If you specify **No** the output queue on the **Work With Output Queues** definition will be used. This option is provided to allow for more complex routing of spool files after Alliance has finished processing the spool file. You can route to a backup output queue based on spool file name, user name, etc. You can even move a spool file to an output queue with an active writer in order to print the spool file.

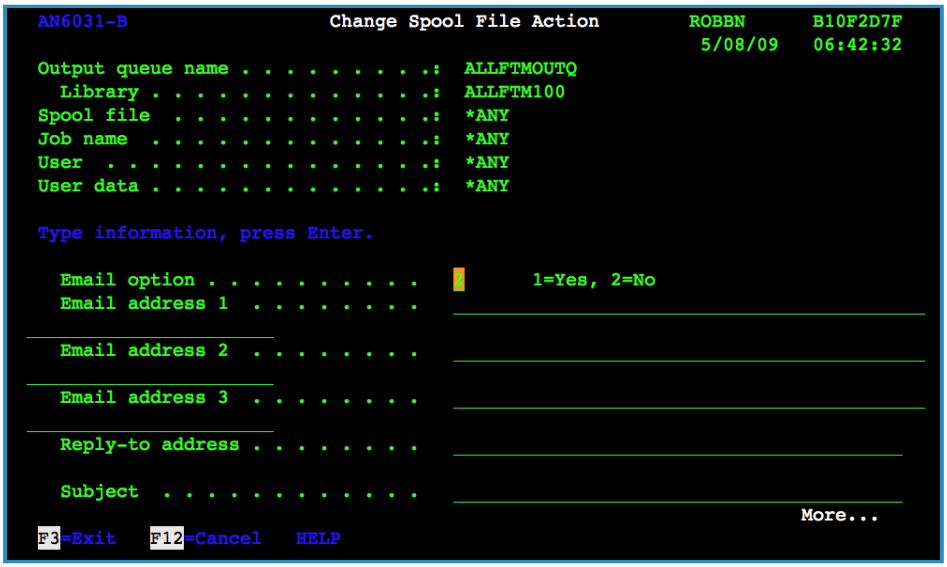
`Output queue and Library': If you specified Yes to use a back up output queue, enter the name of the output queue and library.

FTP sub-directory : If you are using an interface that causes an FTP session to a remote system (Interface types 1, 3, 4 and 5) you can specify a subdirectory for the transferred file. This is a subdirectory from the FTP server's root directory and not necessarily a directory on the remote system. For example, there may be a directory named **C:\MYDIR\MYSUBDIR** on the remote system. But the FTP server may not recognize this as a valid subdirectory. You may need to consult your system administrator for a list of valid subdirectories on the FTP server.

File extension : Enter the file extension to use for the spool file if desired. The default is **.txt** which is commonly used to specify a text file. If you are using **Report.Web** this parameter does not apply. You can leave this field blank to not use a file extension on the transferred file.

Suppress blank headers : Enter **1** for Yes if you want to suppress the leading blank lines on a report. Specify **2** to not suppress leading blank lines. The default is No. Suppressing leading blank lines can produce a file that is more readable.

Press **Enter** to go to the next screen.



Email option : Enter **1** for Yes if you want to send this spool file to an email recipient. Enter **2** for No if you do not want to send the spool file as email.

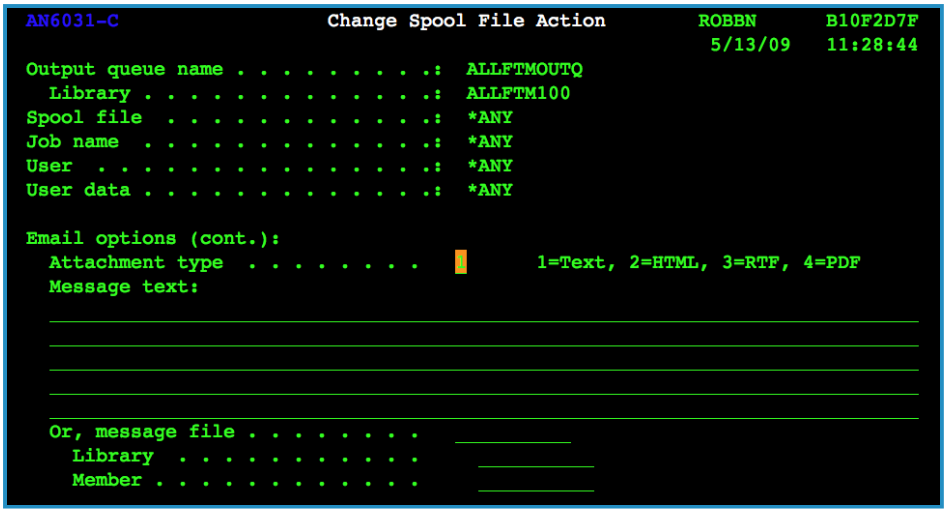
Email address 1, 2, 3 : Enter up to three email addresses to send the spool file to. These must be valid SMTP email addresses. Note that you must first configure Alliance to send email before using spool file distribution. Also, you may need to enable SMTP mail,

router, and gateway options to send email from the IBM i. You can also specify an Email distribution list in the first Email address field. The distribution list must have first been created using the configuration option for distribution lists.

`Reply-to address` : Enter the reply-to address for the email. This may be an IBM i email address, or may be an email address on another mail system.

`Subject` : Enter a subject for the email message.

Press ***Enter*** to go to the next screen.

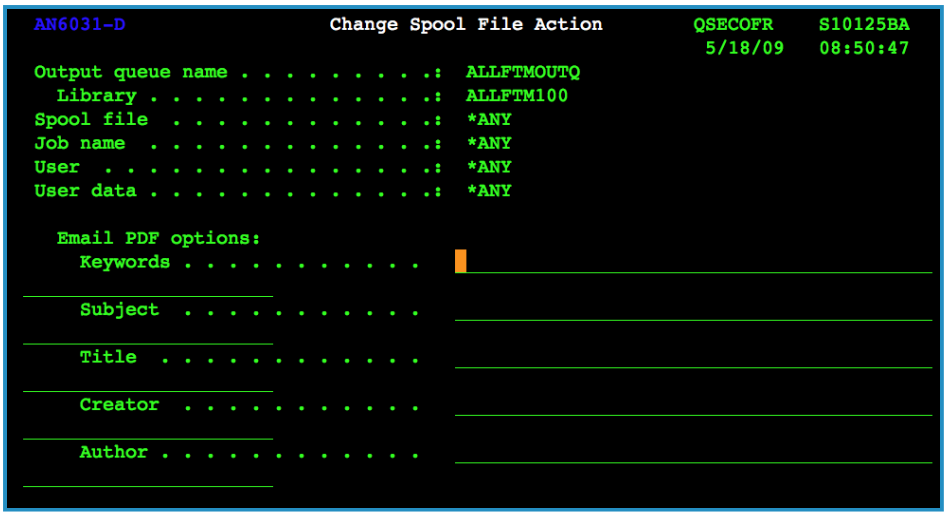


`Attachment type` : Enter **`1`** if you want to send the spool file as a text attachment. Enter **`2`** if you want to send the spool file as an HTML attachment. Enter **`3`** if you want to send the spool file as an RTF attachment. Enter **`4`** if you want to send the spool file as a PDF attachment. Note that if you want to send the spool file as a PDF attachment you must install the PDF files from the **`Alliance Installation`** menu.

`Message text` : Enter up to five lines of message text for the email message.

`Or, Message file, Library, Member` : If you need more than five lines for the message text you can put the message text in a file and enter the file name here. Enter the file, library and member name. You should limit the length of message lines in the file to 75 characters to remain compatible with all email applications.

Press ***Enter*** to go to the next screen.



PDF options:

`Keywords` : Enter keywords to be used with this PDF attachment.

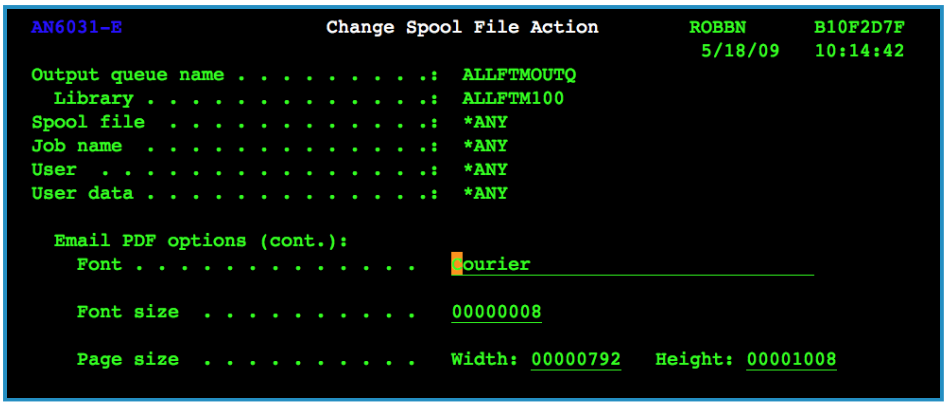
`Subject` : Enter the subject for this PDF attachment.

`Title` : Enter a title for this PDF attachment.

`Creator` : Enter the name of the creator for this PDF attachment.

`Author` : Enter the name of the author of this PDF attachment.

Press ***Enter*** to go to the next screen.



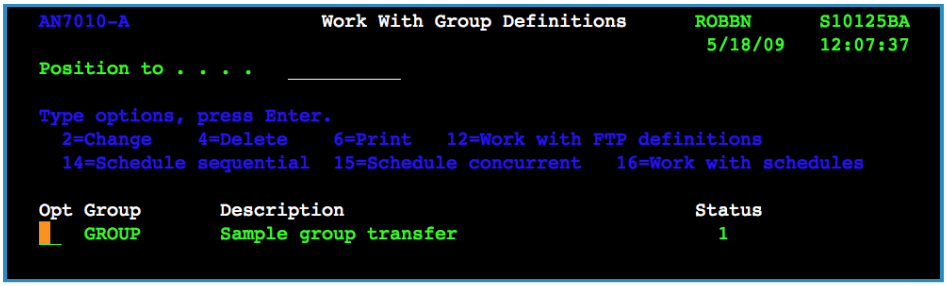
`Font` : Enter the font to be used to create this PDF attachment. You must specify a valid font file installed in the IFS directory.

`Font size` : Enter the font size to be used to create this PDF document.

`Page size` : Enter the page width and height in twips.

Work With Group Definitions

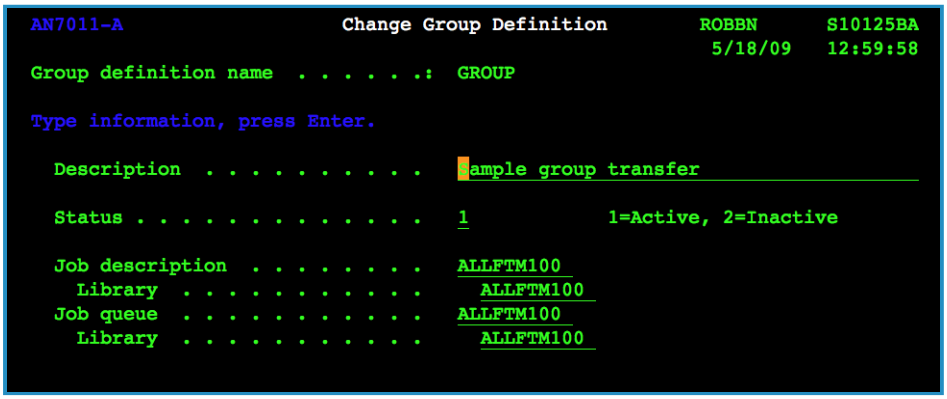
Use this option to define FTP groups. You can group FTP definitions together and perform FTP functions on all members of the group. After creating the group you can use the Work with FTP definitions option to add, change, or delete FTP definitions from the group. You can create schedules for a group, run group FTP sessions interactively, or use the group APIs.



Select an option or function key to begin work. Use ***F6*** to create a new group. Use option **`2`** to change a group. Use option **`12`** to work with FTP definitions. Use option **`14`** to transfer files one after another. Use option **`15`** to send all files at one time. You can use option **`4`** to delete a group and all associated group detail records.

Add/Change group definition

Use this panel to add or change information about a group.

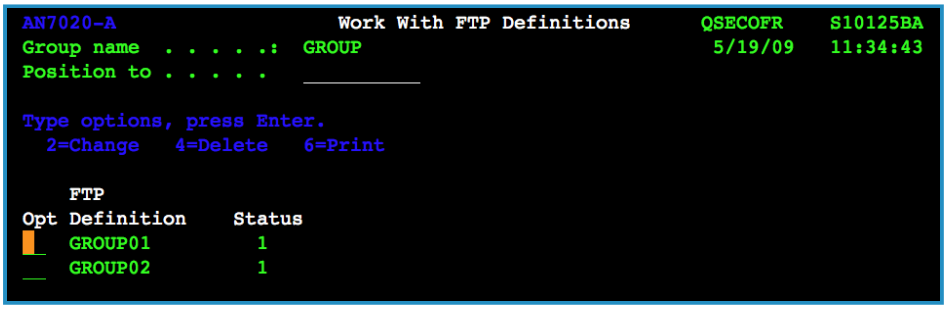


`Description` : Enter a description for this group.

`Status` : Enter **`1`** for Active, or **`2`** for Inactive. If a group is Inactive it will not be run by an interactive option or by an API.

Work With Group FTP Definitions

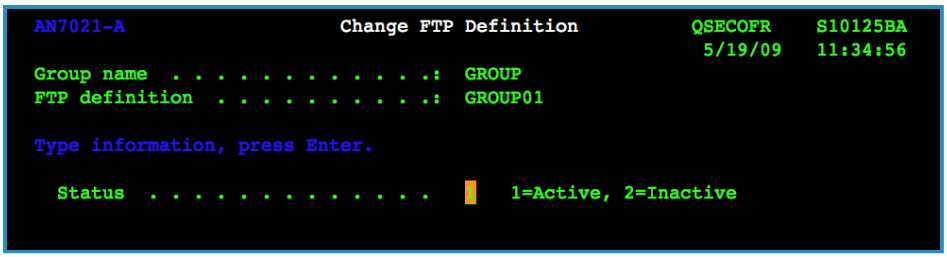
After creating a group definition you can add FTP definitions to it with this option.



Press ***F6*** to add a new FTP definition to this group. Use option **`2`** to change the status of an FTP definition in the group. Use option **`4`** to remove an FTP definition from the group.

Add/Change FTP definition

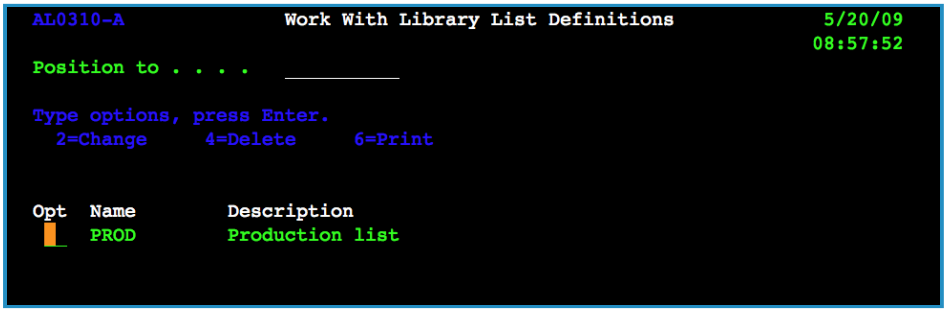
Use this panel change the status of an FTP definition in a group.



`Status` : Enter **`1`** for Active or **`2`** for Inactive. If the status of the FTP definition is Inactive the session will not be processed in the group.

Work With Library Lists

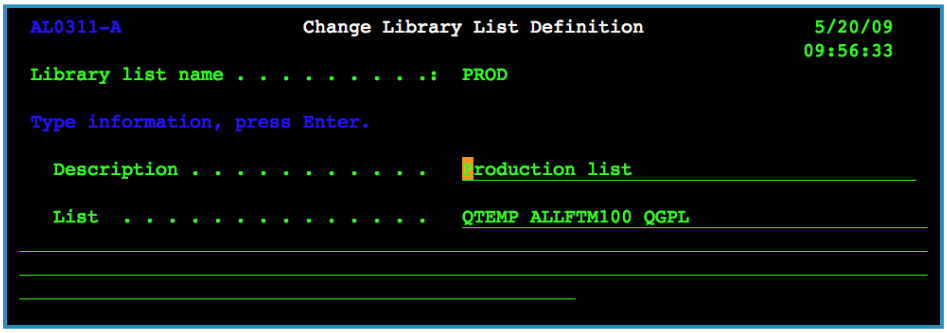
Use this option to define library lists to be used with the library scan function. You can create a library list definition by name and add libraries to it. Then on the Library Scan File Action definition you can name this library list to use.



Option : Enter **2** to change an existing library list definition. Enter **4** to delete a library list definition. Enter **6** to print a definition. Press ***F6*** to add a new definition.

Add/Change library list

Use this panel to add or change a library list definition. You should be sure to include the Alliance product library in the list.



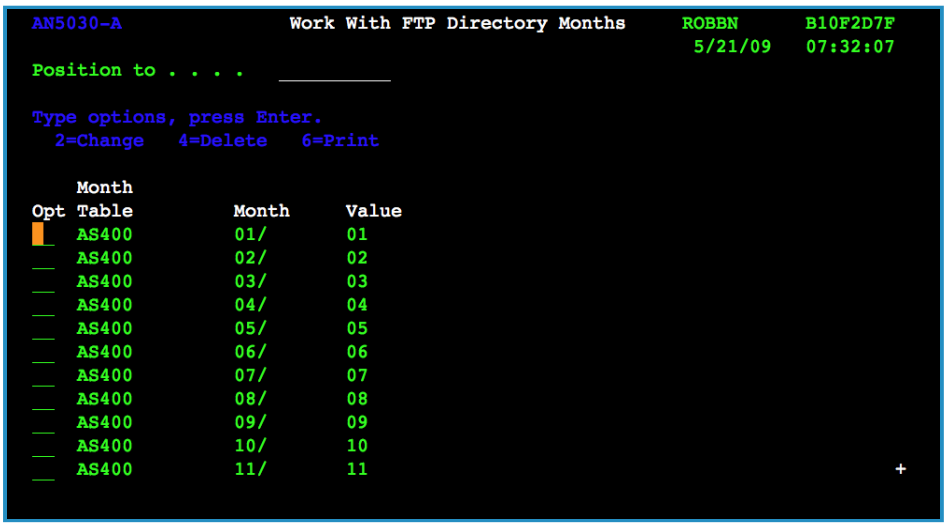
Library list name : Enter a name for this library list. This can be any valid IBM i name up to 10 characters.

Description : Enter a description for this list.

List : Enter a list of libraries separated by a blank. It is good practice to include the library **QTEMP** at the beginning of the list. Be sure to add the Alliance product library in the list.

Work With FTP Directory Months

Use this option to define the month abbreviation used by a remote FTP server. When using Auto Scan to pull files from a remote server, it is sometimes necessary to define how that server displays month text in order to create a list of files on the server. After defining the month table you can create an FTP Directory Definition for the server (option **10** on the configuration menu). Some month definitions have been provided.



Option : Enter **2** to change an existing FTP directory month definition. Enter **4** to delete a month definition. Enter **6** to print a definition. Press ***F6*** to add a new definition.

Add/Change FTP directory month

Use this panel add or change a month definition.

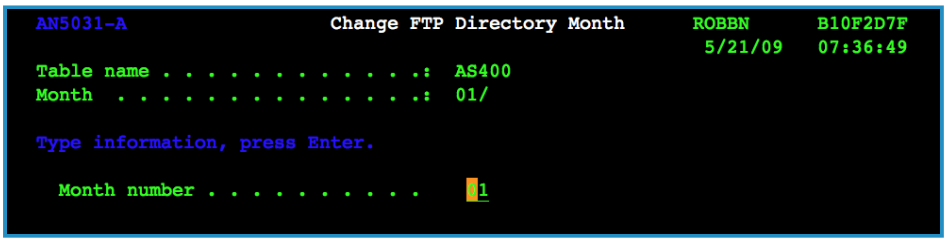


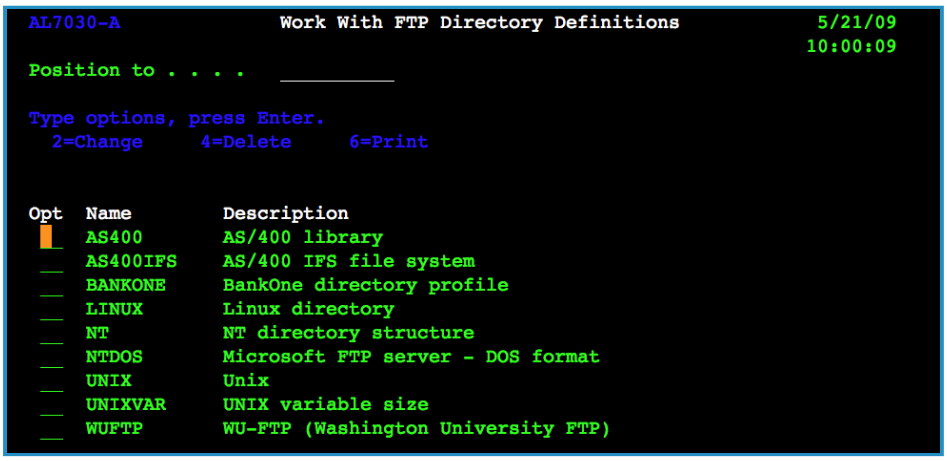
Table name : Enter the name of the table you want to use for this month definition. The name must be a valid IBM i name.

Month : Enter the abbreviation for the month as it is used by the remote FTP server. This field is case sensitive so you should enter the month exactly as the remote FTP server displays it. You can determine how the remote FTP server uses this field by starting an FTP session with the remote server and using the DIR command.

Month number : Enter the number of the month from 01 to 12. For example, you would enter “01” for January, “02” for February, etc.

Work With FTP Directory Definitions

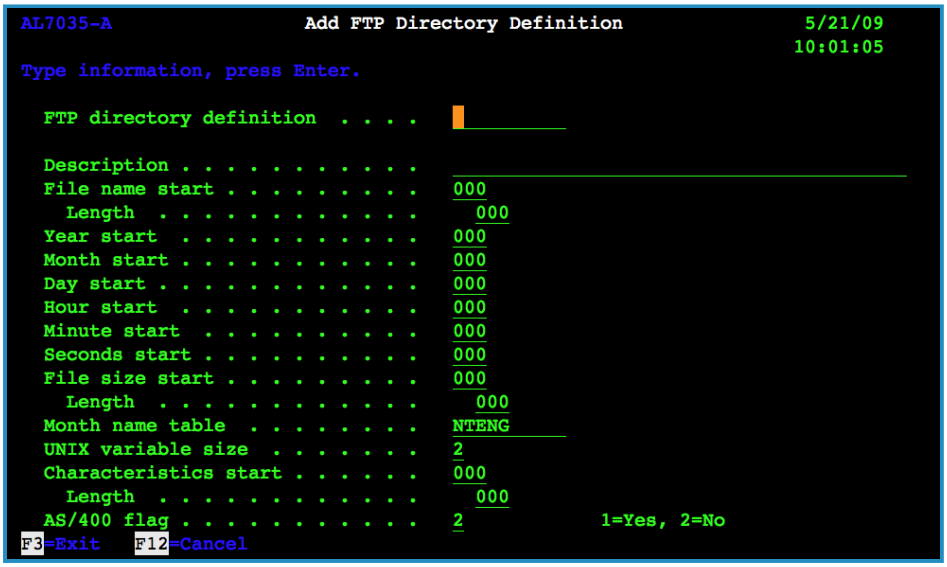
Use this option to define the layout of a remote FTP server’s directory listing. You must define the directory format before you can use any of the Alliance options that list the contents of a remote FTP server, such as AutoScan.



`Option` : Enter **`2`** to change an existing FTP directory definition. Enter **`4`** to delete a definition. Enter **`6`** to print a definition. Press ***F6*** to add a new definition.

Add/Change FTP directory definition

Use this panel add or change a directory definition.



`FTP Directory name` : Enter the name of the table you want to create. This must be a valid IBM i name.

`Description` : Enter a description for this FTP directory definition.

The following positions are based on their position from the left margin of the DIR listing of the FTP server.

`File name start` : Enter the start position of the file name on the FTP server.

`File name length` : Enter the maximum length of a file name on the FTP server. This must not exceed 128.

`Year start` : Enter the start position of the year in the DIR listing of the FTP server. The year is saved as a two byte numeric field.

`Month start` : Enter the start position of the month in the DIR listing of the FTP server. The month will be displayed in character format. For example, January is normally displayed as **`Jan`** on servers hosted in the US. The month will be translated to its numeric equivalent using

the FTP Directory Month table.

`Day start` : Enter the start position of the day in the DIR listing of the FTP server. If you enter the value **`000`** the day will not be captured.

`Hour start` : Enter the start position of the hours portion of the time in the DIR listing of the FTP server.

`Minutes start` : Enter the start position of the minutes portion of the time in the DIR listing of the FTP server.

`Seconds start` : Enter the start position of the seconds portion of the time in the DIR listing of the FTP server. Note that many FTP servers do not store the seconds portion of the file create time. The seconds will not be used if you specify “000” for this value.

`File size start` : Enter the start position of the file size in the DIR listing of the FTP server. File sizes may be right justified in the FTP directory listing. Blanks will be converted to zeros.

`File size length` : Enter the length of the file size field.

`Month name table` : Enter the name of the month table to use for this FTP server.

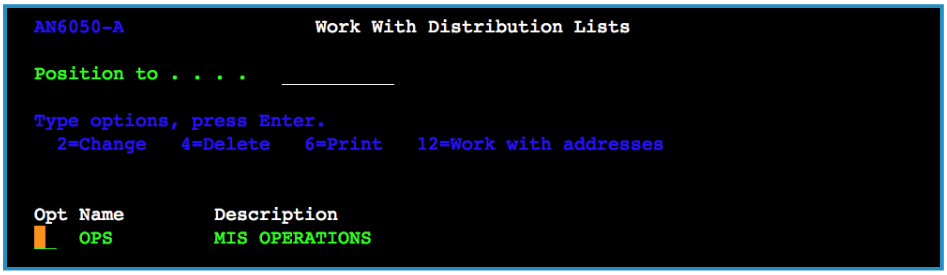
`UNIX variable size` : Enter **`1`** for Yes if the directory listing is for a UNIX system with a variable starting position for the file size. Some UNIX systems shift the starting position of the file size when the size is greater than 7 digits. Enter **`2`** for No if the directory listing is not for a UNIX system with a variable starting position for the file size.

`Characteristics start` : Enter the starting position for the file characteristics. The value must be in the range of **`0`** to **`999`** . If you enter 0 the file characteristics will not be captured. The file characteristics can tell you if the item in the directory list is a file or a sub-directory. On some systems it can also tell you if the file can be read, updated, or executed. There is no standard for reporting file characteristics. Consult you network administrator for information about what characteristics may be available from the FTP server.
`Characteristics length` : Enter the length of the characteristics you want to capture. This must be a value from **`1`** to **`40`** .

`IBM i flag` : Enter **`1`** for Yes if the FTP server is an IBM i. Enter **`2`** for No if the FTP server is not an IBM i.

Work with Email lists

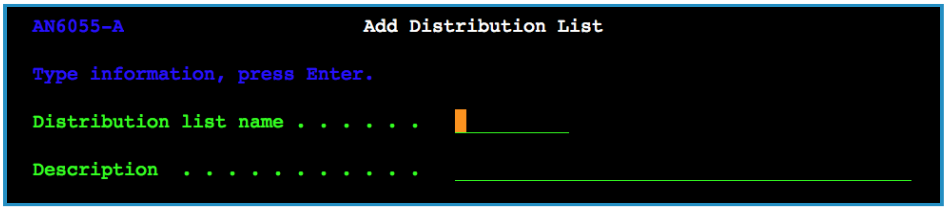
Use this option to work with email distribution lists. An email distribution list is a collection of one or more email addresses that can be used to send error or informational email messages to a group.



Option : Enter **2** to change an existing email list. Enter **4** to delete an email list. Enter **6** to print a list. Enter **12** to work with a list. Press ***F6*** to add a new email list.

Add/Change Email List

Use these panels to add or change an email list.

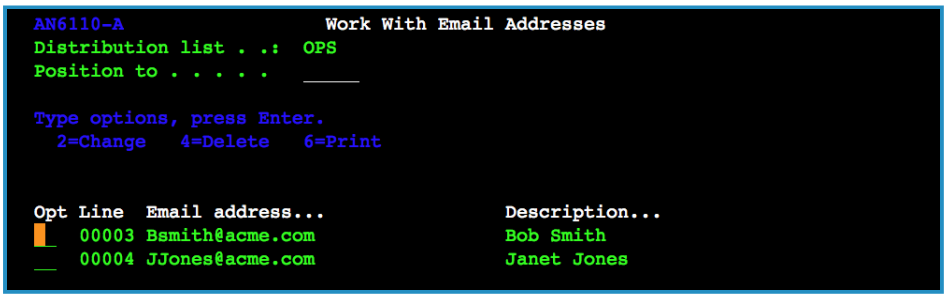


Distribution list name : Enter a name for this distribution list.

Description : Enter a description for this list.

Work with Email addresses

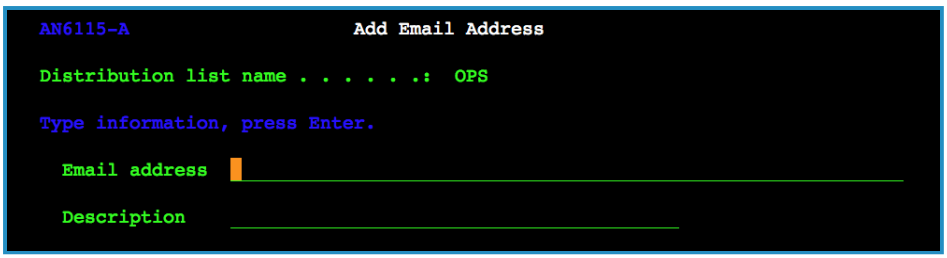
When you use option **12** to work with email addresses a list of addresses is displayed. You can add, change or delete entries in the distribution list.



Option : Enter an option to **Change** , **Delete** or **Print** an email address. Press ***F6*** to add a new email address.

Add/Change email address

Use this option to add or change an email address.

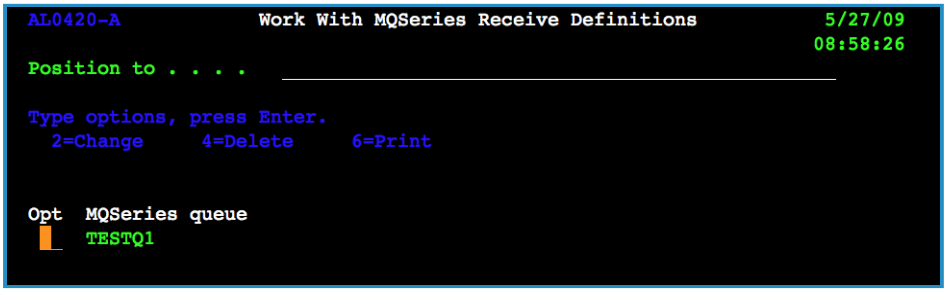


Email address : Enter an email address. This must be a valid SMTP email address.

Description : Enter a description for this email address.

Work with MQSeries receive definitions

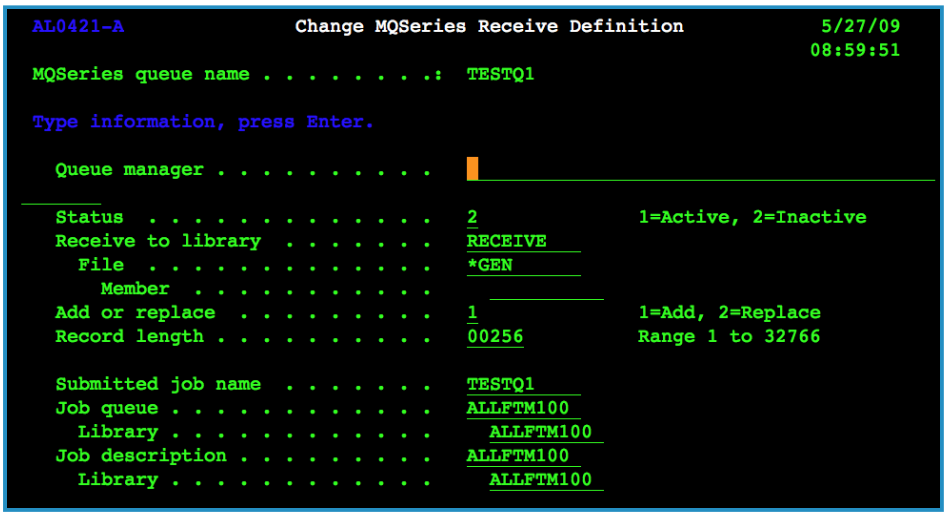
Use this option to work with MQSeries receive definitions. An MQSeries receive definition is used to receive data from an IBM i MQSeries queue and write the data to an IBM i library. The library can be under the control of FTP Manager library scan, or can be a user library.



Option : Enter **2** to change an existing MQSeries receive definition. Enter **4** to delete a definition. Enter **6** to print a definition. Press ***F6*** to add a new definition.

Add/Change MQSeries receive definition

Use this option to add or change an MQSeries receive definition.



MQSeries queue name : Enter the name of the MQSeries queue for receiving data. You must first define the MQSeries queue and start MQSeries support before receiving data. Note that MQSeries queue names are case sensitive and you should enter the name the way it was created.

Status : Enter **1** for Active or **2** for Inactive. If the definition is inactive it will not be automatically started.

Receive to library, File and Member : Enter the name of the library to receive MQSeries data. Enter a file name and member for the data. You can enter the special value of ***GEN** to have Alliance automatically create a file name for you.

Add or replace : Enter **1** to add data to the file, or **2** to replace data in the file. If you specify **2** to replace data the file will be cleared when the receive job starts.

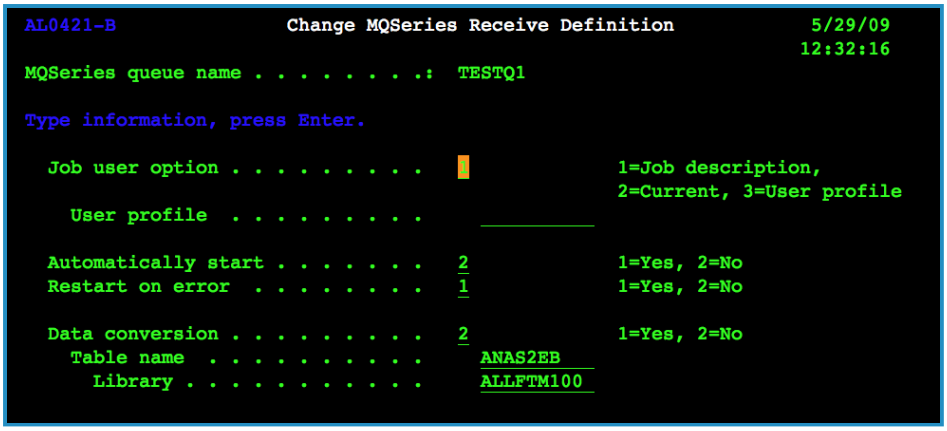
Record length : If the file does not exist in the receive library it will be created with the record length you specify. Enter a record length from 1 to 32766.

Submitted job name : Enter a name for the submitted receive job. This must be a valid IBM i job name.

`Job queue name and Library` : Enter a job queue and library for the submitted receive job. When the job is manually started or auto-started this job queue will be used to submit the job. The default is to use the job queue **`ALLFTM100`** in library **`ALLFTM100`** .

`Job description and Library` : Enter the name of the job description for the submitted receive job. The default is **`ALLFTM100`** in library **`ALLFTM100`** . Note that you may need to use a job description with a user name.

Press ***Enter*** to go to the next screen.



`Job user option` : Enter **`1`** to submit the MQSeries receive job using the user profile on the job description. Enter **`2`** to use the current user for the submitted job. Enter **`3`** to use a specific user profile for the submitted job. If you use option **`3`** you must have authority to the user profile and the user profile must exist.

`User profile` : If you specified option **`3`** for the **`Job User Option`** you must specify a user profiler name in this field.

`Automatically start` : Enter **`1`** for Yes to automatically start the MQSeries receive job when the Alliance subsystem starts. Enter **`2`** for No to not automatically start the job.

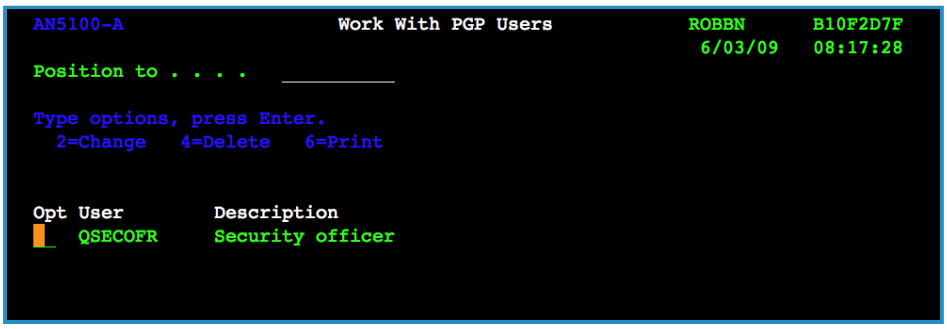
`Restart on error` : Enter **`1`** for Yes to restart the MQSeries receive job after an error. Enter **`2`** for No to not restart after an error.

`Data conversion` : Enter **`1`** for Yes to convert data that is received from the queue. Enter **`2`** for No to not convert data.

`Table name and library` : If you specified Yes to convert received data enter a table name and library in this field. The default is the **`QEBCDIC`** table in library **`QSYS`** to translate ASCII data to EBCDIC. You can specify any valid conversion table.

Work with PGP users

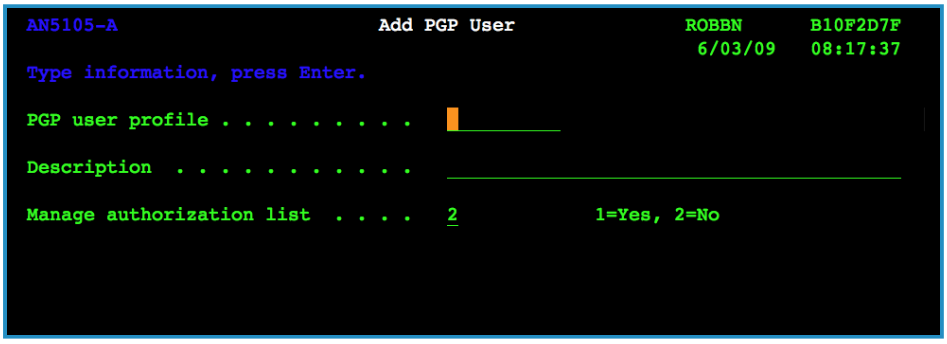
Use this option to define users to PGP. If you have enabled the option for PGP authority management by Alliance you must define IBM i PGP users in this option.



Option : Enter **2** to change an existing PGP user definition. Enter **4** to delete a PGP user definition. Enter **6** to print a definition. Press ***F6*** to add a new PGP user.

Add/Change PGP user

Use this option to add or change a PGP user.



PGP user name : Enter the name of the PGP user. This is the user profile of the PGP user. The user profile must exist.

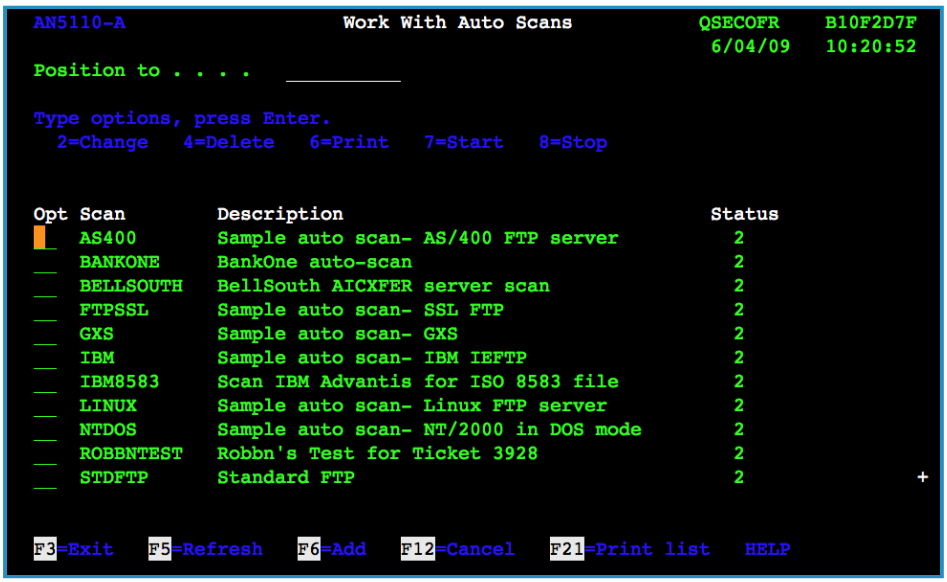
Description : Enter a description of this user.

Manage authorization list : Enter **1** for Yes if you want this user to be able to manage PGP users and the PGP authorization list. Enter **2** for No if you do not want this user to manage PGP lists. The default is **2** for No.

Work with auto scan

Use this option to create an auto scan definition. The auto scan facility is used to periodically connect to a remote FTP server and pull files into a local IBM i library or IFS directory. This gives you the ability to automatically pull any new or changed files to the IBM i and process them into your business applications.

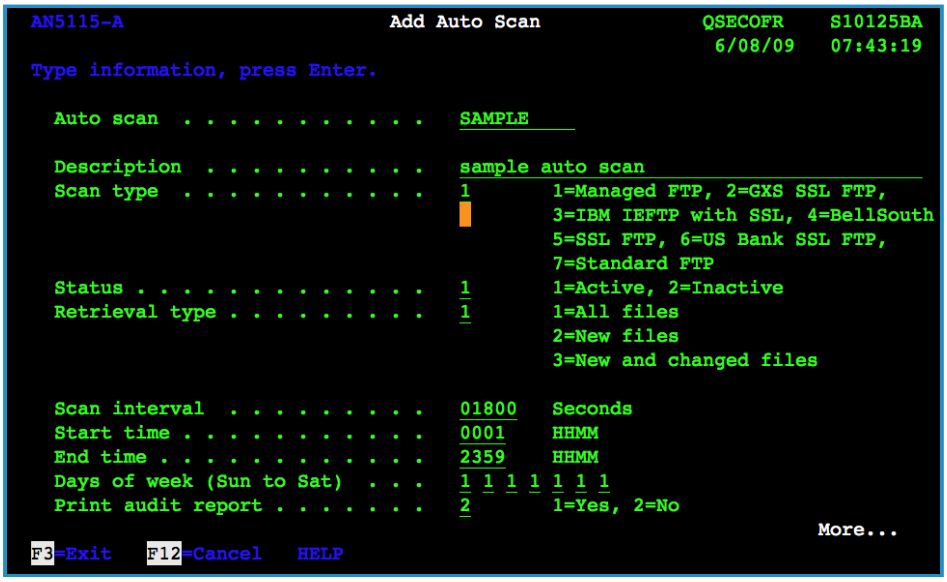
Please read Appendix E for complete instructions on how to set up Auto Scan. It will walk you through setting up 2 scripts and 2 definitions to do the Directory Listing of files (**dir**) and then do the **get** .



Press ***F6*** to add a new auto scan definition. Use option `2` to change an auto scan definition, or option `4` to delete a definition. You can print a definition with option `6` , or print a list of scan definitions with function key ***F21***.

Add/Change auto scan

Use this option to add or change an auto scan definition.



`Auto scan` : Enter a name for this auto scan definition. The name must be a valid IBM i name with no embedded blanks or special characters.

`Description` : Enter a description for this auto scan.

`Scan type` : Enter `1` to use standard FTP with a **`DIR`** listing of files. Enter `2` for an SSL FTP session with GE eXchange Services (GXS). Enter `3` for an SSL FTP session with IBM (Advantis) Information Exchange FTP. Enter `4` for SSL with BellSouth. Enter `5` for SSL. Enter `6` for SLL with US Bank. Enter `7` to use standard FTP with an LS listing of files. Set up for any SSL connection is done on its own panel within this **`Auto Scan`** configuration. SSL does not use definitions or scripts as do the standard FTP sessions.

`Status` : Enter **`1`** for Active or **`2`** for Inactive. If the scan definition is inactive no further connections to the remote FTP server will take place. You do not need to end the Alliance subsystem to make this option effective.

`Retrieval type` : Enter **`1`** to pull all files from the remote FTP server each time auto scan runs. Enter **`2`** to pull only new files from the server. Alliance keeps an historical log of the files and will check to retrieve only files you have not received before. Enter **`3`** to retrieve new and changed files from the remote FTP server. Alliance will retrieve any new files and any file that has changed. Alliance checks the date, time, and file size to determine if the file has changed.

`Scan interval` : Enter the scan interval in seconds. Include leading zeros in the value. For example, if you wanted to scan every 10 minutes you would enter **`00600`** .

`Start time` : Enter the start time in 24-hour format. Alliance will not scan a remote FTP server before this time. It can be helpful to avoid scanning a remote FTP server if you know the server will be down during certain periods.

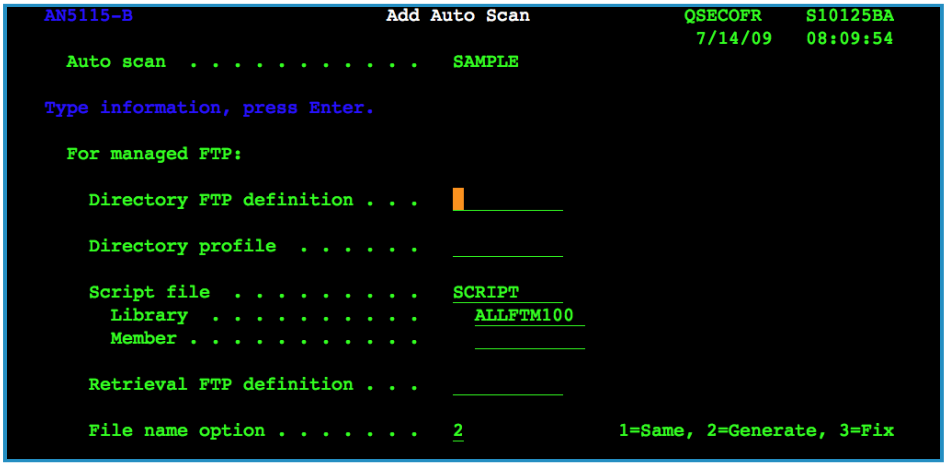
`End time` : Enter the scanning end time in 24-hour format. Alliance will not scan a remote FTP server after this time.

`Print audit report` : Enter **`1`** for Yes to print an audit report. Enter **`2`** for No to not print an audit report. Printing an audit report is recommended for new scan definitions so that you can be sure the files are being retrieved the way you want. It is also recommended for applications that are retrieving sensitive or mission critical data.

`Synchronize access` : Enter **`1`** for Yes to synchronize library access between Alliance applications. If you are using Auto scan to receive to a library, and using library scan to process the same library, you should enable this option and enable the same option on library scan. Enter **`2`** for No to not enable access controls.

`Receive library` : Enter the name of the library that you are receiving files to.

Press ***Enter*** to go to the next screen.



If you choose 1 on the first panel for standard FTP:

`Directory FTP definition` : Enter the name of the Alliance FTP definition to be used to retrieve a list of files on the remote FTP server. This definition is set up in option **`2`** of the **`Configuration`** menu. This must already exist. Do not use any compression or encryption options on the FTP definition.

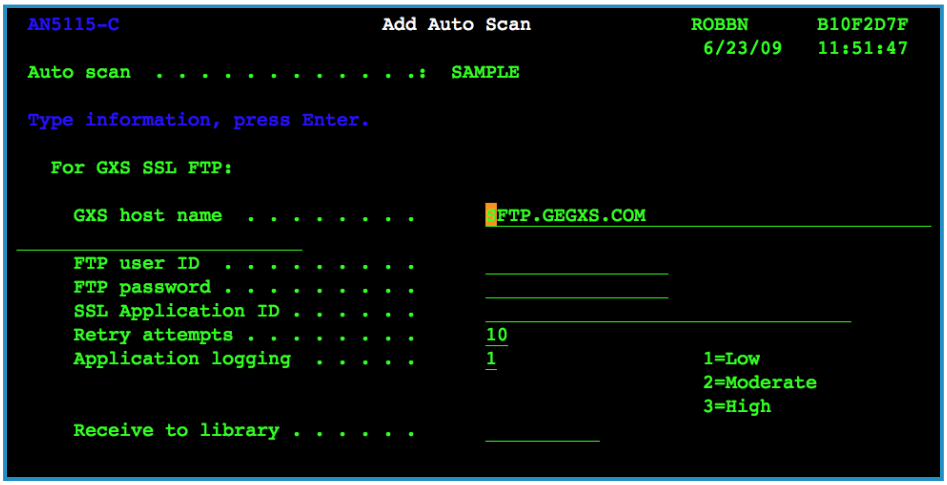
Directory profile : Enter the name of the directory profile for the FTP directory. This is option **10** on the **Configuration** menu and must already exist. Different types of FTP servers format the list of files in different ways. The directory profile defines to Alliance the format of the FTP server's listing. Alliance provides several directory profiles that you can use, and you can create new profiles if you like. We suggest cloning an existing one and naming it to a similar name as this auto scan.

Script file, Library, Member : Enter the name of the FTP script to use for the directory listing. The script contains the logic for creating the directory. You can use the Alliance sample script **SCANDIR** as a starting point. Normally this script only contains the FTP command DIR. This script must already exist.

Retrieval FTP definition : Enter the name of the FTP definition to be used to retrieve files from the remote FTP server. The auto scan function will use this definition for each file it pulls from the server.

File name option : Enter **1** to use the same name on the local IBM i as is used on the FTP server. Note that the IBM i FTP client will change the name if it does not meet IBM i naming conventions. Enter **2** to generate a unique name for the local file. The name will be Fnnnnnnnnn where "nnnnnnnnn" is a unique sequence number. This option will guarantee that you do not overwrite existing files. Enter **3** to have Alliance try to generate a name based on the name of the file on the remote FTP server.

Press ***Enter*** to go to the next screen.



> ***NOTE:*** If you chose option **2** for GXS SSL FTP: The GXS configuration is used primarily to retrieve EDI files from a GXS mailbox. The following options will only be active if you specified option **2** under the **Scan Type** option on the first panel of the **Auto Scan** configuration.

GXS host name : Enter the host name or IP address for the GXS FTP server.
GXS FTP user ID : Enter the FTP user ID for the GXS FTP server.

FTP password : Enter the FTP password for the GXS FTP server.

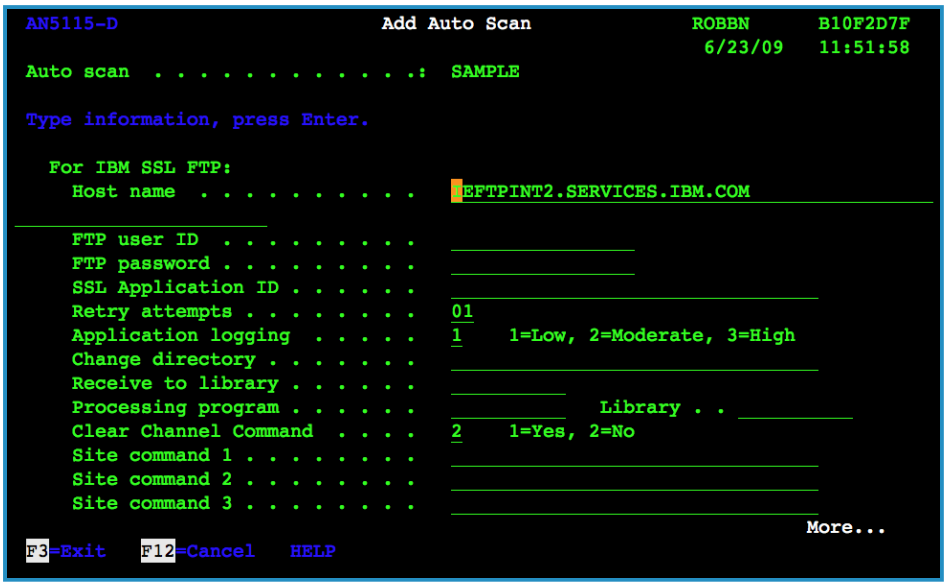
SSL application ID : Enter the SSL application ID to use for the FTP server. The SSL application ID is created by Digital Certificate Manager. Please consult the SSL User's Guide for information on creating an SSL application ID and assigning a certificate to the SSL connection.

`Retry attempts` : Enter the number of retry attempts for the file transfer. It is recommended that you choose a value of 02 or higher. The range of values is from 01 to 99.

`Application logging` : Enter `1` for low application logging. Enter `2` for moderate logging. Enter `3` for high logging. If you are experiencing problems connecting for the first time to GXS use option `3` to print the SSL negotiation to the application log.

`Receive to library` : Enter the name of the library to receive EDI files to. If you are using Alliance EDI Workbench/400 you can receive EDI documents to a scan library in that product. If you are using a third party EDI translator you can receive the EDI documents to an Alliance FTP Manager scan library and use the translation commands in your translator.

Press ***Enter*** to go to the next screen.



> ***NOTE:*** If you chose option `3` for IBM SSL FTP: The IBM configuration is used primarily to retrieve EDI files from an ADVANTIS mailbox. The following options are only active if you specified option `3` under the **`Scan Type`** on the first screen of **`Auto Scan`** .

`Host Name` : Enter the host DNS name to connect to the IBM Advantis network. The default is **`IEFTPINT2.SERVICES.IBM.COM`** . Contact your IBM network support center for the correct DNS name to use.

`FTP user ID` : Enter the FTP user ID for the IBM FTP server. This is in the format Account.UserID. If your account is **`ABCD`** and your user ID is **`ABCD001`** your IBM FTP user ID would be **`ABCD.ABCD001`** .

`FTP password` : Enter the FTP password for the IBM FTP server.

`SSL application ID` : Enter the SSL application ID to use for the FTP server. The SSL application ID is created by Digital Certificate Manager. Please consult the Alliance Installation and User Guide for information on creating an SSL application ID and assigning a certificate to the SSL connection. Please refer to the Alliance documentation on how to receive and install a PKIX certificate from IBM.

`Retry attempts` : Enter the number of retry attempts for the file transfer. It is recommended that you choose a value of 02 or higher. The range of values is from 01 to 99.

`Application logging` : Enter **`1`** for low application logging. Enter **`2`** for moderate logging. Enter **`3`** for high logging. If you are experiencing problems connecting for the first time to IBM use option **`3`** to print the SSL negotiation to the application log.

`Change directory` : Use this field to change the default directory after you log into the IBM network. If you are receiving EDI files you will probably want to change to the #E2 class using a value like this:

```
ACNT.USERID/#E2
```

`ACNT` is your Advantis account name. **`USERID`** is the User ID or mailbox you are using. #E2 is the normal class for EDI files. If you are not receiving EDI files use the actual class name.

`Receive to library` :Enter the name of the library into which you want to receive EDI files.

`Processing Program` : You can enter a user program and library to be called when a file is received from IBM Advantis. If a processing program is defined it will be called with the following parameters:

File name (10, Alpha)

Library name (10, Alpha)

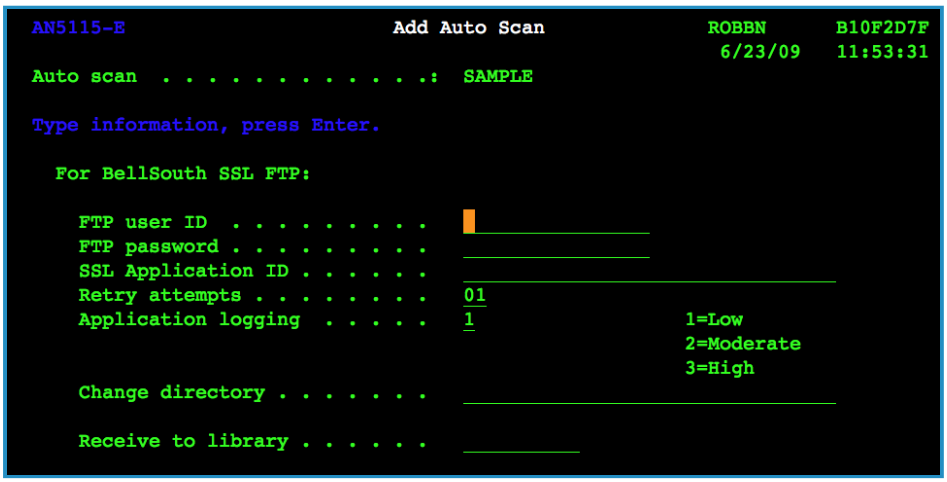
Member name (10, Alpha)

`Clear Channel Command` : Enter **`1`** for Yes or **`2`** for No for using the Clear Channel Command processing with the FTP connection. The Clear Channel Command (CCC) option will remove the SSL encryption from the FTP command channel after sending the FTP user ID and password to the FTP server. This will enable firewalls and routers to properly handle the FTP passive mode commands.

`Site Command 1, 2, 3` : You can enter up to three site commands to be sent to the remote FTP server. For the IBM Advantis Information Exchange network, you can send the **`EDIALIASONLY ON`** command to enable, routing of EDI documents using the ISA header fields. This can be helpful if you are sending to multiple trading partners from one account. Enter the site command without quotes like this:

```
EDIALIASONLY 1
```

Press ***Enter*** to go to the next screen.



If you chose option 4 for BellSouth SSL FTP:

`BellSouth FTP user ID` : Enter the FTP user ID for the BellSouth AICXFER FTP server.

`FTP password` : Enter the FTP password for the BellSouth FTP server.

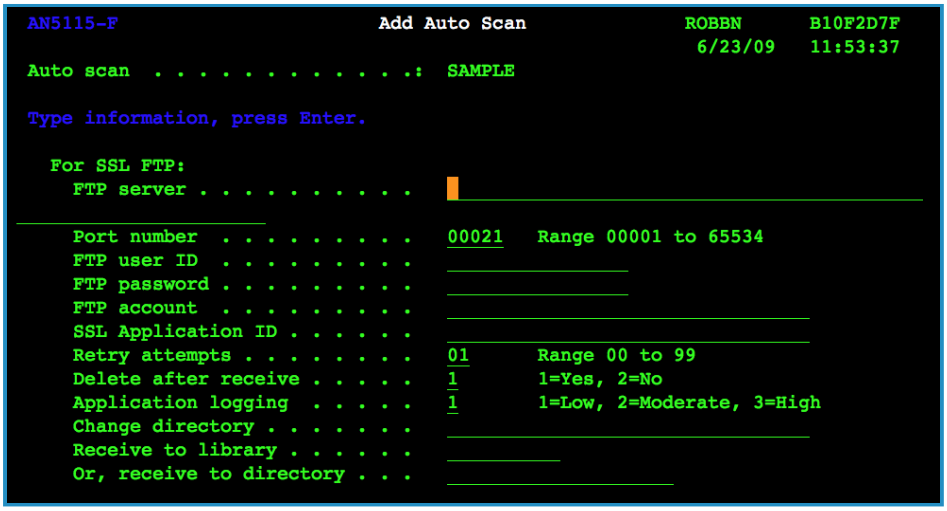
`SSL application ID` : Enter the SSL application ID to use for the FTP server. The SSL application ID is created by Digital Certificate Manager. Please consult the Alliance Installation and User Guide for information on creating an SSL application ID and assigning a certificate to the SSL connection.

`Retry attempts` : Enter the number of retry attempts for the file transfer. It is recommended that you choose a value of 01. The range of values is from 01 to 99.

`Application logging` : Enter **`1`** for low application logging. Enter **`2`** for moderate logging. Enter **`3`** for high logging. If you are experiencing problems connecting for the first time to BellSouth use option **`3`** to print the SSL negotiation to the application log.
`Change directory` : If you enter a directory name in this field the application will change to that directory on the server before receiving files.

`Receive to library` : Enter the name of the library into which you want to receive files. Files will be assigned a unique name.

Press ***Enter*** to go to the next screen.



If you chose option 5 for SSL FTP:

`FTP server` : Enter the IP address or DNS name of the FTP server you want to connect to.

`Port number` : Enter the port number for the FTP server. The default is 21.

`FTP user ID` : Enter the user ID for the FTP server log in.

`FTP password` : Enter the password for the FTP server.

`FTP account` : If the FTP server requires an account name, enter the account name in this field.

`SSL Application ID` : Enter the Digital Certificate Manager Application ID to be used for this SSL connection. The SSL application ID is created by Digital Certificate Manager. Please consult the Alliance Installation and User Guide for information on creating an SSL application ID and assigning a certificate to the SSL connection.

`Retry attempts` : Enter the number of attempts to retry the session in the event the session fails. Enter a number from 01 to 99.

`Delete after retrieve` : Enter `1` for Yes to delete the file from the FTP server after it is received. Enter `2` for No to not delete the file.

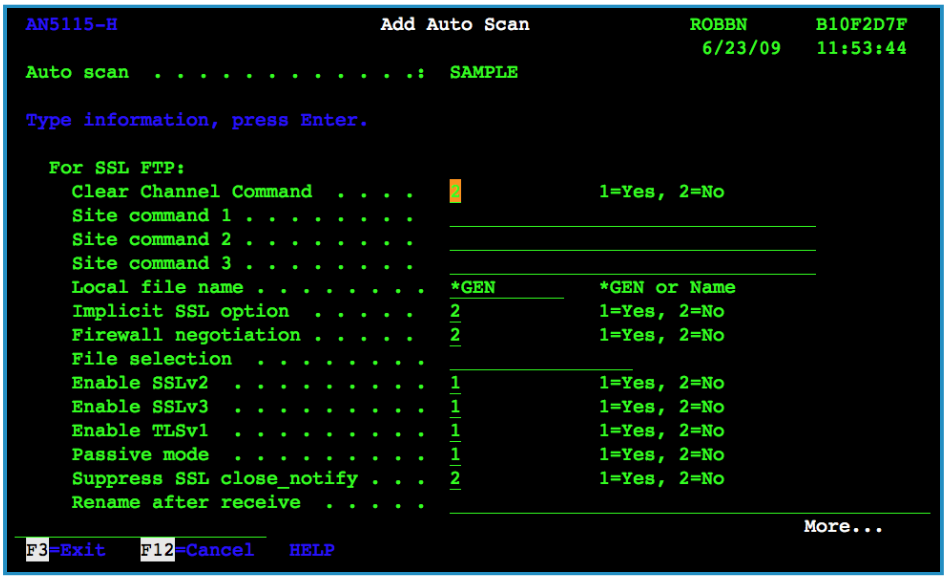
`Application logging` : Enter `1` for minimal FTP session logging. Enter `2` for intermediate level logging. Enter `3` for maximum logging of the FTP session.

`Change directory` : If you want to change to a different directory before transferring the file enter the new directory name in this field.

`Receive to library` : Enter the library name to receive the file.

`Or, receive to directory` : If you want to receive the file to an IFS directory rather than a library, enter the full IFS directory path.

Press ***Enter*** to go to the next screen.



`Clear Channel Command` : Enter `1` for Yes to send the Clear Channel Command (CCC). Enter `2` for No to not send the CCC command. If the remote FTP server is behind a firewall and supports the CCC command, you can enable this option for firewall negotiation.

`Site command 1, 2, 3` : If the remote FTP server needs special **`SITE`** commands you can enter up to three site commands to be sent to the server.

`Local file name` : Specify the local file name as you want it to appear in the library. You can use the special value **`*GEN`** to have Alliance automatically generate a file name (the default), or you can specify your own name.

`Implicit SSL session` : Enter **`1`** for Yes to use implicit SSL. Enter **`2`** for No to not use implicit SSL. The default is to not use implicit SSL. Some older SSL FTP servers use implicit SSL rather than RFC 2228 SSL.

`Firewall negotiation` : Enter **`1`** for Yes to use intelligent firewall negotiation. Enter **`2`** for No to not use firewall negotiation. When enabled Alliance will ignore the IP address sent as a response to the **`PORT`** command and will use the original IP address of the FTP server.

`File selection` : You can enter a selection string to narrow the list of files on the FTP server to receive.

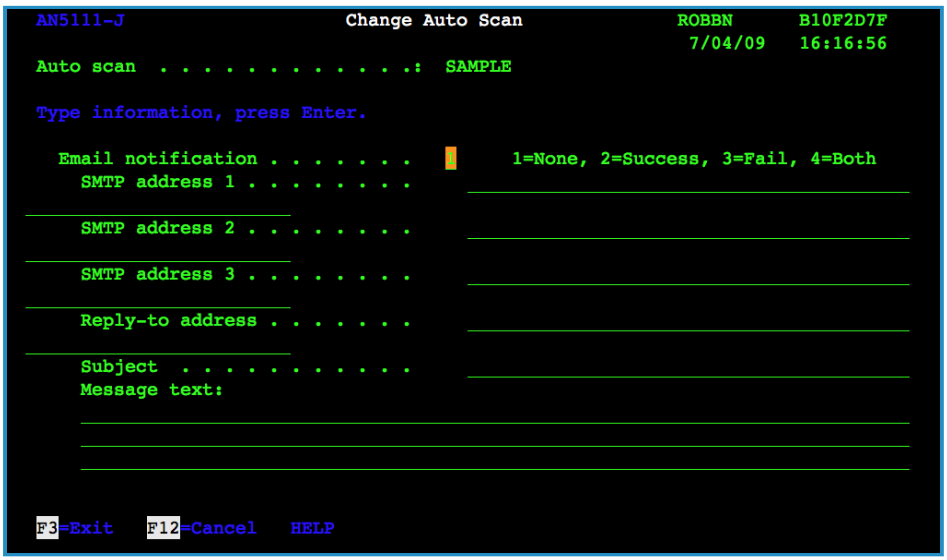
`Enable SSLv2, SSLv3, TLSv1` : By default Alliance SSL FTP will enable all three methods of SSL encryption for the transfer. You can control which methods are actually used with these options.

`Passive mode` : Enter **`1`** for Yes to use passive mode. Enter **`2`** for No to not use passive mode. The default is to use passive mode. Some servers may be set up to not support that

`Suppress SSL close_notify` : On FTP communication sessions that use SSL for encryption, the SSL session may be configured to send a “close-notify” response. While this is not common, it can cause a problem for FTP clients when receiving data. The symptom will be unreadable data on a read operation. The Alliance FTP client can handle these close-notify transactions so that they do not cause a problem. For library scan, auto-scan, and directory scan operations, use the option to enable close-notify handling. If you are using the FTP commands, use the **`FTPCLSNTFY`** command to enable close-notify handling.

`Rename after receive` : You may need to rename the file on the remote server after you retrieve it. Enter the full name here.

Press ***Enter*** to go to the next screen.



This panel is for email notification for an SSL session specifically.

`Email notification` : Enter **`1`** for no email notification on the status of the session. Enter **`2`** for Success to send an email notification if the session was successful. If a session fails a message will not be sent. Enter **`3`** for Fail to send an email notification only if the session fails.

Enter **4** to send an email notification if the session is successful or if it fails. An email notification will always be sent with this option.

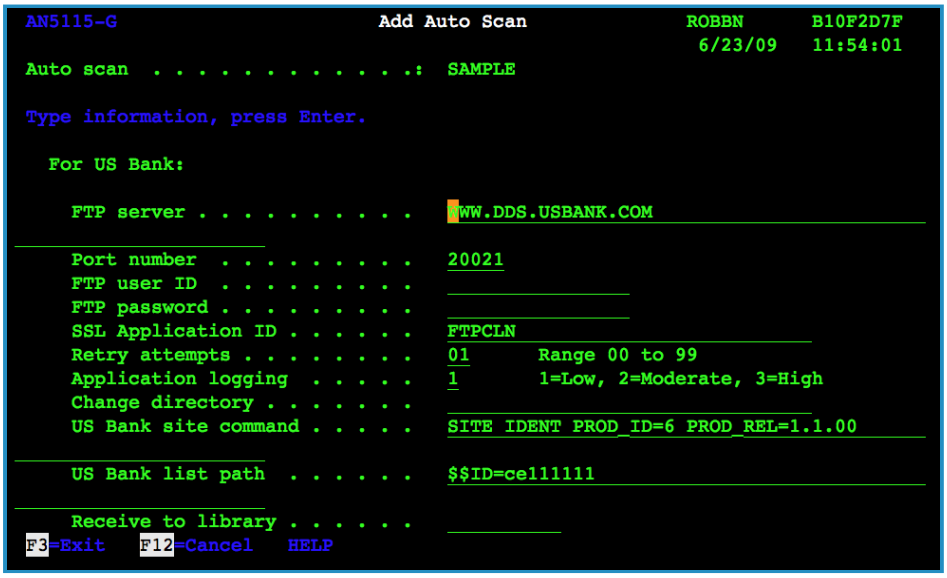
> ***NOTE:*** you must first configure the Email application in Alliance before you attempt to send email notifications.

SMTP address 1, 2 and 3 : Enter up to 3 email addresses for the notification message. You must enter at least one in the first field. The second and third email addresses are optional. Note that you can implement pager notification via email if your pager service supports sending email messages to a pager. Most pager services support this type of interface.

Reply-to address : Enter the Reply-to address for the email message. You should be aware that many ISPs will block messages that have Reply-to addresses that are invalid or not registered to the sending organization.

Subject : Enter a subject for this message. If the FTP session fails and you are using an option that will send a message on failure, the text **ERROR:** will be prefixed to the subject.
Message text : Enter up to three lines of message text. This text will be included in the body of the email message. You can also leave these lines blank.

Press ***Enter*** to go to the next screen.



If you chose option 6 for the US BANK option:

FTP server : Enter the IP address or DNS name of the US Bank FTP server you want to connect to.

FTP port number : Enter the port number of the FTP server. The default is port 20021. You can change the port number if the US Bank FTP server uses a non-standard port.

FTP user ID : Enter the user ID for the FTP server log in.

FTP password : Enter the password for the FTP server.

SSL Application ID : Enter the Digital Certificate Manager Application ID to be used for this SSL connection.

`Retry attempts` : Enter the number of attempts to retry the session in the event the session fails. Enter a number from 01 to 99.

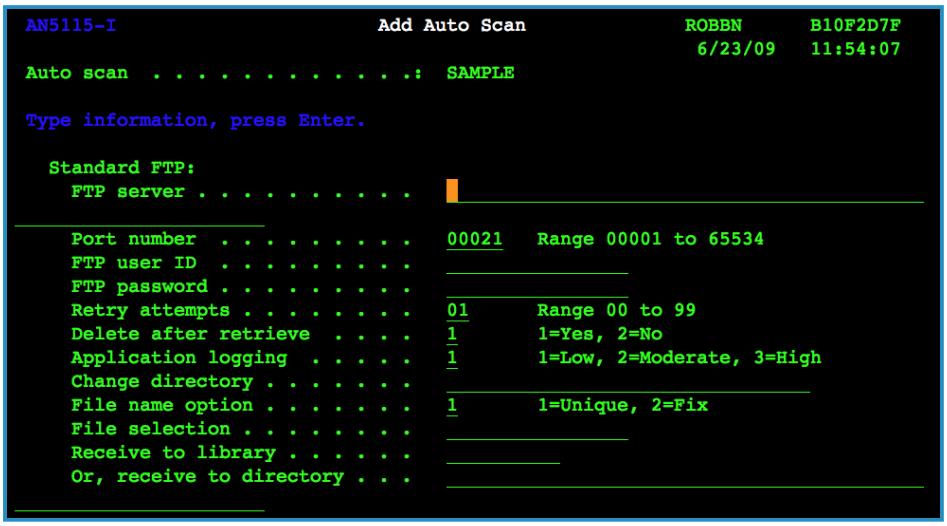
`Application logging` : Enter `1` for minimal FTP session logging. Enter `2` for intermediate level logging. Enter `3` for maximum logging of the FTP session.

`Change directory` : If you want to change to a different directory before transferring the file enter the new directory name in this field.

`Site command` : Enter the site command specified by US Bank.

`Remote file name` : Enter the remote file name format that you received from US Bank.

Press ***Enter*** to go to the next screen.



If you chose option 7 for the Standard FTP:

`FTP server` : Enter the IP address or DNS name of the FTP server you want to connect to.

`FTP port number` : Enter the port number of the FTP server. The default is port 00021. You can change the port number if your trading partner gives you a different one to connect to.

`FTP user ID` : Enter the user ID for the FTP server log in.

`FTP password` : Enter the password for the FTP server.

`Retry attempts` : Enter the number of attempts to retry the session in the event the session fails. Enter a number from 01 to 99.

`Delete after retrieve` : Enter `1` for Yes to delete the file from the FTP server after it is received. Enter `2` for No to not delete the file.

`Application logging` : Enter `1` for minimal FTP session logging. Enter `2` for intermediate level logging. Enter `3` for maximum logging of the FTP session.

`Change directory` : If you want to change to a different directory before transferring the file enter the new directory name in this field.

`File name option` : Enter `1` to create the file on the IBM i with the same name as on the FTP server. Enter `2` to generate a unique name for the file. The name will be in the format Fnnnnnnnn where “nnnnnnnn” is a sequence number. Enter `3` to use the name of the file on the remote FTP server, but make the name compatible with IBM i naming conventions.

`File selection` : If you wish to select files on the FTP server for retrieval, enter the file selection value in this field. For example, if you want to pull files that start with “Ord” enter the value **`Ord*`** in this field.

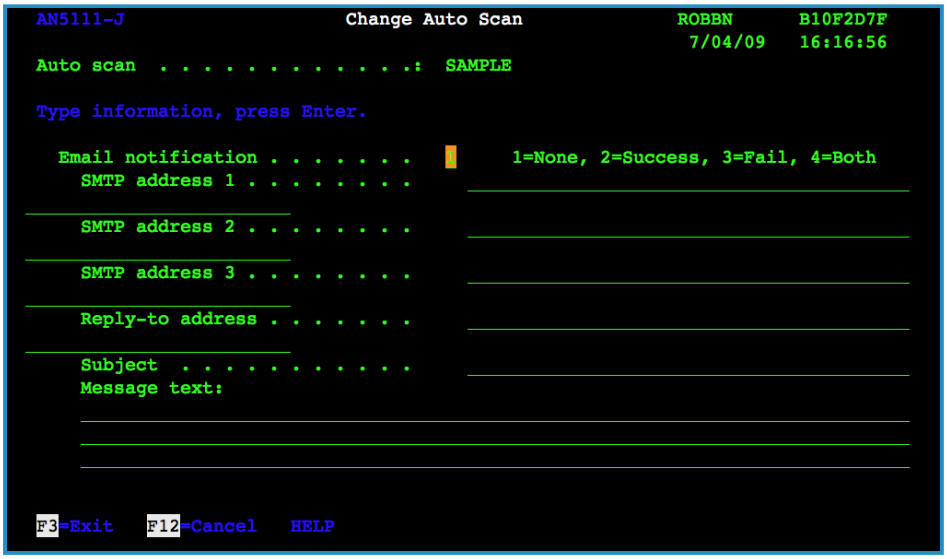
`Receive to library` : Enter the library name to receive the file.

`Or, receive to directory` : If you want to receive the file to an IFS directory rather than a library, enter the full IFS directory path.

Work with auto scan

Use this option to create an auto scan definition. The auto scan facility is used to periodically connect to a remote FTP server and pull files into a local IBM i library or IFS directory. This gives you the ability to automatically pull any new or changed files to the IBM i and process them into your business applications.

Please read Appendix E for complete instructions on how to set up Auto Scan. It will walk you through setting up 2 scripts and 2 definitions to do the Directory Listing of files (**`dir`**) and then do the **`get`** .



If you chose option 7 for the Standard FTP:

`FTP server` : Enter the IP address or DNS name of the FTP server you want to connect to.

`FTP port number` : Enter the port number of the FTP server. The default is port 00021. You can change the port number if your trading partner gives you a different one to connect to.

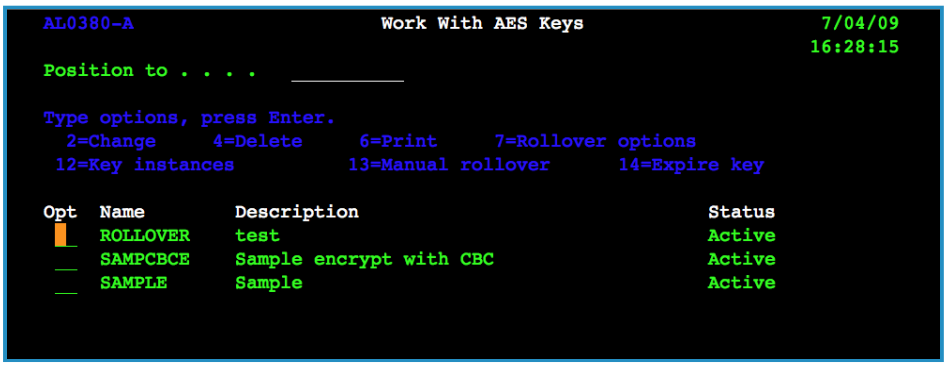
`FTP user ID` : Enter the user ID for the FTP server log in.

`FTP password` : Enter the password for the FTP server.

Work with AES keys

Use this option to create and work with AES keys. You must own a license for Alliance **`AES100`** in order to use this option.

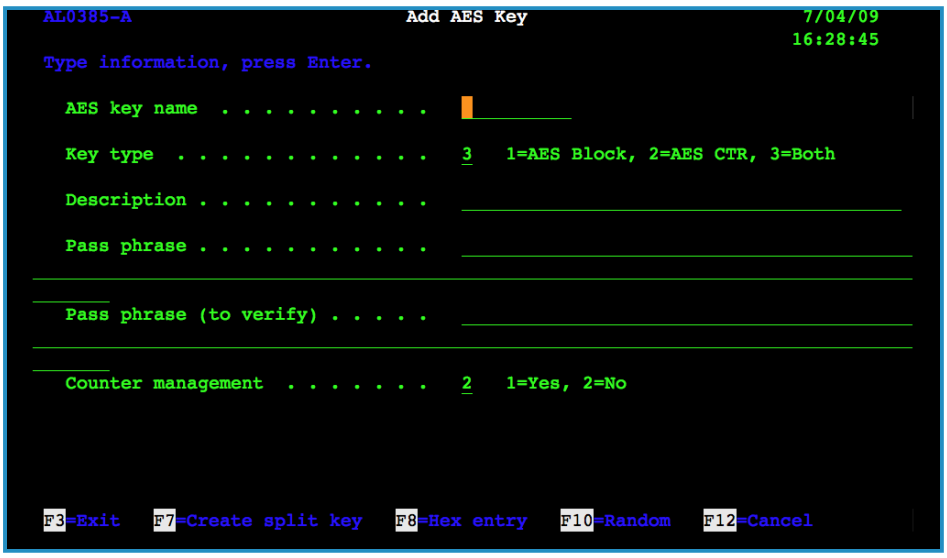
Please read the AES manual for complete instructions on how to work with AES keys.



From this panel you can create a key change a key description or work with rollover options.

Add AES key

When you use function key ***F6*** to add a new AES key the following panel is displayed:



`AES key name` : Give this new AES key a unique name. Do not use embedded blanks or special characters.

`Key type` : Key type `1` indicates a key that is used for AES stream mode encryption. Key type `2` indicates a key that is used for AES counter mode encryption. Key type `3` indicates a key that is used for both AES stream mode and AES counter mode.

`Description` : The description is any text that helps you remember the use of the key, etc. Do not put the pass phrase in the description.

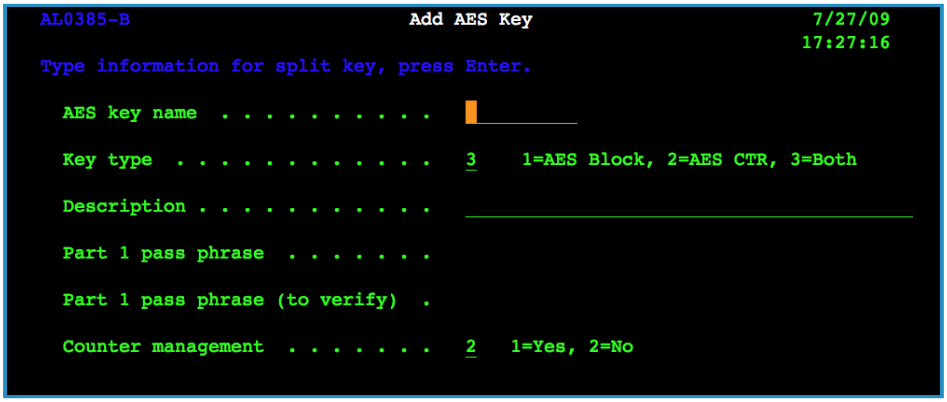
`Pass phrase` : The best pass phrases are from 16 to 32 bytes in length and contain both numbers and letters. Do not use given names, birth dates, or other information that may be easy to guess. Write down the pass phrase and store it in a secure place.

`Pass phrase (to verify)` : Enter the pass phrase again.

`Counter management` : Enter `1` for Yes to enable Counter and Initialization Vector management. Enter `2` for No to not enable Counter and IV management.

`Add split AES key` : On the Add AES key panel you can press ***F7*** to create a split key. An AES split key is a key that is created by two different users who both have security administrator authority. After entering the pass phrase for the split AES key the key will be in

Pending status. Another security administrator must choose to change the key and enter the second part of the pass phrase. Split keys offer more security as no one person will know the entire pass phrase used for encryption and decryption.



`AES key name` : Give this new AES key a unique name. Do not use embedded blanks or special characters.

`Key type` : Enter the key type. A key type of `1` indicates a key that is used for AES stream mode encryption. A key type of `2` indicates a key that is used for AES counter (CTR) mode encryption. A key type of `3` indicates a key that is used for both AES stream mode and AES counter mode.

`Description` : Enter a description for this key. The description is any text that helps you remember the use of the key, etc. Do not put the pass phrase in the description.

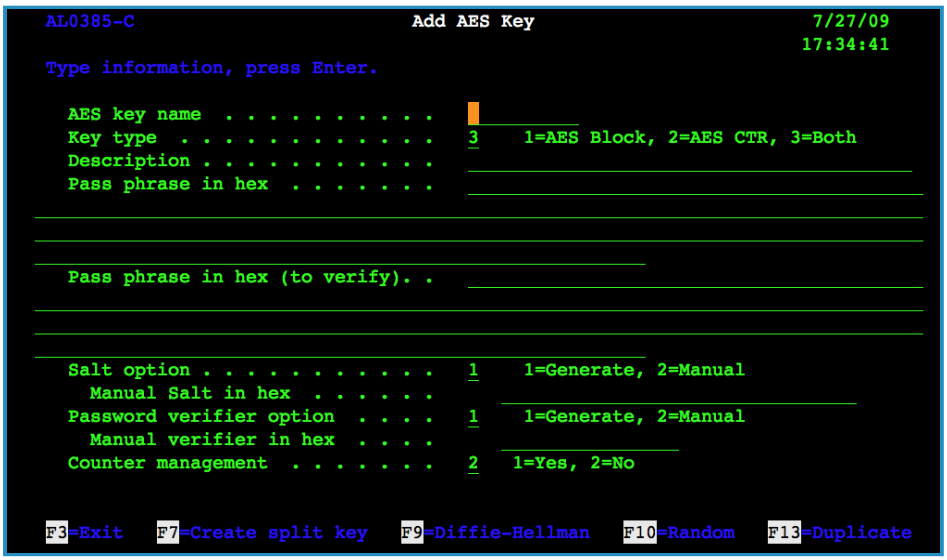
`Part 1 pass phrase and Part 1 pass phrase (to verify)` : Enter the pass phrase twice. Do not use given names, birth dates, or other information that may be easy to guess. Write down the pass phrase and store it in a secure place.

> ***NOTE:*** You can enter up to 16 characters for the pass phrase. Another security administrator will need to change this key (option `2` on the **`Work With Key`** Menu) to enter the second part of the pass phrase before the key is active.

Add hex AES key with Salt and Password Verifier

On the Add AES key panel you can press ***F8*** to enter a key in hex mode. You might want to enter the pass phrase in hex if you created an Alliance AES key on a different platform such as Windows, Linux, or z/OS. Entering the pass phrase in hex also gives you the ability to define a key that contains non-printable characters.

If you are entering a key that you created on a different platform you will want to enter the Salt and Password Verifier from that system. The Pass phrase, Salt, and Password Verifier must all match if you want to decrypt data on the IBM i that was encrypted on a different platform.



AES key name : Give this key a unique name. Do not use embedded blanks or special characters.

Key Type : Enter the key type. A key type of **1** indicates a key that is used for AES stream mode encryption. A key type of **2** indicates a key that is used for AES counter (CTR) mode encryption. A key type of **3** indicates a key that is used for both AES stream mode and AES counter mode.

Description : Enter a description for this key. The description is any text that helps you remember the use of the key, etc. Do not put the pass phrase in the description.

Pass phrase in hex and Pass phrase in hex (to verify) : Enter the pass phrase twice. Enter the pass phrase in hex. That is, use two hex characters to represent each alphameric character in the pass phrase. If you used the Alliance Windows or Linux utility to create the pass phrase salt and verifier you can use the log from that session for these values.

Salt option and Manual Salt in hex : Enter the Salt option. Enter **1** to automatically generate the Salt (random) value for this key. Enter **2** to manually enter the Salt value in hex. The default is to automatically generate the Salt value. If you are entering a key that was created on a different platform you can manually enter the Salt value. Note that the Salt value must match the key pass phrase and must have been created by an Alliance AES key generation routine.

Enter the Salt value in hex. If you selected to manually enter the Salt value, enter the value using hex characters. You must enter 32 hex characters for the Salt field.

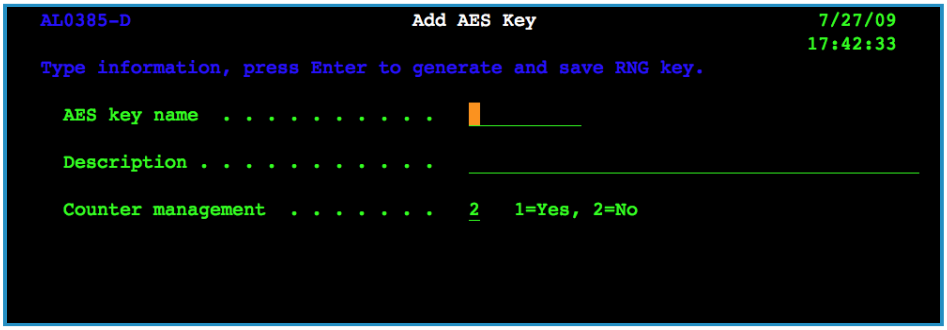
Password verifier option and Manual verifier in hex : Enter the Password verifier option. Enter **1** to automatically generate the Password verifier value for this key. Enter **2** to manually enter the Password verifier in hex. The default is to automatically generate the Password verifier value. If you are entering a key that was created on a different platform you can manually enter the Password verifier. Note that the Password verifier must match the key pass phrase and must have been created by an Alliance AES key generation routine.

Enter the Password verifier in hex. If you selected to manually enter the Password verifier, enter the value using hex characters. You must enter 16 hex characters for the Password verifier field.

Counter management : Enter **1** for Yes to enable Counter and Initialization Vector management. Enter **2** for No to not enable Counter and IV management.

Add AES key from random number generator

On the **`Add AES Key`** panel you can press ***F10*** to generate a key from random numbers.

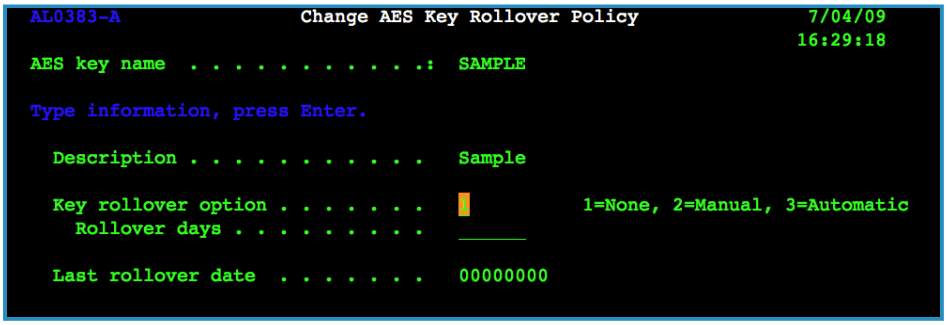


`AES key name` : Enter a name for this key.

`Description` : Enter a description for this encryption key.

`Counter management` : Enter **`1`** for Yes to enable Counter and Initialization Vector management. Enter **`2`** for No to not enable Counter and IV management.

`Change AES Key Rollover Policy` : Use this panel to change rollover options for an AES key. Note that you must use the new encryption APIs if you want to use the rollover options in Alliance AES Encryption. The newer APIs are AESCTR2, AESCBC2, etc.



`Description` : Enter The description of the encryption key.

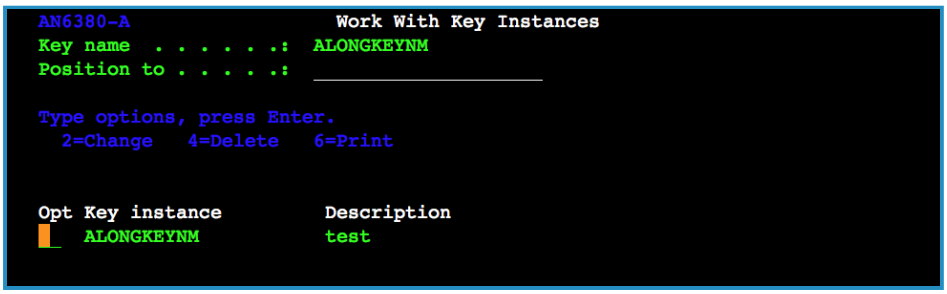
`Key rollover option` : Enter **`1`** to not use key rollover for this encryption key. Enter **`2`** to allow manual key rollover only. Enter **`3`** to enable automatic key rollover. If you specify option **`3`** you must also specify the number of days between key change operations.

`Rollover days` : If you specified option **`3`** for automatic key rollover enter the number of days between key changes. The value must be in the range of 1 to 9999.

`Last rollover date` : The date of the last rollover operation. If this field is all zeros the key has not been manually or automatically changed.

Work With AES Key Instances

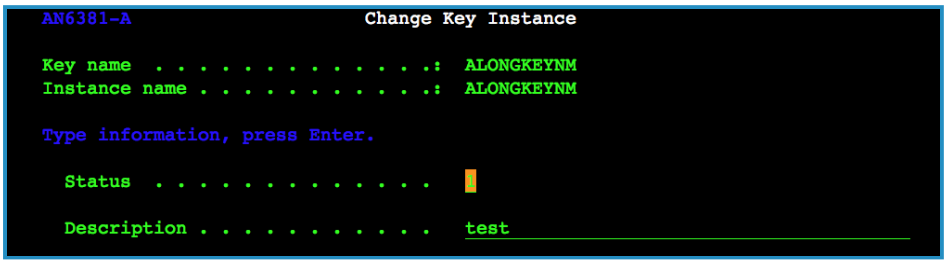
This option displays a list of key instances. You can change the status and description of a key instance, or delete a key instance.



Change : Use this option to change information about the item in the list.

Change Key Instance

Use this panel to change information about an encryption key instance.



Status : Enter **1** for Active or **2** for Inactive.

Description : Enter a description for this key instance.

Manual rollover

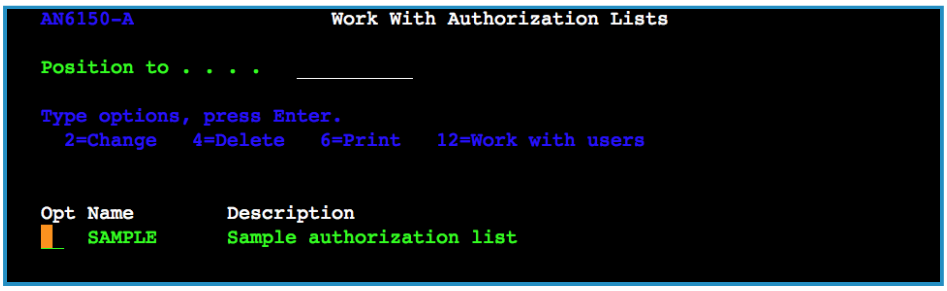
Use this panel to change an encryption key. The current key will be saved as a new key instance and a new key will be generated. The new key will become the default key instance on key retrieval operations.



There are no input fields for this display. Press ***F10*** to change the key.

Work with Authorization Lists

Use this option to create or work with Alliance authorization lists. You can add, change, delete, and print authorization list information. You can also work with the users in the list. Alliance authorization lists may be used in various Alliance applications.



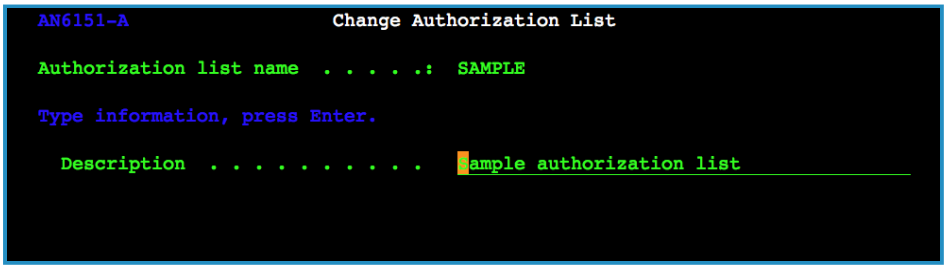
Press ***F6*** to add a new authorization list.



`Authorization list name` : Enter a name for the new authorization list. The list must be a new name.

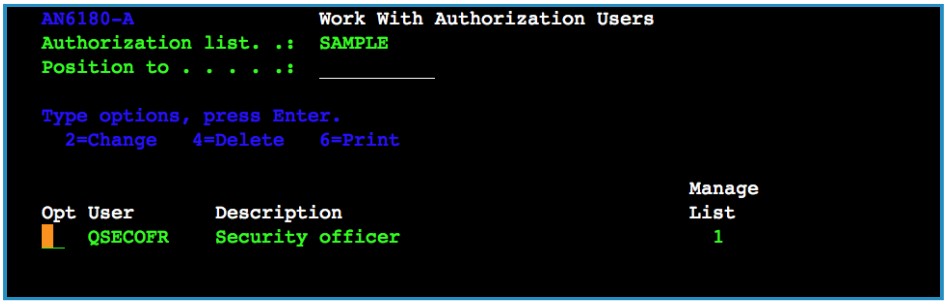
`Description` : Enter a description for this authorization list.

Take option **`2`** to Change an existing authorization list.



`Description` : You can change the description of an existing authorization list.

Take option **`12`** to **`Work With Authorization Users`** .



`Change` : Use this option to change information about the item in the list. **`Delete`** : Use this option to delete an item from the list and all associated detail records.

Press ***F6*** to add a new user.



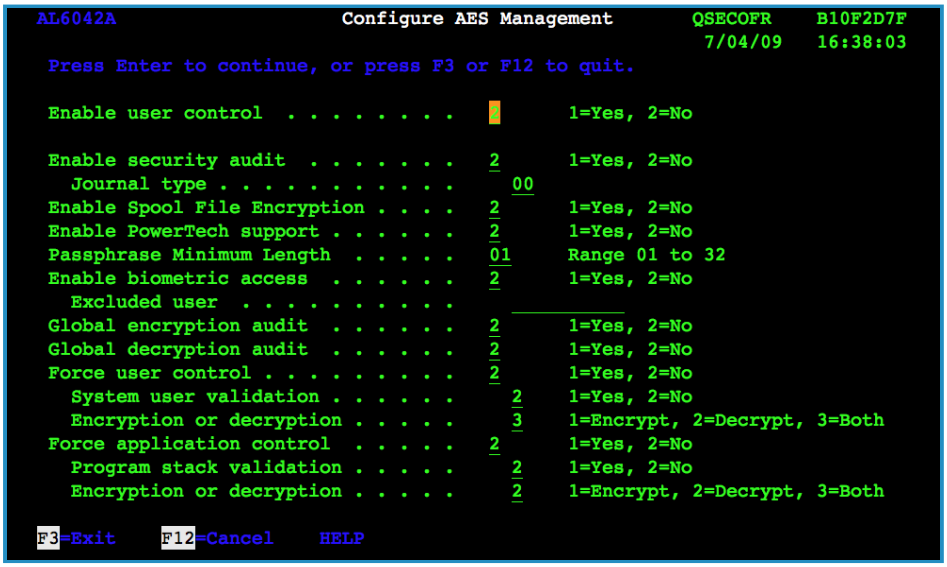
User : Enter the user name to add to the list.

Description : Enter a description for this user.

Manage list : Enter **1** for Yes to allow this user to manage this list. Enter **2** for No to not allow this user to manage this list. The user **QSECOFR** can always manage a list, and any user who is in the group of **QSECOFR** can manage a list.

Work with AES options (must be QSEC)

Use this panel to configure global options for Alliance AES Encryption. You can enable or disable user controls for AES options.



Enable user control : Enter **1** for Yes if you want to enable user controls for accessing AES encryption functions. Enter **2** for No if you do not want to enable user controls.

> ***NOTE:*** If you enable user controlled access to AES encryption functions you must define each user profile to Alliance and the functions they can use. This change takes effect immediately.

Enable security audit : Enter **1** for Yes if you want to capture security audit journal entries from Alliance when security violations occur. Enter **2** for No if you do not want to create Alliance security audit journal entries. Note that you must first enable security audit journaling before you can use this facility. When enabled Alliance will create journal entries

when access is denied to certain encryption options. **Journal type** : Enter the journal type for security audit events. The default is **00** .

Enable Spool File Encryption : Set this to **1** for Yes. This will cause the Encrypt Spool file and Purge Spool file jobs to start automatically in the subsystem. The default is **2** for No.

Enable PowerTech option : Enter **1** for Yes if you want to enable the PowerTech option for user control. Enter **2** for No if you do not want to enable the PowerTech option. Note that you should set up user access rules in the PowerLock application before enabling this option.

Once enabled the Alliance applications will call the PowerTech APIs to determine if a user has authority to the AES encryption function.

Passphrase Minimum Length : Enter a value between 01 and 32 to specify the minimum length required for pass phrases. The default is set to 01, and applies if a pass phrase is used on a Self-decrypting Archive or an Encryption command.

Excluded user : You may exclude one user, such as **QSECOFR** , from biometric access control. Enter the user profile name in this field. If you leave this field blank all users will be under biometric access control for configuration options.

Global encryption audit : Enter **1** for Yes if you want to audit all use of the Alliance encryption APIs. Enter **2** for No if you do not want to audit encryption API use. Note that you must first create the system security journal **QAUDJRN** and related journal receivers. Please refer to the IBM Security Reference manual for more information.

Global decryption audit : Enter **1** for Yes if you want to audit all use of the Alliance decryption APIs. Enter **2** for No if you do not want to audit encryption API use. Note that you must first create the system security journal **QAUDJRN** and related journal receivers. Please refer to the IBM Security Reference manual for more information.

Force user control : Enter **1** for Yes if you want to force user controls in all Alliance encryption APIs that support user control. Enter **2** for No if you do not want to force user controls. When you select Yes for this option the value for user control is ignored and user control validation is always performed.

System user validation : Enter **1** for Yes if you want to use system APIs to retrieve the current user for validation. Any user information provided on the API call will be ignored and the user profile will be retrieved from the system. This option provides an additional level of security. Enter **2** for No if you do not want to force system retrieval of the user.

Encryption or decryption :Enter **1** to implement user controls on encryption only. Enter **2** to implement user controls on decryption only. Enter **3** to implement user controls on both encryption and decryption.

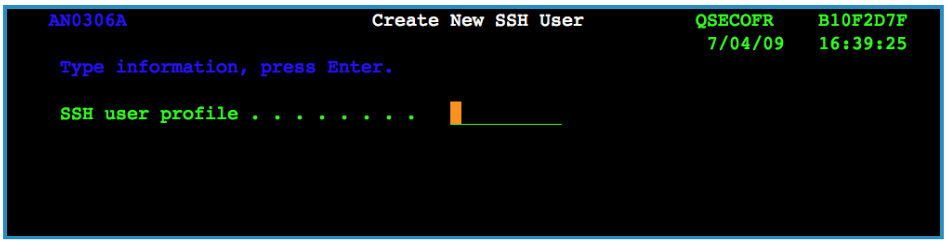
Force application control : Enter **1** for Yes if you want to force application controls in all Alliance encryption APIs that support application control. Enter **2** for No if you do not want to force application controls. When you select Yes for this option the value for application control is ignored and application control validation is always performed.

Program stack validation : Enter **1** for Yes if you want to use system APIs to retrieve the current program stack for validation. Any information provided on the API call will be ignored and the application will be retrieved from the current program stack. This option provides an additional level of security. Enter **2** for No if you do not want to force use of the program stack for application validation.

`Encryption or decryption` : Enter **`1`** to implement user controls on encryption only. Enter **`2`** to implement user controls on decryption only. Enter **`3`** to implement user controls on both encryption and decryption.

Create new SSH user

Use this option to register an SSH user.



This option will create the directory and keys, and the known host table with correct authorities this is the user running the job not the sign on you can create multiple users Automating this process under our subsystem may cause issues, though calling the **`Scanlib`** job through a CL will process the files.

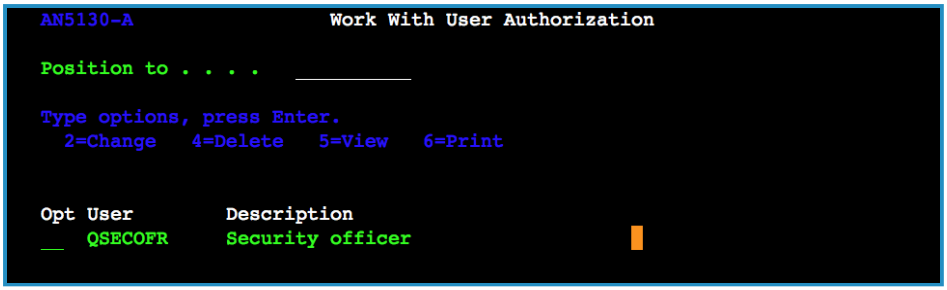
Enroll new SSH server



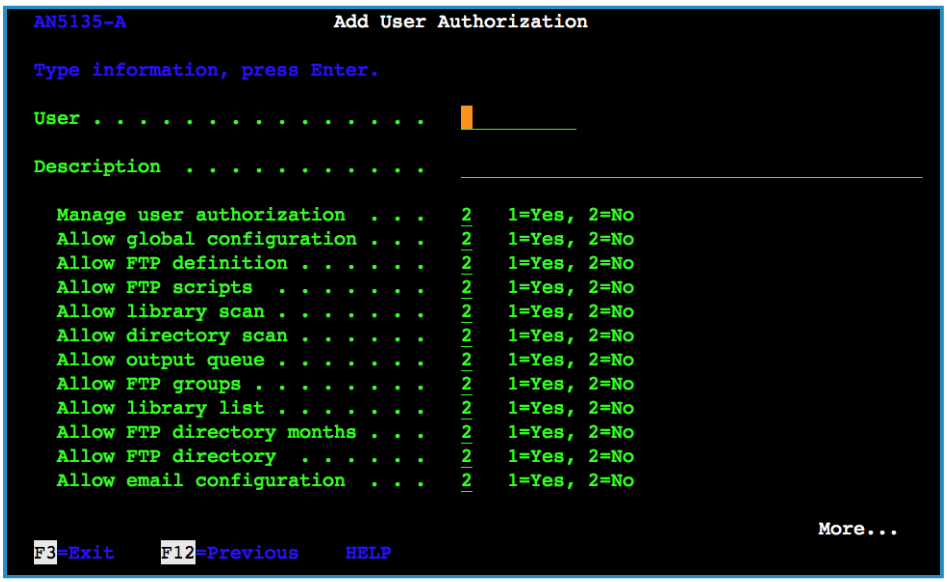
You must retrieve their ssh information into your local known host table. This automates that the ssh profile same as above then add DNS or IP address after running this option:

`wrklnk /home/user/.ssh/known_hosts` Make sure the IP is there.

Work with user access



Use this panel to add authorization information for a user.

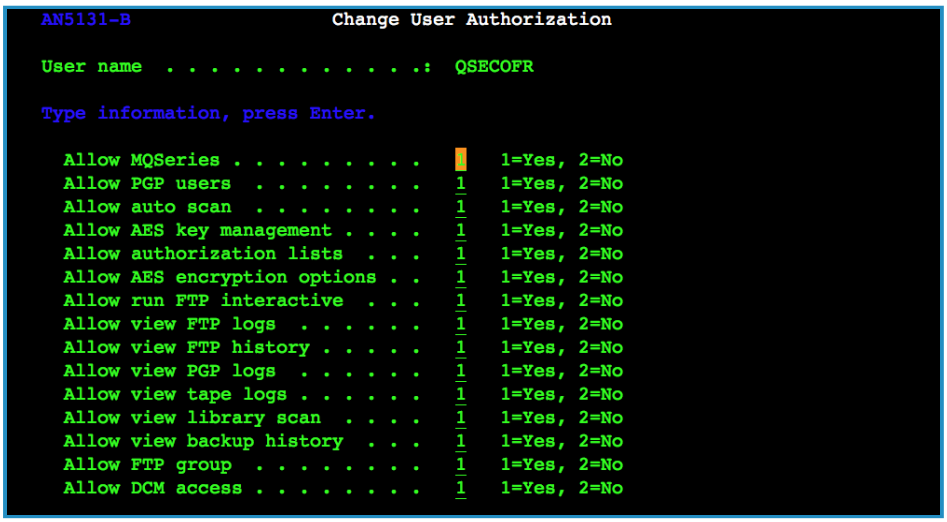


`User` : Enter the user profile name for this new user.

`Description` : Enter a description for this user.

`Manage user authorization` : Enter **`1`** for Yes to allow this user to manage other user authorizations. Enter **`2`** for No to not enable this function for this user.
`Allow global configuration` : Enter **`1`** for Yes to allow this user to configure Alliance FTP Manager. Enter **`2`** for No to not enable this function for this user.
`Allow FTP definition` : Enter **`1`** for Yes to allow this user to add and change FTP definitions. Enter **`2`** for No to not enable this function for this user. **`Allow FTP scripts`** : Enter **`1`** for Yes to allow this user to add or change FTP session scripts. Enter **`2`** for No to not enable this function for this user. **`Allow library scan`** : Enter **`1`** for Yes to allow this user to add or change library scan definitions. Enter **`2`** for No to not enable this function for this user.
`Allow directory scan` : Enter **`1`** for Yes to allow this user to add or change directory scan definitions. Enter **`2`** for No to not enable this function for this user.
`Allow output queue` : Enter **`1`** for Yes to allow this user to add or change output queue defininitions. Enter **`2`** for No to not enable this function for this user.
`Allow FTP groups` : Enter **`1`** for Yes to allow this user to add or change FTP group definitions. Enter **`2`** for No to not enable this function for this user.
`Allow library list` : Enter **`1`** for Yes to allow this user to add or change library list definitions. Enter **`2`** for No to not enable this function for this user.
`Allow FTP directory months` : Enter **`1`** for Yes to allow this user to add or change FTP month definitions. Enter **`2`** for No to not enable this function for this user.
`Allow FTP directory` : Enter **`1`** for Yes to allow this user to add or change FTP directory formats. Enter **`2`** for No to not enable this function or this user.
`Allow email configuration` : Enter **`1`** for Yes to allow this user to add or change email list definitions. Enter **`2`** for No to not enable this function for this user.

Press ***Enter*** to go to the next screen.



`Allow MQSeries` : Enter **`1`** for Yes to allow this user to add or change an MQSeries definition. Enter **`2`** for No to not enable this function for this user.

`Allow PGP users` : Enter **`1`** for Yes to allow this user to add or change PGP user definitions. Enter **`2`** for No to not enable this function for this user.

`Allow auto scan` : Enter **`1`** for Yes to allow this user to add or change FTP auto scan definitions. Enter **`2`** for No to not enable this function for this user.

`Allow AES key management` : Enter **`1`** for Yes to allow this user to add or change AES encryption key definitions. Enter **`2`** for No to not enable this function for this user.

`Allow authorization lists` : Enter **`1`** for Yes to allow this user to add or change authorization list definitions. Enter **`2`** for No to not enable this function for this user.

`Allow AES encryption options` : Enter **`1`** for Yes to allow this user to change AES encryption configuration options. Enter **`2`** for No to not enable this function for this user.

`Allow run FTP interactive` : Enter **`1`** for Yes to allow this user to run an FTP session interactively. Enter **`2`** for No to not enable this function for this user.

`Allow view FTP logs` : Enter **`1`** for Yes to allow this user to view FTP history logs. Enter **`2`** for No to not enable this function for this user.

`Allow view FTP history` : Enter **`1`** for Yes to allow this user to view FTP session history. Enter **`2`** for No to not enable this function for this user.

`Allow view PGP logs` : Enter **`1`** for Yes to allow this user to view PGP session logs. Enter **`2`** for No to not enable this function for this user.

`Allow view tape logs` : Enter **`1`** for Yes to allow this user to view tape session logs. Enter **`2`** for No to not enable this function for this user.

`Allow view library scan history` : Enter **`1`** for Yes to allow this user to view library scan history. Enter **`2`** for No to not enable this function for this user.

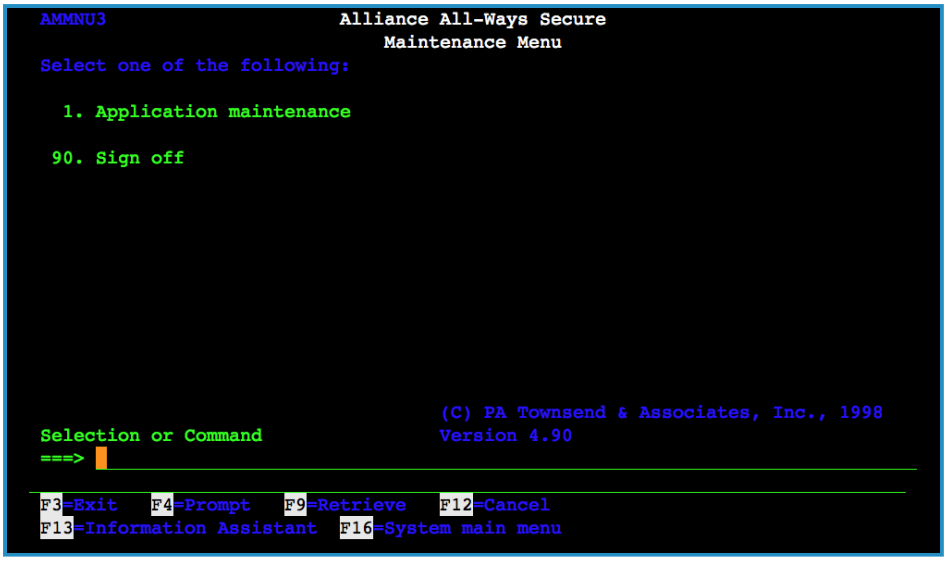
`Allow view backup history` : Enter **`1`** for Yes to allow this user to view file backup history. Enter **`2`** for No to not enable this function for this user.

`Allow FTP group` : Enter **`1`** for Yes to allow this user to run an interactive group FTP session. Enter **`2`** to not allow this function for this user.

Allow DCM authority : Enter **1** for Yes to allow this user to grant authority to DCM configuration. Enter **2** to not enable this function for this user.

Chapter 18: Maintenance

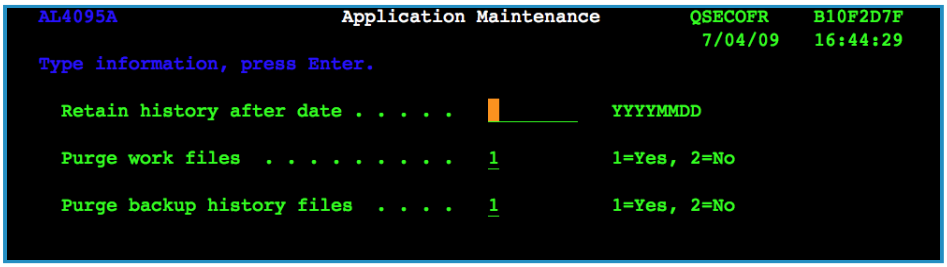
Use this menu to perform routine application maintenance. History and log files will be purged and internal database files will be re-organized. You should perform maintenance weekly or monthly depending on the level of use of the FTP Manager application. Select an option to begin work.



Select an option to begin work.

Application Maintenance

This module is used to remove records from Alliance history and logging files and to reorganize database files to recover database space. It is recommended that you run maintenance weekly or monthly depending on the level of use of the security application.



Retain history after date : Enter the date from which you wish to retain history records. The date is entered in YYYYMMDD format. Records may be removed from **ANHIST** (history file) and **ANLOGA** (application log file).

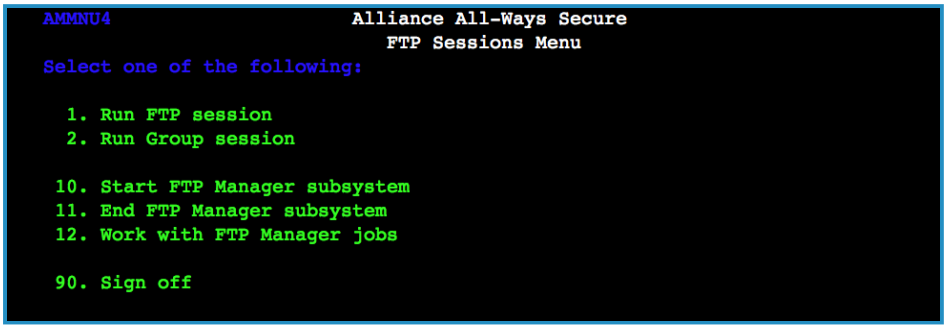
Batch maintenance : You can also run maintenance in a batch job by calling the program **AN4096C** and passing it the date. The date is an 8 byte character field in the format CCYYMMDD where CC is the century, YY is the year, MM is the month, and DD is the day. For example, if you call the program like this:

```
call pgm(an4096c) parm('20090831')
```

All of the records in the history and log files older than August 8th, 2009, will be removed. The batch interface to maintenance is provided so that you can schedule the maintenance in your automated schedule application or perform maintenance as a part of your own maintenance routines.

FTP Sessions Menu

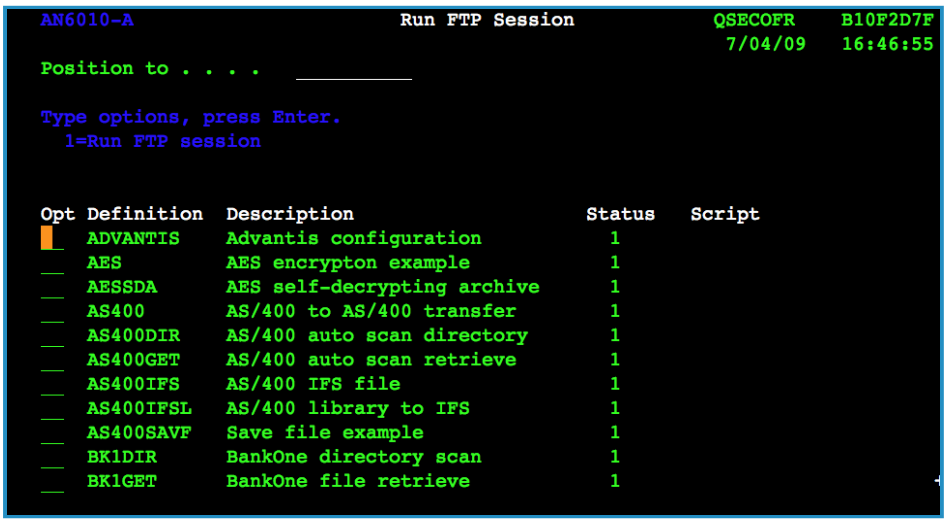
Use this menu to start an FTP session or to start or stop the library scan subsystem.



Select an option to begin work.

Run FTP Session

Use this panel to display a list of FTP definitions that you can use. To run an FTP session place a `1` next to the session name and press ***Enter***. A confirmation panel will be displayed before the session is performed.



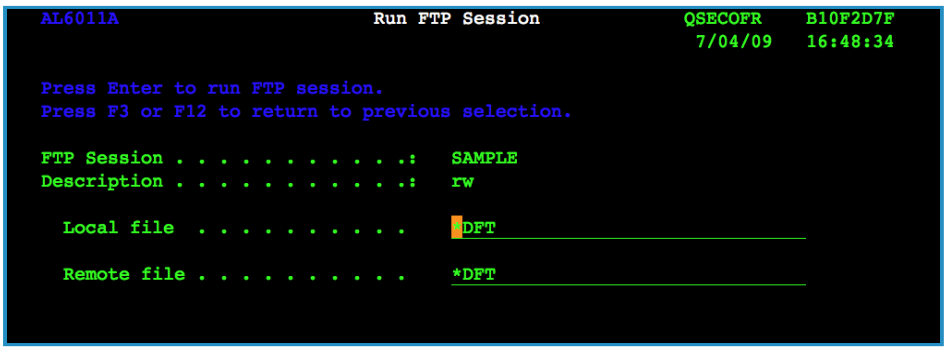
`Position to` : To position the list to a specific record enter the key value in this field and press ***Enter*** or ***F16***.

`Option` : Enter the option corresponding to the function you want to perform. A list of valid options follows:

`1=Run FTP session` : Use this option to run an FTP session using the definition in the list.

Run FTP Session

Use this panel to confirm the FTP session you are about to run. Press ***Enter _** to run this session. Press **_F3*** or ***F12*** to return to the previous display. After the session is complete the results will be displayed on the next panel.

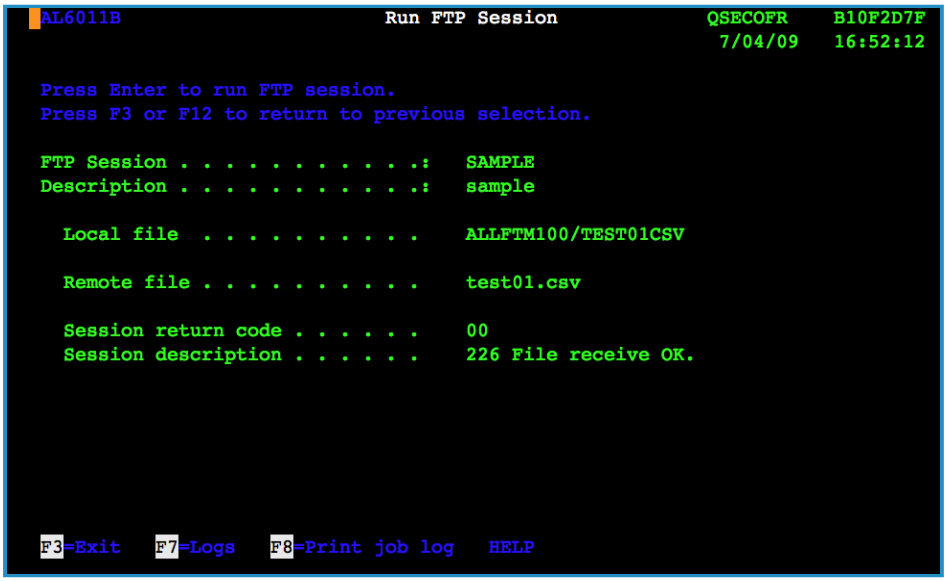


`Local file` : Enter a local file name to send to the remote system, or enter **`*DFT`** for the default file name specified on the FTP definition. If you want to specify a name you must use the standard FTP format: **`Library/file.member`**

`Remote file` : Enter the name of the file for the remote system, or enter **`*DFT`** for the default file name specified on the FTP definition. If you specify a name it must be compatible with the file naming requirements of the remote system.

FTP Session Results

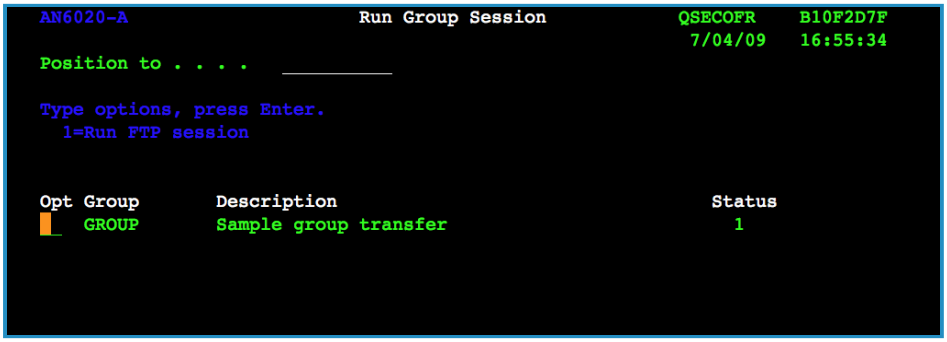
This panel shows the results of running the FTP session you selected. If the return code is **`00`** the session was successful. Any other return code indicates a failure in the session and is shown in red if your display device supports color attributes. If you session failed you can use the History Inquiry option to view the log and more information.



There are no input fields on this display. Press ***Enter*** to continue or ***F3*** or ***F12*** to cancel.

Run FTP Group Session

Use this panel to display a list of FTP Group definitions that you can use. To run an FTP group session place a “1” next to the session name and press Enter. A confirmation panel will be displayed before the session is performed.



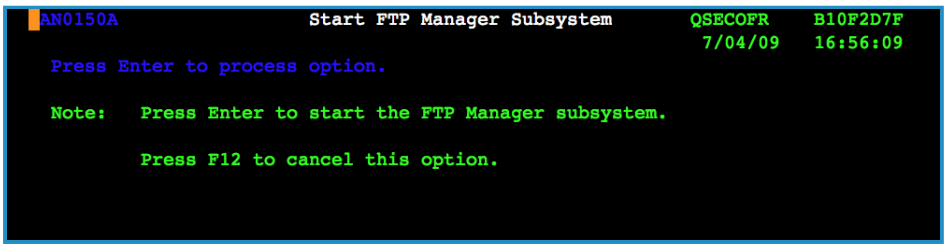
`Position to` : To position the list to a specific record enter the key value in this field and press ***Enter*** or ***F16***.

`Option` : Enter the option corresponding to the function you want to perform. A list of valid options follows:

`1=Run FTP session` : Use this option to run an FTP group session using the definition in the list.

Start FTP Manager Subsystem

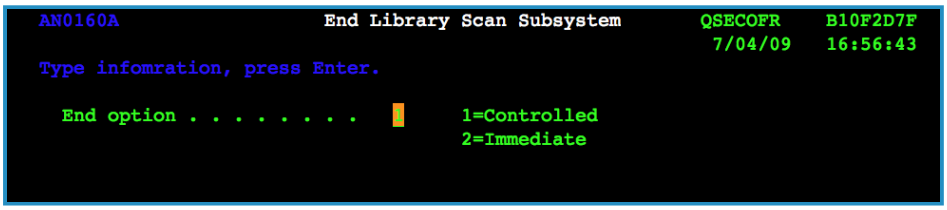
Use this option to start the FTP Manager subsystem named **`ALLFTM100`** . The subsystem runs the library scan functions. If you are not using the library scan functions you do not need to start this subsystem.



There are no input fields on this panel. Press ***Enter*** to start the subsystem, or press ***F3*** to exit. If the subsystem is already active but contains no jobs, use the End FTP Manager subsystem with the **`Immediate`** option to end the subsystem. Then retry this option.

End FTP Manager Subsystem

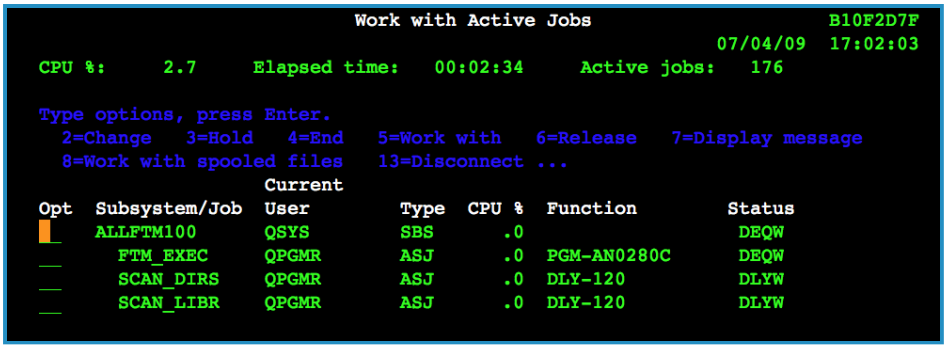
Use this option to end the FTP Manager subsystem named **`ALLFTM100`** . The subsystem runs the library scan`, directory scan, and spool file distribution functions.



Use option `1` to end the jobs in the FTP Manager subsystem in a controlled fashion. The jobs will end when all work and delay intervals are complete. Use option `2` to end the subsystem immediately.

Work With Library Scan Jobs

Use this option to work with active library scan jobs. There will normally only be one job running in this subsystem.



You can view the job log, printed output, and other characteristics of the library scan job with this display. Press ***F3*** or ***F12*** to return to the previous display.

Chapter 19: Inquiry

Use this menu to access the inquiry functions of Alliance FTP Manager. You can display the application log and FTP history log.



Select an option to begin work.

Display Application Log

You can use this option to view a list of user actions and the results as recorded by Alliance. From this panel you can also view the details of an entry and print the log.

The application log is intended to provide the system administrator with diagnostic information about the security module, and should only be created when there is a need for such information. Application log entries will only be created if you enable logging with the Configure

FTP Manager option from the main menu. Entries are removed from the log when you use the Maintenance option on the main menu and specify a date for removing old records. You can also clear the application log with the CLRPFM command:

```
clrpfm anloga
```

`Position to` : To position the list to a specific record enter the key value in this field and press ***Enter*** or ***F16***.

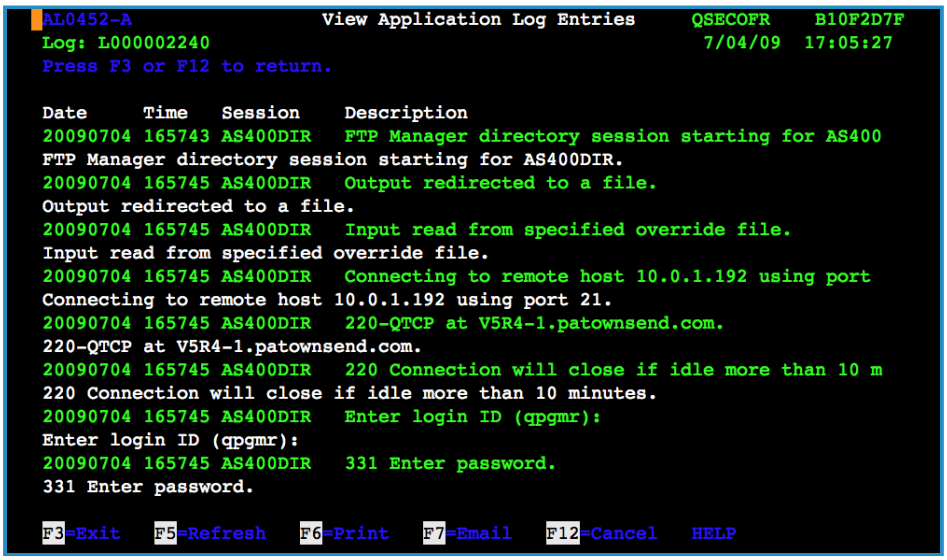
`Option` : Enter the option corresponding to the function you want to perform. A list of valid options follows:

`5=Display` : Use this option to display detailed information about a record.

`6=Print` : Use this option to print a record. The printed report will be placed in the output queue that is the default for your job.

View application log detail

Use this panel to view the detail information for an application log. The detail entries show the results of the FTP session and any errors you might have received.



`Date` : For each line, this is the date the line was created in YYYYMMDD format.

`Time` : For each line this is the time the line was created in HHMMSS format.

`Session` : The name of the FTP session definition used for this transfer.

`Description` : The description provides diagnostic information about the activity and/or errors encountered by the Alliance program. The reasons for rejecting a request will appear in the descriptive text.

FTP session history

This module provides a summary view of the status of all FTP sessions. It shows the date, time, session name, status, and description. The status indicates the success or failure of the session. A value of `00` indicates a successful session. The system administrator can use this

panel to view the status of all FTP sessions.

The history file is **ANHIST** and is an externally described file. The file can be queried using any file utility or query product such as IBM Query, etc. Records are removed from this file only when the Maintenance option is run from the main menu. Records are removed based on the date entered from that panel. You can also use the standard audit trail and exception reports to print information in the history file.



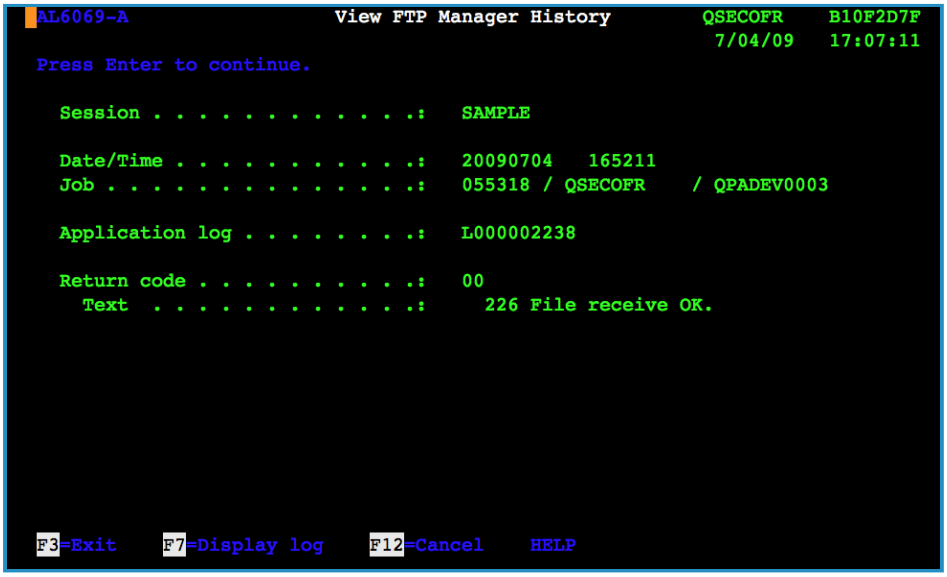
Position to : To position the list to a specific record enter the key value in this field and press ***Enter*** or ***F16***.

Option : Enter the option corresponding to the function you want to perform. A list of valid options follows:

5=Display : Use this option to display detailed information about the history record.

Display history detail

Use this panel to view the details for a history log entry.



Session : The name of the FTP session definition used for this transfer.

`Date / time` : The date and time field records the date and time of the log record. Date is in YYYYMMDD format and time is in HHMMSS format.

`Job` : The job number, user name, and job name are recorded for the history record.

`Application log` : The name of the application log containing the details of the session. You can press ***F7*** to immediately view the log.

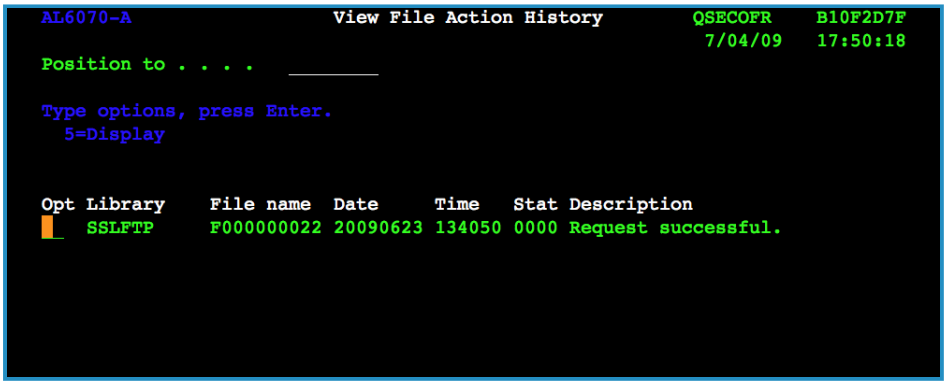
`Return code` : The return code for this session. If the value of the return code is `00` the session was successful. Any other value for the return code indicates a failure.

`Text` : The description of the return code. If the session was not successful this field will contain a description of the problem.

You can also press ***F7*** to view the log for this session.

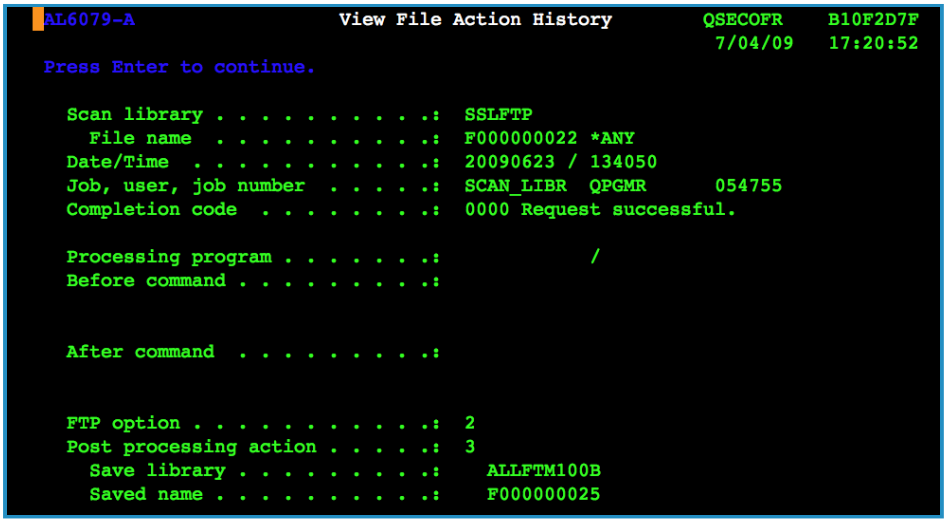
Display library scan history

This module provides a view of the history of library scan activity. The name of the session and the status are displayed. If the status is `0000` the library scan option was successful. Any other value indicates an error. You can use option `5` to display the details for the job.



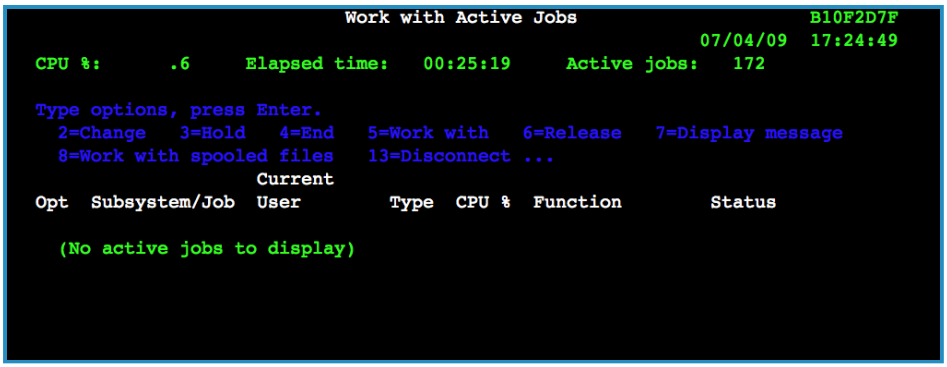
`Position to` : To position the list to a specific record enter the date in YYYYMMDD format and press ***Enter***.

Then `5` to display the history:



Work with jobs

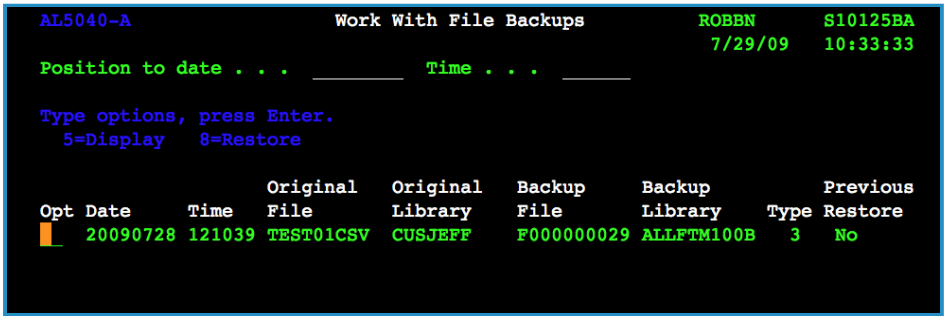
This option displays the jobs running in the **ALLFTM100** subsystem using the Work With Active Jobs (**WRKACTJOB**) command.



Work with file backup history

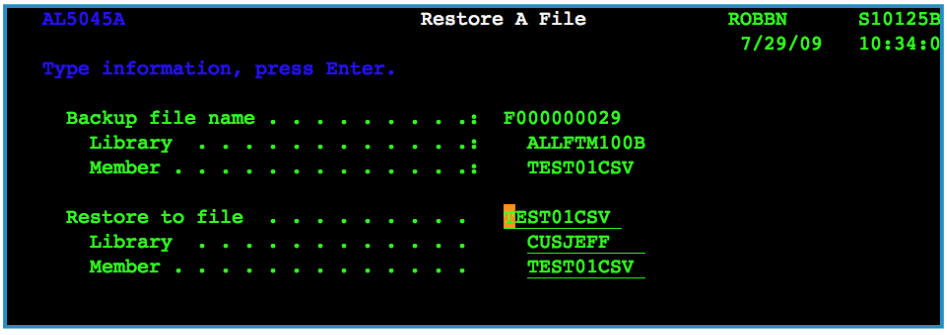
This option provides a view of all files that have been backed up by Library Scan or other file processing options. You can view the details of the back up operation and initiate reprocessing of the file from this display.

Note that some options on Library Scan allow you to process a file and delete or clear it rather than move it to a backup library. If you have specified deletion or clear you will not be able to re-process the file.



You can take option **5** to display the original file and library or option **8** to see the option to restore the original file back into the original library.

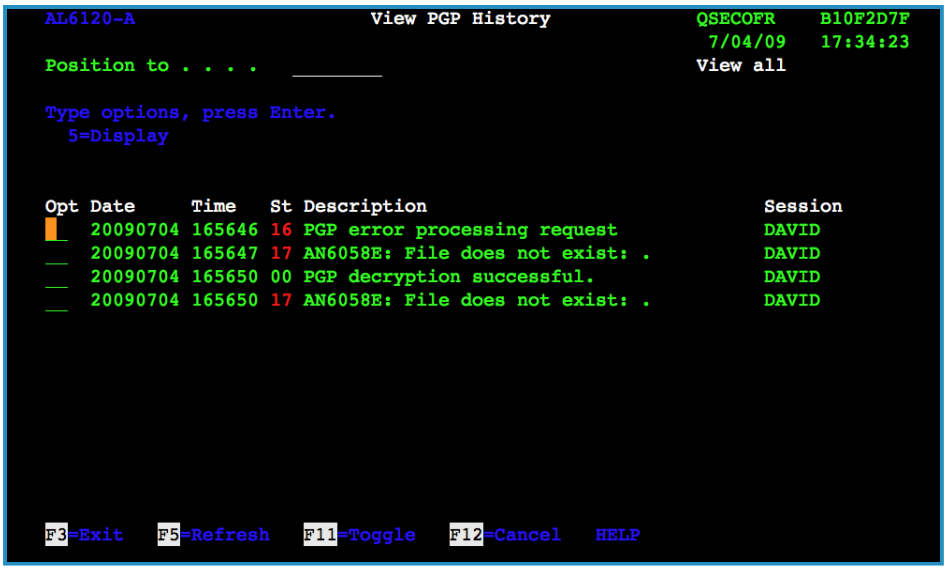
Option **8** to restore a file:



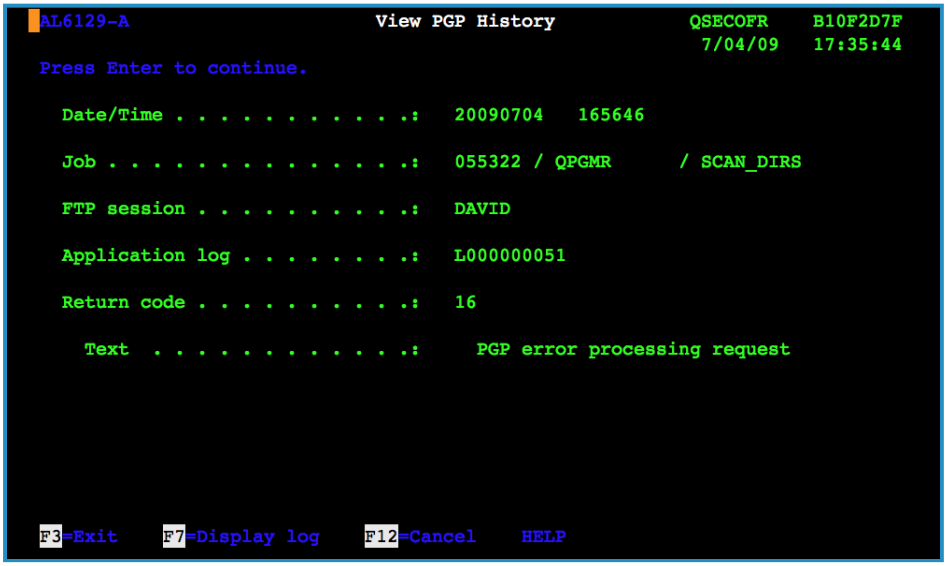
You will notice the “Restore to” information defaults to the original settings. Press ***Enter*** to restore the backup file to the original file name & member, into the original library. You also have the option to change any of the information before restoring the file.

Display PGP history

This option displays the PGP history file.



Choose `5` to display the history of a PGP process.



Then press ***F7*** to display the log.

```
AL0451-A                  View PGP Log Entries          QSECOFR    B10F2D7F
Log: L000000051                                          7/04/09  17:35:55
Press F3 or F12 to return.

Date     Time   Session    Description
20090704 165643 DAVID      PGP decryption started by IFS SCAN in job 055322
PGP decryption started by IFS SCAN in job 055322/QPGMR/SCAN_DIRS
20090704 165644 DAVID      PGP decryption parameters:
PGP decryption parameters:
20090704 165644 DAVID        IFS file:                 /daviddir/add.xml.pg
  IFS file:                 /daviddir/add.xml.pgp
20090704 165644 DAVID        Override output file:     decrypted.xml
  Override output file:     decrypted.xml
20090704 165646 DAVID      ****** PGP Standard Output Log ******
****** PGP Standard Output Log ******
20090704 165646 DAVID      File is encrypted.  Secret key is required to re
File is encrypted.  Secret key is required to read it.
20090704 165646 DAVID      Key for user ID: David PGP9
Key for user ID: David PGP9
20090704 165646 DAVID      2048-bit RSA key, Key ID 0xAF04384C, created 200
2048-bit RSA key, Key ID 0xAF04384C, created 2009/06/05

F3=Exit    F5=Refresh    F6=Print    F7=Email    F12=Cancel      HELP
```

Display IBM IE send history

You can take option 5 to display your Advantis History for files you have sent.

```
AL6160-A                View Advantis IE Send History      ROBBN      S10125BA
                                                            7/29/09    10:35:08
Position to . . . .  _________                              View all

Type options, press Enter.
  5=Display


Opt Log name    Date      Time    St Text
```

```
F3=Exit    F5=Refresh    F11=Toggle    F12=Cancel    HELP
```

Display IBM IE receive history

You can take option 5 to display your Advantis History for files you have received.

```
AL6170-A              View Advantis IE Receive History     ROBBN      S10125BA
                                                            7/29/09    10:36:03
Position to . . . .  _________                              View all

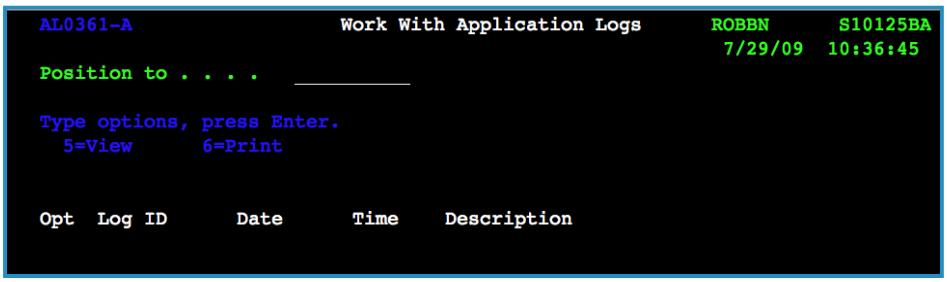
Type options, press Enter.
  5=Display


Opt Log name    Date      Time    St Address
```

```
F3=Exit    F5=Refresh    F11=Toggle    F12=Cancel    HELP
```

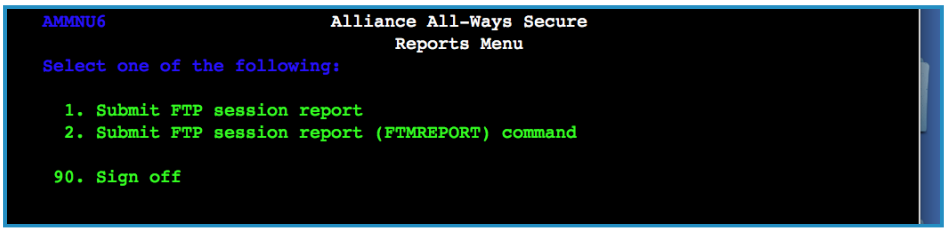
Display tape encryption log

Take this option to display your AES Tape encryption log.



Chapter 20: Application Reports

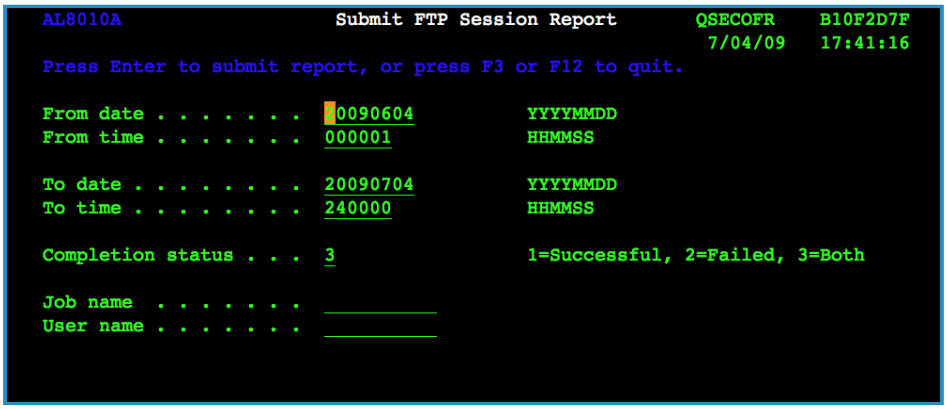
Use this menu to display the report options available in FTP Manager.



Select an option to begin work.

Submit FTP Manager Audit Report

The FTP Manager audit report prints a detailed listing of FTP Manager activity for the period you specify. In addition to selecting the date and time windows for the report, you can select by job name, user name, and ending status of the session.



`From date` : Enter the starting date for records to be included in the audit report. The default starting date is one month prior to today's date. Enter the date in YYYYMMDD format.

`From time` : Enter the from time for the report. The report will start at the time you enter. Enter the time in 24-hour HHMMSS format.

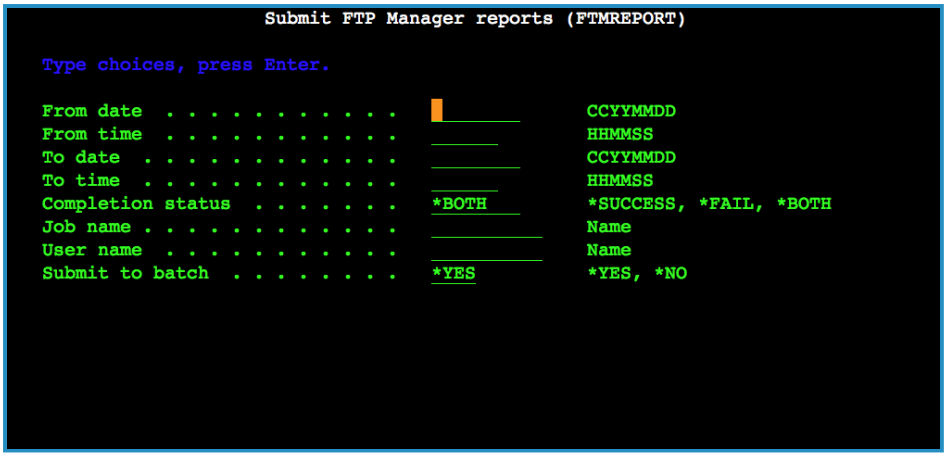
`To date` : Enter the ending date for the records to be included in the report. The default is today's date. Enter the date in YYYYMMDD format.

`To time` : Enter the ending time on the ending date for this report. Enter the time in 24-hour HHMMSS format.

`Completion status` : Enter **`1`** to include only records of FTP Manager sessions where the session was successful Enter **`2`** to include only records of FTP sessions where the session failed. Enter **`3`** to include all records.

`Job name` : Enter a job name if you want to include only records for this job. Leave this field blank to include all jobs in the report.

`User name` : Enter a user name if you want to include only records for this user. Leave this field blank to include all users in the report.



Chapter 21: Support

Use this menu to access all Alliance Support functions. You will normally only use these functions when working with your software vendor to resolve questions or problems on the use of Alliance FTP Manager.

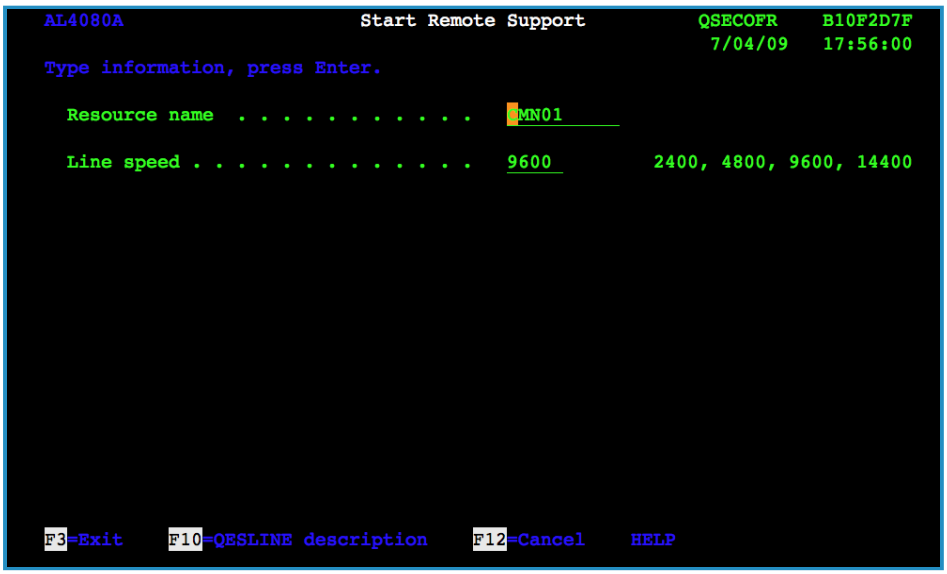


Select an option to begin work.

Start Remote Support

This option is provided to allow your software vendor to perform problem analysis or software updates. When you use this option a line description, controller description, and device description will be created for a dial-in connection to your IBM i. A user profile named PATINC will also be created. The connection will use your normal ECS modem. Your software vendor will request the phone number of your ECS modem to make the connection.

When software support is complete you should run the End Remote Support to disable the user profile PATINC and vary the line, controller, and device off.

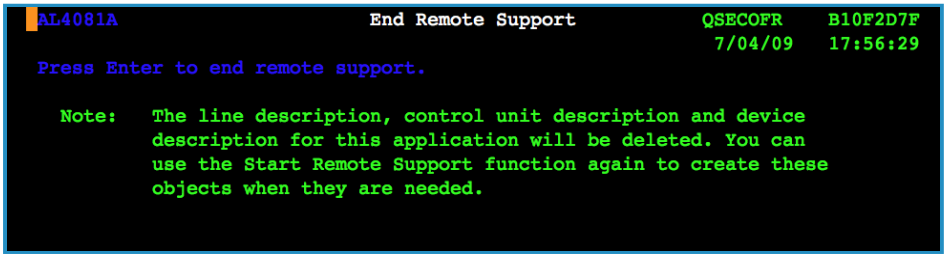


`Resource name` : Enter the resource name of your ECS line. This is usually **`LIN011`** or **`CMN01`** . You can determine the resource name by displaying the line description **`QESLINE`** :

```
dsplind qesline [Enter]
```

`Line speed` : Enter the speed of your ECS modem. This is usually 9600 baud. For an IBM 5853 modem, the speed is 2400. For an IBM 7855 the speed is 9600. For an IBM 7856 modem the speed is 14400.

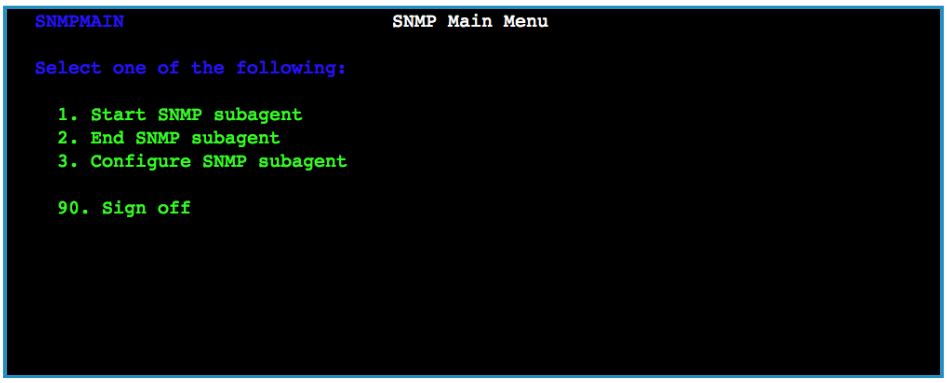
`End Remote Support` : Use this option when your software vendor has finished with remote support. This option will disable the support user profile named PATINC and vary off the line, controller, and device description.



There are no input fields on this panel. Press ***Enter*** to end remote support.

Chapter 22: SNMP Configuration

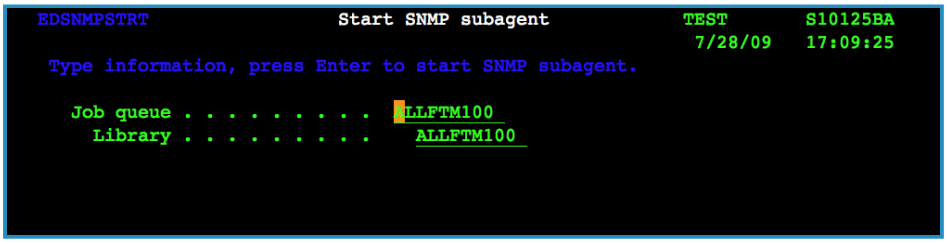
Alliance FTP Manager supports sending error messages to a network manager using SNMP trap messages. You must first configure the IBM i for SNMP support. See the section “Getting Started” for information on how to configure SNMP support on the IBM i. After configuring IBM i SNMP support you can configure Alliance for SNMP message and activate the support here.



Select an option to work with SNMP functions.

Start SNMP subagent

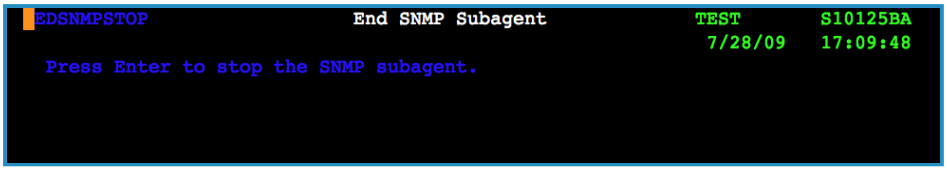
The Alliance SNMP subagent is responsible for sending error messages to your network manager. The network manager might be HPOpenView, IBM SystemView, or any other network manager that supports receiving SNMP trap messages. Note that you may have to configure you network manager to receive traps from the IBM i. Consult you network administrator and network manager documentation to determine if there are configuration steps you need to take to receive trap messages from the IBM i.



`Job queue name and library` : Enter the name of the job queue to submit the SNMP subagent job to. The job queue must be attached to an active subsystem. The default is **`ALLFTM100`** .

End SNMP subagent

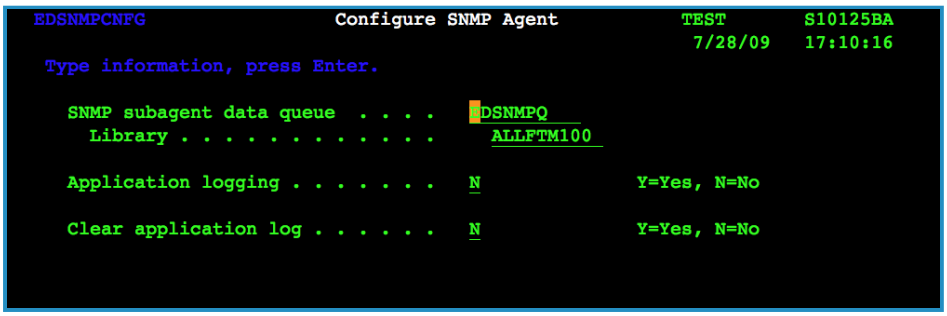
Use this option to end the SNMP subagent. When you use this option a message is sent to the subagent to terminate. This option does not affect the IBM i SNMP agent. It will remain active for other types of SNMP activity.



There are input fields on this panel. Press ***Enter*** to send SNMP support.

Configure SNMP Subagent

Use this option to configure the SNMP subagent. You will not normally need to change this configuration unless instructed by your software provider.



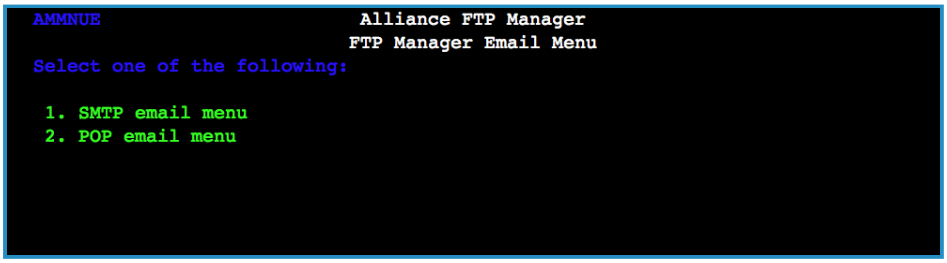
SNMP subagent data queue : Enter the name of the SNMP subagent data queue. This will always be **EDSNMPQ** in library **ALLFTM100** unless your software provider instructs you to use a different name.

Application logging : Enter **Y** for Yes to enable application logging. Enter **N** for No to not use application logging. The default is **N** for No. When logging is enabled diagnostic information is written to file **EDPFLG** .

Clear application log : Enter **Y** for Yes to clear the log when SNMP starts. Enter **N** for No to not clear the log. The default is No.

Chapter 23: Email Menu

When you select the option for email from the main menu the following menu is displayed. You can select to work with SMTP email (outgoing email) or POP email (incoming email).

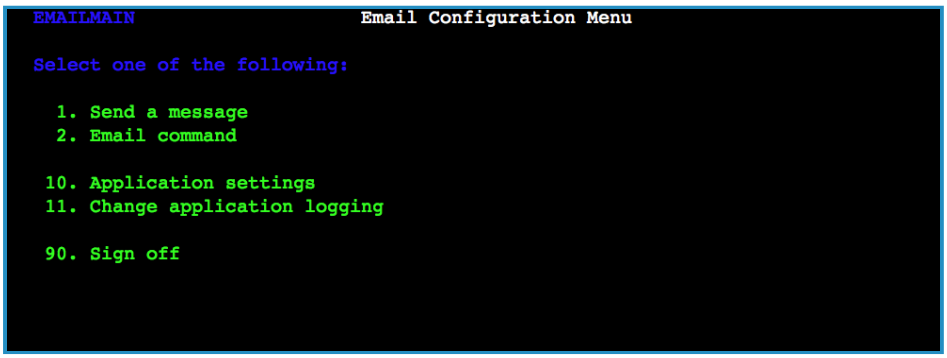


Select an option to begin working with email.

SMTP Email Menu

Alliance FTP Manager contains a full SMTP client application for sending email messages to LAN or Internet recipients. You can even use Alliance to send messages to pagers or PCS devices. If you want to use Alliance to send messages to Alpha pagers you must know the email interface for the pager. This information is available from your pager vendor.

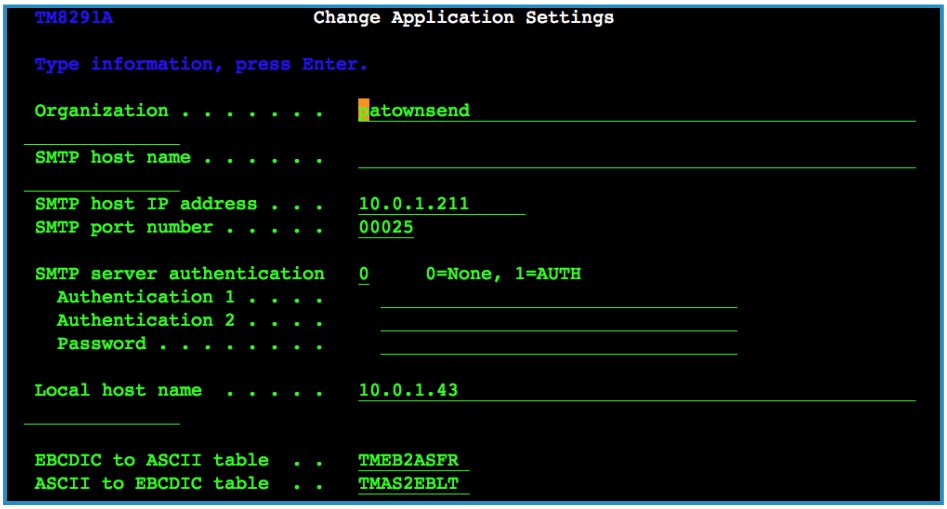
From the main menu select the option for the Email menu.



Select an option to work with SMTP email options. The only option that is required is the option for Application Settings.

SMTP email application settings

Use this option to define the application settings for the Alliance FTP Manager email client. You will define the address of the mail server, port number of the SMTP server, local host name, etc. You should consult your mail administrator before attempting to use the email notification facilities of Alliance. Many mail servers will prevent unauthorized access and you may have to provide the mail administrator with the IP address of the IBM i and the local host name you are using.



`Organization name` : Enter the name of your company or organization. This can be any text description you like.

`SMTP host name` : You must enter either an SMTP host name or an SMTP host IP address. The domain name is in Internet format, for example: acme.com

`SMTP host IP address` : If you do not enter an SMTP host name you must enter an SMTP IP address. This is the IP address of your mail server.

`SMTP port number` : Enter the port number of your mail server for the SMTP service. This is usually port 25, but may be different. Consult your mail administrator for the correct port number.

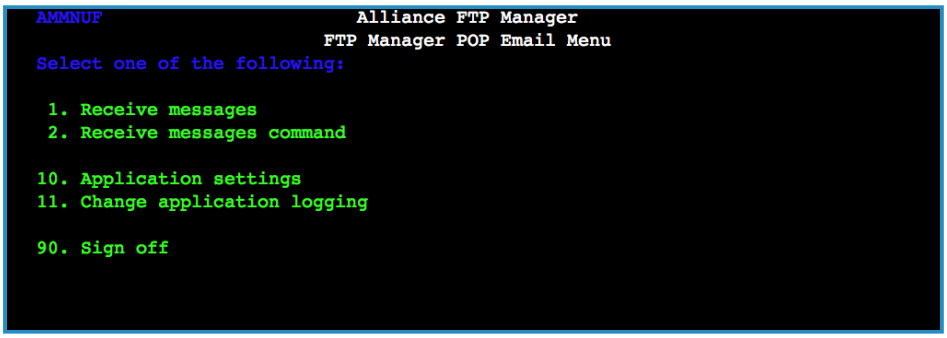
`Local host name` : Enter the name of your IBM i as it will identify itself to the mail server. This is usually the full domain name of the IBM i, for example: “S1015872.ACME.COM”. Give this information to you mail administrator.

EBCDIC to ASCII table : Enter the name of the table to use for EBCDIC to ASCII translation. The table **TMEB2AS** is provided. You may create your own translation tables and register them here. The file **QTBLSRC** in the product library has sample code for creating translation tables.

ASCII to EBCDIC table : Enter the name of the table to use for ASCII to EBCDIC translation. The table **TMAS2EB** is provided. You may create your own translation tables and register them here. The file **QTBLSRC** in the product library has sample code for creating translation tables.

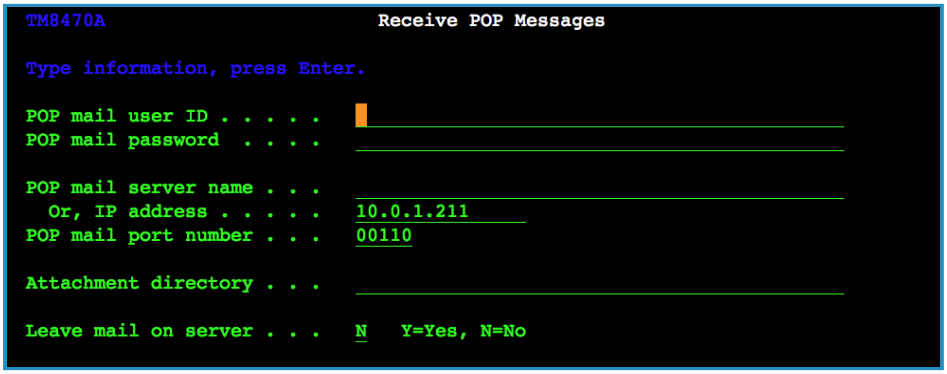
POP Email Menu

Alliance FTP Manager contains a full POP client application for receiving email messages from LAN or Internet recipients. You must first configure information about your mail server before you can use the POP client to receive email. You will need a mail account and password, and you may need to discuss the connection with your mail administrator.



Select an option to work with POP email options. The only option that is required is the option for Application Settings.

Receive messages



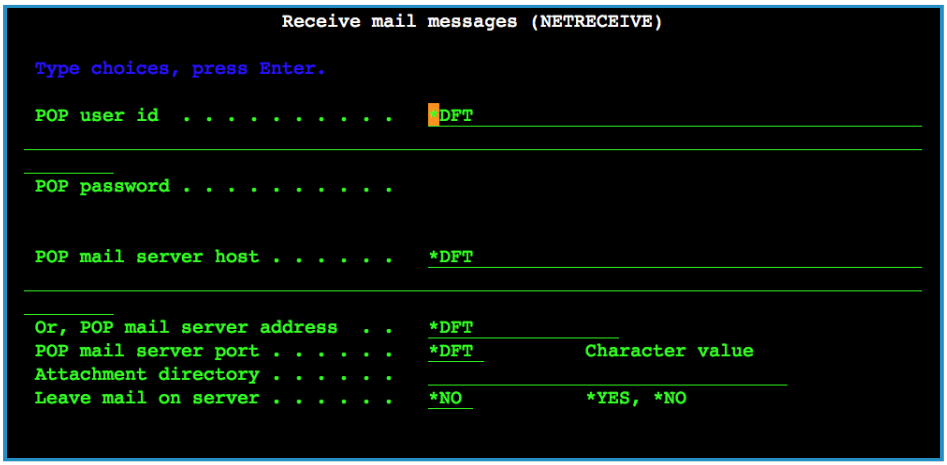
POP mail user ID and Password : Enter the log in ID for the POP mail server and the password for that user.

POP mail server name, IP address, and Port number : Enter either the DNS (server name) OR the IP address of the server. Do not enter both. Also enter the port number through which you are to connect.

Attachment Directory : Enter the directory path in which your attachments are to be placed when you receive them.

`Leave mail on server` : Enter **`Y`** (yes) to leave the mail on the server or **`N`** (no) to pull the mail into the AS400.

Receive messages command



`POP user ID` : Enter the User ID of your mail account.

`POP password` : Enter the password for your user ID. It will not display as you type.

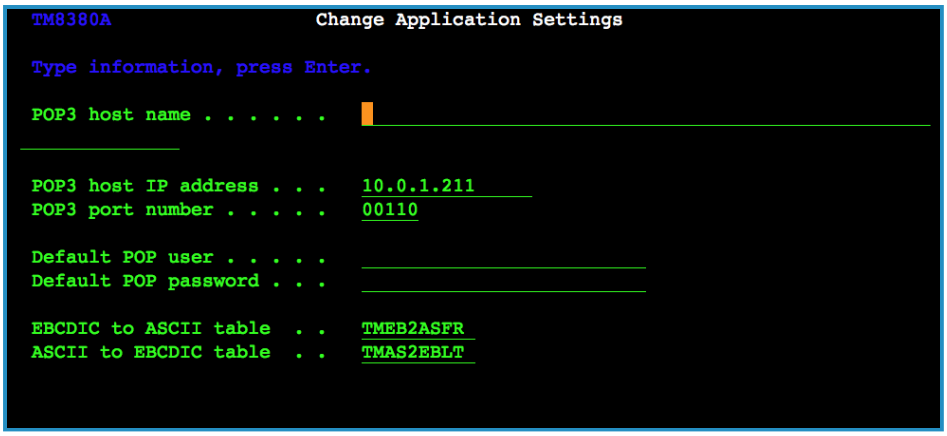
`POP mail server host OR IP address and Port` : Enter the DNS OR the IP address of your mail server. And in the next field enter the port number to which you are to connect.

`Attachment directory` : Enter the full path in the IFS directory to the directory you use to receive attachments.

`Leave mail on server` : Enter **`*YES`** to leave mail on the server or *NO to not leave it on the server..

POP email application settings

Use this option to define the application settings for the Alliance FTP Manager POP email client. You will define the address of the mail server, port number of the POP server, etc. You should consult your mail administrator before attempting to receive mail from the mail server. Most mail servers will prevent unauthorized network connections.



`POP host name` : You must enter either a POP host name or a POP host IP address. The host name is in Internet format, for example: mail.acme.com

POP host IP address : If you do not enter a POP host name you must enter a POP IP address. This is the IP address of your mail server.

POP port number : Enter the port number of your mail server for the POP service. This is usually port 110, but may be different. Consult your mail administrator for the correct port number.

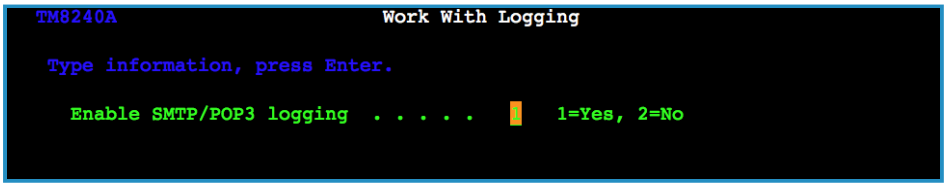
Default POP user : You can specify a default POP user ID to receive mail. This must be a valid POP user ID on the mail server.

Default POP password : If you specified a default POP user you must specify a password. This must be a password that is valid for the POP user ID.

EBCDIC to ASCII and ASCII to EBCDIC tables : Enter the name of the table to use for EBCDIC to ASCII translation. The table **TMEB2AS** is provided. You may create your own translation tables and register them here. The file **QTBLSRC** in the product library has sample code for creating translation tables.

ASCII to EBCDIC table : Enter the name of the table to use for ASCII to EBCDIC translation. The table **TMAS2EB** is provided. You may create your own translation tables and register them here. The file **QTBLSRC** in the product library has sample code for creating translation tables.

Change application logging



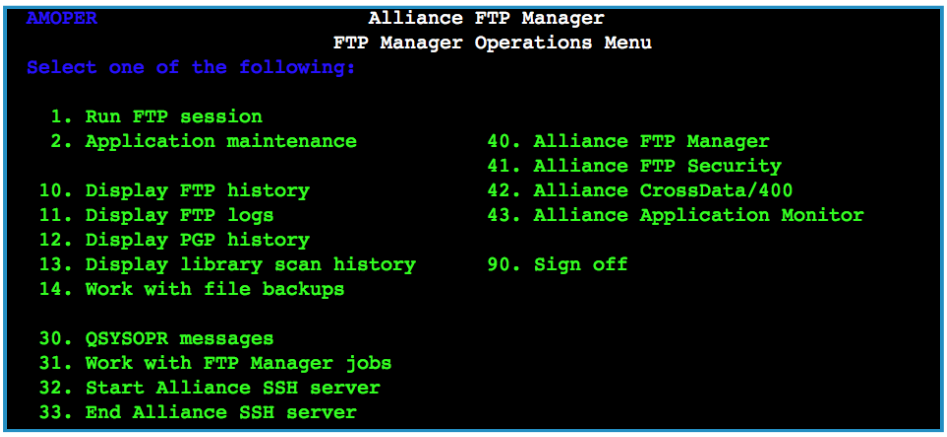
Enter **1** to enable logging or **2** to keep logging turned off.

Chapter 24: Alliance Operations Menu

The operations menu for Alliance FTP Manager is **AMOPER** . You can access this menu by placing the library **ALLFTM100** in your library list and using the **GO** command:

```
addlible ALLFTM100      [Enter]
go amoper                       [Enter]
```

The operations menu is designed to assist IBM i system operators with common FTP Manager tasks. The operator can run FTP sessions interactively, view the status of FTP, PGP, and library scan tasks, and re-queue files for transmission.



The options for the FTP Manager operators menu are described in the previous sections. Please consult the documentation in the previous sections for a description of these functions.

Chapter 25: Problem Determination

Overview

Most problems you will encounter with Alliance will probably have to do with configuring the many options that are available for defining FTP sessions. However, you may encounter actual program problems that require support. Please consider using the following procedures to analyze a problem before reporting the problem to your software vendor.

Using the history log

Alliance records the result of every session that is made using FTP Manager. Use the Display History option on the Inquiry menu to view the results of a session. The history log will provide a short description of any request that failed.

Using the application logs

For a more detailed description of the actions Alliance is taking during FTP processing, you can activate the application log. Use the Configure FTP Manager option to activate application logging. After activating application logging, replicate the problem and view the log with the Display Application Log option on the Inquiry menu. The log will have a detailed list of functions performed during the FTP session. You may be able to determine the cause of the problem by reviewing the log.

QSYSOPR messages

The Alliance applications rarely place messages in the `QSYSOPR` message queue, but you should check there if you are having difficulties. Use the `DSPMSG` command to view this message queue:

```
dspmsg qsysopr
```

Print any error messages from Alliance that may be there. If you have defined a message queue other than `QSYSOPR` to receive messages, please use the `DSPMSG` command to view those queues.

Further action

If you are unable to determine the cause of a problem, please contact your software vendor for support. Be prepared to provide a copy of the history log, application log, source code of your user application, and any `QSYSOPR` messages to your software vendor.

APPENDIX A – IBM 4690 Store Controllers

The following technical notes are extracted from IBM documents for the 4690 store controller. These technical notes may help you set up commands to run via an FTP session. Please refer to the latest technical notes from IBM before making changes to your applications.

FTP server - ADXHSIFL.286

The 4690 ftp server extends file transfer capabilities to an ftp client. Only a single ftp client may be logged on at any given time to the ftp server. Under control of the inetd superserver, a new instance of the ftp server will be started for each client logon request thus allowing multiple clients to be concurrently connected to multiple servers. However, if inetd is not used and the ftp server is currently connected to a client, additional client requests will be rejected until the server concludes its connection and again begins to await incoming connections.

The client must first logon to the server using a userid and (optional) password. The client userid and password is validated by the server when it finds a matching entry in the `TRUSERS` file. This file is named `C:\ADX_SDT1\ADXHSIUF.DAT` and should contain an entry for each client with whom the server will maintain an ftp session. Connection requests from unauthorized clients will be rejected. You should update this file with the ftp clients you wish to grant access.

In addition to userid and password information in the `TRUSERS` file, each entry lists the read/write access privileges for that user. In this way certain users can be limited to more restricted access, e.g. read-only, or more liberal access, e.g. read-write. Access can be granted per drive/directory for physical drives A: C: D: and ram drives T: U: V: W:. In addition, the 4690 printer can be used as a file transfer destination for copying files directly from the client to prn device.

A sample `TRUSERS` file is copied onto the controller during 4690 TCP/IP installation. It contains an entry for user ‘anonymous’ (no password required) and gives read/write access to C:\ and its subdirectories.

> ***IMPORTANT:*** The TRUSERS file must exist in order for the ftp server to successfully initialize. If the server cannot access file `C:\ADX_SDT1\ADXHSIUF.DAT` it will reject the client logon request, log a system message and then terminate.

Information about the format of the **TRUSERS** file can be found in the OS/2 TCP/IP product publications.

An additional command has been added to the 4690 ftp server to support the starting of 4690 background applications from an ftp client. This capability is provided through the use of the "quote" ftp client command with the **ADXSTART** server command provided the user at the client has the necessary execution authorization.

An example of starting background application named **testpgm.286** is:

```
ftp>  quote adxstart testpgm.286
```

If you want to execute a batch file (**.BAT** , etc.) you must use the following format:

```
ftp>  quote adxstart \adx_spgm\command.286 -c mybatfile
```

The previous command would cause the 4690 ftp server to start **testpgm.286** as a 4690 background application. Note that the ftp server returns a message to the ftp client stating that the program has been started. The server does not wait for the background program to complete.

Also, note that the program name must be fully path-qualified and is limited to 22 characters. If using path-qualified program names from a Unix client, you may need to surround the entire quote string in double quotes if the backslash character is required. Optionally you can use forward slash for the path name delimiter.

Program parameters can be specified immediately following the program name, each separated by at least one blank. The length of the parameter list cannot exceed 46 characters. The program will be started with default priority of 200 and cannot be changed.

User authority to start background applications using the 'quote' command is granted through the **TRUSERS** file. A special execution access privilege is granted to clients using the 'ex:' tag for the user. This tag is similar to the 'rd:' and 'wr:' tags used to grant read and write privileges to users. The sample **TRUSERS** file, **ADXHSIUF.DAT** , contains an entry for user=anonymous (no password required) and grants write access (wr:) for drive c:\ (and all subdirectories of the root directory) and read access (rd:) for drive c:\ (and all subdirectories of the root directory) and background application initiation access (ex:). This entry is shown as:

```
user: anonymous
wr: c:\
rd: c:\
ex:
```

Note that the 'ex:' tag does not require additional parameters. Any user without execution authority will be returned a message by the ftp server indicating they are not authorized to start background applications.

FTP Server Timeout

A receive_data timeout has been added to the 4690 FTP server to prevent a server hang condition in the event that either the FTP client has crashed or the physical link between the client and server is disconnected.

The timeout is set by the definition of a User-Defined Logical Filename. To define the logical name, use 4690 Controller Configuration to add the name **ADXFTPTO** . The definition for this name can be a value between 1 and 14400 and represents the maximum number of SECONDS to wait for a client to transmit data to the server. Note that this timeout value is only relevant to the server while it is receiving data. It does not affect either the ftp control connection IDLE time (which is also configurable from the client using the ftp site command) or have any bearing on retransmission of data when the server is sending data to a client that may not be responding.

If the **ADXFTPTO** logical filename is not defined, or it is defined with an invalid value, a default timeout of 2 hours is used. If the timeout expires, the FTP server closes all connections with the client and terminates.

APPENDIX B – Applying an Upgrade

Steps to apply an upgrade for Alliance FTP Manager

> ***IMPORTANT:*** If you are upgrading your Alliance product to a new version you may need a new license code. The Alliance license routines changed at the following release:

Alliance FTP Manager 4.77

If you are on a prior release, contact your software vendor for a new license key. Your old license key will not work. There is no charge to receive a new license key.

Authorization Lists:

If you are upgrading from a version prior to 6.20 you must add users to the Authorization List.

You must sign on as **QSECOFR** and create an Authorization List called **ALLFTMOWN** and a User Profile called **ALLFTMOWN** to secure the product using the following commands:

```
CRTUSRPRF USRPRF(ALLFTMOWN) PASSWORD(*NONE) USRCLS(*SECOFR) INLMNU(

CRTAUTL AUTL(ALLFTMOWN) TEXT('Alliance AUTL for Ownership of FTM Pr
```

Next, you can add Users to the Authorization List by using the following commands:

```
CRTUSRPRF USRPRF(somebody) PASSWORD() AUT(*EXCLUDE)

ADDAUTLE AUTL(ALLFTMOWN) USER(somebody) AUT(ALL)
```

You can add group profiles to the authority lists instead of individual users as well.

Before you install the upgrade make sure you clear the following log files for the products you use:

FTP Manager:

TMTEXT: clrpfm allftm100/tmtext

TMTXOH: clrpfm allftm100/tmtxoh

TMITXT: clrpfm allftm100/tmitxt

TMTXIH: clrpfm allftm100/tmtxih

TMLOGA: clrpfm allftm100/tmloga

ANLOGA: clrpfm allftm100/anloga

MQLOGA: clrpfm allftm100/mqloga

CrossData:

TMLOGA: clrpfm alldta200/tmloga

AFLOGA: clrpfm alldta200/afloga

FTP Security:

AFLOGA: clrpfm allftp130/afloga

AFLOGO: clrpfm allftp130/aflogo

Application Monitor:

TMLOGA: clrpfm allmon100/tmloga

EDPFLG: clrpfm allmon100/edpflg

> ***IMPORTANT:*** It is critical that you save the joblog during this process. This is the only way we can troubleshoot any install errors and validate a good install/upgrade.

Installing the upgrade

If you are installing Alliance as an upgrade to a previous release, make sure you are `QSECOFR` and use these procedures:

1. End the `ALLFTM100` subsystem.
2. Save the library `ALLFTM100` to backup tape or `savefile`.

```
CRTSAVF QGPL/ALLFTMSAVE
SAVLIB LIB(ALLFTM100) DEV(*SAVF) SAVF(QGPL/ALLFTMSAVE)
```

1. Remove the `ALLFTM100` library from your library list using `RMVLIBLE`.

```
RMVLIBLE ALLFTM100
```

1. Rename the existing `ALLFTM100` library to a save name.

```
RNMOBJ ALLFTM100 *LIB ALLFTMOLD
```

> ***NOTE:*** If you are unable to rename the library use the `WRKOBJLCK` command to determine which jobs have a lock on the library and end those jobs, then rename the library.

Restoring an upgrade from a download

You can restore the `ALLFTM.SAV` file to your IBM i using the following steps:

1. Create a save file on the IBM i:

```
CRTSAVF QGPL/ALLFTM
```

1. Move the `allftm.sav`
2. From the PC start an FTP session with the IBM i. After logging on to the IBM i in your FTP session set the transfer type to “Binary” and put the file to the new save file you created:

```
open 1.1.1.1   (use your IBM i IP address)
user myprofile [Enter]
pass mypassword [Enter]
binary           [Enter]
put c:\allftm.sav qgpl/allftm
quit
```

1. After transferring the file to the IBM i you can use the Restore Library command to restore the product. Note that you must have adequate authority to restore objects on the IBM i. You can use the `QSECOFR` or `QSYSOPR` profiles to restore the product:

```
RSTLIB SAVLIB(ALLFTM100) DEV(*SAVF) SAVF(QGPL/ALLFTM) MBROPT(*ALL) ALW
```

Make sure that the restore operation completed successfully. After the product is restored you can delete the save file on the IBM i:

```
DLTF QGPL/ALLFTM
```

1. You can now get to the main menu of the new version:

```
ADDLIBLE ALLFTM100
GO AMMAIN
```

1. From the main menu take option `1` - “installation” and option `20` - “apply release upgrade”.
2. The next screen will ask you for the name of the previous library and this will transfer all of your configurations and license keys from the old version. (In this example, the name of the previous library would be `ALLFTMOLD`).

Please let us know if you have any questions prior to performing the upgrade.

Restoring an upgrade from a CD

After completing steps 1-5 under “Installing the **Upgrade”, follow these instructions

Insert the CD in the IBM i CD Rom drive.

On the IBM i command line use the `LODRUN` command to display the product menu:

```
Lodrun dev(opt01) dir('/')
```

You will see an installation menu. Select the option to install FTP Manager and press ***Enter*** to confirm the name of the library to install to. The installation program will install library `ALLFTM100` to your system.

After the library is restored you are finished with this part of the installation. Press ***F3*** to return to the installation menu.

To complete the upgrade:

```
ADDLIBLE ALLFTM100
GO AMMAIN
```

From the main menu take `1` for install and then option `20` for apply a release upgrade. The next screen will ask you for the name of the previous library and this will transfer all your configurations and licenses. (In this example, the name of the previous library would be `ALLFTMOLD` .)

Please let us know if you have any questions prior to performing the upgrade.

APPENDIX C – Library Scan

LIBRARY SCAN for Processing Files Inbound

Library Scan is designed to work with files in your Library and either send a file to a remote server or process a file into your database.

You will need to create a scan job, and a file action.

Some information you will need for the set up is:

- A Library created for the Scan job to scan
- A processing program to call to process the file. We will pass it the name of the file the library and the member.

Library scan

To set up a scan over this Library, go to the configuration menu and take option **`4`** for Work with Scan Libraries. Press ***F6*** to add a new one. You must name it the same name as the Library. Give it a description, then press ***Enter*** all the way through and you will see the new scan added to the list.

File action

A file action tells FTP Manager what to do with the files it finds in the Library. Put a **`12`** by the Library Scan you've just created and press ***Enter***. Press ***F6*** to add a new file action. There are 3 ways to name a file action. You can name it with a specific file name (**`FILE_NAME`**) if you know that name and it is consistent every time.

You can name it with a generic name (**`ABC*`**) if the first few characters of all the files are the same but the ending of the name is unique (such as a date/time stamp).

You can name the file action **`*ANY`** to process all files the same way.

You can use several file actions in the same Library Scan all doing different things to different files. Library Scan will find the first file and see if there is a specific file action for the first file. If YES, it will process that file using the specific file action, if NO it will see if there is a generic file action for that file. If YES, it will process that file using the generic file action, if NO it will see if there is a **`*ANY`** file action. If yes it will process this file using the **`*ANY`** file action, and if there is no **`*ANY`** Library Scan will not do anything to that file and go look at the next file.

Name your file action and give it a description.

Make Perform FTP Session 2 for no and enter the name of your processing program & the library where it resides in the Processing program & Library fields.

Press ***Enter*** and go to the next screen.

Choose how you want to archive the file after it's processed.

Press ***Enter*** all the way through for it to create this file action and you will see it added to the Library Scan.

If your subsystem is active this job will kick off with the next time the Scan function runs.

If your files are **`csv`** , tab delimited, or Fixed ASCII text, please read the instructions on how to work with a library Scan in Cross Data.

All instructions for setting up scan jobs can be found in the solutions section for support and downloaded as .pdfs.

LIBRARY SCAN for Processing files Outbound

Library Scan is designed to work with files in a library and either send a file to a remote server or process a file into your database.

You will need to create a script, a definition, a scan job, and a file action.

Some information you will need for the set up is:

What is the DNS or IP address of your remote server?

What is the User ID and passphrase for signing on?

Do you need to change directory when you sign on?

Do you have a library created to scan? (we recommend not using a production library but creating one specifically for the scan job)

Script

From the main menu take option **2** for configuration then option **3** for work with scripts.

Find the script called NT. I suggest copying this one then edit it as needed.

You will notice that it only 4 lines:

It signs on using your user ID and password (&uu &pp, and you'll see in the definition where this comes from).

Next it turns off the send passive command. The IBM i default (on) is to send the **PASV** subcommand. When **SENDPASV** is off, then the server does not send the **PASV** subcommand. Some FTP servers may not support the **PASV** subcommand.

Then we put the file (&ll &rr) and quit

Depending on what type of server you are connecting to and if you need to do a change directory on the server- or any other requirements they may have- there may be other commands you need to insert. If you have any questions about how to do that- please contact support (www.townsendsecurity.com, click on the button that says 'Support' then the link to "Start a Support Ticket").

Definition

Your next step is to create a definition that looks to your script for commands and passes it the remote information. Take option **2** from the Configuration menu, Work with Definitions. We suggest making a copy of NT and editing it with the remote information. We suggest naming it the same name as your script.

On the first screen replace the member name for the script file with the script you created in the previous step.

At the bottom of this same screen you'll see the options: "Local file" & "Remote file or ***FILE** "

make those options:

Local file . . . . .LOCAL

Remote file or *FILE* ……FILE

Press ***Enter***.

On the 2nd screen make the necessary changes for the remote server: User ID, password, & DNS or IP address.

Now press ***Enter*** all the way through for it to accept the changes.

Library scan

To set up a scan over this library, go to the configuration menu and take option **`4`** for Work with Scan Libraries. Press ***F6*** to add a new one. You must name it whatever the name is of the library you are scanning. Give it a description.

Press ***Enter*** all the way through and you will see the new scan added to the list.

File action:

A file action tells FTP Manager what to do with the files it finds in the library. Put a **`12`** by your library scan and then press ***F6*** to add a new file action. There are 3 ways to name a file action. You can name it with a specific file name (**`FILE_NAME`**) if you know that name and it is consistent every time. You can name it with a generic name (**`FILE*`**) if the first few characters of all the files are the same but the ending of the name is unique (such as a date/time stamp).

You can name the file action **`*ANY`** to process all files the same way.

You can use several file actions in the same Library Scan all doing different things to different files. Library Scan will find the first file and see if there is a specific file action for the first file. If YES, it will process that file as defined in the file action, if NO it will see if there is a generic file action for that file. If YES, it will process that file as defined in the file action, if NO it will see if there is a **`*ANY`** file action. If yes it will process this file using that file action, and if there is no **`*ANY`** it will not do anything to that file and go look at the next file.

It is for this reason that we suggest not using a production library, as the Scan job will look at every file in the library. We suggest creating a library just for scanning purposes.

Name your file action and give it a description.

Make Perform FTP session **`1`** for yes for sending files to a remote server. The FTP Session name is the name of definition you created in the previous step.

Press ***Enter*** and go to the next screen.

Configure your archive option of the file.

Press ***Enter*** all the way through for it to create this file action and you will see it added to the Library Scan.

If your subsystem is active this job will kick off with the next time the Scan function runs.

LIBRARY SCAN for processing encrypted files outbound

Library Scan is designed to work with files in a library and either send a file to a remote server or process a file into your database.

You will need to create a script, a definition, a scan job, and a file action.

Some information you will need for the set up is:

- What is the DNS or IP address of your remote server?
- What is the User ID and passphrase for signing on?
- Do you need to change directory when you sign on?
- Do you have a library created to scan? (we recommend not using a production library but creating one specifically for the scan job)
- Do you have the trading partner's public key?

Script

From the main menu take option `2` for Configuration then option `3` for Work with Scripts.

Find the script called **`ENCRYPT`** . I suggest copying this one then edit it as needed.

You will notice that it has 6 lines:

It signs on using your user ID and password (&uu &pp, and you'll see in the definition where this comes from).

Next it turns off the send passive command. The IBM i default (on) is to send the **`PASV`** subcommand. When **`SENDPASV`** is off, then the server does not send the **`PASV`** subcommand. Some FTP servers may not support the **`PASV`** subcommand.

The next line sends the file in binary mode (making an image of the file).

The script then states-"locsite namefmt 1" You must set the name format to '1' to work with all IBM i file systems.

Then we put the file (&ll &rr) and quit.

Depending on what type of server you are connecting to and/or if you need to change directory there- or any other requirements they may have- there may be other lines you need to insert. If you have any questions about how to do that- please contact support (www.townsendsecurity.com, click on the button that says 'Support' then the link to "Start a Support Ticket").

Definition

Your next step is to create a definition that looks to your script for commands and passes it the remote information. Take option `2` from the Configuration Menu to Work with Definitions. I suggest making a copy of **`ENCRYPT`** and editing it with the remote information. And I suggest naming it the same name as your script.

On the first screen replace the member name for the script file with the script you created in the previous step. And at the bottom of this same screen you'll see the options: "Local file" & "Remote file or *FILE" make those options:

Local file . . . . . .LOCAL

Remote file or *FILE* ……FILE

Press ***Enter***

On the 2nd screen make the necessary changes for the remote server: User ID, password, & DNS or IP address.

Press ***Enter*** to find the encryption page.

Make the Encryption option **`2`** for PGP.

In the middle of the screen you'll see "Encrypt user ID" this is where you put your trading partner's public key name.

Now press ***Enter*** all the way through for it to accept the changes.

Library scan

To set up a scan over this library, go to the configuration menu and take option **`4`** for Work with Scan Libraries. Press ***F6*** to add a new one. You must name it whatever the name is of the library you are scanning. Give it a description.

Press ***Enter*** all the way through and you will see the new scan added to the list.

File action

A file action tells FTP Manager what to do with the files it finds in the library. Put a **`12`** by your library scan and then press ***F6*** to add a new file action. There are 3 ways to name a file action. You can name it with a specific file name (**`FILE_NAME`**) if you know that name and it is consistent every time. You can name it with a generic name (**`FILE*`**) if the first few characters of all the files are the same but the ending of the name is unique (such as a date/time stamp). You can name the file action **`*ANY`** to process all files the same way.

You can use several file actions in the same Library Scan all doing different things to different files. Library Scan will find the first file and see if there is a specific file action for the first file. If YES, it will process that file as defined by that file action. If NO it will see if there is a generic file action for that file.

If YES, it will process that file as defined by that file action. If NO it will see if there is a **`*ANY`** file action. If yes it will process this file using that file action, and if there is no **`*ANY`** it will not do anything to that file and go look at the next file.

Name your file action and give it a description.

Make Perform FTP session **`1`** for yes for sending files to a remote Server.

And the FTP Session name is the name of definition you created in the previous step.

Press ***Enter*** and go to the next screen.

Configure your archive option for the file.

Press ***Enter*** all the way through for it to create this file action and you will see it added to the Library Scan.

If your subsystem is active this job will kick off with the next time the Scan function runs.

All instructions for setting up scan jobs can be found in the solutions section for support and downloaded as `.pdfs` .

APPENDIX D – Directory Scan

DIRECTORY SCAN for Processing Inbound files

Directory Scan is designed to work with files in your IFS directory and pass them into a library.

You will need to create a scan job, and a file action.

Some information you will need for the set up is:

- A directory created in the IFS directory to scan
- A library to place the file into

Directory scan

To set up a scan over this directory, go to the configuration menu and take option `5` for Work with Scan Directories. Press ***F6*** to add a new one. You can name it anything you like (unlike Library Scan in which you need to name it the name of the library you are scanning). Give it a description.

In the path option put in the entire path to the directory. You must use the forward slash as the first character, it is case sensitive, and do not include the name of the file. You can scan a directory within a directory within a directory.

Press ***Enter*** all the way through and you will see the new scan added to the list.

File action

A file action tells FTP Manager what to do with the files it finds in the directory. Put a `12` by the Directory Scan you've just created and press ***Enter***. Press ***F6*** to add a new file action. There are 3 ways to name a file action. You can name it with a specific file name (`FILE_NAME`) if you know that name and it is consistent every time.

You can name it with a generic name (`ABC*`) if the first few characters of all the files are the same but the ending of the name is unique (such as a date/time stamp).

You can name the file action `*ANY` to process all files the same way.

You can use several file actions in the same Directory Scan all doing different things to different files. Directory Scan will find the first file and see if there is a specific file action for the first file. If YES, it will process that file using the specific file action, if NO it will see if there is a generic file action for that file.

If YES, it will process that file using the generic file action, if NO it will see if there is a `*ANY` file action.

If YES it will process this file using the `*ANY` file action, and if there is no `*ANY` Directory Scan will not do anything to that file and go look at the next file.

It is for this reason we recommend not using the /tmp directory as lots of application use the /tmp directory to place files and Directory Scan will look at every file.

Name your file action and give it a description. Name the library into which you would like to place the file(s) and choose a naming option. Press ***F1*** to get a good description of how the naming options work.

Press ***Enter*** and go to the next screen.

Choose how you want to archive the file after it's processed and put in a path to the directory into which you want to archive the file (or choose 'delete').

Press ***Enter*** all the way through for it to create this file action and you will see it added to the Scan directory.

If your subsystem is active this job will kick off with the next time the Scan function runs.

You can scan IFS or QDLS directories for automatic processing. Be sure to specify adequate public authority on the directory so that FTP Manager can process the files.

Please note that the `/QDLS` file system has special requirements. Before you can access the `QDLS` file system a user must be added to the system distribution directory using the Work With Directory Entries (`WRKDIRE`) command. The `QDLS` file system also has limitations on the length of file names. It is recommended that you use the IFS file system rather than the `QDLS` file system.

If you will be using the directory scan function you should be familiar with authority management on IFS files. The IBM i treats these files differently than library files. For example, programs cannot adopt authority to process files in the IFS file system. Please see the section on IFS File Authority before implementing directory scan.

You can get more information on IFS authorities if you do a search under solutions for "IFS authority".

DIRECTORY SCAN for processing PGP encrypted files Inbound

Directory Scan is designed to work with files in your IFS directory and pass them into a library.

You will need to create a scan job and a file action.

Some information you will need for the set up is:

- A directory created in the IFS directory to scan
- A library to place the file into or the remote information to send the file to another IFS directory or off to a remote server.
- The name and passphrase of your PGP key.

Directory scan

To set up a scan over this directory, go to the configuration menu and take option `5` for Work with Scan Directories. Press ***F6*** to add a new one. You can name it anything you like (unlike Library Scan in which you need to name it the name of the library you are scanning). Give it a description.

In the path option put in the entire path to the directory. You must use the forward slash as the first character, it is case sensitive, and do not include the name of the file. You can scan a directory within a directory within a directory.

Press ***Enter*** and on the 2nd screen choose **`4`** for Decrypt PGP, put in your passphrase twice, then put in your key name in the Encrypt user ID field.

Press ***Enter*** and you will see the scan added to the list.

File action

A file action tells FTP Manager what to do with the files it finds in the directory. Put a **`12`** by your directory and then press ***F6*** to add a new file action. There are 3 ways to name a file action. You can name it with a specific file name if you know that name and it is consistent every time. You can name it with a generic name if the first few characters of all the files are the same but the ending of the name is unique (such as a date/time stamp). You can name the file action **`*ANY`** to process all files the same way.

You can use several file actions in the same Directory Scan all doing different things to different files. Directory Scan will find the first file and see if there is a specific file action for the first file. If YES, it will process that file as defined by that file action. If NO it will see if there is a generic file action for that file. If YES, it will process that file as defined by that file action. If NO it will see if there is a **`*ANY`** file action. If yes it will process this file using that file action, and if there is no **`*ANY`** it will not do anything to that file and go look at the next file.

Directory Scan will, however, attempt to decrypt all files it finds so you would want to set this directory up explicitly for decryption.

Name your file action and give it a description.

If you want to process the file into a library:

Name the library into which you would like to place the file(s) and choose a naming option. Press ***F1*** for more information on the naming formats available. Press ***Enter*** and go to the next screen.

Choose how you want to archive the file after it's processed and put in a path to the directory into which you want to archive the file (or choose **`Delete`**).

Press ***Enter*** all the way through for it to create this file action and you will see it added to the Scan directory.

If your subsystem is active this job will kick off with the next time the Scan function runs.

DIRECTORY SCAN for processing Outbound Files

Directory Scan is designed to work with files in your IFS directory and send a file to a remote server.

You will need to create a script, a definition, a scan job, and a file action.

Some information you will need for the set up is:

- What is the DNS or IP address of your remote server?
- What is the User ID and passphrase for signing on?

- Do you need to change directory when you sign on?
- A directory created in the IFS directory to scan

Script

From the main menu take option `2` for Configuration then option `3` for Work with Scripts.

Find the script called **`NTIFS`** . I suggest copying this one then edit it as needed. You will notice that it has only 6 lines:

It signs on using your user ID and password (&uu &pp, and you'll see in the definition where this comes from). Next it turns off the send passive command. The IBM i default (on) is to send the **`PASV`** subcommand. When **`SENDPASV`** is off, then the server does not send the **`PASV`** subcommand. Some FTP servers may not support the **`PASV`** subcommand.

The next line sends the file in binary mode (making an image of the file). The script then states-"Locsite Namefmt 1" You must set the name format to '1' to work with all IBM i file systems. Then we put the file (&ll &rr) and quit

Depending on what type of server you are connecting to and if you need to change directory there- or any other requirements they may have- there may be other commands you need to insert. If you have any questions about how to do that- please contact support (www.townsendsecurity.com, click on the button that says 'Support' then the link to "Start a Support Ticket").

Definition

Your next step is to create a definition that looks to your script for commands and passes it the remote information. Take option `2` from the Configuration menu to work with definitions. I suggest making a copy of NTIFS and editing it with the remote information. And I suggest naming it the same name as your script.

On the first screen, replace the member name for the script file with the script you created in the previous step. And at the bottom of this same screen you'll see the options: "Local file" & "Remote file or *FILE" make those options:

Local file . . . . . .LOCAL

Remote file or *FILE* ……FILE

On the 2nd screen make the necessary changes for the remote server: User ID, password, & DNS or IP address. Now press ***Enter*** all the way through for it to accept the changes.

Directory scan

To set up a scan over this directory, go to the configuration menu and take option `5` for Work with Scan Directories. Press ***F6*** to add a new one. You can name it anything you like (unlike Library Scan in which you need to name it the name of the library you are scanning). Give it a description.

In the path option put in the entire path to the directory. You must use the forward slash as the first character, it is case sensitive, and do not include the name of the file. You can scan a directory within a directory within a directory.

Press ***Enter*** all the way through and you will see the new scan added to the list.

File action

A file action tells FTP Manager what to do with the files it finds in the directory. Put a **12** by your directory and then press ***F6*** to add a new file action. There are 3 ways to name a file action. You can name it with a specific file name if you know that name and it is consistent every time. You can name it with a generic name if the first few characters of all the files are the same but the ending of the name is unique (such as a date/time stamp). You can name the file action ***ANY** to process all files the same way.

You can use several file actions in the same Directory Scan all doing different things to different files. Directory Scan will find the first file and see if there is a specific file action for the first file. If YES, it will process that file as defined by that file action. If NO it will see if there is a generic file action for that file.

If YES, it will process that file as defined by that file action. If NO it will see if there is a ***ANY** file action. If yes it will process this file using that file action, and if there is no ***ANY** it will not do anything to that file and go look at the next file.

Press ***Enter*** and go to the next screen.

Choose how you want to archive the file after it's processed and put in a path to the directory into which you want to archive the file (or choose 'delete').

The Perform FTP Session option means "send the file to a remote server" so you want that set to **1** for Yes and the Session name is the name of the definition you created in the previous step.

Press ***Enter*** all the way through for it to create this file action and you will see it added to the Scan directory.

If your subsystem is active this job will kick off with the next time the Scan function runs.

You can scan IFS or QDLS directories for automatic processing. It is recommended that you not use the **/tmp** directory as some applications place temporary files in this directory. Be sure to specify adequate public authority on the directory so that FTP Manager can process the files.

Please note that the **/QDLS** file system has special requirements. Before you can access the **QDLS** file system a user must be added to the system distribution directory using the Work With Directory Entries (**WRKDIRE**) command. The **QDLS** file system also has limitations on the length of file names. It is recommended that you use the IFS file system rather than the **QDLS** file system. If you will be using the directory scan function you should be familiar with authority management on IFS files. The IBM i treats these files differently than library files. For example, programs cannot adopt authority to process files in the IFS file system. Please see the section on IFS File Authority before implementing directory scan. You can get more information on IFS authorities if you do a search under solutions for "IFS authority".

DIRECTORY SCAN for processing PGP encrypted files outbound

Directory Scan is designed to work with files in your IFS directory, encrypt them and send them to a remote server.

You will need to create a script, a definition, a scan job, and a file action.

Some information you will need for the set up is:

- What is the DNS or IP address of your remote server?
- What is the User ID and passphrase for signing on?
- Do you need to change directory when you sign on?
- What is the path of the directory created in the IFS directory to scan?
- Have you exchanged public keys with your trading partner?

Script

From the main menu take option `2` for configuration then option `3` for work with scripts.

Find the script called `ENCRYPTIFS` . I suggest copying this one then edit it as needed.

You will notice that it only 6 lines:

It signs on using your user ID and password (&uu &pp, and you'll see in the definition where this comes from).

Next it turns off the send passive command. The IBM i default (on) is to send the `PASV` subcommand. When `SENDPASV` is off, then the server does not send the `PASV` subcommand. Some FTP servers may not support the PASV subcommand.

The next line sends the file in binary mode (making an image of the file)

The script then states-"Namefmt 1" You must set the name format to '1' to work with all IBM i file systems.

Then we put the file (&ll &rr)

and quit

Depending on what type of server you are connecting to and if you need to change directory there- or any other requirements they may have- there may be other lines you need to insert. If you have any questions about how to do that- please contact support (www.townsendsecurity.com, click on the button that says 'Support' then the link to "Start a Support Ticket").

Definition

You next step is to create a definition that looks to your script for commands and passes it the remote information. Take option `2` from the Configuration menu, Work with Definitions. I suggest making a copy of `ENCRYPTIFS` and editing it with the remote information. And I suggest naming it the same name as your script.

On the first screen replace the member name for the script file with the script you created in the previous step. And at the bottom of this same screen you'll see the options:

"Local file" & "Remote file or *FILE "

make those options:

Local file . . . . . .LOCAL

Remote file or *FILE* ……FILE

Press ***Enter***

On the 2nd screen make the necessary changes for the remote server: User ID, password,

& DNS or IP address. Press ***Enter*** and go to the screen with “Encryption option” as the first field. Choose `2` for PGP encryption and in the middle of the screen you’ll see the field “Encrypt user ID.” This is where you enter your trading partner’s public key name. Now press enter all the way through for it to accept the changes.

Directory scan

To set up a scan over this directory, go to the configuration menu and take option `5` for Work with Scan Directories. Press ***F6*** to add a new one. You can name it anything you like (unlike Library Scan in which you need to name it the name of the library you are scanning). Give it a description.

In the path option put in the entire path to the directory. You must use the forward slash as the first character, it is case sensitive, and do not include the name of the file. You can scan a directory within a directory within a directory.

Press ***Enter*** all the way through and you will see the new scan added to the list.

File action

A file action tells FTP Manager what to do with the files it finds in the directory. Put a `12` by your directory, press ***Enter*** and then press ***F6*** to add a new file action. There are 3 ways to name a file action. You can name it with a specific file name if you know that name and it is consistent every time. You can name it with a generic name if the first few characters of all the files are the same but the ending of the name is unique (such as a date/time stamp). You can name the file action `*ANY` to process all files the same way.

You can use several file actions in the same Directory Scan all doing different things to different files. Directory Scan will find the first file and see if there is a specific file action for the first file. If YES, it will process that file accordingly, if NO it will see if there is a generic file action for that file. If YES, it will process that file accordingly, if NO it will see if there is a `*ANY` file action. If yes it will process this file using that file action, and if there is no `*ANY` it will not do anything to that file and go look at the next file.

Name your file action and give it a description.

Press ***Enter*** and go to the next screen.

Choose how you want to archive the file after it’s processed and put in a path to the directory where you want to archive it (or choose ‘delete’).

The Perform FTP Session means “send the file to a remote server” so you want that set to `1` for Yes and the Session name is the name of the definition you created in the previous step.

Press ***Enter*** all the way through for it to create this file action and you will see it added to the Scan directory.

If your subsystem is active this job will kick off with the next time the Scan function runs.

You can scan **IFS** or **QDLS** directories for automatic processing.

It is recommended that you not use the **/tmp** directory as some applications place temporary files in this directory.

Be sure to specify adequate public authority on the directory so that FTP Manager can process the files.

Please note that the **/QDLS** file system has special requirements. Before you can access the **QDLS** file system a user must be added to the system distribution directory using the Work With Directory Entries (**WRKDIRE**) command. The **QDLS** file system also has limitations on the length of file names. It is recommended that you use the **IFS** file system rather than the **QDLS** file system.

If you will be using the directory scan function you should be familiar with authority management on **IFS** files. The IBM i treats these files differently than library files. For example, programs cannot adopt authority to process files in the **IFS** file system. Please see the section on **IFS** File Authority before implementing directory scan.

You can get more information on **IFS** authorities if you do a search under solutions for “IFS authority”.

APPENDIX E – Auto-Scan

AUTO SCAN for Retrieving Files from a Remote Server

Auto Scan is designed to go to a remote server and bring files back to the IBM i. It can bring files back to a library or to the IFS directory.

The first thing Auto Scan does is sign on to the remote server and make a directory listing (**DIR**) of the files to retrieve. You can specify all files, new, new & changed, or specific files by name or generic name.

Then Auto Scan opens another session and passes the names of the files to another script and retrieves the files (**GET**).

Some information you will need for the set up is:

- What type of server you are going to?
- What is the User ID and passphrase for signing on?
- Do you need to change directory when you sign on?
- Are all the files here specifically for you or do you share this ‘mailbox’ ?
- Do you want to get all files, new files, new and changed files, or specific files? If you are to get specific files what are the names or naming convention of those files?
- Are you supposed to delete the files after you retrieve them?
- Do you want to bring them back to a library or a directory?

If you don’t know the type of server you are going to or how it is set up- this part may take a couple of tries before we get it right. You can either ask them what type server it is or you can do a manual FTP session from the 400 and when you sign on you should get a message back

with the server type. (NT, UNIX Type: L8, etc).

Depending on the server and your trading partner requirements you may have to include additional commands.

From the main menu take option **2** for configuration then option **10** for directory definitions. You can see there are several samples to choose from. Find the one that closest suits your needs and note what the positions are for the file name start and length.

Press ***F3*** to exit out of there and now do an ***F6*** to add. We suggest, when creating any scan jobs to name them the same names so if you have to troubleshoot you can track these easier. Enter the file name start position and length. You may have to change these later, but for now this will get you started.

Press ***Enter*** to add the directory definition.

You will be creating 2 scripts and 2 definitions. One script/definition pair will make the directory listing (**DIR**). The other script/definition pair will retrieve the files (**GET**).

Dir script

From the main menu take option **2** for configuration then option **3** for work with scripts.

The first step script will make a directory listing. Find the script called **AS400DIR** . We suggest copying this one then edit it as needed. And I suggest naming it with some sort of reference to it being the **DIR** script- such as **BofADIR** .

You will notice that it only has 3 lines. It signs on using your user ID and password (And you'll see in the definition where this comes from), it makes a directory listing, it quits.

Getting new or new and changed is specified within the auto scan configuration. If you need specific files you can specify that here. For example, If you want to get only the file called **FILE_NAME.txt** , then you directory listing would be:

```
dir FILE_NAME.txt
```

This will list the specific file and pass just that name to your get script.

If you want to get files with the same naming convention, but they are unique (such as **FILE_0001.txt** , **FILE_0002.txt** , etc then you directory listing would be:

```
dir FILE*
```

This will list all files with that name format and pass them all to your get script.

Dir definition

Next, create a definition that looks to your script for commands and passes it the remote information. Take option **2** from the Configuration menu, Work with Definitions. I suggest making a copy of **AS400DIR** and editing it with the remote information. And I suggest naming it the same name as your **DIR** script.

On the first screen replace the member name for the script file with the script you created in the previous step.

Press ***Enter***.

On the 2nd screen make the necessary changes for the remote server: User ID, password, & DNS or IP address.

Now press ***Enter*** all the way through for it to accept the changes.

Get script

From the main menu take option `2` for configuration then option `3` for work with scripts.

Next you will be creating a script that takes the file names from the `DIR` session and signs back onto the server to retrieve the files. Find the script called `AS400GET`. We suggest copying this one then edit it as needed. And I suggest name it with some sort of reference to it being the `GET` script- such as `BofAGET`.

You will notice that it has 5 lines.

It signs on using your user ID and password (And again, this comes from the definition you will create next), it shows how to change your local directory to the library you want to pull the files into, it shows how to change the directory on the remote (and if that step is unnecessary, you may delete it, it gets the files the it quits. Getting new or new and changed is specified within the auto scan configuration.

Get definition

Next, create a definition that looks to your script for commands and passes it the remote information for doing the `GET`. Take option `2` from the Configuration menu to Work with Definitions. I suggest making a copy of `AS400GET` and editing it with the remote information. And I suggest naming it the same name as your `GET` script.

On the first screen replace the member name for the script file with the script you created in the previous step.

Press ***Enter***.

On the 2nd screen make the necessary changes for the remote server: User ID, password, & DNS or IP address.

Now press ***Enter*** all the way through for it to accept the changes.

Autoscan

From the main menu take option `2` for configuration then option `14` for Autoscan. Press ***F6*** to create a new AutoScan. You can name it whatever you like (keep it consistent with the scripts & definitions you just created) and give it a description.

Determine your scan type (It is typically option `1` for an FTP transfer. If you need to do SSL, please call Support to schedule a session to set up DCM).

Your status should be `1` for active.

Determine your retrieval type. Even if you choose new or new and changed, the first time AutoScan runs it retrieves all files because they are all new to it’s listing.

Scan interval is how often we go scan the remote for files to retrieve and that interval is in seconds.

Press ***Enter***

On this next screen- the Directory FTP definition is the definition you created to go make a directory listing (**`AS400DIR`** in the example).

The Directory Profile is the type of box you are going to- which we determined in step 1.

Put the name of that here.

The member of the script file is the script you are using to make the directory listing (**`AS400DIR`**)

The Retrieval FTP definition is your get definition (**`AS400GET`**).

You can press ***Enter*** all the way through for it to accept the changes.

When using Auto Scan you must end and restart the subsystem.

After this is set up and runs the first time, if it does not get files, please start a ticket with support and provide screen shots of your set up & an application log.

APPENDIX F – PGP Certification

PGP Recertification and Set up with a new Trading Partner

Step 1 - adding your trading partner's public key to your keyring

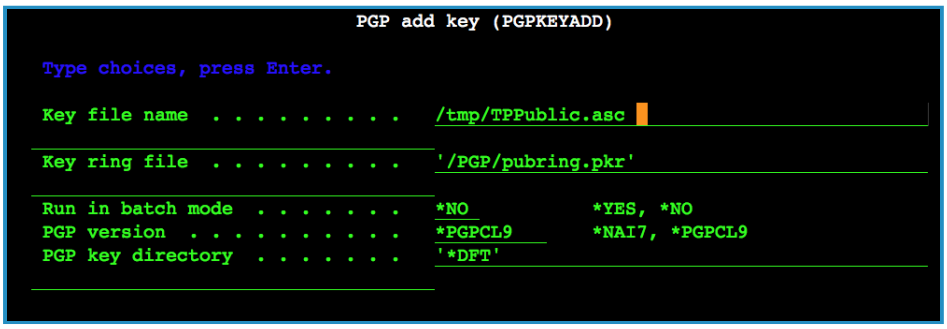
Before you can encrypt a file you must add the recipient's public key to your key ring. You probably received the key in an email as a **`asc`** file. Copy this key to your c:\ drive.

Use FTP from your PC to copy the key in binary mode to your IBM i's IFS directory. It is important that the key is copied in binary mode, as Alliance will expect to find the key in ASCII format. Here is an example of using FTP to transfer the key:

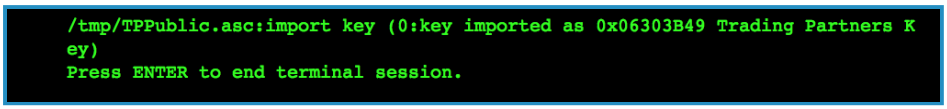
```
Start, Run, FTP
open 1.1.1.1              // use your IBM i IP address
username                          // use your IBM i profile
password                          // use your IBM i password
binary
cd /tmp
lcd c:\
put File.asc
quit
```

You now have the encryption key on your IBM i's IFS file system in the directory **`/tmp`** .

To add the key to your IBM i key ring use the **`PGPKEYADD`** command, or use the "Add key to key ring" option on the Alliance PGP menu. The menu option prompts you with the **`PGPKEYADD`** parameters. Enter the values below:



If the key is successfully added you will see the message:



If you do not see this message, or if you receive any errors when you try to add the key, please stop and contact software support.

> ***NOTE:*** Please note also, that in this example, the name of the file and the name of the key are different.

It is important to pay attention to the key name for your next steps.

You have now added your trading partner's public (encryption) key to your key ring. In the next step you will view the keys on your key ring to verify that it has been added correctly.

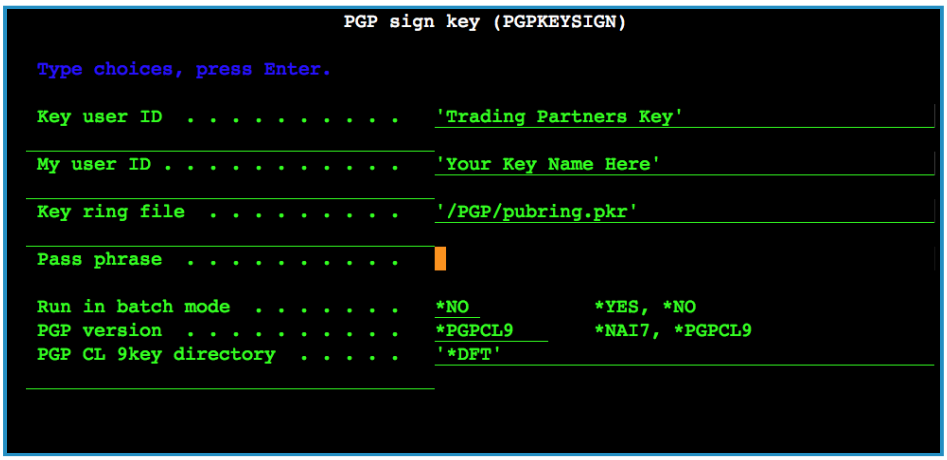
Step 2 - Viewing the public key

After you have added the public key to your key ring, use the **`PGPKEYVIEW`** command to view the keys. You can also use the option "View PGP keys" on the Alliance PGP menu. If you change the option 'PGP version' to **`*NAI7`** you will see a listing of the keys. Keeping the version at **`*PGPCL9`** will give you key details. Verify that the key you imported is added.

If you do not see the key on your key ring please contact technical support before proceeding.

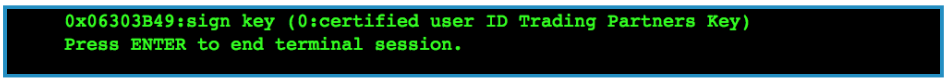
Step 3 - Signing the Trading partner's key

After you add a key to your key ring you should sign the key. The key is considered "untrusted" until you have signed it. When you sign a key PGP will ask you to confirm that you want to sign the key, and you must provide your private pass phrase. You can use the **`PGPKEYSIGN`** command or the option on the Alliance menu to "Sign key". When you take the menu option or prompt the command you will see the following prompts:



Because commands may convert input to upper case values, be sure to enclose the values in single quotes. PGP user IDs, key names and pass phrases are case-sensitive. The pass phrase will not display while you type. The Key user ID and My user ID refer to the actual name of the keys.

When you sign the key you will see something like the following:

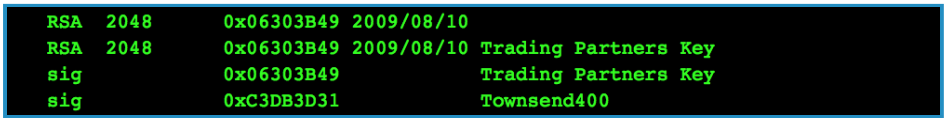


If you receive any errors signing the key be sure you are using the correct pass phrase for PGP. Contact technical support if you have any other errors.

Step 4 – Verifying the key signature

After you have signed the new key you can view the key to see the signatures. You can use the **`PGPKEYVIEW`** command or the Alliance PGP menu option to “View PGP keys”. On the 2nd screen change the signature option to *yes and to see a list of keys you can change the PGP version to **`*NAI7`** .

Press ***Enter*** to start the session. You should see something similar to the following:



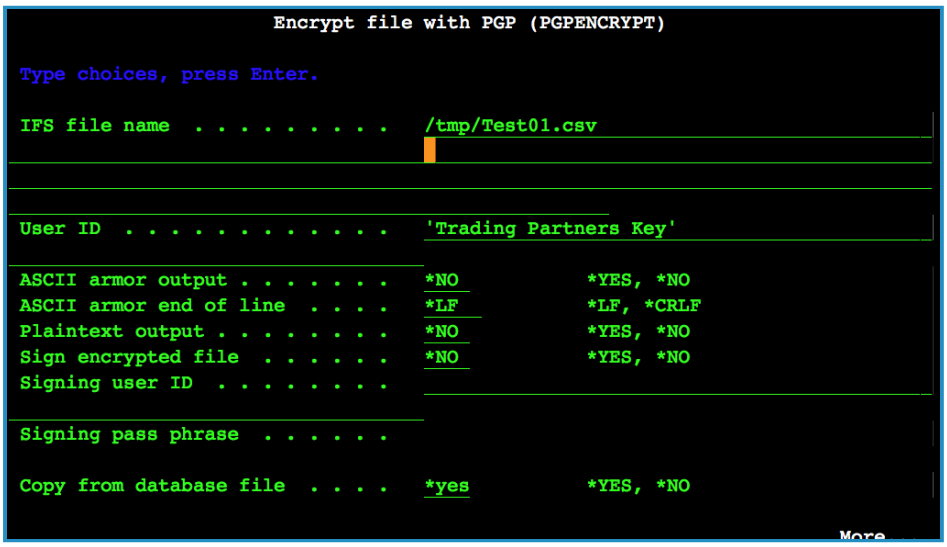
Note that you should observe your PGP user ID (key name) as a signature on the new key.

For every key you sign you should observe a “sig” line with your key name.

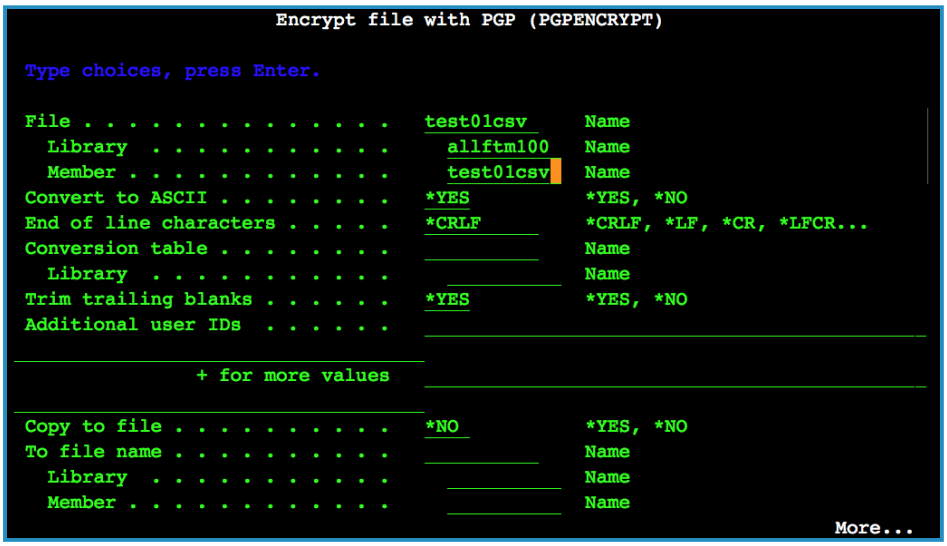
Step 5 - Encrypting the file

After you have verified that the Townsend key has been signed you can encrypt a sample file to send to Townsend. Use the **`PGPENCRYPT`** command or the Alliance menu option to “Encrypt file”. This command supports encrypting both IFS files and IBM i database files. You must supply an IFS file name in both cases. If you are encrypting an IBM i database file it will be automatically copied to the IFS file name you specify. If the file already exists in an IFS directory- you specify the file name and PGP will append the **`.pgp`** extension.

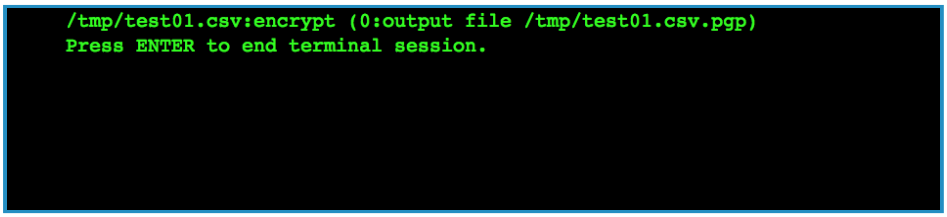
There is a sample file you can use in the product library. Use the following values on the command (you will have to complete the first screen then page down to complete the 2nd screen):



Page down:



When you encrypt the file you should see a session similar to the following:



If the encryption is successful you will see the message `0:output file /tmp/Test01.csv.pgp` . If you do not see this message, contact technical support.

Step 6 – Sending the file to your Trading Partner

You should now have one encrypted file (`test01.csv.pgp`) in the /tmp directory of your IFS system. You can use the `WRKLNK` command to browse the /tmp directory and find the file.

Put this file on your PC in binary mode and send it to Townsend using email. You can use FTP from your PC to transfer the file:

```
Start, Run, FTP
open 1.1.1.1                        // use your IBM i IP address
user username                       // use your IBM i profile
pass password                       // use your IBM i password
binary
cd /tmp
get Test01.csv.pgp
quit
```

Please send an email to your trading partner with this file attached.

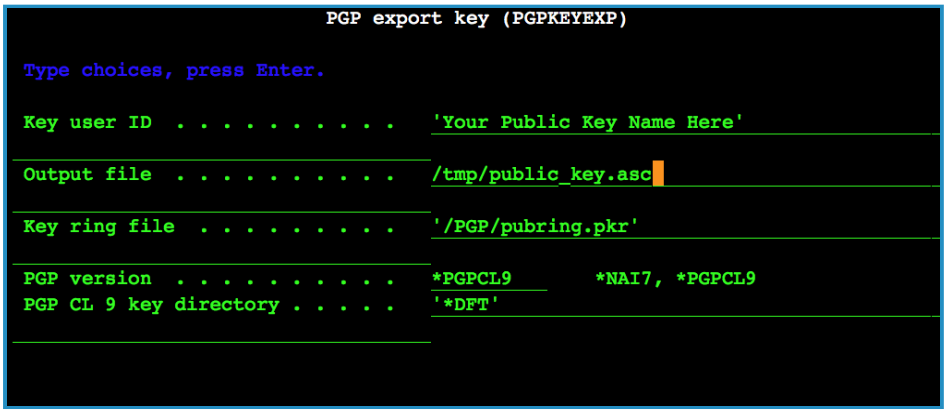
Decrypting a File from your trading partner

This second certification phase involves exporting your public key, sending the key to your trading partner, and then decrypting a test file that you receive from them.

Please use the following steps to create the certification file.

Step 1 – Exporting your public key

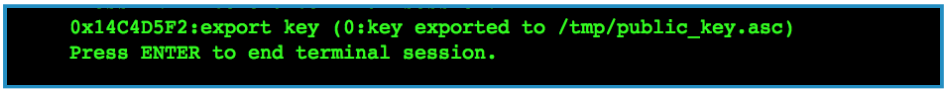
Before anyone can send you an encrypted file you must first provide them your public key. The recipient of your public key will use the key to encrypt a file and will send it to you. Only you will be able to decrypt the file as only you have the matching private key and passphrase. In this step you will export your public key to a file. Use the **`PGPKEYEXP`** command or the Alliance menu option to “Export key”. Use the following values to export your key:



In the Key user ID field use your own public key name.

In the Output file field place of the path and file name as you want it to be to send to your trading partner, for example **`public_key.asc`** .

When you export the key you will see a PGP session similar to the following:



Be sure to observe the completion message
`0:key exported to file '/tmp/public_key.asc` . Your file name may be different. Contact technical support if you observed any errors in the PGP session.

Step 2 – Sending your public key to your trading partner

After you have extracted your public key you must send it to your trading partner via email. FTP the file to your PC in binary mode. Here is an example of how to use FTP to transfer the exported key to your PC, from your PC:

```
Start, Run, FTP
open 1.1.1.1              // use your IBM i IP address
username                          // use your IBM i profile
password                          // use your IBM i password
        binary
        cd /tmp
        lcd c:\
        get public_key.asc
        quit
```

After you transfer the key to your PC send the file to your trading partner. Be sure to attach the exported key file to the message.

Step 3 – Receiving an encrypted file from your trading partner

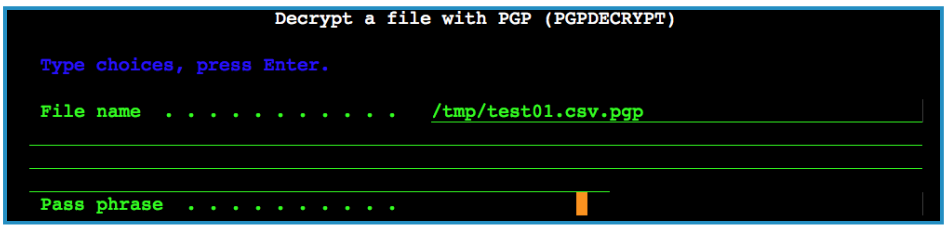
You may receive an encrypted file from trading partner. The file should have been encrypted with your public key. Put the file on your c:\ drive. You should transfer the file to the IBM i using FTP in binary mode. Here is an example of an FTP session to transfer the file:

```
Start, Run, FTP
open 1.1.1.1              // use your IBM i IP address
username                          // use your IBM i profile
password                          // use your IBM i password
binary
cd /tmp
lcd c:\
put test01.csv.pgp
quit
```

On the IBM i you can use the Work With Object Links (**`WRKLNK`**) command to browse the /tmp directory. The file should be in this directory.

Step 4 – Decrypting the file

In this step you will decrypt the file you received from your trading partner. Use the **`PGPDECRYPT`** command or the “Decrypt file” option on the Alliance PGP menu. Enter the following values for the command:



For the pass phrase enter the pass phrase that you used when you created your keys. You must include the pass phrase in single quotes, for example ‘To be or not to be’. You will not see the pass phrase as you type. Press ***Enter*** to start decryption.

You should see a PGP session that looks similar to the following:

```
/tmp/test01.csv.pgp:decrypt (0:output file /tmp/test01.csv)
Press ENTER to end terminal session.
```

You should see the message `0:output file: /tmp/test01.csv` . If you do not see this message, or if you receive an error message, contact technical support.

APPENDIX G – FTP Substitution Values

When using substitution values in a script. You must leave enough room in your script for the substitution value, which in some cases may include the entire path through the file name. The substituted value begins with the '&' symbol in the script.

The following substitution values are allowed in FTP scripts:

User ID - &uu

This will substitute the user ID in the FTP definition for the session. For example, if you have the user ID as `TOWNSEND` in the FTP definition, you could specify substitution like this:

```
&uu              jibsail
```

Password - &pp

This will substitute the password in the FTP definition for the session. For example, if you have the password as `JIBSAIL` in the FTP definition, you could specify substitution like this:

```
townsend &pp
```

To substitute both the user ID and password you would use a line like this:

```
&uu      &pp
```

Local file name - &ll

This will substitute the name of the IBM i file (local file) in your script. The file name must be in standard FTP format (`library/file.member`). Normally you will do file name substitution on an FTP put or get command. If you are sending a file to a remote system the script line might look like this (remember to leave enough room for substitution of the IBM i file name):

```
Put &ll                  /ect/orders/database/ord.dat
```

Remote file name - &rr

This will substitute the name of the remote system file name in your script. The file name must be in standard FTP format and has to be avalid name on the remote system. Normally you will do file name substitution on an FTP put or get command. If you are sending a file to a remote system the script line might look like this (remember to leave enough room for substitution of the remote file name):

```
Put ordprd/orders &rr
```

IP Address - &ip

This will substitute the IP address defined in the FTP definition into your script. The FTP definition can contain either a remote system name or an IP address. If the IP address is blank in the FTP definition the substitution will not take place. Note that the API program `FTMIPADDR` can be used to dynamically change the IP address in an FTP definition prior to use the definition for an FTP session.

```
Open &ip
```

Remote system name - &rs

This will substitute the remote system name defined in the FTP definition into your script. The FTP definition can contain either a remote system name or an IP address. If the remote system name is blank in the FTP definition the substitution will not take place. Note that the API program `FTMIPADDR` can be used to dynamically change the remote system name in an FTP definition prior to use the definition for an FTP session.

```
Open &rs
```

User data 1 - &1

This will substitute any user data you define in the FTP definition into the script file. This can be useful if you are using one script with multiple FTP definitions. An example would be substituting a store number in a file name. For example, if you specify:

```
put mylib/order&1       orders.dat
```

and you have the value `018` in the User data 1 field in the FTP definition, the statement will be modified like this:

```
put mylib/order018   orders.dat
```

You must remember to leave enough room for the entire field to be substituted. There may be blanks in the substitution data. The entire substitution data will replace the line starting with &1.

User data 2 - &2

This will substitute any user data you define in the FTP definition into the script file using the information in the User Data 2 field. This substitution works exactly like &1 above.

File name - &fl

This will substitute the file name into the script file. If the file name is less than 10 characters the trailing spaces will be changed to underline characters. File names are always substituted with a length of 10 characters. For example, if you specify:

```
put mylib/order        orders.dat.&fl
```

and you have a file name of `F020131` the actual PUT statement will be:

```
put mylib/order   orders.dat.F020131___
```

You must remember to leave enough room for the entire field to be substituted.

Library name - &lb

This will substitute the library name of the file into the script file. If the library name is less than 10 characters the trailing spaces will be changed to underline characters. Library names are always substituted with a length of 10 characters. For example, if you specify:

```
put mylib/order orders.dat.&lb
```

and you have a library name of `PRODLIB` the actual PUT statement will be:

```
put mylib/order   orders.dat.prodlib___
```

You must remember to leave enough room for the entire field to be substituted.

Member name - &mb

This will substitute the member name of the file into the script file. If the member name is less than 10 characters the trailing spaces will be changed to underline characters. Member names are always substituted with a length of 10 characters. For example, if you specify:

```
put mylib/order orders.dat.&mb
```

and you have a file with a member name of `M020131` the actual PUT statement will be:

```
put mylib/order   orders.dat.M020131___
```

Date and Date/Time Substitution:

You must leave enough room in your script for the date substitution. An example of date substitution would be to append the date to a file name for a date/time stamp. If you are getting a file and placing it into a library with only the date as the name, you must prefix the name with an alpha character as IBM i names in the **QSYS** library system cannot start with a number. Here is one possible way of specifying a name on the IBM i:

```
Get /etc/inventory/inv.dat  D&d6
```

This will create a file on the IBM i with a date embedded in the name. For example “D081128”.

You can use the date in a file name you are placing on a remote server. Here is one possible way of specifying a name in the script:

```
Put ORDER&d6
```

This will create a file on the target system with a date embedded in the name. For example “ORDER050115”.

Another is one possible way of specifying a name on the IBM i:

```
put mylib/order&md      orders.dat
```

This will send a file from the IBM i with the month and day in the file name. For example, if the date is April 24, 2008 the substitution will create the following statement:

```
 put mylib/order0424  orders.dat
```

You can append a file extension after any substitution value if you keep in mind that you must leave enough spaces to hold the entire substitution value (with the ‘&’ as the first character of that value). Here is one example of using this format in an FTP script:

```
put mylib/orders        orders.&d6    .dat
```

This will send a file from the IBM i to a remote file with the date embedded in the name. For example, if the date is September 11, 2008 the substitution will create the following statement:

```
put mylib/orders  orders.080911.dat
```

Keep in mind If you need a 4 digit year it will be represented as CCYY (century, year)

Date (2 digit DD) - &d2

This will substitute the day of the system date.

Time (4 digit MMSS) - &ms

This will substitute the minutes and seconds of the system date.

Date (4 digit MMDD) - &md

This will substitute the month and day of the system date. It does NOT specify a year.

Date (4 digit MMYY) - &my

This will substitute the month and year of the system date.

Date (5 digit YYDDD) - &jd

This will substitute the year and 3 digit day of the system date. This substitution is provided for users who must have the Julian Date capability.

Date (6 digit CCYYMM) - &de

This will substitute the year and month of the system date. This is the same as &c1.

Date (6 digit CCYYMM) - &c1

This will substitute the century, year and month of the system date. It does not specify the day of the month. This is the same as &de.

Date with previous month (6 digit CCYYMM) - &c2

This will substitute the century, year and month of the system date after subtracting one from the month.

Date (6 digit MMDDYY) - &c3

This will substitute the month, day, and year of the system date.

Date with previous day (6 digit MMDDYY) - &dm

This will substitute the month, day, and year of the system date after subtracting one from the day.

Date (6 digit YYMMDD) - &d6

This will substitute the year, month, and day of the system date.

Time (6 digit HHMMSS) - &tt

This will substitute the hour, minutes, and seconds of the system time.

Date (8 digit CCYYMMDD) - &d8

This will substitute the century, year, month, and day of the system date.

Date and Time (8 digit DDHHMMSS) - &dh

This will substitute the day, hour, minutes, and seconds of the system date.

Date with Underscore (8 digit date with underscore MM_DD_YY) - &ud

This will substitute the month, day, and year of the system date with an underscore between the values

Date and Time (10 digit MMDDYYHHMM) - &mh

This will substitute the month, day, year, hour, and minutes of the system date.

Date with dashes (10 digit CCYY-MM-DD) - &da

This will substitute the year, month, and day of the system date with dashes between the values.

Date and Time (12 digit MMDDCCYYHHMM) - &ch

This will substitute the month, day, century, year, hour, and minutes, of the system date.

Date and Time (12 digit CCYYMMDDHHMM) - &dd

This will substitute the century, year, month, day, hour, & minutes of the system date.

Date and Time (12 digit YYMMDDHHMMSS) - &dc

This will substitute the year, month, day, hour, minutes, and seconds of the system date.

Date and Time (14 digit CCYYMMDDHHMMSS) - &df

This will substitute the century, year, month, day, hour, minutes, and seconds of the system date.

ISO Date (26 character IS date) - &is

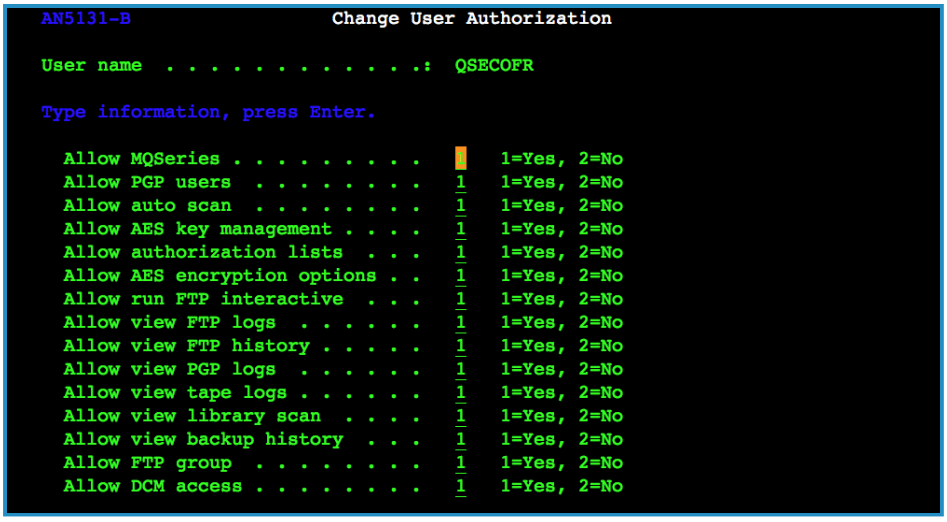
This will substitute the system date in ISO timestamp format. This substitution is provided for users who must have a 26 digit ISO date capability. Here is one example of specifying an ISO date:

```
Put &ll          myfile&is
```

This will create a file on the target system with a name like this (the correct date will be inserted):

```
Myfile2008-04-22-13.42.51.653000
```

Press ***Enter*** to go to the next screen.



Allow MQSeries

Enter `1` for Yes to allow this user to add or change an MQSeries definition. Enter `2` for No to not enable this function for this user.

Allow PGP users

Enter `1` for Yes to allow this user to add or change PGP user definitions. Enter `2` for No to not enable this function for this user.

Allow auto scan

Enter `1` for Yes to allow this user to add or change FTP auto scan definitions. Enter `2` for No to not enable this function for this user.

Allow AES key management

Enter `1` for Yes to allow this user to add or change AES encryption key definitions. Enter `2` for No to not enable this function for this user.

Allow authorization lists

Enter `1` for Yes to allow this user to add or change authorization list definitions. Enter `2` for No to not enable this function for this user.

Allow AES encryption options

Enter `1` for Yes to allow this user to change AES encryption configuration options. Enter `2` for No to not enable this function for this user.

Allow run FTP interactive